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Princeton Plasma Physics Laboratory – RF Roof Replacement

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SECTION 010010 – GENERAL REQUIREMENTS

PART 1 GENERAL

1.1 PURPOSE

A. The following specification provides the project-specific requirements that are supplementary to the requirements of the PPPL General Provisions for Non-Commercial Subcontracts, Part G - Provisions & Conditions for Fixed Price Construction Subcontracts

B. The Subcontractor is required to plan, control, and deliver the final project according to the project subcontract documents. The project must possess complete functional systems and be assembled safely and in quality fashion following industry standards, regulatory requirements, and best practices. The Subcontractor is required to complete the project within the specified time.

1.2 SUMMARY

A. The Subcontractor shall provide in the performance of the work under this subcontract all labor, materials, equipment, transportation, services, occupational exposure monitoring and supervision required to maintain work sites that meet the ES&H requirements found in Section 013100.

B. All major elements of the work are believed to be adequately described herein. The Subcontractor is expected, as a requirement of the Subcontractor's qualifications, to utilize his or her knowledge and experience to anticipate, and include in the cost of the work any incidentals of work which may be required, but are not specifically expressed herein, in order to provide a complete and fully functional facility and/or system.

C. Definitions and roles and responsibilities can be found in Section 013100.

1.3 REFERENCES

A. The publications listed below form a part of this subcontract to the extent referenced.
1. Subcontract Agreement
2. PPPL General Provisions for Non-Commercial Subcontracts Part G - Provisions & Conditions for Fixed Price Construction Subcontracts

B. The publications listed below form a part of this specification to the extent referenced.
1. 013100 Environment, Safety, and Health Requirements
2. 013119 Meetings
3. 013200 Schedule and Progress Documentation
4. 013300 Submittals
5. 013514 Building Information Modeling (BIM) Requirements
6. 014100 Quality Requirements
7. 015000 Temporary Facilities and Controls
8. 017419 Construction Waste Management and Control

9.	017700	Closeout Procedures
10.	017823	Operation and Maintenance Data
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13.	018113	Sustainable Design Requirements
14.	018119	Indoor Air Quality During Construction
15.	019113	General Commissioning Requirements

C. Drawings:
   1. Please reference 3.0 Drawings section of the Statement of Work.

1.4 SITE LOCATION

A. The Subcontractor shall confine all operations (including storage of materials) on PPPL premises to areas authorized or approved by PPPL. The Subcontractor shall hold and save PPPL and the Government, its officers and agents, free and harmless from liability of any nature occasioned by the Subcontractor's performance.

B. The project is located on the Princeton Plasma Physics Laboratory (PPPL) Site in Princeton, NJ. PPPL is located on Princeton University's Forrestal Campus in Plainsboro Township, Middlesex County, New Jersey.

1.5 DEFINITION OF ROLES

A. Design Coordinator: An PPPL person who is assigned to the project and works with the Construction (Project) Manager and Construction Coordinator to assist in the technical coordination of the project.

B. Procurement Specialist: An PPPL person responsible for and specifically assigned to the project, who is responsible for negotiating and administering the subcontract terms and conditions. All modifications to the subcontract shall come from the Procurement Specialist or designee, in writing. The Procurement Specialist or designee is the sole entity that can modify the subcontract or initiate change orders.

C. Princeton Technical Representative (PTR): A PPPL person responsible for managing, phasing, and ensuring the project is meeting key milestones. Also referred to as the Project Manager.

1.6 DESCRIPTION OF WORK

A. The Subcontractor shall furnish all required supervision, labor, materials, tools, plant and appurtenances necessary to perform diligently and fully all work as described in the drawings and technical specifications. All work includes transportation, overhead, bonding costs, safety oversight, quality control oversight, and supervision as required to construct the project described in the drawings, technical specification and the detailed description contained herein.

B. The Subcontractor's scope of work is outlined in the Statement of Work document. The Subcontractor shall perform all construction work in accordance with the drawings and/or specifications.

1.7 MATERIALS & EQUIPMENT FURNISHED BY PPPL

A. No materials or equipment is anticipated to be furnished for this project.

1.8 BUY AMERICAN ACT

A. PPPL maintains a preference for domestic construction material. Materials shall be purchased in accordance with PPPL General Provisions for Non-Commercial Subcontracts Part G - Provisions & Conditions for Fixed Price Construction Subcontracts.

B. There are no exemptions for this project.

1.9 SERVICES FURNISHED BY PPPL

A. PPPL existing utility and building services shall be coordinated with the Princeton Technical Representative (PTR):
   1. PPPL will de-energize and re-energized existing utilities as required to support the project. The subcontractor shall provide as much notice as possible for utility shut-downs and tie-ins. Refer to section 015000 for additional information regarding Temporary Facilities & Controls.

B. PPPL furnished services shall be coordinated with the PTR.
   1. The PTR coordinates issuance of permits, reference Section 013100 for additional information;
   2. The PTR or designee requests for and provides disablement of existing systems; Availability and use of existing PPPL utility services will comply with Section 013100 and 015000

1.10 ITEMS AFFECTING WORK PLANNING

A. Safety Officer/Representative: Subcontractor Safety Representative is required on the project. Reference Section 013100, Article 1.15 for further information.

B. Additional Training: Project specific training required to be completed by the subcontractor (administered by PPPL). See Section 013100, Article 1.17 for further information.

Access Area or Task	Training Required
General Site	1. General Employee Training (GET) – 2 hours 2. COVID-19 Training – 1 hour
Hot Work	1. Fire Watch/Fire Extinguisher – 3 hours (1.5 hour annual refresher)

C. Erosion Control: Erosion Control is required on this project in accordance with Section 013100 and 015000.

D. Metal Recycling: Metal is acceptable for commercial recycling.

E. Waste Disposal: The laboratory will dispose of existing regulated waste or laboratory generated regulated waste. The Subcontractor is permitted to dispose of any regulated waste that is generated during the performance of this subcontract. However, the subcontractor is required to notify the PTR prior to the commencement of any regulated waste generation and off-site shipments of any waste materials destined for disposal.

1. Existing asbestos elements shall be mitigated by PPPL. Coordination of this work shall be completed in concert with all other construction activities. The Subcontractor shall be responsible to coordinate this work with PPPL.

2. The Subcontractor shall remove any non-asbestos regulated materials, (e.g., fluorescent lamps and ballasts) and PPPL will recycle or dispose of the laboratory generated regulated materials. The Subcontractor shall be responsible to coordinate this work with PPPL.

3. Any bricks with lead-based paint or glazing shall not be recycled and shall be disposed of in accordance with PPPL requirements. The Subcontractor shall be responsible to coordinate this work with PPPL.

4. Refer to section 017419 for further details.

F. Badging: Subcontractor will be required to have PPPL Picture ID badges. Reference Section 013100 for further details.

G. Reporting: Subcontractor shall submit recycling data from vendor, reference Section 017419 for additional information.

H. Coordination: PPPL may award other subcontracts at or near the site of the work. The Subcontractor shall fully cooperate and coordinate with other subcontractors and with PPPL's employees during the execution of the work. This may include Owner's Representatives, Special Inspections Representatives, Commissioning Agents, etc. for this project or subcontractors for other projects.

I. PPPL On-Going Operations: Subcontractor must accommodate on-going and planned laboratory operations. All actions that affect laboratory operations must be coordinated through the PTR.

1.11 ELECTRONIC PROJECT DRAWINGS

A. PPPL may, at the Subcontractor's request and in accordance with technical specifications, release "electronic" project drawings for use in preparation of shop drawings by the Subcontractor or lower-tier subcontractors or vendors.

B. These electronic drawings will be distributed to the Subcontractor utilizing PPPL's current file and media format.

C. The project design drawings reflect the intent and scope of the project and general routing. The routing of utilities and building/facility services indicated on the design drawings is for bid purposes only. The Subcontractor and/or lower-tier subcontractors are expected to make a site visit and make allowances for all necessary adjustments for the actual system installation and to examine physical conditions which may be material to the performance of his or her work, and coordinate the actual routing as necessary to accommodate the existing conditions, obstructions, and the work of other trades.

1.12 PARKING AND STAGING AREA

A. Parking of the Subcontractor's and the lower-tier subcontractor's vehicles shall be confined to the Subcontractor's designated construction area or in a specific area designated by PPPL.

B. The Subcontractor shall provide and maintain temporary barricades, fences, or other means to delineate the construction limits, as required for the protection of public and PPPL employees. The Subcontractor shall provide signage at all entrances to the construction site with the Subcontractor's company's name and contact information. Signage shall include: Danger Construction Area, Do Not Enter. Visitors shall sign-in with the Subcontractor's representative and must be aware of the current hazards and conditions and be escorted by the Subcontractor.

C. No material shall be stored beyond the construction limits unless prior written arrangements have been made through the PTR.

D. Refer to Section 015000 for additional information.

1.13 SITE ACCESS AND HAULING

A. The Subcontractor shall, under regulations prescribed by PPPL, use only established roadways and use such temporary roadways constructed by the Subcontractor when and as may be authorized by PPPL. Where materials are transported during the course of the work, vehicles shall not be loaded beyond the loading capacity recommended by the manufacturer of the vehicles or prescribed by any Federal, State or local law or regulation. When it is necessary to cross curbings or sidewalks, the Subcontractor shall protect them from damage. The Subcontractor shall repair or pay for the repair of any damaged curbs, sidewalks or roads at no cost to PPPL or the Government.

B. Construction equipment and deliveries to the site are restricted to the main entrance gate. Access must be arranged at a minimum of 48 hours prior to the use of this access point through the PTR.

C. PPPL site access and hauling shall be subject to the following conditions:
   1. Electronic communication devices, such as cell phones, texting devices, laptops, etc. must not be used while driving any motor vehicle on PPPL's site;

2. All roads shall remain open to emergency traffic at all times;
3. All equipment and vehicles shall be confined to operating along defined construction roads and approved access routes;
4. No overland hauling or off-road travel shall be permitted without prior approval of the PTR;
5. Interruption of normal traffic patterns or temporary road closings necessitated by movement of equipment or delivery of materials or utility installations shall require advance notice and shall require proper barricades, signage and flag persons to divert normal traffic safely;
6. Traffic on paved roads shall be restricted to rubber-tired vehicles. Where crawler mounted equipment is required to cross paved roads or areas, the pavement shall be suitably protected from damage to the satisfaction of the PTR;
7. Dust, debris mud, and litter on any PPPL roads caused by the Subcontractor's operations shall be removed by the Subcontractor in a manner as directed by the PTR;
8. NJDot Standards shall apply to the Subcontractor’s use of all existing roads.

1.14 TRANSPORTATION OF EQUIPMENT AND MATERIALS

A. Transportation of equipment and materials shall be subject to the following conditions:
   1. Transportation of equipment and materials used by the Subcontractor at the job site shall be furnished by the Subcontractor at his/her own expense;
   2. The Subcontractor shall be responsible for minimizing any interference with local traffic, other Subcontractors and PPPL operations;
   3. The Subcontractor shall coordinate the anticipated schedule for major material deliveries and site hauling of excavated materials with the PTR

1.15 TEMPORARY SERVICES AND FACILITIES

A. Temporary services and facilities shall be subject to the following conditions:
   1. Temporary services and facilities required during the construction period shall be furnished, installed, and paid for by the Subcontractor (refer to Section 015000 for information on specific services and facilities);
   2. All installations shall be subject to PPPL approval;
   3. Temporary lighting shall be installed throughout the project to provide safe access and exit conditions and adequate lighting for the various work operations. The installation shall comply with the National Electrical Code (NFPA 70);
   4. Telephone/Internet service for the project will be provided by and paid for by PPPL;
   5. Temporary fire protection shall be in accordance with the OSHA (29 CFR 1926);
      a. An approved fire extinguisher shall be provided by the Subcontractor on all trucks and similar equipment, at all enclosures, and at on-site construction offices;

b. Each extinguisher shall be inspected monthly and a date tag certifying adequacy of the charge and workability of the extinguisher shall be affixed;
c. The Subcontractor shall remove the extinguishers at the conclusion of the job.

B. Temporary ventilation shall be sufficient to provide a safe working environment for construction personnel, in accordance with OSHA 29 CFR 1926.57. Subcontractor shall provide exhaust and supply air fans, ducting and other equipment as needed.

C. Temporary heating shall include heating devices, protective coverings and temporary enclosures as necessary to protect the work and to provide a safe working environment for personnel.
1. Coal or kerosene type salamanders, pots or open fires shall not be permitted;
2. Where permanent heating equipment has been installed and made operational prior to completion of the project, the Subcontractor may request temporary use of such equipment, at no cost, provided it is properly maintained by the Subcontractor and that all required warranties are extended to include the period of use by the Subcontractor prior to Final Acceptance of the project by PPPL. Such equipment shall be turned over to PPPL in like-new condition, including new filters and similar consumables.

D. Temporary Drainage shall be sufficient to remove standing water and prevent flooding.
1. Subcontractor shall furnish pumping equipment and other dewatering equipment as required for proper operation;
2. Pump discharges from dewatering activities shall be intercepted by silt removal or sedimentation basins before being directed to natural drainage courses and away from adjacent work limits of other subcontractors.

E. Temporary buildings (e.g., storage sheds, shops, offices) and utilities may be erected by the Subcontractor only with PPPL approval and shall be built with labor and materials furnished by the Subcontractor without expense to PPPL. The temporary buildings and utilities shall remain the property of the Subcontractor and shall be removed by the Subcontractor at its expense upon completion of the work. With the written consent of PPPL, the buildings and utilities may be abandoned and need not be removed.

F. Refer to Section 015000 for additional information.

1.16 PPPL CLOSURE

A. The PPPL site is closed for major holiday and no construction activities shall occur, unless specifically approved by the PTR, on these days:
1. New Year's Day – January 1st
2. Martin Luther King Jr. Day – 1st Monday after January 15th
3. Memorial Day – Last Monday in May
4. Juneteenth – June 17th

5. Independence Day – July 4th
6. Labor Day – 1st Monday in September
7. Thanksgiving Day – 4th Thursday in November
8. Day After Thanksgiving – Friday after Thanksgiving
9. Christmas Eve – December 24th
10. Christmas Day – December 25th
11. New Year's Eve – December 31st

B. If any of these holidays occur on a weekend day, a weekday will be used for the holiday;

C. PPPL holidays are subject to change based on Princeton University. Any additional PPPL site closures shall be communicated by the PTR.

1.17 MATERIAL SUBSTITUTIONS

A. In accordance with PPPL General Provisions for Non-Commercial Subcontracts Part G - Provisions & Conditions for Fixed Price Construction Subcontracts, products or materials which are equivalent to those specified will be considered for approval by PPPL. The submittal procedure described elsewhere will be followed, with the following additional conditions:
   1. In addition to the required information for the proposed substitute material, the submittal shall contain the same information pertaining to the original specified product for purposes of comparison;
   2. The submittal shall explain fully the differences, if any, between the original specified product and proposed substitute product;
   3. Any change to the drawings or specifications for related work required for proper installation of the proposed substitute product shall be indicated in the submittal;
   4. If the proposed substitute product requires alterations of any kind to other equipment or construction, or necessitates any engineering design changes for its proper installation, such alterations and engineering design changes shall be accomplished at no cost to PPPL;
   5. The substitute submittal must contain a statement detailing the cost and schedule impact of the proposed substitution;

B. PPPL is the sole judge of the acceptability of the proposed substitution.

1.18 DOCUMENTATION

A. The Subcontractor shall prepare weekly progress meeting minutes including on-going punch list items, ESH issues, PPPL corrective actions, and status of these items. These minutes will become part of the PPPL project file. At the discretion of PPPL, the weekly progress meeting may be reduced or eliminated depending on safety, quality, and schedule of the project.

B. Quality Control Documentation reference article 3.3.

C. On-going punch list: During the duration of the project, the PTR, AE, and/or CM shall identify deficient items that do not adhere to the technical specifications/drawings. This list will be kept current through the duration of the

project by the Subcontractor's Project Manager and/or Field Superintendent and made part of the Weekly Progress Meeting Minutes. At a minimum, the subcontractor shall, within five (5) working days, submit a proposed corrective action plan that details the mitigation method including a schedule for the correction of the noncompliance.

D. Notification of punch list items: The PTR, AE and/or CM will notify the Subcontractor of deficiencies and/or discrepancies in the quality of the construction. These notifications will include:
   1. Date Identified;
   2. Identifier;
   3. Location;
   4. Description, including specification or drawing reference.

E. As-Built Drawings:
   1. The Subcontractor shall maintain a set of prints of the Subcontract drawings in the construction office at the project site. Current set of as-builts may be electronic with prior approval by PPPL;
   2. A daily record, in red, shall be kept on these prints of the work installed with all modifications or changes thereon. This set of prints shall be available to PPPL for inspection at all times and print copies provided upon request to PPPL within 24 hours of the request being made;
   3. The PTR and the Subcontractor's Project Manager/Field Superintendent should review the as-built drawings prior to the Monthly Construction Schedule Updates described elsewhere, in order to determine if the as-built mark-up set is current and accurate. The processing of the monthly payment request is contingent of the up-to-date status of the as-built documentation;
   4. Prior to Final Acceptance described elsewhere in article 3.4, the Subcontractor shall transmit the complete set of marked-up prints to PPPL.
   5. The Final Acceptance submittal shall include a statement from the Subcontractor that indicates that the work was installed as shown thereon;
   6. Final Acceptance of the work is contingent on the receipt and approval of the complete as-built documentation.
   7. Refer to Section 01300 for more information on electronic as-built requirements.

F. Final Acceptance Documentation: Final Acceptance of the work is contingent on the receipt and approval of the following documentation:
   1. As-built documentation as described elsewhere in the subcontract documents;
   2. Shop drawing record set as described in (Specifications and Drawings for Construction) in accordance with PPPL Subcontract General Provisions and Special Terms and Conditions for Construction Subcontracts;
   3. Warranty documentation;
   4. Attic stock, if required by the technical specifications/drawings;
   5. Operations and Maintenance manuals.

1.19 SUBCONTRACTOR'S REQUESTS FOR INFORMATION (RFI)

A. When the Subcontractor is unable to determine from the Subcontract Documents, the exact material, process or system to be installed; or when the items of construction are required to occupy the same space (interference); or when an item of work is described differently at two places in the Subcontract Documents; or when other clarification is required; PPPL (or the AE if applicable) shall be requested to make a clarification of the indeterminate or interfering item. The RFI procedure given shall be the method for requesting the clarification.

B. RFIs shall be submitted on a PPPL approved Subcontractor's form. Each RFI shall be given a discrete, consecutive number. Each page of the RFI and of attachments to the RFI shall bear the RFI number and Project number. Resubmitted RFIs with additional information shall be RFI numbers with a .1 prefix for the first resubmitted RFI, .2 for the second, and so on.

C. RFIs from lower-tier subcontractors shall be submitted through, reviewed by, given a number, and signed by the Subcontractor prior to submittal to PPPL (or AE). The Subcontractor shall keep a log of the RFIs. RFIs submitted directly by lower-tier subcontractors would be returned unanswered to the sender and the Subcontractor. Subcontractor shall be responsible for delays resulting from the necessity to resubmit an RFI.

D. Subcontractor shall carefully study the subcontract documents to assure that the requested information is not available therein.

E. In all cases where RFIs are issued to request clarification of coordination issues (e.g., pipe and duct routing, clearances, specific locations of work shown diagrammatically, apparent interference and similar items) the Subcontractor shall furnish all information required for PPPL (or AE) to analyze and/or understand the circumstances causing the RFI in order to prepare a clarification or direction as to how Subcontractor shall proceed.

F. RFIs shall not be used for the following purposes:
   1. To request approval of submittals;
   2. To request approval of substitutions;
   3. To request changes which entail additional cost or credit;
   4. To request different methods of performing work than those indicated on the drawings and/or specified.

G. In the event the Subcontractor believes that a clarification by PPPL (or AE) results in additional cost, Subcontractor shall not proceed with the work indicated by the RFI until authorized by PPPL to proceed by formal direction in writing and/or a Subcontract Modification.

H. PPPL (or AE) will return RFIs to the Subcontractor within 5 working days of receipt. RFIs received after 12:00 noon shall be considered received on the next regular work day for the purpose of establishing the start of the 5 working day response period.

PART 2 PRODUCTS - Not Used

PART 3 EXECUTION

3.1 NOTICE TO PROCEED

A. The Notice to Proceed (NTP) represents the basis of the start of the project duration. The following describes the requirements for the Notice to Proceed:
   1. The NTP is issued by the PPPL Procurement Specialist;
   2. The issuance of the NTP establishes the start of work on the PPPL Site.

B. Submittals Required Prior to Notice to Proceed. Within ten (10) working days after subcontract award, the Subcontractor shall submit the following to PPPL for acceptance. These items must be submitted and accepted by PPPL prior issuance of Notice-To-Proceed (NTP).
   1. Performance and Payment bonds
   2. Insurance certificate(s), for an acceptable level of insurance and from an acceptable insurance carrier
   3. Construction Schedule
   4. Schedule of values
   5. Job Hazard Analysis (JHA)
   6. Lower-tier subcontractors (of applicable)
   7. Quality Assurance Program (QAP)

3.2 CONSTRUCTION SCHEDULE

A. The Subcontractor’s scope of work is outlined in the Statement of Work document.

B. This Project Complete milestone marks the completion of the project including punch list items, clean-up and acceptance of as-built drawings and if required, submittals.

C. Monthly construction schedule updates shall accurately reflect the execution of the project. This is required for payment.

D. Refer to Section 013200 for additional information.

3.3 QUALITY CONTROL AND CONTROL OF INSTALLATION

A. The Subcontractor is responsible for all activities necessary to manage, control, and document that work complies with the Subcontract documents. The Subcontractor’s responsibility includes ensuring adequate Quality Control services are provided for work accomplished on- and off-site by his/her organization, suppliers, lower-tier subcontractors, technical laboratories, and consultants.

B. The work activities include safety, submittal management, testing and inspection, and all other functions relating to the requirement for quality construction. Subcontractor is responsible to respond to Non-Compliance Notices which may

be issued by the Subcontractor, PPPL or the AE. Further, the Subcontractor shall contain procedures for tracking construction deficiencies from identification through acceptable corrective action. The Subcontractor shall track and log deficiencies on a form subject to PPPL approval.

C. The Subcontractor shall perform periodic inspection for potential suspect counterfeit materials. The most common suspect counterfeit materials found at US Department of Energy facilities have been threaded fasteners fraudulently marked as high strength bolts. This includes electrical circuit breakers with false certifications. If the Subcontractor suspects any counterfeit material being utilized, contact the PTR immediately.

D. Refer to Section 014100 for additional quality requirements.

3.4 ACCEPTANCE

A. Beneficial Occupancy: An interim inspection process may be used when PPPL assumes responsibility for portions of the work prior to substantial completion. The following details concern Beneficial Occupancy:
1. The Subcontractor shall provide a twenty (20) working day notice to the PTR before this inspection is performed;
2. The Subcontractor's deficiency list with status noted shall be attached to the Beneficial Occupancy documentation;
3. The PTR will coordinate the completion of the Beneficial Occupancy form;

B. Transition to Operations: refer to the following sections for additional information:
1. Section 017700 Closeout Procedures
2. Section 017823 Operation and Maintenance Data
3. Section 017836 Warranties
4. Section 017900 Demonstration and Training

C. Final punch list Inspection: An inspection of the uncompleted items of the subcontract documents. The following describes the punch list process:
1. When the Subcontractor believes the work is substantially complete, the Subcontractor shall update the deficiency list for presentation to the PTR;
2. The Subcontractor shall provide a twenty (20) working day notice to the PTR in order to make arrangements for the final inspection of the construction project with the AE and PPPL personnel;
3. The PTR will arrange for a final inspection visit of PPPL stakeholders to verify the accuracy of the Subcontractor's deficiency list and to add, as necessary, those items of work that are not complete;
4. The AE and/or CM will develop a comprehensive punch list and issue it to the Subcontractor after the final inspection;
5. The AE, CM, and/or PTR will sign off on the completion of each punch list item.

D. Final Acceptance: This inspection will document the completion of the project scope. The following describes this process:
1. The Subcontractor shall provide a ten (10) working day notice prior to the Final Acceptance inspection;
2. The Subcontractor shall submit documentation that the work associated with the punch list is complete;

3. The PTR will coordinate the final inspection walkthrough with PPPL stakeholders;
4. The PTR will develop and issue the final acceptance documentation when the punch list items have been completed.
5. All punch list items will be completed within 60 days of Substantial Completion.

END OF SECTION 010010

SECTION 012200

UNIT PRICES

PART 1 GENERAL

1.1 GENERAL REQUIREMENTS

A. Work of this Section, as shown or specified, shall be in accordance with the requirements of the Contract Documents.

1.2 SUMMARY

A. General: The requirements of this Section are procedural, administrative or explanatory in nature.

B. Definition: A unit price is an amount proposed by Bidders and stated on the Bid Form as a price per unit of measurements for materials or services that may be added to or deducted from the Contract Sum by Change Order in the event the estimated quantities of Work required by the Contract Documents are increased or decreased, note that one price shall apply to increase or decrease in work.

1. Unit prices shall be considered all-inclusive and include all necessary material, labor, supervision, equipment, transportation, handling, insurance, bonds, rentals, overhead, profit, applicable taxes, and incidentals.

2. Methods of measurement and payment for unit prices are specified in this Section.

C. Measurement: The Owner reserves the right to reject the Contractor's measurement of work-in-place that involves use of established unit prices, and to have this Work measured by an independent surveyor acceptable to the Contractor at the Owner's expense.

PART 2 PRODUCTS

(Not Applicable)

PART 3 EXECUTION

3.1 UNIT PRICE SCHEDULE

Unit Price 1: Provide additional custom tectum to match the spacing between existing sub-purlins, where directed by the architect per specification section 035114 Cementitious Roof Deck.

1. **Description:** Refer to detail C3/C40-A803 for typical installation requirements/details.
2. **Unit of Measurement:** Per board

Unit Price 2: Provide additional cost for the removal of existing condensing unit, CU-U1, and replacement/installation of new condensing unit, Liebert Model DCDF-104.

1. **Description:** Refer to Mechanical Drawing C40-M001 for additional information.
2. **Unit of Measurement:** Per Unit

Unit Price 3: Provide additional cost for the removal of existing exhaust fan, EF-6, and replacement/installation of exhaust fan, Greenheck Model Cube-161HP.

1. **Description:** Refer to Mechanical Drawing C40-M001 for additional information.
2. **Unit of Measurement:** Per Unit

Unit Price 4: Provide additional cost for the removal of existing exhaust fan, EF-U2, and replacement/installation of exhaust fan, Greenheck 18-CSW.

1. **Description:** Refer to Mechanical Drawing C40-M001 for additional information.
2. **Unit of Measurement:** Per Unit

Unit Price 5: Provide additional cost for the removal of existing exhaust fans, EF-7 and EF-8, and replacement/installation of exhaust fan, (2 quantity) Howden Model ACL-BI-540-CL2-ARG10.

1. **Description:** Refer to Mechanical Drawing C40-M001 for additional information.
2. **Unit of Measurement:** Per Unit

END OF SECTION

SECTION 012900 - PAYMENT PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. This Section specifies administrative and procedural requirements necessary to prepare and process Applications for Payment. These procedures include project-specific requirements that are supplementary to the requirements of the PPPL General Provisions for Non-Commercial Subcontracts Part G - Provisions & Conditions for Fixed Price Construction Subcontracts

B. See Division 01 Section 013200 "Construction Progress Documentation" for submitting the construction schedule.

1.2 SCHEDULE OF VALUES

A. The schedule of values shall be prepared in conjunction with the development of the performance schedule. Installed components of work and their associated tasks listed in the Schedule of Values shall match the activities in the construction schedule. Once approved the activity values listed in the Schedule of Values shall not be changed without PPPL's approval of the change.

B. Format & Content:

1. Arrange the Schedule of Values in tabular form with separate columns to indicate the following for each activity listed. The Subcontractor shall provide a MS Excel Spreadsheet electronic file.

    a. ID number – The identification or activity number shall match the identification used on the performance schedule.
    b. Activity Description – The activity description shall match the description used on the performance schedule.
    c. Number of Units
    d. Units – The units identified for an activity shall be suitable to determine the activity's percent complete on a monthly basis.
    e. Unit Value for each activity
    f. Total Value for each activity

2. Round amounts to the nearest whole dollar; total shall equal the Subcontract Sum.

3. Provide a separate line item in the Schedule of Values for each part of the Work where Applications for Payment may include materials or equipment purchased or fabricated and stored, but not yet installed.

4. Each item in the Schedule of Values and Application for Payment shall be complete. Include total cost and proportionate share of indirect costs and profit for each item.

    a. Temporary facilities and other major cost items that are not direct cost of actual work-in-place may not be shown as separate line items in the Schedule of Values. Distribute the cost as indirect costs expense across actual work-in-place activities.
    b. The amount of premiums paid for performance and payment bonds may be a line item of the Schedule of Values.
    c. Mobilization line items are acceptable provided there is an equal value demobilization line item.
    d. Items indicated as lump sum for the unit will be paid only when the Item is 100% complete.

5. Allowances: Provide a separate line item in the schedule of values for each allowance. Show line-item value of allowances, as a product of the unit cost, multiplied by measured quantity. Use information indicated in the Subcontract Documents to determine quantities.

6. Closeout Costs. Include separate line items for the subcontractor and lower-tier subcontractor (if applicable) for Project closeout requirements.

7. Updating: Update and resubmit the Schedule of Values before the next Application for Payment when Subcontract Modifications change the Subcontract Sum.

C. Submit a draft of the Schedule of Values to Princeton Technical Representative (PTR) at earliest possible date. No Applications for Payment will be accepted until after the PTR has approved the Schedule of Values. Applications for Payment shall be submitted to the Procurement Specialist.

D. Schedule of Values will not be approved without an approved construction schedule.

1.3 PAYMENT FOR STORED MATERIALS

A. Typically, PPPL doesn't pay for material received but not installed. Upon the Subcontractor's request and the PTR's concurrence, the Procurement Specialist may consider payment for materials received but not installed. The authorized payment for stored material will be based on the lesser of the following: (1) The total value of all invoices submitted for the activity or (2) the material value listed in the Schedule of Values.

    1. Material invoices shall be legible and clearly document the type, quantity, and cost of the materials covered in the Application for Payment.

    2. Material invoiced must include accompanying Bill of sale, certificate of insurance for bonded storage facility, and photos of materials in storage location.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 APPLICATIONS FOR PAYMENT

A. Applications for Payment may be submitted no more than monthly, 8 working days prior to the end of the month or as agreed by PPPL. Each Application for Payment shall be consistent with previous applications and payments as certified by the Procurement Specialist and paid for by PPPL.

B. Prior to preparation and submission of an Application for Payment, the Subcontractor's representative and the PTR shall review work complete to reflect a previous agreed upon scheduled milestone.

C. Application Preparation: All Applications for Payment shall be submitted electronically. Submit a signed Application for Payment to the Procurement Specialist. Incomplete applications will be returned without action. With each application, submit the following items:

1. Application for Payment Form (in conformance with the standard AIA G702)
2. Monthly voucher identifying the work complete.
3. The Monthly Voucher shall be validated by percent or milestone complete on the performance schedule.
4. Copies of stored material invoices.
5. Certified payroll for all subcontractor labor.
6. Technical Submittals up-to-date. Work without approved submittals will not be paid.
7. Quality Control documentation demonstrating up to date record keeping and compliance with the project's approved quality control plan.
8. As-Builts up-to-date.
9. Waste Reduction Progress Reports up-to-date.
10. Daily Reports up-to-date.
11. Quality Control Report up-to-date.

D. Administrative actions and submittals that must precede or coincide with submittal of Application for Payment include the following:

1. List of Lower-tier Subcontractors
2. Approved Schedule of Values including updates.
3. Performance Schedule.
4. Technical Submittals up-to-date. Work without approved submittals will not be paid.
5. Satisfactory Inspection and Test Reports for work billed to 100%
6. As-Builts up-to-date.
7. Waste Reduction Progress Reports up-to-date.
8. Daily Reports up-to-date.
9. Payroll submissions up-to-date.

E. Certification: Each Application for Payment shall be certified in accordance with the requirements of the PPPL Subcontract General Provisions and Special Terms and Conditions for Construction Subcontracts.

1. The amounts requested are only for performance in accordance with the specifications, terms, and conditions of the subcontract;
2. All payments due to subcontractors and suppliers from previous payments received under the subcontract have been made, and timely payments will be made from the proceeds of the payment covered by this certification
3. Request for progress payments does not include any amounts which the subcontractor intends to withhold or retain from a lower-tier subcontractor or supplier.

3.2 SUBCONTRACT MODIFICATIONS

A. Each subcontract modification shall be added to the end of the approved Schedule of Values. The modification shall be broken into a sufficient number of activities to adequately identify the percent complete of each activity. Activities shall be reflected in an updated schedule to monitor completion of added work.

END OF SECTION 012900

SECTION 013100 – ENVIRONMENT, SAFETY, AND HEALTH REQUIREMENTS

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SECTION 013100 – ENVIRONMENT, SAFETY, AND HEALTH REQUIREMENTS

PART 1 GENERAL

1.1 LABORATORY'S ES&H POLICY

A. At Princeton Plasma Physics Laboratory (PPPL), safety and environmental protection are of the highest importance. The Laboratory's policy is to protect the environment and the safety and health of all persons, may they be laboratory personnel, subcontractor personnel, scientific visitors, or visiting members of the public, from accident or injury while they are present on the PPPL Site or other location managed by PPPL. Nothing shall have a higher priority.

1.2 SUMMARY

A. This section describes the requirements, responsibilities and expectations for the environment, safety and health (ES&H) aspects of the project.

B. Safety, as used in this document, encompasses environment, safety, and health, including pollution prevention and waste minimization.

C. The Subcontractor shall provide all labor, materials, equipment, services, occupational exposure monitoring, and supervision required to maintain work sites that meet the ES&H requirements of all applicable federal, state, local and PPPL regulations. In addition, the Subcontractor shall protect the environment and the safety and health of its personnel, the personnel of its lower-tier subcontractors, lower-tier vendors, PPPL's personnel, and the general public.

1.3 REFERENCES

A. The Subcontractor and lower-tier subcontractors shall comply with the following reference documents. The publications referenced herein, form a part of this Section and Subcontract documents.

1. PPPL ESHD 5008 http://bp-pub.pppl.gov/ESHD_MANUAL/sm.html
2. 10 CFR 851 Worker Safety and Health Program https://www.energy.gov/gc/10-cfr-851-worker-safety-and-health-program
3. 29 CFR 1910, Occupational Safety and Health Standards https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910TableofContents
4. 10 CFR 708, DOE Contractor Employee Protection Program
5. 29 CFR 1904, Record Keeping Guidelines for Occupational Injuries and Illnesses
6. 29 CFR 1926, Occupational Safety and Health Standards for Construction
7. 40 CFR Protection of the Environment (USA EPA)
8. 49 CFR Transportation
9. DOE Order 442.1B, Department of Energy Employee Concerns Program
10. National Fire Protection Association (NFPA) codes and standards
11. NFPA 70E, Standard for Electrical Safety in the Workplace (2015)

12. American Conference of Governmental Industrial Hygienists - ACGIH (2016), “Threshold Limit Values for Chemical Substances and Physical Agents and Biological Exposure Indices”
13. American Society of Mechanical Engineers (ASME) Codes and Standards:
    a. ASME B30 Series, Crane Safety
    b. ASME B31 Standards of Pressure Piping
14. American National Standards Institute (ANSI) Standards:
    a. ANSI A10, Construction Package
    b. ANSI Z136.1 Safe Use of Ladders
15. PPPL Lab wide Procedures and Policies
    a. http://bp-pppl.gov/procedures.html
    b. http://bp-pppl.gov/policy.html

B. Subcontractor and its lower-tier subcontractors working on PPPL site are subject to United States of America - Department of Energy (DOE) civil penalties or subcontract fee reductions for noncompliance, in accordance with 10 CFR 851 “Worker Safety and Health Program” and 10 CFR 835 “Occupational Radiation Protection”. Subcontractor(s) shall also comply with 10 CFR 851 “Worker Safety and Health Program” which defines worker safety and health requirements for the Department of Energy contractors and their subcontractors. The Worker Safety and Health Program acknowledgment is part of the Construction ES&H Plan.

C. Subcontractor shall comply with 10 CFR 835 “Occupational Radiation Protection”. This Regulation defines the requirements pertinent to work where potential exposure to ionizing radiation hazards occur. The protective measures to be taken where this hazard is present shall be communicated to the Subcontractor through the PPPL PTR. PPPL has an extensive level of expertise in management of ionizing radiation hazards that will be utilized to identify the work locations where such hazards are present and establish these protective measures.

D. The publications listed below form a part of this specification to the extent referenced.
1. PPPL General Provisions and Special Terms and Conditions for Construction Subcontracts
2. 010010 General Requirements

1.4 DEFINITIONS

A. Competent Person: One who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate the identified hazardous. Duties related to ES&H shall take precedence over other duties.

B. Construction: The combination of erection, installation, assembly, demolition, or fabrication activities involved to create a new facility or to alter, add to, rehabilitate, dismantle, or remove an existing facility. It also includes the alteration and repair (including dredging, excavating, and painting) of buildings,

structures, or other real property, as well as any construction, demolition, and excavation activities conducted as part of environmental restoration or remediation efforts.

C. Construction Environmental Safety & Health Plan (CESHP): An official, binding document prepared by the Subcontractor, bearing the signature of a responsible manager of the subcontracting company that defines the safety and health practices and responsibilities necessary to conduct operations on PPPL property in a safe manner.

D. Construction Manager: An individual or firm responsible to DOE or its subcontractor, PPPL, for the supervision and administration of a construction project to ensure the construction subcontractor's compliance with construction project requirements.

E. Construction Management Office: An office consisting of a Construction (Project) Manager, PTR, Procurement Specialist, and ES&H Safety Coordinator. This office is responsible for supervising and administrating the construction project to ensure the construction subcontractor's compliance with technical specifications and ES&H requirements.

F. Construction Worksite: The area within the limits necessary to perform the work described in this Subcontract. Reference Attachment 2 - Site Location Map for Site Access and Hauling, & Parking and Staging.

G. Design Coordinator: An PPPL person who is assigned to the project and works with the Construction (Project) Manager and PTR to assist in the technical coordination of the project.

H. ES&H Construction Safety Specialist (CSS): An PPPL person responsible for ES&H guidance, periodic construction site visits, support for the PPPL PTR, and provide oversight of the Subcontractor's safety program. The ES&H CSS will review job hazard analysis and training documentation for on-going work activities. Any deficiencies noted shall be brought to the attention of the PPPL PTR for follow up with the Subcontractor. The ES&H CSS has authority to stop work activities for imminent danger, fatality, or major environmental release, but does not have authority to direct changes in the work scope of the project or the Subcontractor's means and methods of construction.

I. Imminent Danger: A hazard which, if allowed to persist, is likely to cause an accident that will result in death, injury, property damage, or environmental impairment.

J. Integrated Safety Management (ISM): ISM is defined as a system for performing work safely and in an environmentally responsible manner. The term "integrated" is used to indicate that the Environment, Safety and Health (ES&H) management systems are normal and natural elements of accomplishing work. The intent is to integrate the management of ES&H with the management of the other primary elements of construction: quality, scope, cost, and schedule. PPPL subscribes to the philosophy of Integrated Safety Management (ISM) by following the program outlined in this section. The ISM core functions are implemented by robust daily

work planning of construction activities. The ISM core functions include 1) define the scope of work, 2) identify and analyze the hazards, 3) develop and implement hazard controls, 4) perform work with controls, and 5) provide feedback for continuous improvement. PPPL also requires this of subcontractors and lower-tier subcontractors.

K. Princeton Technical Representative (PTR): An PPPL person responsible for ensuring the work being performed is in conformance to the subcontract technical requirements. The PPPL PTR is the primary point of contact with the Subcontractor.

L. Procurement Specialist: An PPPL person responsible for and specifically assigned to the project, who is responsible for negotiating and administering the subcontract terms and conditions. All modifications to the subcontract shall come from the Procurement Specialist or designee, in writing. The Procurement Specialist or designee is the sole entity that can modify the subcontract or initiate change orders.

M. Project Manager: A PPPL person responsible for managing, phasing, and ensuring the project is meeting key milestones.

N. Qualified Person: One who, by possession of a recognized degree, certification, or professional standing, or who by extensive knowledge, training, and experience, has successfully demonstrated his or her ability to solve or resolve problems relating to the subject matter, the work, or the project.

O. Subcontractor's Safety Officer/Representative: The Subcontractor's Safety Representative shall have completed the OSHA 30-hour construction safety course and have a minimum of 10 years of construction experience consistent with the type of activities included in the scope of work.

P. Subcontractor's ES&H Program: Subcontractor's (corporate) company policies and procedures to ensure operations comply with applicable safety and occupational health laws and regulations, and protect the safety and health of personnel and members of the public.

Q. Stop-Work Order: A definitive statement made openly to another individual that an imminent danger situation exists and therefore, all related work must stop immediately.

R. Work Hours: All work should be completed during normal working hours of 7:00 AM to 3:30PM, Monday through Friday. Off hours shall be communicated and approved with PPPL prior to the start of work, and in accordance with "Notification Requirements Off-Normal Events and Issues" (Attachment 1). Noise restrictions may be placed on the subcontractor where off hour work is necessary.

1.5 SUBMITTALS

A. Submittals with Subcontractor's proposal:

1. Construction Environmental Safety and Health Plan, Part 1 that includes acknowledgement regarding Worker Safety 10 CFR 851;
2. Experience Modification Rating (EMR) Rates and Occupational Safety Health Administration (OSHA) 300 log for last three years. Reference the PPPL Subcontractor Safety Information Questionnaire;
3. Subcontractor's ES&H Program (if required by Article 1.9); and
4. Resume' of the Subcontractor's Safety Representative

B. ES&H Submittals Required Prior to Notice to Proceed but not limited to. Within ten (10) business days after subcontract award, the Subcontractor shall submit the following to PPPL for acceptance. These items must be submitted and accepted by PPPL prior issuance of Notice-To-Proceed (NTP):
   1. Job hazard analysis (if NTP is waived, a project job hazard analysis is still required prior to commencement of work);

C. Submittals after Notice to Proceed include, if applicable:
1. Initial and Annual Certification of Subcontractor provided Training
2. The completed Construction Environmental Safety & Health Plan
3. Individual Trade, Craft, or Task Specific job hazard analysis
4. Safety Data Sheets (SDS)
5. Tabulation of On-site Work Hours on a Weekly basis as part of Weekly Progress Meetings
6. Incident Investigation Reports
7. Lift Plans
8. Welding Plans
9. Environmental, Erosion Control Reports; and
10. Excavation Plans, etc.

1.6 STOP WORK AUTHORITY

A. Any PPPL employee or Subcontract personnel has the authority to stop a work activity if there is imminent danger or major environmental release. If the hazard cannot be abated in a timely manner, the work activity shall be stopped through the use of a Stop Work Order by the PPPL PTR or the PPPL Procurement Specialist.

B. Work shall not be permitted to continue until the hazardous situation has been eliminated and PPPL has issued a Restart Work Order.

C. The Subcontractor is solely responsible for safe working methods, Subcontractor shall ensure that Subcontractor and their lower-tier subcontract personnel, PPPL and other personnel or visitors are not exposed to safety hazards inside their designated work sites.

D. If unsafe behavior is observed, any PPPL worker shall stop the task activity, inform the PTR and Subcontractor's Field Superintendent, and refer the worker(s) to the PPPL approved Construction ES&H Plan (CESHP), Job Hazard Analysis (JHA), and/or 10 CFR 851 regulations, or the appropriate project documents (for example, the JHA) for the violation.

1.7 ENFORCEMENT

A. The US Department of Energy has the authority to exercise enforcement actions on any subcontractor who violates any requirement set forth in 10 CFR 851 and 10 CFR 835. The Subcontractor may be subject to civil penalties up to $80,000 for each such violation, in accordance with 10 CFR 851 and civil penalties up to $110,000 for each such violation, in accordance with 10 CFR 835.

B. If any violation is a continuous violation, each day of the violation shall constitute a separate violation for the purpose of computing the civil penalty.

1.8 CONSTRUCTION ES&H PLAN

A. In accordance with 10 CFR 851 the Subcontractor shall submit a written project Construction ES&H Plan encompassing all pertinent aspects of the Subcontractor's ES&H Program and addressing how the requirements of this specification section will be implemented. The Construction ES&H Plan shall include site-specific information of the Subcontractor's activities for this project.

B. The Construction ES&H (CESH) Plan shall be submitted with the Subcontractor proposal and reviewed by PPPL prior to the award of the subcontract.

C. The CESH Plan shall encompass the work of any and all lower-tier subcontractors involved in activities under this Subcontract, and it shall include the Subcontractor's methods to enforce the elements of the Safety Program for all personnel on the construction worksite.
   1. The Subcontractor's CESH Plan shall include the following, at a minimum:
      a. A statement of the subcontractor's commitment to provide a safe and healthful construction worksite for all personnel including Subcontractor and their lower-tier subcontract and PPPL personnel;
      b. A signature of a responsible manager of the subcontracting company;
      c. Provide Name, title and qualifications of the designated Site Safety and Health Representative and designated alternates;
      d. Documented evidence of Off-Site Training for the Site Environment, Safety and Health Representative;
      e. Occupational Medicine Program, including identifying the qualified occupational medicine services provider;
      f. Description of the inspections performed by the Subcontractor (frequency and format for documentation of inspections);
      g. Procedures for coordinating safety and health with lower-tier subcontractors and with PPPL personnel on the construction worksite;
      h. Procedures for communicating and coordinating safety and health requirements to non-English speaking subcontractor personnel;
      i. Preliminary assessment of hazards and a list of the activity hazard analyses that will be performed for the project;
      j. Description of activity job hazard analysis process, including:

1) How workers are informed of hazards and protective actions; and
2) How workers will acknowledge being provided the information;

k. A disciplinary process for non-compliance with protective measures identified in the JHA;
l. How objective evidence (i.e. Industrial Hygiene (IH) exposure monitoring results) is to be used for establishing control measures, including exposure assessments to verify adequacy of control, if necessary as dictated by the hazards expected to be encountered;
m. Identify safety and health training requirements and procedures including but not limited to:
1) New Worker Orientation shall occur prior to beginning work on the construction worksite; and
2) Each worker acknowledging work hazards;
n. Frequency of tool-box safety meetings;
o. Updates or changes on safety practices relevant to the construction worksite, discussions how corrective actions and lessons learned from incidents at the construction worksite and elsewhere will be incorporated into the project's Safety Program.

D. The Subcontractor's incident reporting and occupational medical program and the specifics for the construction worksite shall include:
1. Identification of the activities on the construction worksite that will require personnel to be under an occupational medical or exposure-monitoring program. PPPL can request the Subcontractor to submit documentation of compliance.
2. Discussion of how medical personnel will be made available for advice and consultation on matters of occupational health.
3. An emergency response plan that sets forth the procedures to be followed upon the occurrence of serious injuries, illnesses, fatalities, fires, structural failures, or other emergencies, including procedures for the administration of first aid and/or other necessary medical treatment including:
a. Identified provision for prompt medical treatment.
b. Process for reporting and investigating recordable injuries for possible cause and corrective action in accordance with 10 CFR 1904.
c. Specific designation of management persons responsible for reviewing injury and illness reports.
4. Procedures for recording and reporting safety incidents and maintaining safety and health records in accordance with Occupational Safety and Health Administration (OSHA) requirements and in accordance with Article 3.3.
a. Procedures for the investigation of job-related incidents to determine possible cause and corrective action.
b. Specific designation of management persons responsible for review of injury and illness reports

1.9 SUBCONTRACTOR'S ENVIRONMENT, SAFETY, AND HEALTH PROGRAM

A. The Subcontractor shall have an effective Environment, Safety and Health (ES&H) program (corporate safety manual) incorporating the philosophy of Integrated Safety Management (ISM). It shall be submitted to the PPPL Procurement Specialist prior to Notice to Proceed.

B. On all subcontracts that require performance bonding, the Subcontractor shall have an ES&H Program that is commensurate with the complexity and nature of the work. This ES&H Program will describe the Subcontractor's overall commitment to safety and measures that will be taken specific to this project work scope and site. The following describes the ES&H Program requirements.

1. The Subcontractor shall submit to PPPL one (1) printed copy and one (1) electronic copy in Adobe portable document format (PDF);
2. The ES&H Program will address the Subcontractor's commitment to each of the following ISM principals. A brief explanation and key elements to be addressed follows each:

a. Line Management Responsibility for Safety: Line management shall be responsible and accountable for the protection of the personnel, the public, and the environment. Examples of expected items to support this statement are:

1) Statement of ES&H policy and goals;
2) Workforce accountability for strict compliance with subcontractor's ES&H program;
3) Policy statement concerning substance abuse on the construction worksite;
4) Process for progressive discipline;
5) Means of holding lower-tier subcontractors accountable for compliance with ES&H requirements;
6) Evidence of worker participation;
7) Participation of management in safety meetings, inspection, and documentation;
8) Process for personnel to identify and help resolve ES&H issues quickly, including stop work authority; and
9) Management support without hint of retribution or harassment.
10) On-going status and compliance verification reporting to PPPL.

b. Clear Roles and Responsibilities: The roles and responsibilities, and authority at all levels of the organization, including potential lower-tier subcontractors are clearly identified. Examples of expected items to support this statement are:

1) ES&H and Quality Assurance Plan (QAP) responsibilities for principals, field superintendent, foremen, competent person, ES&H officer, and workforce are documented; and
2) Stop work authority.

c. Competence Commensurate with Responsibility: Personnel possess the experience, knowledge, skills, and abilities that are necessary to discharge their responsibilities. Examples of expected items to support this statement are:

1) Identification of required training and experience of field superintendent, foremen, competent person, ES&H personnel, and workforce;
2) Identification of process for documenting completion of training;
3) Process for assuring lower-tier subcontractors are adequately skilled to perform their work activities; and
4) Training for personnel and lower-tier personnel on Integrated Safety Management and job hazard analysis.

d. Balanced Priorities: Resources are effectively allocated to address safety, programmatic, and operational considerations. Protecting the public, the workers, and the environment shall be a priority whenever activities are planned and performed. Examples of expected items to support this statement are:

1) Management commitment of resources to adequately implement their ES&H program;
2) Selection process for lower-tier subcontractors that include cost, quality, schedule adherence, and safety performance; and
3) Process for the Subcontractor to authorize start of work by lower-tier subcontractors.

e. Identification of Safety Standards and Requirements: Before work commences, the associated hazards are evaluated and an agreed upon set of safety standards and requirements are established which will provide adequate assurance that the public, the workers, and the environment are protected from adverse consequences. Examples of expected items to support this statement are:

1) Subcontractor ES&H Program Plan, by reference;
2) Subcontractor Quality Assurance Plan (QAP), by reference; and
3) Job hazard analysis process which includes defining scope of work, analysis of hazards, identification of hazard controls, requirement to perform work within these controls, and means to provide feedback and improvement.

f. Hazard Controls Tailored to Work Being Performed: Administrative and engineering controls, tailored to the work being performed, are present to prevent and mitigate hazards. Examples of expected items to support this statement are:

1) Job hazard analysis process;

2) Subcontractor ES&H Program Plan, by reference;
3) Planning and selection of appropriate and effective protective measures;
4) Active regimen of workplace inspections and prompt abatement of identified hazards; and
5) Occupational exposure (industrial hygiene) monitoring to verify adequacy of controls and compliance with occupational exposure limits;
6) Inspections, assessment, and audits of lower-tier subcontractor's adherence to ES&H and QAP; and
7) Daily work planning and hazard reviews at the worker level documented in form found in Attachment B.

g. Operations Authorization: The conditions and requirements to be satisfied for operations to be initiated and conducted are clearly established and understood by all. Examples of expected items to support this statement are:

1) Process to assure workers are informed of hazards and required protective measures before work is allowed to begin;
2) Process to assure workers, including lower-tier subcontractors are appropriately trained to do their job safely;
3) Process to assure that when an incident occurs, the scene is secured until the incident investigation is complete; and
4) Investigation process includes analysis, examination of trends and lessons learned, and a means to report to PPPL in a timely manner.
5) Process to assure that applicable PPPL permits are in place prior to allowing work to commence.

C. The ES&H Program should describe the following:

1. Basic Safety and Health Provisions including Emergency Action/Response Plan, Accident Investigation Program, Recording and Reporting of Injuries, Housekeeping, Hazard Communication Plan, Personal Protective Equipment and Fire Protection and Prevention.
2. Job Hazard Analysis Process including how hazards are identified and analyzed, preventive controls and the periodic inspection program. How workers are informed of hazards and protective actions. How objective evidence, i.e., monitoring results, is to be used for establishing controls measure, including exposures assessments to verify adequacy of control, e.g., verifying adequacy of hearing protection while monitoring equipment noise that is expected to be greater than 85dba exposure.
3. Waste Handling and Disposal including characterization of waste, packaging and labeling requirements and assurance that appropriate transportation and handling facilities will be used;
4. Erosion Control and Environmental Protection including Soil Erosion and Sediment Control Plan(s);

5. Other Programs (as dictated by the scope of this work) including the following:

    a. Control of Hazardous Energy (Lockout/Tagout);
    b. Confined Space;
    c. Concrete Cutting;
    d. Hearing Conservation;
    e. Ionizing Radiation;
    f. Non-ionizing Radiation;
    g. Lead, Beryllium, or Other Metals;
    h. Electrical (including Power Transmission and Distribution);
    i. Welding and Cutting;
    j. Scaffolds;
    k. Fall Protection;
    l. Excavations;
    m. Signs, Signals, And, Barricades;
    n. Tools - Hand and Power;
    o. Ladders & Stairways;
    p. Commercial Diving Operations;
    q. Motor Vehicles, Mechanized Equipment, and Marine Operations;
    r. Cranes, Derricks, Hoists, Elevators, and Conveyors;
    s. Concrete and Masonry Construction;
    t. Steel Erection;
    u. Underground Construction, Caissons, Cofferdams, & Compressed Air
    v. Demolition; and
    w. Blasting and the Use of Explosives

D. Changes and Updates: The ES&H Program is a living program. Updates that reflect changes to processes and programs shall be submitted as changes are made. PPPL may require changes for the program acceptance prior to Subcontract award. Once accepted by PPPL, the Subcontractor shall be required to comply with the requirements set forth in their program.

E. All lower-tier subcontractors employed by the Subcontractor must agree in writing to follow the Subcontractor's ES&H Program. If not, the lower-tier subcontractor will submit for acceptance one (1) electronic version in Adobe portable document format (pdf) of their ES&H Program and provide their own CESHP acknowledging the 10 CFR 851, reference paragraphs 1.3, B and Article 1.8.

1.10 JOB SITE ORIENTATION

A. The Subcontractor shall ensure and demonstrate; through a documented job site orientation program that lower-tier subcontractor personnel are aware of the ES&H requirements of the job.

B. The lower-tier subcontractors working for the Subcontractor shall follow and perform all required ES&H programs defined by the Subcontractor's approved and accepted ES&H program for the job site.

1.11 JOB HAZARD ANALYSIS

A. The job hazard analysis document details the specific hazards associated with the work activities and mitigating actions (including personal protective equipment) that the Subcontractor and lower-tier subcontractors will take to reduce or eliminate the risk of injury.

B. The initial job hazard analysis shall be submitted as part of the CESHP and accepted by PPPL prior to notice to proceed (NTP).

C. As the project progresses, task specific job hazard analysis shall be prepared and submitted. The Subcontractor shall prepare a job hazard analysis for all trade work. The Subcontractor may submit their own JHA format for PPPL approval, or use the PPPL format provided.

D. The following criteria shall be addressed, if applicable, when developing the job hazard analysis:

1. A PPPL accepted job hazard analysis shall be required for all work activities;
2. All subcontractor and lower-tier subcontractor personnel are required to sign the analyses affecting their work thereby acknowledging understanding of the hazards and the mitigation activities. The signature list shall be available for review by the PPPL Construction Management Office. As the JHA is updated, the subcontractor and lower-tier subcontractor personnel shall be advised of the new information and re-sign the document;
3. Prior to the start of subsequent new work activities, the Subcontractor shall review and revise the job hazard analysis, or develop a new job hazard analysis, as necessary to incorporate new hazards. Each revision must be submitted and accepted by PPPL before the associated element of work is begun;
4. Safety Data Sheet (SDS) of products that may significantly impact the safety or environment of PPPL or subcontractor personnel are to be submitted as part of the job hazard analysis process;
5. The name of Competent Persons shall be included on the job hazard analysis and communicated to all affected workforces;
6. Specific procedures in the areas of fall protection, excavation, confined space, hoisting and rigging, and Lockout/Tagout may be required as job conditions dictate; and
7. Identifying silica exposure such as cutting masonry products, installing, cutting, or removing concrete, installation of or removal of sheetrock compound or dealing with other silica constraining products. Reference Attachment A for recommended mitigations.

E. The PPPL Construction Management Office will provide informal job hazard analysis training for Subcontractor and lower-tier subcontractor personnel upon request.

1.12 EXISTING UTILITIES, EQUIPMENT, AND STRUCTURES

A. Utility Identification and Location:

1. Structures and utilities shown on the drawings represent the best information available. Their number and exact locations are not guaranteed. Excavation during construction may reveal the presence of underground drainage tiles, culverts, utilities, and other obstructions. The Subcontractor shall request from PPPL direction for rerouting, sealing or otherwise modifying underground obstructions not shown on the drawings.
2. The Subcontractor shall preserve and protect all structures, equipment, and vegetation (such as trees, shrubs, and grass) on or adjacent to the work site, which are not to be removed and which do not unreasonably interfere with the work required under this subcontract. The Subcontractor shall only remove trees when specifically authorized to do so, and shall avoid damaging vegetation that will remain in place. If any limbs or branches of trees are broken during subcontract performance, or by the careless operation of equipment, or by workmen, the Subcontractor shall trim those limbs or branches with a clean cut and paint the cut with a tree-pruning compound as directed by PPPL.
3. The Subcontractor shall protect from damage all existing in PPPL structure (1) at or near the work site and (2) on adjacent property of a third party, the locations of which are made known to or should be known by the Subcontractor. The Subcontractor shall repair any damage to those facilities, including those that are the property of a third party, resulting from failure to comply with the requirements of this subcontract or failure to exercise reasonable care in performing the work. If the Subcontractor fails or refuses to repair the damage promptly, PPPL may have the necessary work performed and charge the cost to the Subcontractor.
4. Electrical cables, duct banks, fiber optic cables, and gas service in the area of the excavation, that is, 18 inches either side or crossing, must be de-energized/de-pressured and lock-out/tagged-out. If the services cannot be de-energized/de-pressured, then vacuum excavation methods, consisting of air or water to break up the soil and a vacuum device to collect the spoil, shall be used to locate fiber optic cables, electrical cables, duct banks, and gas lines prior to the excavation activity.

B. Work on Existing Utilities:

1. PPPL will identify through drawings, notations and field locates, the approximate location of known utilities and underground structures;
2. No work shall be performed on existing in-service utility systems without prior approval and coordination of the system outage by the PPPL PTR;
3. Pressure shall be relieved on all piping systems before opening system and starting the work;
4. Lockout/Tagout shall be used by the Subcontractor for all valves, blank-offs and relief lines;

5. "Hot Tap" connections on utility services shall not be permitted unless specified by the subcontract documents and specific procedures have been submitted to and accepted by PPPL.
6. Existing fire and life safety systems shall be re-activated at the end of each business work day. If systems are unable to be re-activated at the end of the business work day, notification must be made to the PPPL PTR.

1.13 ENVIRONMENTAL ISSUES AFFECTING THE WORK

A. Environmental issues affecting the work planning specific to this project can be found in Section 010010 General Requirements.

B. Soil Erosion and Sediment Control shall be employed on all projects involving excavations, the following requirements apply:

1. Subcontractor shall have all required erosion control devices required by the Soil Erosion and Sediment Control Plan (SESCP) and as shown on the drawings, in place prior to commencing any work for which they are required.

    a. As the work evolves, additional interim control structures may be required in order to protect waterways and/or comply with permit terms and conditions.
    b. Costs for installation and maintenance of these structures shall be considered incidental to the project and included in the original proposal.
    c. The Subcontractor shall install all such structures within 24 hours of notification by PPPL.
2. Maintenance of Erosion Control Structures:
    a. Subcontractor shall be required to perform inspections of all control structures as specified in the drawings and SESCP, and to maintain all control devices until final stabilization of all disturbed areas.
    b. Subcontractor shall perform at a minimum, weekly erosion control inspections and after ½-inch or more precipitation events.
3. Temporary and Permanent Seeding and Stabilization
    a. The Subcontractor shall be required to follow seeding dates and requirements as specified and in accordance to the State of New Jersey Soil & Erosion Control Manual.
    b. The Subcontractor shall be responsible for providing appropriately vegetated surfaces as outlined in the technical specifications or specified on the drawings.
    c. The Subcontractor shall be responsible for establishing sufficient final vegetation required for stabilization as accepted by PPPL to remove the erosion control structures. Any permanent seeding operations completed at the end of the growing season shall be carried out per dormant seeding requirements (State of New Jersey Soil and Erosion Control Manual). Subcontractor shall be

responsible for establishing final vegetation over all disturbed areas.

1.14 ADVANCE NOTICE OF WORK ACTIVITIES

A. The following activities require the Subcontractor to provide written notification to the PPPL PTR a minimum of seven (7) business days prior to the commencement of work:

1. Interruption of road traffic;
2. Closure of any roads;
3. Connection to or interruption of any existing underground utility;
4. Intended use of ICW from any hydrant;
5. Connection to temporary electric power sources;
6. Request for disablement of fire alarms or related safety devices;
7. Request for disablement of fire protection systems;
8. Intended activity beyond the specified construction limits;
9. Intended access to or work within a confined space;
10. Connection to or interruption of any existing power system;
11. Saw cutting or core drilling at manholes, foundations and paved areas;
12. Excavation activities;
13. Backfilling of underground utilities;
14. Delivery of a radioactive source on the PPPL site.

B. Subcontractor must receive PPPL's approval prior to proceeding on the above listed work activities.

1.15 SUBCONTRACTOR'S PROJECT TEAM & PERSONNEL

A. Field Superintendent - The Subcontractor shall, at all times during the progress of the work, provide a competent superintendent. In addition, the following requirements for the Field Superintendent are described below:

1. The Subcontractor shall provide a competent Field Superintendent, who is the Subcontractor's representative designated for the duration of the project to the running of the day-to-day operations of the work including safety, quality control and lower-tier subcontractor coordination responsibilities.
2. The Field Superintendent shall have knowledge and experience of Occupational Safety and Health Administration (OSHA) and other related safety standards, and has the authority to enforce such standards in the field.
3. The Field Superintendent must be present on the PPPL project site whenever work activities are ongoing.
4. In the absence of the designated Field Superintendent, the Subcontractor shall identify an alternate individual with similar qualifications acceptable to PPPL.
5. Should more than one work shift be required on this project, the Subcontractor shall identify and assign a designated individual meeting the above requirements for each work shift.

6. In the event that excavations are part of the project scope, the Subcontractor shall provide a competent person for excavation activities who meets the requirements of OSHA 29 CFR 1926.32 (f).
7. In the event that scaffolding is to be utilized during the execution of the project, the Subcontractor shall provide a competent person for scaffolding who meets the requirements of OSHA 29 CFR 1926.32 (f).

B. Subcontractor's Safety Representative - If a Subcontractor's Safety Representative is required, the Subcontractor shall employ a Safety Representative who acts as the authorized agent of the Subcontractor, responsible for safety activities of all work sites under this subcontract. Reference Section **010010.**

C. The Subcontractor's Field Representative or if required by Section **010010** the Subcontractor's Safety Representative as identified in the Construction Environmental Safety & Health Plan (CESHP) shall interface with the PPPL PTR and PPPL ESH CSS on all safety matters, and assure the Subcontractor does the following:

1. Act as the Competent Person
2. Provide Input to Daily Task/Work Planning Meeting
3. Interface with PPPL PTR on all safety matters;
4. Prepare and submit Hazard Analyses including revisions and updates;
5. Review and accept lower-tier safety plans and hazard analyses;
6. Assure that all lower-tier subcontractors have accepted the ES&H Plan;
7. Update the ES&H Plan as required;
8. Maintain a list of Competent and Qualified Persons;
9. Assure that Hazard Analyses are understood and signed by all workers;
10. Inspect work in progress;
11. Identify and reports and corrects deficiencies;
12. Assure that personal protective equipment is available;
13. Conducts tool box meetings;
14. Maintain all safety records including minutes, training records, inspections, etc.
15. Maintain safety related signage;
16. Assure that equipment inspections are performed;
17. Attend weekly construction meetings;
18. Coordinate permit applications with PPPL PTR;
19. Investigates all incidents.

D. The Subcontractor's Safety Representative or the alternate shall be present at all meetings between the Subcontractor and PPPL at which changes in construction methodology are discussed. The Subcontractor's Safety Representative shall approve these changes.

E. Drug Free Work Place - All personnel of the subcontractor and lower-tier subcontractors are prohibited from engaging in the unlawful manufacture, distribution, dispensing, possession, or use of controlled substances. All subcontractors and subcontractor's personnel shall adhere to the Federal Controlled Substances Act (12 USC 8120 and further defined in Federal Regulations 29 CFR 1308.11-3808.15) and stated in the subcontract documents.

F. Any damage to property or environment released by the subcontractor or lower-tier subcontractor's personnel are subject to drug test and results shall be submitted to the PTR. All Subcontractor vehicles and equipment must be in good working order with no leaks of any kind. Any spills must be reported immediately to PPPL and work stopped until the spill is addressed. The Subcontractor is responsible for remediation costs caused by negligence and or faulty equipment. PPPL Environmental Services will determine the proper method of remediation.

1.16 IDENTIFICATION BADGING

A. The Subcontractor personnel and lower-tier subcontract personnel will be required to obtain identification badging for access onto the PPPL site, unless otherwise stated in Section **010010**.

1. The PPPL PTR will assist in the process of identification badging;
2. Once identification badges are obtained, they shall be available at all times while on and entering the PPPL site; and
3. Deliveries, incidentals, and escorted work activities under forty (40) hours are exempted from the badging requirement.
4. The PTR will collect all identification badges at the close of the project and return them to Site Protection.

1.17 PPPL TRAINING

A. Subcontractors working at PPPL may be required to complete the following training depending on the specific areas accessed. Please note that certain D-Site areas not referenced in the table below will require a full-time PPPL escort for access. Areas include the D-Site Count Room, the Mock-up Clean Room, the NBPC Machine Shop, the NSTX-U Storage Area, the Neutral Beam Compressor Room, and the Radioactive Waste Handling Facility. Refer to Section 010010 to determine which of the trainings from the table below are required for the project. PPPL may also chose to not require training below to specific areas and provide a full-time PPPL escort for access in cases where scope of work if is short in duration. This must be included in the bidder's clarifications and assumptions and approved by PPPL prior to award.

Access Area or Task	Training Required
General Site	1. General Employee Training (GET) – 2 hours 2. COVID-19 Training – 1 hour
Hot Work	1. Fire Watch/Fire Extinguisher – 3 hours (1.5 hour annual refresher)
D-Site Grounds	1. General Employee Training (GET) – 2 hours 2. COVID-19 Training – 1 hour
D-Site General Building	1. General Employee Training (GET) – 2 hours 2. COVID-19 Training – 1 hour 3. D-Site Orientation – 2 hours 4. Procedure D-OP-AD-09, "D-Site Work Permits" – 1 hour 5. Procedure D-OP-AD-56, "Control of D-Site Equipment and System Status (Chain of Command)" – 1 hour
D-Site NSTX-U Test Cell	1. General Employee Training (GET) – 2 hours 2. COVID-19 Training – 1 hour 3. D-Site Orientation – 2 hours 4. Lock-out/Tag-out (LOTO) Awareness – 1 hour 5. NSTX-U Accelerator Safety Order Awareness – 1 hour 6. Radiation Safety – 4 hours 7. NSTX-U Radiation Safety Training Video – 1 hour 8. Basic Electrical Safety (BES) – 1 hour 9. Oxygen Deficiency Hazard (ODH) – 8 hours 10. Procedure D-OP-AD-09, "D-Site Work Permits" – 1 hour 11. Procedure D-OP-AD-24, "Control of Workplace Cleanliness Around D-Site Experimental Areas" – 1 hour 12. Procedure D-OP-AD-39, "D-Site Conduct of Operations" – 1 hour 13. Procedure D-OP-AD-56, "Control of D-Site Equipment and System Status (Chain of Command)" – 1 hour 14. Procedure D-OP-AD-117, "Access to NSTX-U Experimental Areas" – 1 hour 15. Procedure ENG-036, “Control of Temporary Modifications”
D-Site TFTR Test Cell and Tritium Areas (includes Upper & Lower DARMs, & Elevator)	1. General Employee Training (GET) – 2 hours 2. COVID-19 Training – 1 hour 3. D-Site Orientation – 2 hours 4. Radiation Safety – 4 hours 5. Procedure D-OP-AD-09, "D-Site Work Permits" – 1 hour

	6. Procedure D-OP-AD-56, "Control of D-Site Equipment and System Status (Chain of Command)" – 1 hour
D-Site FCPC & MG Building	1. General Employee Training (GET) – 2 hours 2. COVID-19 Training – 1 hour 3. D-Site Orientation – 2 hours 4. Lock-out/Tag-out (LOTO) Awareness – 1 hour 5. NSTX-U Accelerator Safety Order Awareness – 1 hour 6. Procedure D-OP-AD-09, "D-Site Work Permits" – 1 hour 7. Procedure D-OP-AD-56, "Control of D-Site Equipment and System Status (Chain of Command)" – 1 hour

1.18 WASTE DISPOSAL/RECYCLING

A. No trash burning, dumping, or disposal is permitted on PPPL property. The disposal of regulated waste generated from the use of materials provided by PPPL (Government property) shall be the responsibility of the designated PPPL representative. Disposal of all other trash and waste materials generated during the performance of this subcontract shall be the responsibility of the Subcontractor and must be performed in compliance with all applicable federal, state and local laws and regulations.

B. The governing statutes include, but are not limited to, the Resource Conservation and Recovery Act, the Toxic Substances Control Act, the Hazardous Materials Transportation Act, the State of New Jersey Environmental Protection Act/Solid and Special Waste Management Regulations, and the laws and regulations of any other state receiving regulated waste material generated during the performance of this Subcontract.

C. PPPL remains subject to a DOE-wide suspension of recycling metal originating from radiological areas. Refer to Section **017419** for requirements pertaining to recycling metals. Off-site disposal of recycled materials, trash, debris, demolished material, pallets, crates, packing materials, rubbish and all waste material shall be the responsibility of the Subcontractor. The goal for recycling construction and demolition waste is 80% based on weight, excluding metals restricted in Section **017419**.

D. The Subcontractor shall furnish all necessary dumpsters or containers to prevent dispersion of debris both within and outside of the construction site.

E. PPPL Approved Recycling:

1. The Subcontractor shall utilize a recycling waste hauler, obtain a report on percentage recycled by weight from the vendor and submit the report to the PPPL PTR;

2. The minimum amount of recycled material is 50% as measured by weight excluding metals restricted by above in paragraph B.
3. The Subcontractor shall submit a report that details the percentage, by weight, of recycled materials.

F. Regulated Waste:

1. Where regulated waste is generated (waste regulated by the Resource Conservation and Recovery Act Hazardous, Toxic Substances Control Act, State of New Jersey Special Waste, etc.) the Subcontractor shall immediately notify the PPPL PTR; and
2. Unless specified above in paragraph B, all regulated waste shall be disposed through the PPPL Hazard Control Technology Team.

PART 2 PRODUCTS - Not Used

PART 3 EXECUTION

3.1 PERMITS

A. PPPL has its own Authority Having Jurisdiction. No municipality permits are required. However, PPPL conducts work through the use of on-site permits. All PPPL required permits will be identified to the Subcontractor by the PPPL PTR, who will arrange for all necessary permits at no cost to the Subcontractor.

B. No work activity shall be performed without the required permits.

C. Activities requiring permits include, but are not limited to:
1. Work notification;
2. Excavation (Dig Permit see below);
3. Electrical work;
4. Existing Structure/Concrete Cutting & Coring
5. Hot Work
6. Modification to drinking water or sanitary sewer systems;
7. Radioactive sources on site;
8. Working with/on radioactive material, working in radiological areas;
9. Moving government or PPPL property off site.

D. The Subcontractor shall comply with all restrictions or provisions listed on permits.

E. All requests for permits shall be made a minimum of two (2) working days prior to the need for the permit, except excavation permit.

F. Dig Permit: A Dig Permit issued by PPPL via the PPPL PTR is required before any excavation/digging can begin at the construction site. The following requirements are associated with the Dig Permit:

1. Subcontractor shall coordinate the preparation of the excavation permit application with the PPPL PTR;
2. Dig permits may require a minimum of fifteen (15) working days to process;
3. Subcontractor shall be responsible for New Jersey One Call notification;
4. No excavation shall proceed without an approved Dig Permit, signed by the Subcontractor Competent Person and the PPPL PTR.

G. Confined Space Permit: The minimum acceptable requirements for confined space work at PPPL include:

1. The PPPL PTR shall identify all existing confined work spaces to the Subcontractor;
2. If a Subcontractor is required to enter a permit-required confined space as part of their contract with PPPL, the subcontractor shall provide the PPPL PTR with the following information at the pre-construction meeting or at least five (5) working days prior to entry:
   a. A written copy of the Subcontractor's confined space entry program;
   b. An example of the Subcontractor's entry permit;
   c. Training records for potential entrants, attendants, and entry supervisors;
   d. Evidence that all air monitoring equipment is properly calibrated within the calibration period specified by the subcontractor's program or manufacturer's instructions. This may be in the form of a calibration log, certification indicator on the instrument, or other means. (It is imperative that the equipment used by the Subcontractor be capable of monitoring for the contaminants associated with the confined space to be entered.
3. It will be the Subcontractor's responsibility to provide all of their own personal protective equipment (PPE), such as lifelines, harnesses, respirators, tripods, ventilators, etc., as specified by the entry permit;
4. In addition to complying with the permit space requirements listed above, each Subcontractor retained to perform permit-required confined space entry operations shall:
   a. Obtain any available information regarding permit space hazards and entry operations from the PPPL PTR;
   b. Coordinate entry operations with PPPL, when both PPPL personnel and Subcontractor personnel will be working in or near permit spaces;
   c. Prior to entry, inform the PPPL PTR of the specific permit space procedures the Subcontractor will follow;
   d. Inform the PPPL PTR who will inform the PPPL Emergency Services Unit (ESU) prior to entering the space;
   e. Inform the PPPL PTR of any unanticipated hazards encountered during confined space entry;
   f. Provide the PPPL PTR with a copy of the Subcontractor's confined space permit, reclassification form or written certification once the work has been completed.

H. Hot Work Permit:

1. Information concerning the Hot Work permit is listed below:
   a. The PPPL PTR will contact the PPPL Emergency Services Unit (ESU) and secure the Hot Work Permit;
   b. Members of the ESU will meet with the PPPL PTR and the Subcontractor's Field Superintendent and examine the proposed operation, prescribe precautions, assure appropriate instructions are understood, and then issue a written Hot Work Permit;
   c. The Subcontractor must arrange for fire watches during burning, welding, or other fire or spark generating work. This fire watch must continue for a minimum of sixty minutes after work is complete;
   d. It is the Subcontractor's responsibility to furnish the proper number and type of fire extinguishers for any welding, cutting, or brazing activities as specified in the Hot Work Permit;
   e. The extinguishers must be located in clear sight and no farther than 50 feet from the work areas;
   f. All welding shall be in accordance with the requirements of the American Welding Society (AWS) Standard: Safety in Welding, Cutting, and Allied Process (ANSI/ASC Z49.1-94);
   g. UL or FM listed check valves shall be installed on oxygen-fuel torch cutting equipment
2. The Subcontractor shall provide approval drawings and/or other documentation of protective measures for which applicable OSHA standards require preparation by a Professional Engineer or other qualified individual. These should be included in the ISM Plan and include but are not limited to Scaffolding and Fall Protection barriers and anchors.

3.2 PRECONSTRUCTION MEETING

A. Preconstruction Meetings may be required for Subcontracts which require bonding. This meeting, chaired by the PPPL Procurement Specialist and PTR, will typically be held after the Subcontract award and before Notice to Proceed is issued.

B. The Subcontractor's Project Team is expected to attend this meeting.

3.3 REPORTING REQUIREMENTS

A. The following requirements concern Subcontractor reporting requirements:

1. Emergencies: All emergencies occurring at the PPPL site must be reported immediately by dialing extension **3333** from a PPPL phone or **(609) 243-3333**. The types of emergencies to be reported include: injury or illness requiring emergency care, fire, explosion, security incident, vehicle accident, radiation incident, utility failure, tornado sighting, and hazardous material spill or release;

2. Non-Emergencies (or Off-Normal Events See attachment 3): All incidents, including any injury/illness, any non-emergency incident, uncontrolled hazardous energy sources, spills and near misses must be reported immediately to the PPPL PTR. PPPL shall determine if the incident scene shall be preserved and secured by the Subcontractor to enable PPPL and DOE to conduct any necessary investigations. After any necessary emergency response is made, the scene shall be left unchanged and protected until the PPPL PTR is notified and releases the incident site for work to continue; and
3. Investigation and Reporting: The Subcontractor must investigate all incidents (including near misses). The Subcontractor shall submit, within 48 hours of an incident, a written report of an investigation. The investigation must include root causes, corrective actions and preventive measures.

3.4 SUBCONTRACTOR TRAINING

A. The following requirements concern Subcontractor training:

1. The Subcontractor shall be responsible for assuring that their personnel and lower-tier subcontractor personnel, understand all ES&H requirements provided by PPPL, and commit to abiding by these requirements, regardless of communication barriers. The Subcontractor must be able to communicate any necessary instructions to those personnel to ensure safe operations;The Subcontractor shall provide any accommodation or language translation requested, to ensure full understanding of PPPL requirements.
2. All Subcontractors performing work at PPPL shall provide to their personnel any necessary ES&H training as may be required by federal/state regulations and as appropriate for their Subcontract activities at PPPL. Exceptions involve hazards, which are unusual for the trade of the Subcontractor's personnel. In particular, PPPL normally provides appropriate training for Subcontractors working in radiation areas or oxygen deficient hazard areas, and expected emergency response.

B. ES&H training that was provided by the Subcontractor or others and received by Subcontractor personnel performing subcontractor activities at PPPL shall be maintained on-site and available for review by PPPL.

C. Subcontractors shall maintain on-site and provide to PPPL upon request, any and all occupational safety and environmental records. Such records include, but are not limited to, the records required to be maintained by federal/state regulation. Such records include OSHA injury/illness logs, training records, inspection records, safety meetings, and incident investigations. The subcontractor is responsible for submitting a letter, prior to commencing work on-site and annually thereafter while working under this subcontract, certifying each of the workers has received the required training. Additional records appropriate for the Subcontractor's activities shall also be maintained and provided to PPPL upon request. Examples include but not limited to:
   1. Excavation

2. Scaffolding
3. Fall Protection
4. Confined Space
5. Welding
6. Crane Inspections
7. NFPA 70E
8. Respiratory Protection

D. If the Subcontractor intends to administer first aid or Cardio Pulmonary Resuscitation (CPR), the Subcontractor must comply with 29 CFR 1926 and have available the list of names of any employee who will administer first aid or CPR, along with current certifications.

3.5 JOB SITE ES&H INSPECTIONS/MEETINGS

A. After the start of construction and throughout the entire construction period, the Subcontractor shall monitor and inspect the construction area and operations for compliance with the Subcontractor's accepted ES&H and/or CESH plan on a daily basis. The Subcontractor's Project Team is expected to conduct these inspections and correct any deficiencies found.

B. These inspections shall be documented by the Subcontractor and maintained on-site for the duration of the project. Records shall be available for review upon request by PPPL.

C. The following requirements concern Job Site ES&H meetings:

1. Daily Work Planning Meetings in the form of daily briefings shall be conducted by the Subcontractor Field Superintendent and attended by all subcontractors on site that day. The daily planning meeting will discuss the planned work activities, review the applicable job hazard analysis, and allow for employee questions and feedback regarding the work activity.
2. Weekly Toolbox Meetings shall be conducted at the job site by the various area/job foreman or superintendents for their specific crafts. These meetings shall emphasize the current construction operations and provide an opportunity for inspection of tools and personal protective equipment.

D. The Subcontractor will document the daily and weekly toolbox meetings (date, topic, attendance, etc.) and provide a copy to PPPL.

E. The PPPL PTR will be notified of all job site ES&H meetings and may attend.

3.6 PERSONAL PROTECTIVE EQUIPMENT

A. 100% eye, head, and foot protection – All construction workers and other personnel on the construction worksite shall wear at all times eye, head, and foot protection that complies with the applicable ANSI Standards. The type of protective eyewear shall be selected as appropriate for the hazard. Gloves shall be worn when handling sharp objects.

B. The Subcontractor shall furnish personal protective equipment (PPE) as required to reduce employee exposure to hazards when engineering and administrative controls are not feasible or effective in reducing hazard exposures. The minimum acceptable PPE for work on the PPPL site are:

1. Hard hats shall be furnished by the Subcontractor and shall be worn in the construction work areas as designated in the job hazard analysis and/or applicable OSHA standards. Personnel working on construction activities or in the field shall also wear hard hats, brim facing forward or full brim style hard hat. Hard hats shall meet the ANSI Z89.1 standard as required by 29 CFR 1926.100 and bear the “Z89.1” designation. High voltage exposure work requires hard hats that meet the ANSI Z89.2 standard and bear the “Z89.2a” designation;
2. Safety glasses with side shields shall be furnished by the Subcontractor and shall be worn in the construction work areas as designated in the job hazard analysis and/or applicable OSHA standards. Eye protection must meet the requirements of 29 CFR 1926.102. Safety glasses shall be ANSI approved and shall be marked with the ANSI “Z87.1” designation;
3. Clothing suitable for the work and weather conditions is required. In construction areas, the minimum shall be short (1/4 length) sleeve shirts, long trousers, and hard sole leather work boots providing ankle protection. In addition, any work that presents a greater hazard to the feet or toes requires the use of safety toed or metatarsal guards, meeting ANSI Z41. Canvas, tennis, or deck shoes are not permitted within the construction work area;
4. Hand protection, such as gloves shall be readily available for the task suited for the hazards on the individual subcontractor’s personnel and lower-tier personnel;
5. All personnel working on construction activities outside and involving heavy equipment in the field shall wear a reflective, high visibility garment, minimum ANSI Class II, as designated in the job hazard analysis and/or applicable OSHA standards.

3.7 ELECTRICAL WORK

A. The Construction Management Office will determine whether electrical work permits are required.

B. The following sets forth the minimum acceptable requirements for work on electrical systems at PPPL.
1. All electrical work shall be performed in accordance with NFPA 70E, Standard for Electrical Safety in the Workplace;
2. The Subcontractor personnel must be trained in Lockout/Tagout (LOTO) prior to participating in LOTO of hazardous energy sources and working on LOTO systems or equipment;
3. The Subcontractor shall provide ground fault circuit interrupter protection for electric hand held tools, portable generators, temporary electrical extension cords and other wiring, etc. The assured equipment-grounding program is <u>not</u> an acceptable alternative at PPPL.

3.8 OXYGEN DEFICIENT HAZARDS

A. PPPL has policies and procedures governing work in ODH areas. The PPPL PTR will communicate specific requirements and work practices to the Subcontractor.

B. All Subcontractor and lower-tier subcontractor personnel who must enter designated ODH areas must have and display a level of medical fitness acceptable to PPPL prior to entering those areas.

C. PPPL will assess the need for ODH training for Subcontractor personnel. If ODH training is necessary PPPL will provide it free of charge.

D. Oxygen monitoring equipment will be supplied to the Subcontractor personnel, as necessary. The Subcontractor is responsible for returning this equipment upon request or upon completion of the work.

E. PPPL will furnish emergency evacuation equipment. Care, use, and the return of such equipment will be the responsibility of the Subcontractor.

3.9 RADIATION PROTECTION

A. PPPL has policies and procedures governing radiological work. The PPPL PTR will advise the Subcontractor of the requirements and work practices, if potential for radiation affects the work scope.

B. PPPL will assess the need for radiological training for Subcontractor personnel. If radiological training is necessary, it will be provided free of charge by PPPL.

C. Radiation dosimeters will be supplied to the Subcontractor personnel, as necessary. The Subcontractor is responsible for returning this equipment upon request or upon completion of the work.

D. PPPL will furnish protective clothing. Disposal of such clothing will be the responsibility of PPPL.

E. Prescribed procedures for material handling and segregation shall be followed explicitly. Potentially radioactive material must be surveyed prior to removal from site. The PPPL PTR shall coordinate this survey.

F. The use of industrial radioactive testing sources is subject to monitoring and oversight by PPPL based on the following:
   1. Nuclear density meters will be inspected at the construction site by PPPL ES&H and Health Physics personnel. Review for Department of Transportation compliance with survey and inspection requirements will be completed at that time;
   2. When required by specification, Subcontractor use of radiography sources will require five (5) business days' advance notice to the PPPL PTR. During this time, the Subcontractor shall submit to the PPPL PTR documentation showing the Subcontractor's Nuclear Regulatory Commission or Agreement State license for the material;

3. When the source is brought to the PPPL site, PPPL Health Physics staff will meet the subcontractor, escort him/her to the construction site, and monitor the use of the source during the testing activity;
4. Any work with radiography will occur outside normal business hours.

3.10 ENVIRONMENTAL PROTECTION

A. All construction work on the PPPL site shall comply with all applicable environmental executive orders, laws, regulations, and permits. All Subcontractors and lower-tier subcontractors shall conduct their activities in an environmentally sound manner that limits the risks to the environment and protects the public health. The following sets forth the minimum acceptable requirements for environmental protection at PPPL:
   1. Refer to Section 1.12 of this document for the Soil Erosion and Sedimentation Control (SESC) requirements for this project;
   2. If required, the Subcontractor shall install all erosion control in accordance with SESC plan prior to the start of excavation activities;
   3. Excavation at or adjacent to streams' tributaries, wetlands, or other surface waters shall be done only after notification to the PPPL PTR;
   4. The PPPL PTR will inform the Subcontractor if any wetlands are present in the work area and what protective measures are necessary;
   5. Unexpected environmental impacts shall be immediately reported to the PPPL PTR and mitigated by the Subcontractor;
   6. Flammable and/or combustible liquids, fuels, and oils shall be provided with containment and shall not be stockpiled beyond one day's usage. Storage of these materials, plus maintenance and fueling areas used by the Subcontractor, shall be properly graded and maintained and shall be located a minimum of 100 feet away from a wetland or water body boundary so that adverse effects on the environment are eliminated;
   7. The Subcontractor shall make routine inspections to assure that all motorized equipment is free of leaks of petroleum and other toxic or hazardous materials. The Subcontractor shall keep sufficient cleanup supplies on hand (e.g. oil dry, absorbent booms, etc.) to contain/absorb any spill or leak of fuels, oils, etc. that could potentially leak from their equipment. If a spill or leak should occur, the Subcontractor should immediately take appropriate steps to contain spills, move equipment out of sensitive areas (near wetland or water body) and immediately notify the PPPL PTR;
   8. At the close of each workday, the Subcontractor's Field Superintendent shall inspect the complete construction site to ensure that all erosion controls, drainage patterns, excavations and staging areas are in environmentally sound condition for the weather conditions anticipated.

3.11 TEMPORARY HEATING DEVICES

A. Open burning, fire barrels, coal or kerosene type salamanders, or open flame heating devices that have exposed fuel below the flame are not allowed on the PPPL site.

B. Temporary heating devices shall be coordinated through the PPPL PTR

3.12 SMOKING

A. Smoking including electronic smoking devices are prohibited in locations where flammable and/or combustible materials are stored. “No smoking” signs shall be posted in these areas.

B. Smoking including electronic smoking devices are prohibited in all PPPL buildings, and is acceptable only in designated outdoor areas.

3.13 FUEL STORAGE TANKS

A. Above ground fuel storage tanks for construction vehicles shall not be permitted on the PPPL site.

B. Fuel tanks mounted on pick-up trucks shall conform to the requirements of the State of New Jersey Fire Marshal's Office.

C. Fuel tanks mounted on pick-up trucks shall be removed from the PPPL site at the end of each workday.

D. Refueling of equipment while the motor is running is prohibited.

E. During refueling from truck-mounted fuel tanks or with portable fuel cans, etc., a 20-pound (minimum) A-B-C dry chemical fire extinguisher must be present.

F. All fueling trucks or dispensing stations shall be properly grounded and bonded.

G. Maintenance and fueling areas used by the Subcontractor shall be properly graded and maintained and shall be located a minimum of 100 feet away from a wetland or water body boundary to avoid adverse effect on the environment.

3.14 EXPLOSIVES

A. The use of explosives is not permitted without prior written approval of PPPL.

3.15 VEHICLES AND EQUIPMENT

A. The following sets forth the minimum acceptable requirements for vehicles and equipment at PPPL:
   1. Operators must have an appropriate, valid driver’s license when operating vehicles on site. Seat belts are required to be provided and worn for the operators and passengers of all vehicles;
   2. All vehicles and mobile powered equipment, except automobiles and pickup trucks, shall have reverse signal alarms (a.k.a. backup alarms) audible above the surrounding noise level. If backup alarms are not present on the equipment, a spotter (other than the driver of the vehicle) must be present to warn pedestrians and the drivers of other moving equipment;
   3. If required by the equipment manufacturer, roll over protection structures shall be provided;

4. Personnel lifts must be equipped with audible motion alarms. These alarms must be in operation and audible over the surrounding ambient noise when the lift is in use. Additionally, all lifts require two distinct actions in order to make the lift move in a forward or backward direction or in an upward or downwards direction. A foot pedal is considered one of the actions if independent of the other controls;
5. The equipment manufacturer must approve any modifications to lifting and hoisting equipment;
6. All hand and power tools must be checked prior to use on each shift to assure that they are maintained in a safe condition. Any deficiencies shall be repaired, or defective parts replaced, before continued use.

B. RIGGING, HOISTING AND CRANES

1. Pre-approval is required before special purpose vehicles or equipment such as mobile cranes, forklifts, man-lifts, etc. can be delivered and accepted onsite. The Subcontractor must submit to PPPL operator training qualifications, as well as, equipment maintenance and inspection records for all special purpose equipment. At a minimum, signed and dated annual inspection records (check-list type) shall include model and serial number for each specific piece of equipment. In addition, the equipment manual shall be submitted for review and provided on site along with the equipment. Upon delivery and prior to use, PPPL reserves the right to perform equipment inspections to verify that the equipment is acceptable for usage onsite. For additional requirements not mentioned refer to PPPL Engineering Standards ES-MECH-007, ES-MECH-010, ES-MECH-011, ES-MECH-012. PPPL Engineering Standards may be found online at http://bp.pppl.gov/EngStds.html.
2. For all hoisting and rigging, a Lift Plan shall be submitted prior to all activity. Multiple lift plans may be required depending on change of conditions, loads or equipment used. The lift plan shall include:
    a. Product data for all applicable hoisting & rigging equipment (with serial numbers for major components)
    b. Periodic (minimum annual) equipment inspection reports (with serial numbers) for all equipment
    c. Monthly hook and wire rope inspection reports
    d. Lift data sheet detailing loads, center of gravity, attachment points, component capacities, calculations, lift angle, and swing location. Requires signature by the Subcontractor's Lift Engineer.
    e. Qualifications of the Rigger and Signal Person. If qualifications are not available, an officer from the Subcontractor's firm shall sign that the stated Rigger and Signal Person are qualified per OSHA 1926.1400.
    f. Equipment manual and load charts.

C. Equipment inspection and modification (The subcontractor shall comply with 29 CFR 1926.600, Subpart O):
   1. The Subcontractor must inspect all heavy equipment before use on site, prior to use on each shift, and during use to make sure it is in safe operating condition. Defective equipment shall be removed from service;
   2. The Subcontractor is to assure that regulatory inspection records are complete and up-to-date and that operating manuals are available;
   3. In no case shall the original safety factor of the equipment be reduced.
   4. All tools and equipment brought on site by the Subcontractor are subject to inspection by PPPL. Items found to be damaged or out of compliance shall be repaired or immediately removed from service, and tagged out of service.
   5.

3.16 CONCERN REPORTING PROCESS

A. Whistleblower Protection for Subcontractor Personnel
   1. The Subcontractor shall comply with the requirements of the "DOE Contractor Employee Protection Program" at 10 CFR Part 708.
   2. The Subcontractor shall insert or have inserted the substance of this clause, including this paragraph (b), in lower tier subcontracts, at all tiers, with respect to work performed on any construction worksite at a DOE-owned or leased facility, as provided for at 10 CFR part 708.
   3. Concern Reporting Processes - Subcontractor personnel on the construction worksite are entitled to use any of the means available to communicate concerns about ES&H conditions and practices. Information about concern reporting is available on ES&H bulletin boards throughout PPPL, and shall be included with PPPL provided materials for the construction worksite postings for this project. The options for reporting concerns include:
      a. DOE Concern Reporting:
         1) Telephone: (609) 243-3700
         2) Email: EmployeeConcerns@science.doe.gov

3.17 PROJECT BULLETIN BOARD

A. Subcontractor is responsible for installing and maintaining a safety bulletin board at the location where the majority of the subcontractor's personnel, lower-tier subcontractors, and lower-tier vendors report to work. Information shall include:

   1. DOE Worker Safety and Health Poster DOE-F 5480.2 (Worker Rights), furnished by PPPL. Spanish versions of the DOE Safety and Health Poster are also available from PPPL upon request.

   2. DOE Occupational Safety and Health Complaint Form 5480.4, available on-line.

3.18 JOBSITE SAFEGUARDS

A. The Subcontractor shall be responsible for providing and implementing the necessary precautions to safeguard material and equipment at the project site.

B. In the event of theft or damage to Subcontractor property, PPPL property, and/or Government property, the Subcontractor shall immediately notify Emergency Services Unit (ESU) by telephone (609) 243-3333 and PPPL PTR.

3.19 EMERGENCY EGRESS AND SEVERE WEATHER

A. The following sets forth the minimum acceptable requirements for emergency egress and severe weather protection at PPPL:
   1. All emergency egress routes shall be kept clear at all times;
   2. Severe weather shelter locations and specific evacuation procedures will be provided by the PPPL PTR;
   3. The Subcontractor shall communicate egress routes and severe weather shelter to their personnel and all lower-tier subcontractors.

3.20 WORK COMPLETION AND CLEAN UP

A. All work and clean-up operations shall be in compliance with the Subcontractor's ES&H Plan.

B. Requested documentation for all aspects of the ES&H program shall be complete and submitted prior to Subcontract close-out.

END OF SECTION

Attachment 1

Construction Environmental, Safety & Health Plan (CESHP)

INSTRUCTIONS

This Construction Environmental, Safety & Health Plan (CESHP) is required for each construction project at DOE's Princeton Plasma Physics Laboratory (PPPL). See Technical Specification 013100 for further information.

This document is a template for the CESHP designed to assist construction subcontractors in describing project conditions and developing project-specific hazards and controls information.

The prime subcontractor may *either* flow down this requirement to each of its subcontractors; or serve as a control and coordination point, requiring all subcontractors' activities to be conducted under the prime subcontractor's solitary CESHP.

Subcontractors must submit the completed CESHP to the PPPL Procurement Office, which will distribute the document for review and concurrence by the PPPL ES&H and Project Management teams prior to work commencing. The CESHP is intended to be a living document, which may be updated as necessary throughout a project as information changes or as the project progresses (e.g. – as additional Task/trade Hazard Analyses are developed, subcontractors added, etc.).

All of the sections in Part 1 are required to be completed for each construction project, regardless of the size or complexity. Enter information in all of the fill-in blocks that are applicable. For those that are not applicable, enter "N/A" or other suitable explanation.

PROJECT GOVERNANCE / EMERGENCY INFORMATION

Section 1 – Project Description and Emergency Contacts

Fill in the names and telephone numbers of the contact personnel for this particular project. In accordance with Title 10 of the *Code of Federal Regulations*, Part 851 (10 *CFR* 851), the Subcontractor must have a designated on-site safety representative who is knowledgeable of the project's hazards and has the authority to correct unsafe conditions or behavior. This is either the Field Superintendent or a designated Subcontractor's Safety Representative as delineated in Technical Specification 010010. Attach the qualifications of your safety representative's resume' for this project. If you have subcontractors performing work on this project, list their contact information. Update as necessary throughout the project.

Section 2 – Subcontractor Policy Statement

Enter your Company's health and safety policy statement. At minimum, your policy should include:

- Specify if Subcontractor's ES&H Program is required and if a dedicated Subcontractor Safety Representative is required.
- Specify if a Quality Control Plan (Reference Technical Specification 010010 if a dedicated Quality Control Representative is required.
- Specific statement of intent to comply with Title 29 of the *Code of Federal Regulations*, Part 1910, "General Industry Safety and Health Standards", and Part 1926, "Safety and

Health Standards for the Construction Industry"; 10 *CFR* 851, "Worker Safety and Health"; and other applicable codes and standards.

- A statement if occupational exposure monitoring is required and if so, that it complies with 10 CFR 851.
- Specify what occupational (industrial hygiene sampling) monitoring may be required.

Section 3 – 10 CFR 851 Acknowledgement

Because Princeton Plasma Physics Laboratory is a Department of Energy site, your company must meet the Department of Energy requirements of Title 10, *Code of Federal Regulations*, Part 851, "Worker Safety and Health Program" (Link to 10 CFR 851). It is your responsibility to ensure you have read and understand the actual regulatory requirements. Specify medical surveillance and qualifications. This may include Occupational exposure monitoring.

Section 4 – Safety Briefings and Inspection

The Subcontractor must conduct periodic safety briefings and inspections, based upon the duration and complexity of the project. Describe the frequency and initiation of safety briefings and inspections on your project at Princeton Plasma Physics Laboratory.

ATTACHMENT 1

(Project Name)

Construction Environmental, Health & Safety Plan

SECTION 1 - PROJECT DESCRIPTION & EMERGENCY CONTACTS		
Subcontract / Project Number	Enter subcontract and/or task order number	
Project Start / End Dates	Start:	Complete:
Project Location	Enter the work location(s) at PPPL	
Scope of Work	Enter breakdown and description of work activities	

FOR ALL EMERGENCIES CALL:

PPPL Emergency Contact number: x3333 - or - (609) 243-3333 from a cell phone.

For all incidents, injuries, property damage, near-misses, work-induced illness or chemical over-exposures, the following personnel MUST be immediately contacted upon scene stabilization, but in all cases within one hour:

PROJECT TEAM & PERSONNEL CONTACT INFORMATION			
Subcontractor Project Manager	Enter the name of subcontractor Project Manager	Enter number: xxx-xxx-xxxx	Enter: user@domain
Subcontractor Site Superintendent	Enter the name of subcontractor's Site Superintendent	Enter number: xxx-xxx-xxxx	Enter: user@domain
Subcontractor Health & Safety Representative **	Enter the name of subcontractor health & safety representative	Enter number: xxx-xxx-xxxx	Enter: user@domain

Subcontractors - Company Name	**Name of Designated Safety Representative ****	**Phone Number**
Enter Subcontractor company name	Enter name of individual**	Enter number: xxx-xxx-xxxx
Enter Subtier company name	Enter name of individual**	Enter number: xxx-xxx-xxxx
Enter Subtier company name	Enter name of individual**	Enter number: xxx-xxx-xxxx

***** If required, attach a description of qualifications, or resume', for each Safety Representative per Technical Specification 010010.***

CESHP REVIEWS		
Reviewed & Approved by: (Subcontractor)	**CESHP Reviewed & Accepted by: (PPPL)**	**CESHP Reviewed & Accepted by: (PPPL)**
Enter Subcontrator - Company Officer	Enter PPPL Name	Enter PPPL Project ESH&Q Name
Signatures and dates		

SECTION 2a – Subcontractor's ES&H Program	
2a Subcontractor's ES&H Program Required See Technical Specification 013100, Article 1.8	☐ Yes ☐ No
2b Dedicated Subcontractor Safety Representative required, see Technical Specification 010010	☐ Yes ☐ No

SECTION 2b – Subcontractor's Quality Control Plan (Program) if Section 014100 is required	
2c Subcontractor's Quality Control Plan (Program) Required	☐ Yes ☐ No
2d Dedicated Subcontractor Quality Control Representative required	☐ Yes ☐ No

SECTION 2c - SUBCONTRACTOR POLICY STATEMENT
Enter here the Subcontractor's policy on occupational safety & health (see instructions):

Section 2d - Occupational Exposure Monitoring and Qualifications			
Occupational Exposure Monitoring	Will you be using materials or engaging in an activity that potentially may produce occupational exposures at 10 to 50% of the occupational exposure limits? (e.g., silica, hexavalent chromium, lead, etc.) Reference Section 10 for additional information.	Yes ☐	No ☐
	If yes, you will need to: 1. Comply with 10 CFR 851, Appendix A, Section 6. 2. Provide your industrial hygiene provider contact information.		
Certified Industrial Hygienist (CIH) or IH Technician under the supervision of a CIH.	.Enter the name and address of your company's IH provider for this project		

Section 2e - Industrial Hygiene Sampling		
Yes	No	
☐	☐	Asbestos
☐	☐	Asphalt fumes
☐	☐	Diesel Exhaust
☐	☐	Fiberglass Insulation
☐	☐	Hexavalent Chromium
☐	☐	Lead
☐	☐	Naphtha (Coal Tar)
☐	☐	Noise
☐	☐	Non-Ionizing Radiation
☐	☐	Silica
☐	☐	Solvents
☐	☐	Temperature Extremes (Heat or Cold)
☐	☐	Welding, Cutting & Brazing
☐	☐	Other: (List...)

SECTION 3 - ACKNOWLEDGMENT of 10 CFR 851			
As a subcontractor to PPPL, while your workers are physically located at PPPL you must meet the requirements of Title 10, *Code of Federal Regulations*, "Energy", Part 851, "Worker Safety and Health Program" (10 CFR 851). As such, you must be aware of, and comply with, the requirements of this regulation. (Link to 10 CFR 851)			
Acknowledgment	I, (the author of this CEHSP), certify that that I have read the requirements of 10 CFR 851 and attest that my firm and its sub-tier contractors will comply with the requirements of 10 CFR 851.	Yes ☐	No ☐
MEDICAL SURVEILLANCE AND QUALIFICATION			
Occupational Medicine	Will you have any employees that will work on-site at PPPL for 30, eight-hour days in a 12-month period, or are enrolled for any length of time in a medical or exposure monitoring program required by federal, state, or local regulations (including hearing conservation, respiratory protection, lead exposure)?	Yes ☐	No ☐
	If yes, you will need to: 1. Comply with the occupational medicine requirements of 10 CFR 851, Appendix A 2. Provide your occupational medicine provider contact information		
Clinic / Physician	Enter the name and address of your company's occupational medicine provider for this project	Enter telephone number: xxx-xxx-xxxx	Enter e-mail address: user@domain

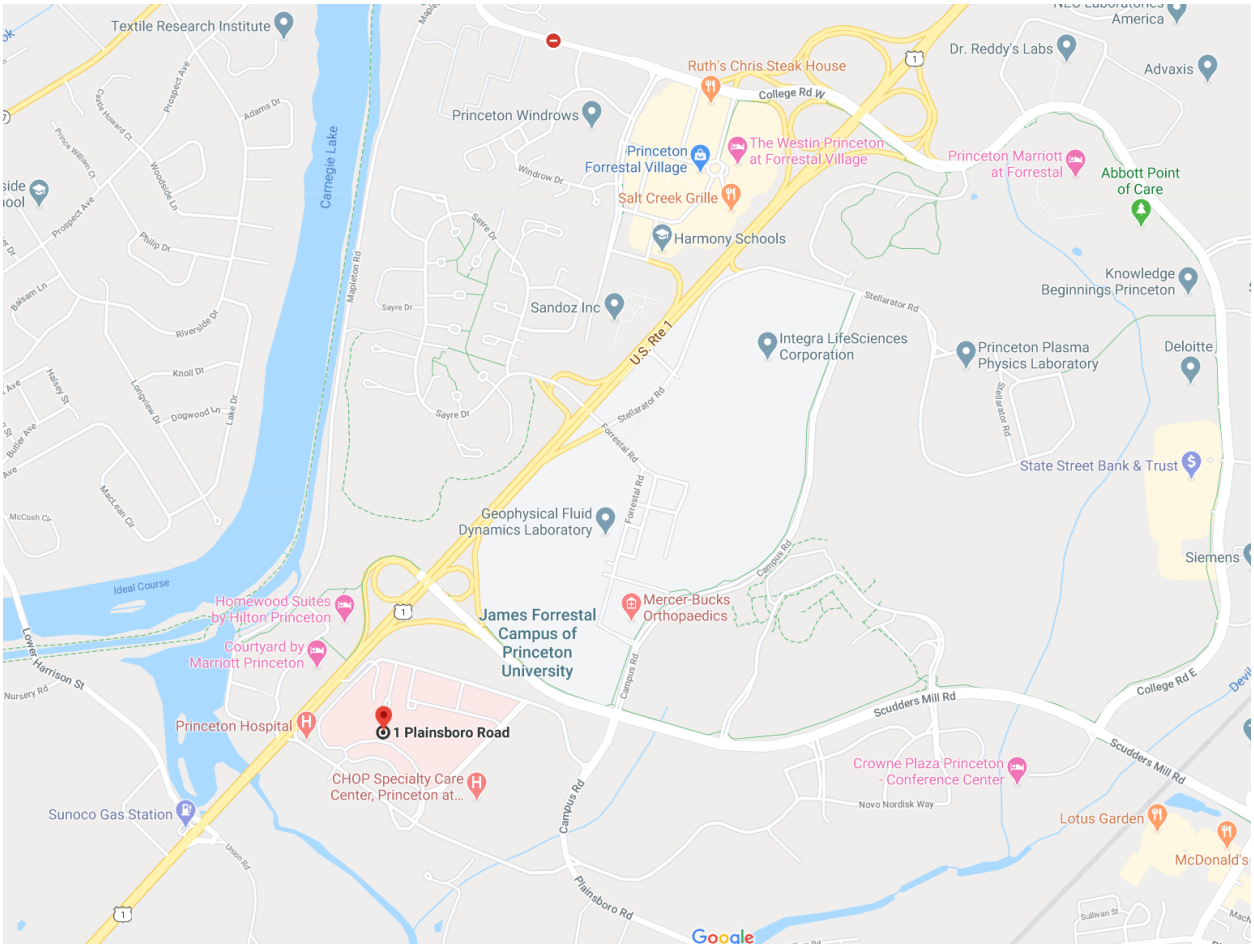
Required Medical Surveillance		Task-specific medical testing
☐ DOT/Commercial Vehicle	☐ Blood Lead	List specific task(s) requiring medical surveillance
☐ Hearing Conservation	☐ Respirator User	
☐ Fit for Duty	☐ Other(s) List other(s)	
☐ Substance Abuse Testing		

SECTION 4 - SAFETY DAILY WORK PLANNING AND INSPECTIONS
Safety Briefings: Discuss the conduct of safety briefings on your project at PPPL, See Section 3.6
Safety Inspections: Discuss your conduct of safety inspections during this project at PPPL, See Section 3.6
Job Site Orientation: Discuss how job site orientation, See Section 1.10

ATTACHMENT 2
Site Location Map and Information

PPPL has identified Princeton Hospital as the closest medical facility. The Subcontractor shall provide a highlight on the map the appropriate subcontractor designated facility. If the subcontractor would like to use another facility, please include a map and address location of the facility.

University Medical Center of Princeton
1 Plainsboro Road
Plainsboro, NJ 0857
(609) 853-6500



ATTACHMENT 3

Notification Requirements Off-Normal Events and Issues

Note: Contacting the PPPL Emergency Services Unit at X3333 to obtain emergency assistance and taking immediate actions to protect workers, guests and visitors takes precedence over the notification requirements described herein.

The Subcontractor shall notify PPPL within 15 minutes of becoming aware of off-normal events and issues; off-normal events and issues include:

- Personnel injuries of any kind
- Near misses that could have resulted in significant (recordable injury or worse) worker injuries
- Uncontrolled or unforeseen personnel exposure to chemical, biological or physical hazards (e.g., noise, laser, ultraviolet light, heat, etc.)
- Fire emergency of any kind
- Any unexpected discovery of an uncontrolled hazardous energy source (e.g., live electrical power circuit, etc.) not including discoveries made by zero-energy checks and other precautionary investigations made before work is authorized to begin
- Any failure to follow a prescribed hazardous energy control process (e.g., lockout/tagout)
- Damage of any kind to Laboratory property
- Damage of any kind to personal property owned by PPPL personnel, guests and visitors
- Any spills of chemicals, fuels, lubricants, etc.
- Other off-normal events and issues deemed by the Subcontractor to require notification

In the event of an off-normal event or issue, the Subcontractor and its lower-tier subcontractors SHALL:

- Notify PPPL Emergency Services Unit at X3333 for emergency response, if required
- Take immediate action to evacuate and post affected area, if required
- Cease related work or work in affected area
- Take actions to post and safe-off affected area
- Within 15 minutes: Contact Subcontractor Site Supervisor, PPPL PTR, ESH Construction Safety Engineer and ESH Head

Regarding these off-normal events or issues, no recovery actions (other than those immediately required to preserve life safety) or repairs are to take place until they are reviewed by the Subcontractor and PPPL, and specifically authorized by the PPPL PTR or ESH Department Head. Resumption of work that has been stopped by PPPL due to off-normal events and issues must follow the provisions of PPPL Policy P-012.

SECTION 013119 – MEETINGS

PART 1 GENERAL

1.1 SUMMARY

A. General:
   1. The following applies to this section:
      a. PPPL General Provisions for Non-Commercial Subcontracts Part G - Provisions & Conditions for Fixed Price Construction Subcontracts
      b. Related requirements in other Subcontract Documents including, but not necessarily limited to the Drawings and the Specification Sections in the Project Manual.
   2. Review these documents for coordination with additional requirements and information that apply to work under this Section.
   3. Related Sections:
      a. SECTION 013200 - SCHEDULES AND PROGRESS DOCUMENTATION
      b. SECTION 013514 - BUILDING INFORMATION MODELING REQUIREMENTS
      c. SECTION 014100 - QUALITY REQUIREMENTS
      d. SECTION 015000 - TEMPORARY FACILITIES AND CONTROLS
      e. SECTION 018113 - SUSTAINABLE DESIGN REQUIREMENTS

B. Section Includes:
   1. Requirements for the Subcontractor’s scheduling and administering of pre-construction meetings and periodic progress meetings.
   2. Specially-called meetings throughout progress of the work.
   3. Meetings for other purposes are covered elsewhere in the Subcontract Documents.

1.2 REQUIREMENTS INCLUDED:

A. Schedule and administer pre-construction/preparatory, weekly progress, field engineering, safety, and specially called meetings throughout progress of the Work.
   1. Prepare and distribute agenda for meetings.
   2. Distribute written notice and agenda of each meeting four (4) working days in advance of the meeting date.
   3. Make physical arrangements for meetings.
   4. Preside at meetings.
   5. Subcontractor will conduct the meetings, prepare minutes, and provide presentation materials; include significant proceedings and decisions.

Items not concluded will be retained on the agenda and in the minutes until conclusion is recorded in subsequent minutes.

a. Unless otherwise specified, the subcontractor shall distribute electronic copies of minutes within two (2) working days of the date of each meeting or teleconference to PPPL to review and comment on, then modify meeting notes accordingly, and distribute to participants in the meeting, to parties affected by decisions made at the meeting, to the PPPL procurement specialist, project manager, construction manager, field construction manager, safety officer, design team, and third party consultants. Subcontractor shall upload all agendas and meeting minutes to the Construction Project Management (CPM) software.

b. Minutes shall include:

i. The title, date, time, and location of the meeting
ii. A list of attendees and contact information
iii. The agenda
iv. A list of major issues, and documentation of resolutions and exchanges
v. Identification of action items, owners, and need by dates,
vi. The effect of the meeting's conclusions relative to subcontract schedule or cost.
vii. All relative attachments reviewed during the meeting

B. Representatives of Subcontractors, lower-tier subcontractors, and suppliers attending meetings shall be qualified and authorized to act on behalf of the entity each represents.

C. PPPL representatives will attend meetings to ascertain that Work is expedited consistent with Subcontract Documents and construction schedules.

1.3 PRE-START SAFETY WALKTHROUGH

A. Immediately after subcontract award and prior to submitting a safety plan, the subcontractor shall attend a pre-start job walkthrough at PPPL to identify any hazards at the job site. All of these hazards, at a minimum, shall be addressed in the Site-Specific Safety Plan Content and Approval Form and the Job Safety Analysis unless otherwise directed by PPPL. The Subcontractor's attendees at this walkthrough shall include, but not necessarily be limited to: the Subcontractor's superintendent, the Subcontractor's manager responsible for Subcontractor's safety program and each of the principal foremen and/or lower-tier Subcontractor superintendents.

1.4 PRE CONSTRUCTION MEETING: PRE-KICK-OFF/KICK-OFF

A. PPPL will schedule a meeting after notice of award and prior to "Notice to Proceed" chaired by the PPPL Procurement Specialist and attended by PPPL PTR.

B. Attendance Required: Subcontractor Key Personnel (Project Manager, Superintendent, Safety Officer, Quality Assurance Professional at a minimum), PPPL identified lower tier subcontractors, and others required.

C. PPPL will record minutes and distribute copies for this meeting.

D. Agenda:
   1. Designation of personnel representing the parties in subcontract
      a. Org Charts
      b. Contact Lists
   2. Distribution of subcontract documents
   3. Review of Notice to Proceed/Kick-Off Deliverables:
      a. Performance and Payment bonds
      b. Certificate of Insurance
      c. Construction Schedule
      d. Schedule of values
      e. Integrated Safety Management (ISM) Plan
      f. Job Hazard Analysis (JHA)
      g. List of chemicals that will be brought on site along with the SDS
      h. Lower-tier subcontractors (of applicable)
      i. OSHA qualified on-site Supervisor
      j. Quality Assurance Program (QAP)
   4. Discussion of procedures and processing of field decisions, submittals, substitutions, applications for payments, proposal request, change orders, request for information, and project closeout procedures
   5. Scheduling, sequence of construction, and scheduling of inspection and testing
   6. Submittals required for mobilization & temporary facilities and controls
   7. Safety overview including site access, training/permit requirements, documentation/planning requirements, meeting requirements, etc.
   8. Procedures for maintaining project record documents (as-builts)
   9. Requirements for start-up of equipment & inspection/acceptance of equipment put into service during construction

1.5 SITE MOBILIZATION MEETINGS

A. Subcontractor shall schedule and preside over meetings prior to mobilizing onto the construction site. This is a readiness review to discuss all permits, sequence and schedule for mobilizing equipment, materials and other facilities necessary for the completion of the Work. The Subcontractor shall prepare a Logistics Plan to discuss construction operations and the impact to the surrounding site. Also see paragraph 1.10.

1.6 PPPL ES&H ORIENTATION

A. See Section 013100 ESH Requirements for information on orientation / training.

1.7 COORDINATION MEETINGS

A. Weekly coordination meetings will be arranged and conducted by the Subcontractor. The purpose of these meetings will be to discuss progress of the work, jobsite safety, coordination issues, and other pertinent project concerns. Subcontractor shall be represented by its project manager and site superintendent. The Subcontractor's

safety professional, quality assurance professional, and MEP Coordinator (and periodically lower-tier subcontractors) shall be available if required by the project or requested by PPPL. Other attendees may include the AE, CM, the PPPL Procurement Specialist, Project Manager, Construction Manager, Field Construction Manager, Field Safety representative, site inspectors, and other PPPL personnel who may be affected by the work.

B. The Subcontractor shall prepare the agenda with coordination from PPPL and preside at meetings.

C. Preparation includes preparation of the 6 week look-ahead by the Subcontractor according to Section 013200 – Schedules and Progress Documentation.

D. Agenda:
   1. Project safety
   2. Review work completed, 6 week look-ahead schedule and project milestones
   3. Review of RFIs, submittals and material lead times
   4. Quality Assurance/Quality Control
   5. Sustainability Requirements
   6. Old Business
   7. New Business

E. CM/GC shall record and prepare minutes of the meetings and distribute copies.

F. End-of-Month Coordination Meeting
   1. The agenda for the last meeting of the month shall include review of overall project CPM schedule (including progress photos), significant issues of the Subcontractor (including safety update), significant issues of PPPL, Change Orders, and the Subcontractor's Application for Payment, a summary of any unresolved disputes and a summary of any new Cost Proposals.

1.8 DAILY TAILGATE/TOOLBOX

A. Subcontractor's Safety Plan shall include daily Tailgate/Toolbox meetings. And the subcontractor shall preside over and record/provide record documents for daily tailgate/toolbox meetings.

B. All workers are required to attend the daily meetings and confirm attendance by signing the daily work plan. No workers shall be allowed to perform any work without signing the daily work plan and prior to completion of the daily Tailgate/Toolbox meeting.

C. Lower-tier subcontractors are expected to participate in the general daily tailgate/toolbox meeting and also have their own daily tailgate/toolbox coordination meeting.

D. Refer to PPPL ESHD 5008 Manual for information on Work Planning and Hazard Analysis. See Section 013100 ESH Requirements for further details.

1.9 SAFETY STEWARDSHIP MEETINGS

A. Weekly meeting lasting approximately 30 minutes to discuss project safety issues from the past week and project safety issues for the coming weeks. In addition, project-to-date trends will be reviewed, including compliance with daily tailgate/toolbox meetings, safety performance review, problem areas, lower-tier subcontractor concerns, shutdown interfaces.

B. Required attendees include subcontractor project manager, superintendent, and safety representative. Additionally, any lower-tier superintendent or foreman as necessary or requested by PTR, PPPL Field Safety representative, PPPL Field Construction Manager, and PPPL Project Manager.

C. Preparation includes provision of the 6 week look-ahead by the Subcontractor according to Section 013200 – Schedules and Progress Documentation.

1.10 LOGISTICS IMPACT MEETING

A. A weekly meeting lasting approximately 30 minutes to discuss site logistics as they relate to ongoing activities at the PPPL. The Subcontractor shall prepare a summary of related project activities, such as roadway or building access disruptions, utility tie-ins/shutdowns, crane locations, etc which impact the overall PPPL site. The Subcontractor shall prepare an updated Logistics Plan to review as required.

B. Required attendees include subcontractor, superintendent, & project manager, and safety representative. Additionally, as requested by PTR, PPPL Field Safety representative, PPPL Field Construction Manager, PPPL Project Manager, PPPL Security, and other impacted PPPL department representatives as necessary.

C. Preparation includes Site Logistics Plan by the subcontractor according to Section 015000 – Temporary Facilities & Controls.

D. The Subcontractor shall be responsible for preparing all documentation for review and shall record and distribute minutes to all appropriate parties.

1.11 COST MEETING

A. A biweekly (or weekly if required) meeting to review summary and status of allowances, forecasted changes, Potential Change Orders, Field Orders and Subcontract Modifications. This meeting may also be utilized to review work in place for monthly requisitions.

B. Required attendees include the subcontractor PM, PPPL PTR/PM, AE, and CM.

C. The Subcontractor shall be responsible for preparing all documentation for review and shall record and distribute minutes to all appropriate parties.

1.12 PRE-INSTALLATION MEETINGS

A. When required in individual specification sections, in the Quality Control Plan, or as determined by PPPL, Subcontractor shall convene a pre-installation/pre-mobilization meeting at the site prior to commencing work of specified section including:

1. BIM Coordination meetings to coordinate utility routing, clash detection, access requirements, etc. Refer to Section 013514 for additional information related to BIM Coordination meetings.
2. Dedicated coordination meetings for any and all utility outages and work planning.
3. Dedicated coordination meetings for Audiovisual installation.
4. Dedicated coordination meetings for Security component installation (cameras, card readers, and door hardware).
5. Dedicated coordination meetings for IT and network cabling component installation.

B. Attendance is required by all personnel directly affecting, or affected by, work of the specific section including sub-tier foremen and superintendents.

C. The Subcontractor shall prepare agenda and preside at meetings to:

1. Review conditions of proposed installation, preparation, and installation procedures.
2. Review coordination with related work of other Subcontractors on site.
3. The Subcontractor shall record and distribute minutes.

1.13 GUARANTEES, BONDS AND SERVICE MAINTENANCE SUBCONTRACT REVIEW MEETINGS

A. Refer to SECTION 017700 CLOSEOUT PROCEDURES

PART 2 PRODUCTS – NOT USED

PART 3 EXECUTION – NOT USED

END OF SECTION 013119

SECTION 013200 – SCHEDULES AND PROGRESS DOCUMENTATION

PART 1 – GENERAL

1.1 SUMMARY

A. The work under this subcontract will be planned, scheduled, executed, and reported using Critical Path Method scheduling (hereinafter referred to as CPM). The Subcontractor's schedule will be developed using a CPM schedule demonstrating complete fulfillment of all subcontract requirements. The schedule shall be kept up-to-date in accordance with the requirements of this section and utilized in planning, coordinating, and performing the work under this subcontract (including all activities of lower-tier Subcontractors, equipment vendors, and suppliers). The schedule will be in precedence format, computer generated and updated, and the controlling schedule document utilized for managing work and measuring performance.

B. It is expressly understood and agreed that the time of beginning, the rate of progress, and the time of completion of the work are of the essence to this Subcontract. Therefore, the primary objectives of CPM scheduling are to insure the adequate planning, scheduling, and execution of the work so it may be performed in an orderly and expeditious manner within the Schedule of Dates stipulated by the Subcontract, to provide optimum coordination of the work, to establish the basis for measuring and monitoring Subcontractor progress, to detect problems for the purpose of taking corrective action, to provide a mechanism or tool for determining and monitoring such corrective actions, and to be used as a tool to assess the Subcontractor's schedule of values, payment schedule, and progress payments.

1.2 DEFINITIONS

A. Activity - Any portion or element of work, action, and/or reaction that is precisely described, readily identifiable, and is a function of a logical sequential process.

B. Subcontractor's Schedule - Represents the approved plan against which the Subcontract's actual performance – with respect to accomplishing the work scope contained within each detail activity – is measured. Once the schedule is approved by PPPL, it will be considered baselined. At this time, the dates, resource costs, activity codes, work breakdown structure, and activity descriptions are no longer subject to any changes unless these changes are part of a documented and approved change control process (see Section 1.12 Schedule Revisions).

C. Float Time - Float is defined as the amount of time between the early start date and the late start date, or the early finish date and the late finish date, of any activity in the project schedule. Total float is defined as the amount of time any given activity or path of activities may be delayed before it will affect the project completion time.

D. Network Diagram - Graphical representation of relationships that exist between different activities of the schedule drawn to time scale.

E. Percentage Complete - The physical percentage of completion, NOT the resource or duration percentage complete.

F. Precedence Method - Scheduling system where relationships between activities are expressed from start or finish of an activity to start or finish of the succeeding activity, with minimal use of lags. Leads shall not be used, and if lags are used they shall be documented in a log. Preference is to create lag activities in lieu of lags.

G. Remaining Duration - Shall be equal to the factor of the remaining percentage of completion times the original as-planned duration, or as approved by PPPL.

H. Supplemental Schedules - Those schedules that are not part of the baseline schedule hierarchy (working and baseline). They are often used for day-to-day operational planning and management, and as the name suggests, simply supplement the baseline and working schedules, but are not under configuration control. This category would include “what-if” schedules generated to evaluate potential alternate management options.

I. Recovery Schedule - A recovery schedule may be required when the critical activities are not being completed on the dates set forth in the schedule. Such a schedule must indicate how the Subcontractor intends to complete the project within the allotted time and what resources will be used to accelerate the work so that it can be completed on time.

J. Critical Path - The critical path is simply all the tasks that determine the end date in your project schedule. PPPL/Project defines the critical path as any activity with 0 days of total float.

K. Schedule of Values - Detailed schedule apportioning of the Subcontract sum among all activities in Subcontract Schedule or PPPL approved equivalent. The Schedule of Values is the basis for all Invoices.

L. Schedule of Dates - The dates that correspond to the relevant Schedule of Values

M. Payment Schedule - The Payment Schedule is the schedule of payments to the Subcontractor in accordance with the approved Schedule of Values

1.3 REQUIREMENTS

A. Software: The Subcontractor shall utilize Oracle Primavera P6

1.4 SUBMITTALS

A. SUBCONTRACTOR’S SCHEDULE: As soon as practical, but no later than fifteen (15) calendar days following Subcontract Award, the Subcontractor shall submit to PPPL a Near Term schedule showing the first ninety (90) days of activity with the remainder of the plan resource loaded in summary activities. In addition, as soon as is practical, but no later than six weeks or forty-two (42) calendar days following Subcontract Award, the Subcontractor shall submit to PPPL a baseline CPM schedule in precedence format for the entire Subcontractor's work scope. In CM/GC project delivery method, schedules shall be submitted with design packages as required by the Statement of Work. The Subcontractor's schedule is a necessary tool for the proper execution of the Subcontractor's work and against which all progress shall be measured. The schedule shall be used by the Subcontractor and PPPL as the basis for evaluating changes and claims and for evaluation of Application for Payment. The Schedule of Values and the

Payment Schedule shall be developed supporting the values indicated in the baseline schedule and organized in accordance with the PPPL WBS.

B. CONTENT: The Near Term and CPM schedules shall meet the following criteria:

1. Subcontractor's schedule shall be cost loaded.
2. The following should be depicted for each activity in the schedule:
    a. Activity descriptions and identification codes shall be clear and concise.
    b. The beginning and end of each activity shall be readily verifiable.
    c. All activity starts and finishes must be tied into the schedule by logical relationships.
    d. Activity numbers shall be unique and shall not change as revisions are made to the schedule.
    e. Activities shall be assigned to a Work Breakdown Structure (WBS) consistent with the PPPL WBS to be provided to the Subcontractor.
    f. Activity codes requirements will be provided to the Subcontractor and may be used to:
        i. Define specific performance responsibility by discipline or Subcontractor;
        ii. Define concise work areas;
        iii. Identify phase of each activity;
        iv. Identify the Specification Section for the activity.
    g. The durations of activities shall be expressed in whole working days.

3. The level of detail of the schedule shall be a function of the complexity of the work involved. Generally, no activity shall have a duration of less than five (5) work days or longer than twenty (20) work days without prior acceptance by PPPL. All durations shall be the result of definitive manpower and resource planning by the Subcontractor and lower-tier Subcontractors to perform the work in consideration of the requirements of the Subcontract. The twenty (20) work day limitation is not intended to impede or restrict Subcontractor's flexibility to properly plan and schedule the work.

4. The schedule shall be based on and incorporate the WBS and provide a complete and detailed sequence of operations of the work within the time limits specified in the subcontract including specified mandatory milestone dates.

5. Schedules shall indicate the relationship and time frames in which the various phases of the work will be made substantially complete and turned over to PPPL as indicated in the Subcontract Documents.

6. Schedules shall depict major components of the work and the sequenced relationships between major components and subdivisions of major components.
7. For all major equipment and materials fabricated or supplied for this project, the schedule shall show a sequence of activities including:

    a. Preparation & submission of all submittals (NTP, project specific plans, safety, product data, shop drawings, samples, etc.)
    b. Review and approval of all submittals (allow for one resubmittal)
    c. Material lead-times
    d. Shop fabrication
    e. Testing of equipment and materials

f. Delivery
g. QA/QC Inspections (including mock-ups)
h. Preparation, submission, review, and approval of Close-out documents (warranties and O&M manuals)
i. Training

8. Responsibility Codes for each activity shall be identified with a single performing organization including interfaces with the work of other Subcontractors or agencies (i.e., Subcontractor, lower-tier Subcontractors, suppliers, utilities, PPPL agencies, etc.).

9. Total planned value shall equal the total subcontract value. Activities shall have an assigned dollar value (cost loading) that shall cumulatively equal the subcontract Amount and be consistent with the Schedule of Values.

10. The Subcontractor may be required to provide Resource requirements (labor by craft, commodity, type, equipment, services, etc.) for activities at or near the critical path. However, at a minimum the Subcontractor shall be prepared to discuss Resource requirements for activities at or near the critical path.

11. All scheduled activities shall utilize a five (5) day, one shift, work week calendar including established holidays, unless approved by PPPL otherwise. All work to be scheduled on a different calendar, to account for weather – for example, must be pre-approved by PPPL and the calendar(s) must be provided to PPPL for incorporation into PPPL's Primavera P6 system.

12. The schedule shall indicate a completion date for the project that is no later than the project's required completion date. All activity durations shall be given in work days.

13. The schedule shall contain constraints only for Subcontract milestones including the completion date. No other constraints shall be used.

14. Monthly anticipated adverse weather days and holidays shall be considered and included in the planning and scheduling of all weather affected work activities in order to complete all work within the Subcontract Dates. Projected days lost shall be based on historical data available from the National Oceanic and Atmospheric Administration (NOAA) for the region.

15. Subcontractor shall ensure that the schedule represents an accurate, efficient, reasonable, and feasible plan and method for accomplishing the work throughout the time of performance. While PPPL will review the Subcontractor's Schedule, the schedule itself is the Subcontractor's, who has full responsibility for its preparation, content, revisions, and updating in accordance with the requirements of the Subcontract Documents.

16. Abbreviations shall be fully described in a legend attached to the schedule.

1.5 FORM:

A. The Subcontractor shall submit the schedule in native electronic data form, utilizing Primavera P6 scheduling software. The Subcontractor shall have the capability of furnishing data reports or sorts using formats/layouts acceptable to PPPL.

1.6 REVIEW AND ACCEPTANCE

A. PPPL will review the Subcontractor's schedule submittals, for sound logic, constructability, and adherence to plans and specifications. PPPL will assume that the Subcontractor has accurately correlated the schedule in its entirety and its individual elements to the Subcontractor's and lower-tier Subcontractor's as-bid project execution plans including but not limited to subcontracts, purchase orders, delivery dates, crew strategies, etc. Non-conformance of any of the Subcontractor or lower-tier Subcontractor project execution plans to the Subcontract Documents shall not relieve the Subcontractor or his lower-tier Subcontractor of any of the requirements of the Subcontract Documents. Acceptance by PPPL of the Subcontractor's schedule is advisory only and shall not relieve the Subcontractor of the responsibility for accomplishing the work within all of the subcontract requirements, including but not limited to, each and every subcontract-required Milestone and Completion date. Omissions and errors in the approved schedule shall not excuse performance which is not in compliance with the subcontract. Approval by PPPL in no way makes PPPL an insurer of the schedule's success or liable for time or cost overruns flowing from its shortcomings. PPPL hereby disclaim any obligation or liability by reason of PPPL’s approval of or acquiescence to the schedule.

1.7 WEEKLY PROGRESS MEETING

A. As necessary, normally no more than once every week, a meeting will be held to assess the progress achieved by the Subcontractor during the previous work period. The Subcontractor shall submit a bar chart indicating progress for the previous period (the Weekly Schedule Report) and the activities scheduled for the succeeding 3 periods [the Short Interval Schedule (SIS)].

1.8 PROGRESS REPORTING, CHANGES AND THE USE OF FLOAT

A. Once each month, one week prior to the end of the month or as agreed by PPPL, prepare and submit to PPPL a working schedule file (XER or other approved schedule software format) showing current progress and all approved changes in network logic, including but not limited to changes as the result of change orders and any changes in subcontract completion dates which have been approved by PPPL within this section since the last revision of the baseline schedule. If the Subcontractor desires to make changes in its method of operating and scheduling, the Subcontractor shall follow the procedures set out in Paragraph 1.12, Schedule Revisions of this Section.

B. In addition to the foregoing, submit a typewritten narrative report once each month with the updated schedule describing the actual work accomplished during the reporting period and causes of any variances from the baseline (positive or negative).

C. In addition to the above, the Subcontractor may be required to submit at PPPL’s request, a Critical Items Report - The Critical Items Report identifying items by cause and impact that are or will seriously affect the Subcontractor's progress or ability to perform work in accordance with the current schedule.

D. Float is not time for the exclusive use or benefit of either PPPL or the Subcontractor, but must be used in the best interest of completing the project on time. Once an activity has been started the Subcontractor shall continue the work on that activity as scheduled until it is completed unless discussed with and approved by PPPL. PPPL reserves the right to request a report on such circumstances that delay the start of activities or require their

suspension. PPPL may require the Subcontractor to proceed utilizing the early dates if the circumstances are not justified in PPPL's opinion. PPPL will not be responsible for delays caused by the Subcontractor not proceeding per the early start dates or suspending the work if such actions were within Subcontractor's control. If in PPPL''s opinion the delay/suspension is justifiable, PPPL shall authorize the Subcontractor to prepare a supplemental or 'what if' schedule for PPPL's approval including all activities in the Subcontractor schedule necessary to clearly represent the impact. Pursuant to the float sharing requirements as set forth above, (1), use of float suppression techniques such as preferential sequencing, special lead lag logic restraints, extended Activity durations or imposed dates and (2), use of float time disclosed or implied by the use of alternate float suppression techniques shall be shared to the proportionate benefit of PPPL, PPPL's CM, Architect, Engineer, Subcontractors, and lower-tier Subcontractors. The Subcontractor shall revise the Subcontractor's schedule as reasonable to mitigate the impact of delays to the project with no change in subcontract price. Extensions of time for performance required under the General Requirements pertaining to equitable time adjustment will be granted only to the extent that the equitable time adjustment exceeds total float in the activity or path of activities affected at the time notice to proceed was issued for the change.

1.9 PAYMENTS TO SUBCONTRACTOR

A. If the Subcontractor fails to submit the interim or baseline schedule within the time prescribed, or the updated monthly schedule within the time requested, PPPL may withhold approval of progress payment estimates until such time as the Subcontractor submits the required schedules.

1.10 RESPONSIBILITY FOR COMPLETION

A. If, in the opinion of PPPL, the Subcontractor falls behind the progress schedule, the Subcontractor shall take any and all steps necessary to improve Subcontractor's progress at no additional cost to PPPL, including cost impacts to other Subcontractors, agencies or PPPL directly caused by Subcontractor's delay, including but not limited to the increase manpower in such quantities and crafts as will substantially eliminate the lag in scheduled progress, increase the number of working hours per shift, shifts per working day, working days per week, or the amount of equipment, or any combination of the foregoing, sufficiently to substantially eliminate lag in scheduled progress; and/or resequencing of activities to achieve maximum practical concurrent accomplishment of work activities.

B. PPPL may also require the Subcontractor to submit for approval and at no additional cost to PPPL such supplementary Recovery Schedules as may be deemed necessary to demonstrate the manner in which the approved schedule will be regained during the immediate subsequent pay period or, if PPPL agrees, subsequent pay periods.

C. PPPL and the Subcontractor will meet for the normal monthly update to determine the effectiveness of the Recovery Schedule and to determine whether the Subcontractor has regained compliance with the schedule. If, in the opinion of PPPL, the Subcontractor is still behind schedule, the Subcontractor, in conjunction with PPPL will prepare another Recovery Schedule at the Subcontractor's expense, to take effect during the immediate subsequent pay period or, if PPPL agrees, subsequent pay periods.

D. Failure of the Subcontractor to comply with the requirements of PPPL under this paragraph shall be grounds for determination by PPPL that the Subcontractor is not

executing the work with such diligence as will ensure completion within the time specified. Upon such determination PPPL may terminate the Subcontractor's right to proceed with the work, or any separable part thereof, in accordance with the applicable provisions of the subcontract.

1.11 SCHEDULE REVISIONS

A. Should the Subcontractor, desire to change the Subcontractor's plan, the Subcontractor shall submit the Subcontractor's requested revisions to PPPL along with a written statement of the revisions including a description of the logic for rescheduling the work, methods of maintaining adherence to intermediate milestones, specific dates, and the reasons for the revisions. If the requested changes are acceptable to PPPL, they will be incorporated into the schedule in the next reporting period, unless a subcontract modification is required. No change to historical (previous period) data is allowed.

B. When PPPL orders changes which have the potential to substantially impact the schedule of Dates, a supplemental ‘what if’ schedule of the new and existing activities directly affected by the change will be prepared by the Subcontractor and provided with the Subcontractor's price proposal or extra work tabulation as required to PPPL for concurrence or revision as PPPL deems necessary. After the schedule has been approved by PPPL, it will be incorporated into the schedule by the Subcontractor. Change Order logic will affect only those activities and performance dates directly concerned. Adjustments in scheduled intermediate completion dates or for the Subcontract as a whole, will be considered only to the extent that there is insufficient remaining float to absorb these changes. No changes will be made to historical (previous period) data.

C. Neither the updating or revision of the Subcontractor's schedule, nor the submission, updating, change or revision of any report or schedule submitted to PPPL by Subcontractor under this Section, nor PPPL’s review or nonexistence of any such report or schedule shall have the effect of amending or modifying, in any way, the Schedule of Dates or modifying or limiting in any way Subcontractor's obligations under this subcontract.

D. Should the Subcontractor desire to change logic links in Subcontractor's plan, Subcontractor will clearly notify PPPL of each individual change.

1.12 TIME OF COMPLETION

A. If the Subcontractor should desire or intend to complete the work earlier than the Schedule of Dates, PPPL shall not be liable to the Subcontractor for any costs or other damages should the Subcontractor be unable to complete the work before the early milestone completion dates or before the Schedule of Dates. Approval by PPPL of a schedule which indicates completion prior to Schedule of Dates shall be for the convenience of the Subcontractor and shall not change any of the subcontract requirements including but not limited to the Schedule of Dates. All subcontract requirements shall be consistent with and applicable only to the completion of the work in accordance with the Schedule of Dates required by the subcontract unless PPPL and Subcontractor otherwise agree by issuance of the appropriate Change Order.

B. The updated schedule submitted by Subcontractor shall not show a completion date later than the Schedule of Dates, subject to any time extensions approved by PPPL. If Subcontractor believes Subcontractor is entitled to an extension to the Schedule of Dates under the Subcontract Documents; the Subcontractor shall submit to PPPL a

formal time extension request as required by the subcontract and a detailed narrative justifying the time extension requested. The time extension request shall include schedule forecasts that predict the actual Project Completion Date, and any separable portions thereof specified by PPPL plus a forecast of the actual achievement of any milestones listed in the Schedule of Dates. PPPL shall have no obligation to consider any time extension request unless the Subcontractor is in full compliance with the requirements of the Subcontract Documents. PPPL shall not be responsible or liable to Subcontractor for any constructive acceleration due to failure of PPPL to grant time extensions under the Subcontract Documents or should the Subcontractor fail to substantially comply with the submission requirements and the justification requirements of this subcontract for time extension requests. The Subcontractor's failure to perform in accordance with the schedule shall not be excused, nor be chargeable to PPPL because Subcontractor has submitted time extension requests.

1.13 COORDINATION

A. The Subcontractor shall coordinate the Subcontractor's work and that of lower-tier Subcontractors in maintaining orderly progress toward completion of the work as scheduled. Coordination shall include entities which may be under subcontract with PPPL such as the A/E, C/M, Independent Testing Agency, etc.

B. The Subcontractor shall coordinate the Subcontractor’s work and that of lower-tier Subcontractors with other subcontractors on PPPL’s site (separate projects, maintenance, etc.) in maintaining orderly progress toward completion of the work as scheduled.

C. Failure of lower-tier Subcontractors to meet their schedule shall not be justification for an extension of time, except where such failure causes, in the opinion of PPPL, an unreasonable delay in the Subcontractor's Work, in which case the provisions regarding extensions of time and extra work shall apply.

1.14 FAILURE TO PERFORM SCHEDULING TASKS

A. Should the Subcontractor fail to substantially comply with the provisions of the Subcontract Documents relating to planning, scheduling, reporting, and execution of the work as called for in the schedule, PPPL shall have the right, at PPPL’s option, to retain the services of scheduling consultants or experts (including attorneys, if necessary in PPPL's opinion) to prepare a schedule in accordance with the Subcontract Documents. This schedule will be used to allow PPPL to evaluate the program of the Work by the Subcontractor and to determine whether the Subcontractor is substantially complying with the Subcontract Documents. This schedule will be used to direct such action on the part of the Subcontractor as permitted by the Subcontract Documents, and as required to ensure that the Subcontractor will comply with such schedule. All costs incurred by PPPL in preparing this schedule shall be charged to the Subcontractor's account and/or deducted from any monies due to the Subcontractor. If the Subcontractor fails to substantially comply with the scheduling and execution of the work requirements of the Subcontract Documents, the Subcontractor hereby agrees in such instance to comply with such schedules as PPPL develops or direct activity sequences and durations as PPPL may reasonably require, without additional cost to PPPL (subject only to cost adjustments for such changes in the work as PPPL may direct), to ensure completion within the Schedule of Dates.

1.15 SUBCONTRACTOR COVENANTS AND GUARANTEES

A. Subcontractor covenants and guarantees that Subcontractor will not:
   1. Misrepresent to PPPL its schedule and all of its components or Subcontractor's actual execution of the work.
   2. Utilize schedules materially different from those made available by Subcontractor to PPPL, any lower-tier Subcontractor or separate Subcontractors for the direction, execution, or coordination of the Work.
   3. Prepare schedules, updates, revisions or reports for the work which are not feasible or realistic; or which do not accurately reflect the actual intent or reasonable and actual expectations of the Subcontractor and its lower-tier Subcontractors.

1.16 PAYMENT FOR SCHEDULING

A. The cost of performing all work incidental to preparing the Project Construction Schedule, including preparation, execution, updates, maintenance, revisions, and adjustments shall be included in the Bid Proposal.

PART 2 – PRODUCTS
(Not Applicable)

PART 3 – EXECUTION
(Not Applicable)

END OF SECTION 013200

SECTION 013300 – SUBMITTALS

PART 1 GENERAL

1.1 PURPOSE

A. This section specifics general methods and requirements of project work related submittals. Additionally detailed submittal requirements may be found in the bid design and/or technical specification sections.

1.2 REFERENCES

A. The publications listed below form a part of this subcontract to the extent referenced.
1. PPPL General Provisions for Non-Commercial Subcontracts Part G - Provisions & Conditions for Fixed Price Construction Subcontracts
2. 010010 General Requirements
3. 013100 Environment, Safety, and Health Requirements

1.3 SHOP DRAWINGS, MATERIAL SAMPLES, AND MOCKUPS

A. Subcontractor shall provide submittals for all materials and assemblies used on the project which are normally required in the construction industry.

B. The Subcontractor shall submit a submittal log prior to the start of construction. This log shall include all submittals required for construction of the project and show anticipated submission dates that allow sufficient time for review and approval. The subcontractor will re-submit this log if any changes are necessary during construction.

C. The Subcontractor shall supply three (3) copies of all operation and maintenance manuals and an electronic PDF version for equipment furnished by the Subcontractor or their lower-tier subcontractor prior to final acceptance of the project by PPPL.

D. The Subcontractor shall complete the "Submittal for Review Transmittal" (or similar using Construction Management Software) form for each submittal, reference Attachment A.

E. Shop drawings of all trades shall be submitted by the Subcontractor, who shall indicate by signed stamp on the drawings, that he has checked the shop drawings and that the work shown is in accordance with the Subcontract requirements and that it has been checked for dimension and relationship with the work of all trades. These drawings shall be submitted not later than the approved schedule for shop drawing transmittal, shall contain all required detailed information, and shall be submitted in logical order to facilitate proper review.

F. Shop drawings which relate to a complete assembly, structure or system with various parts shall not be submitted until all drawings required to completely detail that assembly, structure or system are complete, so that they may be checked in relation to each other.

G. Mockups: Before installing portions of the Work requiring mockups, build mockups for each form of construction and finish required to comply with the following requirements, using materials indicated for the completed Work:

1. Scope and sizes of mock-ups are shown on the Subcontract Drawings and Technical Specifications;

2. Construct mockups in location and orientation at Project site approved by PPPL;

3. Mockups shall be complete in all respects and shall represent the final completed assembly as required by the Work. This includes structural framework assemblies as required to support and display the mockups;

4. Do not release materials for fabrication for the remainder of the Work until mockups are approved;

5. The subcontractor shall be responsible for revisions to the mockup as directed by the AE or PPPL;

6. Mockups shall remain in place until during construction in an undisturbed condition as a standard for judging the completed work;

7. Demolish and remove mockups and restore areas as directed by PPPL.

1.4 AS-BUILT DRAWINGS

A. As specified in PPPL General Provisions and Special Terms and Conditions for Construction Subcontracts, the Subcontractor shall furnish As-Built drawings as outlined below.

B. The Subcontractor shall maintain a set of prints of the Subcontract drawings in the construction office at the job site. An electronic set of prints may be utilized with PPPL approval. A daily record, in red, shall be kept on these prints of the work installed with all modifications or changes thereon. This set of prints shall be available to PPPL for inspection at all times and print copies provided upon request to PPPL within 24 hours of the request being made. PPPL (and/or the CM) and the Subcontractor's superintendent shall review the as-built drawings, and the scheduling of a progress payment update will be contingent on whether the as-built mark-up set is current and accurate.

C. Upon completion of all work, the Subcontractor shall transmit the above set of marked-up prints to PPPL (or the AE). Prior to submittal, the Subcontractor shall sign them to indicate that the work was installed as shown thereon.

1.5 PHOTOGRAPHIC DOCUMENTATION

A. Submittals

1. Identification: Label each print with the following:

    a. Name of Project;

    b. Name of Subcontractor;

    c. Date photograph was taken;

    d. Description of vantage point, indicating location, direction (by compass point) and elevation or story of construction

2. Digital Copies: Submit a complete set of electronic digital files monthly. Identify date photographs were taken.

3. Digital Images: Provide images in JPG format, color, minimum 2 Mb file size, produced by a digital camera with minimum sensor size of 8 megapixels, and at an image resolution of not less than 1600 by 1200 pixels and 400 dpi, in unaltered original files, with same aspect ratio as the sensor, uncropped, date- and time- stamped.

B. General: Take photographs using the maximum range of depth of field, and that are in focus, to clearly show the Work. Photographs with blurry or out-of-focus areas will not be accepted.

1. Provide a key plan with each set of construction photographs that identifies each photographic location

C. Digital Images: Submit digital images exactly as originally recorded in the digital camera, without alteration, manipulation, editing, or modifications using image-editing software.

1. Date and Time: Include date and time in file name for each image.

2. Field Office Images: Maintain one set of images accessible in the field office at Project site, available at all times for reference. Identify images in the same manner as those submitted to the Program Manager.

D. Preconstruction Photographs: Before commencement of excavation, demolition or starting construction, take photographs of Project site and surrounding properties, including existing items to remain during construction, from different vantage points, as directed by the PTR

1. Flag construction limits before taking construction photographs.

2. Take sufficient number of photographs, but not less than 20, to show existing conditions adjacent to property before starting the Work.

3. Take sufficient number of photographs, but not less than 20, of existing buildings either on or adjoining property to accurately record physical conditions at start of construction.

4. Take additional photographs as required to record settlement or cracking of adjacent structures, pavements, and improvements.

5. The Subcontractor is responsible for photographically documenting all conditions of the project site(s) prior to construction. Any claim made by PPPL or other parties that damage was caused as a result of construction or the Subcontractor's activities will be assumed to be valid unless there is photographic documentation that the condition existed prior to construction.

E. Periodic Construction Photographs: Take a sufficient number of photographs, but not less than 20, weekly, with timing each month adjusted to coincide with the cutoff date associated with each Application for Payment. Select vantage points to show status of construction and progress since last photographs were taken. In addition to the weekly periodic construction photographs, photographs should be taken at the following events:

1. Commencement of the Work.

2. Discovery of any unforeseen conditions or obstructions, such as utilities, boulders, debris, differing soil conditions, water, etc.

3. Completion of excavation.

4. Completion of subgrade construction.

5. Completion of placement of base courses.

6. Completion of forming.

7. Completion of reinforcement placement.

8. Completion of placement of asphalt paving or concrete.

9. Above-grade structural framing

10. Exterior building enclosure.

11. Prior to enclosing or sealing any element of work from view.

12. Interior Work, through date of Substantial Completion

F. Final Completion Construction Photographs: Take a sufficient number of photographs, but not less than 20, after date of Substantial Completion for submission as Project Record Documents. PTR will inform photographer of desired vantage points.

1.6 PPPL'S ROLES

A. PPPL shall be the final judge of the acceptability of any and all drawings. Final acceptance of the work is contingent on the receipt and PPPL's approval of the complete set of As-Built prints.

B. The Following distribution of drawing and technical specifications will be made by PPPL to the Subcontractor:
   1. After the award of the Subcontract and prior to Notice to Proceed, PPPL will make available one electronic set of conformed Drawings and Technical Specifications;
   2. For any change as defined under PPPL General Provisions and Special Terms and Conditions for Construction Subcontracts, PPPL will make available one electronic set of revised Drawings and/or Technical Specifications;
   3. Hard copies of either the conformed bid documents or any revised documents issued under the "Changes" provision shall be the responsibility of the Subcontractor.

1.7 USING THE SUBMITTAL FOR REVIEW FORM

A. Submittals shall be submitted to PPPL (or the AE).

1.8 SUBMITTAL PROCESS

A. PPPL (or the AE) will review the submittals and return one copy of all submittals within 10 (ten) working days with one of the following actions:

   1. "No Exceptions Taken" response on the Subcontractor's submittal or on the Submittal for Review Transmittal form indicates the Subcontractor may proceed with procurement, fabrication, manufacture and installation of the material and/or product without any comments to address

   2. "Make Corrections Noted" response on the Subcontractor's or on the "Submittal for Review Transmittal" form indicates the Subcontractor may proceed with procurement, fabrication, manufacture and installation of the material and/or product assuming the noted items on the submittal are incorporated into the final design and/or product;

   3. "Revise & Resubmit" on the Subcontractor's submittal or on the Submittal for Review Transmittal form indicates the Subcontractor may not proceed with procurement, fabrication, and manufacture of the material and/or product. The Subcontractor will revise the submittal and resubmit to PPPL (or the AE) for approval and will not be able to erect and/or install any material until he has received either the "No Exceptions Taken" or the "Make Corrections Noted" action by PPPL (or the AE);

   4. With the "Rejected" response on the Subcontractor's shop drawing submittal or on the "Submittal for Review Transmittal" form, the reasons for the disapproval will be stated on the submittal. The Subcontractor will revise the submittal to conform to the drawings and specifications and resubmit them to PPPL (or the AE) for approval. No procurement, fabrication, manufacture or installation shall be performed by the Subcontractor until one of the above actions listed above is received;

   5. "For Record Only" response on the Subcontractor's shop drawing submittal or on the "Submittal for Review Transmittal" form acknowledges

receipt of such items as test results, professional engineering calculations, welding certificates and inspection reports.

1.9 MATERIAL SUBSTITUTIONS

A. Products or materials which are equal to, or the equivalent of, those specified will be considered for approval by PPPL (or the AE). The submittal procedure will be the same as above, with the following additional conditions: In addition to the required information for the substitute material, the submittal shall contain the same information pertaining to a product named in the specification for purposes of comparison.
   1. The submittal shall explain fully the differences, if any, between the proposed product and those named in the specification.
   2. Any change to the drawings or specifications for related work required for proper installation of the proposed substitute product shall be indicated in the submittal.

B. If the proposed substitute product requires alterations of any kind to equipment the building, or necessitates any engineering design changes for its proper installation, such alterations and engineering design changes shall be accomplished at no cost to PPPL.

PART 2 PRODUCTS – Not Used

PART 3 EXECUTION – Not Used

END OF SECTION 013300

PPPL PRINCETON PLASMA PHYSICS LABORATORY

ATTACHMENT A

PPPL PRINCETON PLASMA PHYSICS LABORATORY

Submittal for Review Transmittal

TO: Princeton Plasma Physics Laboratory
P.O. Box 451
Princeton, NJ 08540

Submittal No.:<XX-XX-XXX-XXX>

FROM: <Subcontractor's Name>
<Subcontractor's Address>
<Subcontractor's City and State>

Date:	Project No.: <XX-XX-XXX>
Attn:	
Project Name:	

Submittal Item No.	Specification Section or Drawing No. (If Applicable)	Description
01		
02		
03		
04		
05		
06		
07		
08		
09		
10		
11		

Comments:

SECTION 013514 – BUILDING INFORMATION MODELING (BIM) REQUIREMENTS

PART 1 GENERAL

1.1. PURPOSE

A. Section Includes:
   1. Use of the Building Information Model (BIM) by the Subcontractor during the preliminary, final design and construction phases.
   2. BIM's administrative requirements.
   3. Procedures for incorporating project information to the BIM.
   4. Procedures for transferring the BIM to PPPL at Project Closeout.

1.2. REFERENCES

A. General: The following document forms part of the subcontract. Bring conflicts between subcontract documents to the attention of the Project Manager, in writing, for resolution before taking any related action. Where differences exist between codes and standards, the one affording the greatest protection shall apply.
   1. Comply with standards in effect as of date of the Subcontract Documents unless otherwise indicated.

B. National Institute of Building Sciences:
   1. National BIM Standard – United States (NBIMS-US), Version 3, 2015

1.3. DEFINITIONS

A. BIM: Combined BIM Model prepared and developed to levels of detail as specified below by the A/E. Buildings and structures including all Objects shall be modeled as specific systems or assemblies with accurate quantities, recognizable configuration, spatial location, and an accurate relationship to datum and grids. Each enclosed space shall be identified as a unique Room with associated parameters.

B. BIM Coordinator – The individual to host and manage the BIM coordination process assisting the BIM Manager.

C. BIM Manager - The individual responsible for managing all aspects of the A/E's 0BIM processes, including managing design BIM preparation, the A/E's BIM Staff, lower-tier subcontractor use of BIM, clash detection, and all other aspects of the A/E's BIM requirements.

D. LOD300 - This is the "design" level of development (LOD) and shall be applicable for preliminary and final design submittals. Buildings and structures including all Objects shall be modeled as specific systems or assemblies with accurate quantities, recognizable configuration, spatial location, and an accurate relationship to datum and grids. Each enclosed space shall be identified as a unique Room with associated parameters. Each equipment item shall be identified as a unique item with associated

parameters.

E. LOD400 - This is the “construction” level of development (LOD) and shall be applicable for final design submittals. Buildings and structures including all objects would be modeled as specific systems or assemblies with accurate quantities, recognizable configuration, spatial location, and orientation, with complete fabrication, assembly, and detailing information to provide shop drawing level of information. Each enclosed space shall be identified as a unique Room with associated parameters. Each equipment item shall be identified as a unique item with associated parameters.

F. LOD500 - This is the “as-built” level of development (LOD) . Buildings and/or structures including all objects shall be modeled as constructed systems or assemblies with accurate quantities, shape, spatial location, and orientation, with complete fabrication, assembly, and detailing information. Each enclosed space shall be identified as a unique room with associated parameters. Objects shall include the data needed to operate and maintain the facility without the need for traditional drawings and operation and maintenance manuals when developed as RECORD models. Each equipment item shall be identified as a unique item with associated parameters.

G. Definitions of other terms used in this section shall be the ones in the National BIM Standard – United States, Version 3, 2015.

1.4. CLOSEOUT SUBMITTALS

A. BIM Operation and Maintenance Data.

B. All model files for the as-built or record documents, on magnetic media or compact disk.

1.5. QUALITY ASSURANCE

A. Architect's Digital Data Files: Electronic digital data files of the Subcontract Drawings shall be provided by the A/E for Subcontractor) used in preparing submittals, shop drawings, and their other uses. This use is conditional to the signing of a data licensing agreement between the A/E and Subcontractor defining the rights and responsibilities of the parties.
    1. Agreement Form: AIA Document E202, Building Information Modeling Protocol Exhibit, 2008
    2. Digital Data Software Program: Drawings in Revit 2017
    3. Architect makes no representations as to the accuracy or completeness of digital files as they relate to subcontract Drawings

B. Use of the BIM for Submittals: At the discretion of the Subcontractor, construction lower-tier subcontractors will develop and incorporate Shop Drawing and other submittal information files into the Building Information Model established for the Project.
    1. Prepare Shop Drawings in the same digital data software program, version, and operating system as the original Drawings. If such software

is not available for a specific discipline, the software used shall be compliant with IFC 2x3 standards.

C. Use of the BIM Delegated Design Services: Incorporate delegated-design drawing and data files into Building Information Model established for Project.
   1. Prepare delegated-design drawings in the same digital data software program, version, and operating system as the original Drawings. If such software is not available for a specific discipline, the software used shall be compliant with IFC 2x3 standards.
   2. The entity identified in the data licensing agreement will incorporate the changes into the Model.
   3. The A/E will furnish the Subcontractor one set of digital data files of Drawings for use in construction lower-tier subcontractors preparing coordination digital data files.

D. Use of the BIM for the preparation of Design Coordination Drawings:
   1. Coordination Digital Data Files: Prepare design coordination digital data files according to the following requirements:
      a. File Preparation Format: Same digital data software program, version, and operating system as original Drawings.
      b. BIM File Incorporation: Develop and incorporate coordination drawing files into the Building Information Model established for the Project.
      c. Perform three-dimensional component conflict analysis as part of preparation of coordination drawings. Resolve component conflicts prior to any submittal package to PPPL. Indicate where conflict resolution requires modification of design requirements by PPPL.
      d. A/E will assist the Subcontractor in resolving construction lower-tier subcontractors' component conflicts during the design period.

1.6. BIM PROJECT MEETINGS

A. BIM Preconstruction Conference: Schedule and conduct a pre construction BIM conference before construction at a date to be determined and at a time convenient to PPPL, A/E, Subcontractor and other attendees.
   1. Conduct the conference to review responsibilities and personnel assignments.
   2. Attendees: Authorized representatives of PPPL, PPPL's Commissioning Authority, Subcontractor and their consultants; A/E's BIM Coordinator and BIM Manager; and other concerned parties shall attend the conference. Participants at the conference shall be familiar with the Project and authorized to conclude matters relating to the Work.
   3. Agenda: Discuss items of significance that could affect progress, including the following:
      a. Tentative BIM progress schedule.
      b. Review of the BIM level development and quality of the information at the time of the transfer of the BIM from the A/E to the Subcontractor.
      c. Phasing.

d. Designation of key personnel and their duties.
e. Lines of communications.
f. Procedures for recording and incorporating the following information to the BIM during the construction process.
    1) Field conditions.
    2) Changes to the Work.
    3) Testing and inspecting.
    4) Specialties, equipment, furniture, and other items incorporated in the work.
    5) Maintenance building requirements, such as clearances.
    6) Commissioning requirements.
    7) Sustainability requirements.
g. Procedures for recording and incorporating the following information to the BIM at project closeout.
    1) Maintenance and operation data.
    2) Demonstration and training data
    3) Incorporation of record documents data.
    4) Commissioning requirements.
    5) Operational sustainability requirements.
h. Procedures for RFIs regarding BIM maintenance and updating.
i. Procedures for processing information by PPPL, A&E, Subcontractor, and other entities.
j. BIM Submittal procedures.
k. Preparation of BIM information.
l. BIM restrictions.
m. Procedures for delivering the BIM to PPPL.
n. Coordination of information input with Project Schedule.
o. Progress maintenance and updating of the BIM.

4. Reporting: Record meeting results and distribute copies to everyone in attendance and to others affected by decisions or actions resulting from each meeting.

B. BIM Coordination Meetings: Schedule and conduct BIM coordination meetings at monthly intervals at a minimum. Project BIM coordination meetings are in addition to specific meetings held for other purposes, such as Project progress meetings and Project conferences.

1. Attendees: The Subcontractor shall require the BIM Manager and lower-tier subcontractors regularly attend all coordination meetings. In addition the Subcontractor shall call upon representatives of PPPL's Facility Engineering and Subject Matter Expert (SME) staff, PPPL's Maintenance and Operations (M&O), PPPL's Commissioning Authority, A/E's BIM Coordinator and BIM Manager, and all entities concerned with current progress or involved in planning, coordination, or performance of future activities which affect the incorporation or use of the BIM shall be represented at these meetings. All participants at the meetings shall be familiar with Project, the Project BIM, and authorized to conclude matters relating to the Work.
2. Agenda: Review and correct or approve minutes of the previous coordination meeting. Review other items of significance that could affect progress. Include topics for discussion as appropriate to the status of the

Project.

a. Coordination with Subcontractor's Construction Schedule: Review progress since the last coordination meeting. Determine whether information is incorporated on time, ahead of schedule, or behind schedule, in relation to combined Subcontractor's construction schedule. Determine how information incorporation behind schedule will be expedited; secure commitments from parties involved to do so. Discuss whether procedure revisions are required to ensure that current and subsequent activities will be incorporated so that the Project BIM schedule is not delayed.
b. Schedule Updating: Revise Project BIM schedule after each coordination meeting where revisions to the schedule have been made or recognized. Issue revised schedule concurrently with report of each meeting.
c. Review present and future needs of each contractor present, including the following:
    1) Resolution of BIM component conflicts.
    2) Status of information incorporation.

3. Reporting: Record meeting results and distribute copies to everyone in attendance and to others affected by decisions or actions resulting from each meeting.

C. BIM Project Closeout Conference: Schedule and facilitate a project closeout conference, at a time to be determined and convenient to PPPL, the A/E, and the Subcontractor.
1. Conduct the conference to review requirements and responsibilities related to the BIM completion and delivery to PPPL at Project closeout.
2. Attendees: In addition to representatives of PPPL's SME, PPPL M&O, PPPL's Commissioning Authority, A/E's BIM Coordinator and BIM Manager, the Subcontractor and their lower-tier subcontractors; suppliers; and other concerned parties shall attend the meeting. Participants at the meeting shall be familiar with Project and authorized to conclude matters relating to the BIM.
3. Agenda: Discuss items of significance that could affect or delay Project closeout, including the following:
    a. Preparation of BIM record information.
    b. Procedures required prior to inspection of the BIM for Substantial Completion and for final inspection for acceptance.
    c. Requirements for incorporating the operations and maintenance data.
    d. Requirements for incorporating the material samples, attic stock, and spare parts data.
    e. Requirements for incorporating the demonstration and training data.
    f. PPPL's maintenance requirements data.
4. Reporting: Record meeting results and distribute copies to everyone in attendance and to others affected by decisions or actions resulting from each meeting.

PART 2 PRODUCTS

1.1. BIM FILES

A. The A/E will furnish the Subcontractor one set of digital data files of the Subcontract Drawings for use in recording information.
   1. Review the BIM with PPPL and the A/E.
   2. Report to PPPL issues of discrepancies, incomplete information, and other conflicts for resolution before using or modifying the BIM.

B. Unless otherwise directed by PPPL, include information and modifications to the BIM model using the same electronic program.

C. A/E to provide PPPL and Subcontractor with symbol libraries and families used in the development of the completed BIM model.

D. BIM model shall utilize Autodesk's REVIT 2017 software platform. For systems where REVIT 2017 software is not available, IFC 2x3 compatible software may be used after PPPL review and approval.

PART 3 EXECUTION

2.1. PROJECT RECORD DOCUMENTS DATA

A. Record Digital Data Files: Immediately before inspection for Certificate of Substantial Completion, review marked-up record prints with PPPL and the A/E. A full set of corrected digital data files of the Subcontract Drawings shall be prepared, as follows:
   1. Format: Same digital data software program, version, and operating system as the original Subcontract Drawings.
   2. Incorporate changes and additional information previously marked on record prints. Delete, redraw, and add details and notations where applicable.
      a. Record markups in separate layers.
   3. Final BIM model submission to PPPL shall incorporate all changes and additional information as reflected on the marked-up record prints.
   4. Refer in instances of uncertainty to PPPL for resolution.
   5. Also refer to PPPL A/E Drawings Deliverables requirements

END OF SECTION 013514

SECTION 014100 QUALITY REQUIREMENTS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. This Section specifies the requirements for quality control, including those for tests, documentation of quality activities, and providing certificates of compliance. These requirements are in addition to the requirements specified elsewhere in the Subcontract.

B. The Subcontractor is responsible for quality control and shall establish, document, implement, and maintain an effective Subcontractor quality control system, hereafter referred to as the Quality Assurance Program (QAP). The QAP shall apply to all Subcontractor on-site and off-site activities.

C. The Subcontractor's Project Manager in addition to requirements elsewhere in the Subcontract shall be the highest-level manager responsible for the overall activities on and off of the site, including quality, safety, and production, is subject to removal by PPPL for non-compliance with the quality requirements.

D. The Subcontractor's Quality Control Coordinator (SQCC) will be held responsible for the implementation and management of the QAP and is subject to removal by PPPL for failure to perform the aforementioned. Inherent in the responsibility is the authority to stop work, and to accept or reject any or all work, materials, or equipment associated with the job including work-in- progress. Subcontractor Quality Control (SQC) staff including Quality Engineers, Quality Inspectors, Laboratory and Field Technicians, or other personnel will be hired or assigned as determined by the SQCC and as required by the Subcontract. The SQCC shall be responsible for the implementation of the QAP for all quality related construction activities on and off site.

E. No separate payment will be made for work covered under this specification. The work associated with this specification is considered incidental to the completion of work in which it pertains. The cost of this work shall be accounted for in accordance with the general Provisions.

F. Failure of the Subcontractor to comply with the requirements of this Section shall be grounds to stop work, for a credit assessed against the Subcontractor for quality control work not performed, and to withhold progress payments, in part or in total, until the Subcontractor is in compliance. In addition, no non-conforming work will be approved for payment.

G. Subcontractor's Responsibility for Conformance: Neither review and/or approval of Subcontractor's documents nor inspection of Subcontractor's items or services shall relieve the Subcontractor of responsibility for full compliance with requirements of the subcontract. The Subcontractor is responsible for assuring that all requirements and restrictions are imposed on any lower-tier suppliers. The Subcontractor is responsible for assuring that all requirements and restrictions are imposed on any lower-tier suppliers.

1.2 DEFINITIONS

A. Construction Quality Management is the performance of tasks that ensure that construction is performed according to Subcontract Drawings and Specifications. Quality

is the attainment of a predetermined level of excellence established by the Subcontract Drawings and Specifications. The Subcontractor, in building to the quality standards in the Subcontract Drawings and Specifications, controls the quality of the Work.

B. Subcontractor Quality Control (SQC) is the successful execution of the Subcontractor Quality Program, providing certification that the required standards of quality construction are met and to preclude problems resulting from poor quality or lack of quality. The Subcontractor, for each definable feature of work, shall conduct at least three phases of control as described in Article 1.8.

C. Construction Quality Assurance (CQA) involves the means by which PPPL protects its interests. Through reviews, inspections, tests, Quality Assurance surveillance and audits, PPPL assures that QAP is being implemented and is working effectively, and that the end product complies with the level of quality established by the Subcontract.

D. A definable feature of work is a particular work task that is separate and distinct from other work tasks, has separate control requirements delineated in the Specifications, and may be performed by different trades or disciplines, or it may be work by the same trade in a different environment. Although each Section of these Specifications may generally be considered as a definable feature of work, there is frequently more than one definable feature under a particular Section. For example, Division 03 – Concrete Specifications may encompass multiple definable features of work such as cast-in-place structural concrete for walls, slabs, walkways, precast one-pass segmental liners and shotcrete.

E. Witness Points are points in the sequence of work where PPPL will review, witness, or test parts and processes of the Work. The SQCC is required to notify PPPL, who may attend, at least 48 hours in advance of the required Witness Point or as otherwise agreeable to PPPL.

F. Hold Points are mandatory verification or inspection points in the sequence of work beyond which work shall not proceed until the SQCC notifies the organization establishing or mandating the Hold Points and the organization performs the necessary verification or inspection to confirm that specified conditions have been met, and gives authorization to the Subcontractor to then proceed to the next sequence of work. Hold Points may be established by the Subcontractor, PPPL, or approving authorities. The Subcontractor is required to notify PPPL at least 72 hours in advance of the required Hold Point or as otherwise agreeable to PPPL. The Subcontractor is responsible for notifying and coordinating with other approving authorities.

1.3 OBSERVATIONS BY PPPL

A. All work (which includes but is not restricted to materials, workmanship, testing, and manufacture and fabrication of components) shall be subject to inspection, including Hold Points, Witness Points, and tests by PPPL and by other authorized personnel designated by PPPL. Such inspection or test is for the sole benefit of PPPL and shall not relieve the Subcontractor of the responsibility or provide quality control measures to assure that the work strictly complies with the Subcontract requirements. No inspection, test, surveillance or audit by PPPL shall be construed as constituting or implying acceptance. Inspections, tests, surveillance or audits by PPPL shall not relieve the Subcontractor of responsibility for damage to or loss of the material prior to acceptance, or in any way affects the continuing rights of PPPL after acceptance of the completed work.

B. The Subcontractor shall make available all offsite fabrication facilities for inspection, surveillance, and/or audit of fabricated items by PPPL.

C. If the Subcontractor does not promptly replace rejected material or correct rejected workmanship, PPPL may, by Subcontract or otherwise, replace such material or correct such workmanship and deduct the cost thereof from subsequent progress payments.

D. The Subcontractor shall be held responsible for delay of inspection or test due to material or work not being acceptable or ready at the time specified by the Subcontractor for such inspection or test, and due to re-inspection or retest necessitated by prior rejection.

E. The Subcontractor shall not cover, or allow to be covered, any of the Work installed under this Subcontract without PPPL's approval. Should any of the work be covered prior to such approval, PPPL shall have the authority to require the work to be uncovered for PPPL's inspection and approval, re-covered, and all resultant damage repaired, all at the Subcontractor's expense. Work that is found to be unacceptable shall be documented and dispositioned as required by the QAP

F. Should it be necessary or advisable by PPPL, at any time before Acceptance, to make an examination of work already completed, the Subcontractor shall, if requested, promptly furnish all necessary facilities, equipment, labor, and material required. If such work is found to be defective or nonconforming, the Subcontractor shall be responsible for all the expenses of such examination and for satisfactory reconstruction. If, however, PPPL was allowed ample opportunity to inspect said work and work is found to meet the requirements of the Subcontract, the Subcontractor will be compensated for the additional services involved in such examination and reconstruction under Force Account, and, if completion of the Work has been delayed thereby, the Subcontractor shall, in addition, be granted a suitable extension of time.

G. Neither inspections nor approvals by PPPL or by others shall relieve the Subcontractor from the obligation to perform the Work in accordance with the Subcontract.

1.4 SUBMITTALS

A. The Subcontractor shall submit the Quality Assurance Program (QAP), in written form and in one manual, for review and acceptance by PPPL, not later than ten (10) working days after subcontract award as part of the Notice to Proceed (NTP) documents.

B. PPPL will consider an interim QAP to be utilized as part of the Notice to Proceed (NTP) documents for submission of Submittals and RFIs, and for Mobilization activities as described in the Item descriptions of the Payment Section of the Subcontract Specifications.

C. Construction will be permitted to begin only after approval of the final QAP or as otherwise approved by PPPL.

1.5 QUALITY ASSURANCE PROGRAM (QAP)

A. The Subcontractor shall establish, implement and maintain an effective Quality Assurance Program (QAP) that describes how the Subcontractor is to manage, control and document the Work to assure that it complies with the requirements of the Subcontract. All lower-tier subcontractors, fabricators, Suppliers, and Inspection and Testing Laboratories shall comply with the QAP.

B. The QAP shall identify personnel, procedures, controls, instructions, tests, certifications, records, and forms to be used, and shall include, as a minimum, the following to cover all construction operations, both onsite and offsite, including Work by lower-tier subcontractors, fabricators, Suppliers, Inspection and Testing Laboratories:

1. A description of the Subcontractor's organization for the Subcontract (including Subcontractor's senior management, lower-tier subcontractors, and Suppliers) and an organizational chart showing names, titles, responsibilities, lines of authority, and interrelationships among those responsible for managing the Subcontractor Quality Program and the rest of the Subcontractor's staff. Additionally, the Subcontractor shall identify staff supporting the SQCC in the implementation of the QAP, hereafter referred to as Subcontractor Quality Control staff.
2. The Subcontractor shall also acknowledge that the Subcontractor Quality Control (SQC) staff shall implement the three-phased control system defined in Article 1.8 herein for all aspects of the Work specified. The staff shall include a Subcontractor's Quality Control Coordinator (SQCC) who shall report to the Subcontractor's Management at an organization level superior to the Subcontractor's Project Manager. This reporting structure shall provide the SQCC with direct access to the Subcontractor's Principal Officers.
3. In resume format, the names, qualifications, certifications, duties, responsibilities, and authorities of the SQCC, Subcontractor Quality Control staff, and other Subcontractor lead staff such as foremen, project and office engineers, and other staff involved in coordinating and monitoring testing.
4. Document Control Procedures and Records
    a. For submitting, processing, and managing documents such as Submittals, RFIs, and other communications including tracking and verifying that appropriate personnel are reviewing documents and working with correct information.
    b. To ensure the use of the most current approved documents, including but not limited to Submittals, RFIs, drawings, specifications, procedures, and directives.
    c. To provide for control of documents and changes to preclude use of inappropriate documents.
    d. To establish records control logs and a systematic filing system for ease of retrieval.
    e. To establish format for forms and documents, including logs, Quality Control and inspection reports, test reports, procedures, Quality Control Inspection Checklists and Subcontractor Non-Conformance Reports.
5. Procurement/Purchasing/Receipt Procedures
    a. To ensure the Supplier/manufacturer has an established quality program that verifies and documents the quality process.
    b. For receipt inspection and approval of procured product to ensure products comply with the Subcontract requirements, and to assure materials, equipment, and parts are clearly identified (plates, tags, and markings) for intended use and have proper supporting documentation.
    c. To establish measures for handling, storing, and shipping, including cleaning, packaging, and preserving the product in accordance with Supplier/manufacturer's instructions, and assuring non-compliant materials, equipment, and parts are not used and are segregated.
6. Process Control Procedures
    a. To ensure all work processes and activities affecting quality include documented instructions, procedures, planning sheets, job tickets, or

drawings appropriate for the work operations.

b. To ensure special processes such as welding, heat treating, soldering, cleaning, non-destructive testing, and prestressed and post stressed concrete activities are accomplished using qualified personnel and procedures, and in accordance with applicable codes and Subcontract requirements.

c. Provide current Certificates of welders' qualifications in accordance to ANSI/AWS D1.1. Certificates shall include the date of certification, qualification, grade and rating, and signature of the AWS Certified Welding Inspector. Welders shall be retested and recertified, at no expense to PPPL, when their work creates a reasonable doubt as to proficiency.

d. To establish methods for assuring and documenting that activities have been completed in accordance with Subcontract requirements.

7. Quality Control Inspection and Testing Procedures (including those for the Independent Testing Laboratories)

a. For the performance, control, verification, certification, and acceptance of each specific test.

b. To verify the Work processes, products and systems comply with documented instructions, codes, and Subcontract Documents.

c. For documenting inspection activities, testing activities, and test results.

d. To assure testing and measuring equipment are properly calibrated and maintained in accordance to nationally recognized standards, and that the equipment utilized for inspection, measuring, or testing is of the proper range and type.

e. For evaluating the validity of previous inspection and test results and acceptability of materials and equipment so checked if the test equipment is found to be out of calibration.

f. For maintaining records and suitable markings of test equipment to indicate calibration status, including the entity that performed the last calibration, date that the calibration was performed, and date on which re-calibration is due.

g. Refer to Article 1.9 herein for additional requirements.

8. Inspection and Test Status Procedures

a. For ensuring the use of only material, parts, systems, and equipment that have acceptable documented receipt inspection.

b. For indicating the quality status as a result of required inspection or testing performed upon furnished and installed materials, parts, systems, or equipment. Subsequent work shall not progress until the non-acceptable status as a result of inspection of furnished and installed material, parts, systems, or equipment has been removed by the SQCC.

9. Procedures for tracking and documenting preparatory, initial, and follow- up control phases.

10. Non-Conformance Procedure

a. For controlling, tracking and reporting of construction deficiencies, non-conforming items, QAP failures, or shortcomings, and/or inadequate administration of the QAP from identification through acceptable corrective action. These procedures shall establish

measures to identify, segregate, and remove from work operation the non-conforming items to prevent inadvertent use/incorporation into the Subcontract, identify root cause, and provide documented verification that identified deficiencies have been corrected and approved by PPPL. Refer to Article 1.10 herein for additional requirements of Non-Conformances.

b. The Subcontractor shall have Subcontractor Non-Conformance Report (SNCR) forms and a SNCR log for identifying, correcting, documenting, and controlling non-conformances. The format and content of the SNCR form and SNCR log shall be approved by PPPL.

11. Corrective Action Procedure:
    a. For ensuring that QAP failures or shortcomings, inadequate administration or implementation of the QAP, or series of related or repeating Non-Conformances are identified and documented through the use of separate Corrective Action Requests, or may be included in the Non-Conformance Report and that rigorous steps are implemented by the SQCC and Subcontractor's Principal Officers to prevent recurrence.
    b. Additionally, subsequent work may not progress on work that is the subject of a Corrective Action Request, until conditions adverse to quality are corrected as documented by the SQCC via the Corrective Action Request. Refer to Article 1.10 herein for additional requirements of corrective actions.
12. Audit and Surveillance Procedure:
    a. For ensuring that the Quality Control Program is being effectively implemented through scheduled or spontaneous Audits or Surveillances performed by qualified SQC Staff.
13. A training program to assure that suitable proficiency is achieved and sustained by personnel performing activities that affect and ensure quality. Additionally, the Subcontractor shall provide awareness training in the general requirements of the QAP for the Subcontractor and lower-tier subcontractor's Quality Control staff and management teams prior to work being performed and upon changes to the QAP. All training shall be documented with an agenda, related information, minutes of the training, attendance sheet, date, and time. PPPL shall be notified a minimum of 2 Days in advance of training which it may attend.
14. Provisions for identification of all types of quality records as evidence that all activities comply with the requirements of the Subcontract Documents and QAP.
15. A list of definable features of the work and the Subcontract quality requirements for each feature, and how the Subcontractor intends to furnish that quality, including control testing, certifications, and records.
16. An organized list of tests that includes the test name, frequency of test, specifications Section and Article requiring test, feature of work to be tested, person and organization responsible for each test, and test type, including Factory Tests, Construction Inspection Tests, Installation Tests, Acceptance Tests, and other tests.
17. An inventory of the materials, equipment, in-process work, and systems subject to testing and certification.
18. The current qualifications and certifications of all the proposed Independent

Inspections and Testing Laboratories/Agencies and their personnel to be used in the work by the SQCC and as required by Article 1.8 herein.

19. Other elements required of the QAP include lower-tier Subcontractor/Supplier/ Vendor Controls; Start-Up Testing Control; and Facility Turnover.

C. Acceptance of the Subcontractor Quality Program (QAP) by PPPL is required prior to the start of construction. Acceptance is conditional and will be predicated on satisfactory performance during construction. PPPL reserves the right to require the Subcontractor to make changes in its QAP and operations including removal of personnel, as necessary, to obtain the quality specified at no cost to PPPL.

1. After acceptance of the QAP, the Subcontractor shall notify PPPL in writing of proposed changes, including changes to Subcontractor Quality Control staff. Proposed changes are subject to approval by PPPL.

1.6 SQC COORDINATION CONFERENCE

A. After the Pre-construction Meeting and prior to acceptance by PPPL of the Subcontractor Quality Program (QAP), the Subcontractor shall convene a SQC Coordination Conference with PPPL or authorized representative and discuss the QAP. The QAP shall be submitted for review a minimum of 7 Days prior to the Coordination Conference. PPPL will return comments on the QAP submittal within 15 Working days unless otherwise noted after the Conference. During the Conference, a mutual understanding of the system details shall be developed, including the forms for recording the Subcontractor’s Quality Control operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Subcontractor's Management and control with PPPL's Quality Assurance. Minutes of the meeting shall be prepared by the SQCC and signed by the Superintendent, the SQCC, and PPPL. The minutes shall become a part of the Subcontract file. Subsequent conferences can be called by either party to reconfirm mutual understandings and address deficiencies in the QAP.

1.7 SUBCONTRACTOR QUALITY CONTROL ORGANIZATION

A. Personnel Requirements: The SQC organization shall include the SQCC and a sufficient number of additional qualified personnel to ensure compliance with the Subcontract Documents. Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly will also be included as part of the SQC organization. The Subcontractor SQCC and QC staff shall maintain a presence at the site at all times during progress of the work and shall not simultaneously perform any functions other than quality, and have complete authority and responsibility to take any action necessary to ensure Subcontract compliance. The SQC staff shall be subject to acceptance by PPPL. The Subcontractor shall provide adequate office space, filing systems, and other resources as necessary to maintain an effective and fully functional SQC organization. Complete records of all letters, material submittals, shop drawing submittals, schedules, and all other Subcontract documentation shall be promptly furnished to the SQC organization by the Subcontractor. The SQC organization shall be responsible to maintain these documents and records at the site at all times.

B. Subcontractor’s Quality Control Coordinator (SQCC) and Subcontractor Quality Staff qualifications:

1. The Subcontractor shall identify as SQCC an individual within the onsite work organization who shall be responsible for overall management of QAP and have the authority to act in all quality matters for the Subcontractor.

2. The SQCC shall have a minimum of 10 years of construction quality control experience on construction similar to that of this Subcontract, or as otherwise approved by PPPL.
3. Each SQC Staff shall have a combination of education and experience, with a minimum of 3 years of experience in quality control on construction similar to that of this Subcontract, as determined and documented by the SQCC, sufficient to perform in their assigned Quality capacities. All SQC Staff are subject to review and acceptance by PPPL.
4. The SQCC shall be on the site as deemed necessary by the Subcontractor to manage, supervise, implement , and perform the Work in accordance with the Subcontract and its incorporated documents.
5. An Assistant SQCC shall be identified to serve in the in the event of the SQCC's absence. The requirements for the Assistant SQCC are the same as for the SQCC.

1.8 CONTROL

A. Subcontractor Quality Control is the means by which the Subcontractor ensures that the construction, including that of lower-tier subcontractors and Suppliers, complies with the requirements of the Subcontract. The SQCC and Superintendent, for each definable feature of work, shall conduct at least three phases of control as described in the following:
   1. Preparatory Phase: This phase shall be performed at least 15 Working Days, except as otherwise acceptable to PPPL, prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted by PPPL, and after copies of said documents are at the work site. This phase shall include:
      a. A review of each paragraph of applicable Subcontract requirements, reference codes, and standards. The Subcontractor at the preparatory phase meeting shall make a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field available. These copies shall be maintained in the field and available for use by PPPL until Final Acceptance of the work.
      b. A review of the relevant Subcontract Documents, Submittals, RFIs, and other instructions, including verification that PPPL's approval has been obtained for all Submittals or RFIs, subsequent to the start of work for each definable feature of work.
      c. A physical examination of required materials, equipment, and sample work to assure that they have been tested, submitted, and approved are on hand, conform to approved shop drawings or submitted data, and are properly stored.
      d. A review of manufacturers' instructions, including that for handling, storing, and installing.
      e. Review of all relevant inspection and tests requirements, including start-up testing, and procedures including:
         1) Stating the reason for conducting the individual tests.
         2) Identifying pre-requisite tests and providing documentation of the results.
         3) Identifying test facilities and equipment (and model numbers) and their valid current certifications and calibrations.

4) Describing the setup of the test equipment with aid of drawings and sketches and the test sequence that must be taken to safely conduct the test.
5) Identify the test Pass/Fail criteria, as required by the Subcontract, for each test.
6) Identifying personnel involved and needed in the tests, including PPPL Staff, their responsibilities, and their certifications and qualifications as appropriate.

f. Review of relevant permits and their requirements including governing jurisdiction inspection requirements.

g. Examination of the work area and documentation to assure that all required preliminary work and previous definable features of work have been completed and are in compliance with the Subcontract, and all relevant Subcontractor Non-Conformance Reports, and Non-Compliance Notices have been properly dispositioned and closed, including acceptance by PPPL. Subcontractor shall provide organized and compiled documents and records of the previous definable feature of work, including but not limited to QC reports, test results, and Special Inspection reports to PPPL as evidence of compliance.

h. A review of the safety activity hazard analysis to assure safety requirements are met.

i. Review of quality control procedures and activities including inspections and tests for controlling quality of the work including the prevention of repetitive deficiencies.

j. A check to ensure that PPPL has accepted the portion of the plan for the work to be performed.

k. Review of Subcontractor's Quality Control inspection checklists. The checklists, based on the above requirements, shall be developed for the definable work feature to control and document the work quality. Each checklist shall be in a format that field personnel and Subcontractor Quality Control staff, can monitor, implement and document results, and include but is not limited to work sequence, work location, inspection and test requirements, acceptance criteria (including allowable tolerances, Pass/Fail criteria and level of workmanship), test equipment identification number and calibration date, date of test and inspection, inspection and test Hold Points and Witness Points, and as agreed upon or established by PPPL and as required by approving agencies, to ensure the work complies with Subcontract requirements. Multiple checklists may be required for each definable feature of work because work occurs at multiple locations and on different days.

l. PPPL shall be notified at least 3 Working Days in advance of beginning the preparatory control phase; 18 Working Days prior to the start of work on each definable feature of Work. This phase shall include a meeting, jointly conducted by the Superintendent and SQCC, attended by PPPL, other SQC personnel (as applicable), and the supervisors, including foreman responsible for the definable feature. The results of the preparatory phase actions shall be documented by separate minutes prepared by the SQCC and attached to the daily SQC report. The Superintendent shall instruct applicable workers as to the

acceptable level of workmanship required in order to meet Subcontract Specifications.

2. Initial Phase: The purpose of this Initial Phase, which is separate from the Preparatory Phase, is to prepare for work, refine the controls and close out open items from the Preparatory Phase. This phase shall include a meeting, jointly conducted by the Superintendent and SQCC, and attended by PPPL, other SQC personnel (as applicable), Subcontractor Safety Engineer and the supervisors, including foreman responsible for the definable feature. The Initial Phase meeting shall occur after the Preparatory Phase and a minimum of 2 Working Days, or as approved by PPPL, prior to initiating the field activity for the particular definable feature of work. The following shall be accomplished:
   a. Review minutes of the preparatory meeting and ensure open items are properly closed out.
   b. SQCC shall update and finalize the Quality Control inspection checklists as required.
   c. Subcontractor's Safety Engineer shall verify that safety considerations are adequate or if it is determined that upgrading of the safety plan and activity hazard analysis, is required. The Safety Engineer shall review the activity analysis with each worker.
   d. Separate minutes of this phase shall be prepared by the SQCC and attached to the daily SQC report.
   e. Exact location of items reconciled during the initial phase, such as acceptable architectural finishes or quality of craftsmanship, shall be indicated/documented for future reference and comparison with follow-up phases.
   f. The initial phase shall be repeated for each new crew to work onsite, or any time acceptable specified quality standards are not being met as determined by the SQCC, Superintendent, or PPPL.
3. Follow-up Phase: Daily quality control activities, including control inspection and testing, shall be performed to assure work complies with Subcontract requirements, until completion of the particular feature of work. All testing and inspections shall be made a matter of record in the SQC documentation. Final follow-up inspection and testing, shall be conducted and all non-conforming work is to be corrected as required by the QAP. The Subcontractor shall correct all non-conforming work prior to the start of additional features of work that may be affected by the non-conforming work. The Subcontractor shall not build upon or conceal non-conforming work.
4. The Subcontractor shall refer to Article 1.9 herein for additional requirements.
5. All Subcontractor quality control documentation, including minutes of the Preparatory and Initial Phases, checklists, and inspection and test report shall be organized, compiled, and archived by the relevant definable feature of work.

1.9 ADDITIONAL PREPARATORY AND INITIAL PHASES

A. Additional preparatory and initial phases shall be conducted on the same definable features of work if: the quality of on-going work is unacceptable; if there are changes in the applicable SQC staff, onsite production supervision or work crew; if work on a definable feature is resumed after a substantial period of inactivity; or if problems develop.

1.10 INSPECTIONS AND TESTINGS

A. The Subcontractor shall perform specified or required quality control activities including inspections and tests by qualified personnel to verify that control measures are adequate to provide a product, which conforms to Subcontract requirements. Upon request, the Subcontractor shall furnish to PPPL duplicate samples of test specimens for possible testing by PPPL. The Subcontractor shall perform the following activities and record and provide supporting documentation:

1. Verify that personnel qualifications, work processes, products, systems, and quality control activities comply with documented instructions, codes, standards, Subcontract requirements, and the QAP.
2. Document quality control inspection activities, testing activities, and test results.
3. Verify testing and measuring equipment are properly calibrated and maintained in accordance with nationally recognized standards, and are of proper range and type. The equipment shall have suitable markings to indicate calibration status including the entity that performed the last calibration, date that calibration was performed, and date on which calibration is due.
4. Verify that recording forms and test identification control number Systems, identified in the QAP, including all of the test documentation requirements, have been prepared. Test reports shall include the type of test (e.g. factory tests, construction tests, acceptance tests, and/or Special Inspection tests), date and location of the test, reason for the test (e.g., reference permit number, applicable codes and Specifications Sections and Articles(s)), duration of test, the test results, name and signature of person(s) conducting the test, all data and results obtained during the tests, analysis of the data, conclusions relating to the test pass/fail criteria outlined in the test procedure, sign-offs and certification by authorized staff, and identify follow-up and corrective actions that are required if re-testing is required.
5. Copies of each completed and signed-off test report, including related quality reports, shall be provided to PPPL after approval by the SQCC for acceptance within 5 Days after completion of each test.
6. Results of all tests, both passing and failing, shall be recorded on the SQC report for the date the test was conducted. Specifications Section and Article reference, location where tests were performed, and the sequential control number identifying the test shall be provided. An information copy of tests performed by an offsite or commercial test facility shall be provided directly to PPPL within 2 Days of completion of each test. Failure to submit timely test reports as stated may result in nonpayment for related work performed and disapproval of the test facility for this Subcontract.

B. Testing and Inspecting include but is not limited to the following:

1. Factory Tests: Factory Tests include Design Qualification Tests and Production Verification Tests. Design Qualification Tests verify the performance of new engineering designs as specified. Examples include smoke and flammability tests of new material and switchgear design tests. Production Verification Tests confirm that equipment and materials are manufactured in accordance with accepted workmanship standards and the specifications. Examples include composition tests of lubricants, equipment chassis wiring continuity and Megger tests, and relay rack factory acceptance tests.

2. Installation Verification Tests: Installation Verification Tests are the responsibility of the Subcontractor and include In-Process Acceptance Tests conducted at the subsystem or assembly level during the installation of each element as specified. Examples include shop electrical power distribution wiring continuity tests, local field acceptance tests of communications subsystems, and hydrostatic testing.
3. Acceptance Tests: Acceptance Tests are the responsibility of the Subcontractor and are conducted at the subsystem level to verify that the performance of each element and subsystem/assembly is in compliance with specification requirements. These tests are performed on-site and are prerequisites to the System Integration Tests where specified.

C. PPPL will engage a 3rd Party special inspection testing agency to cover all testing and inspection work as required by the Subcontract and the governing jurisdictions. This may include, but is not limited to the following:
1. Independent Certified Testing Laboratory to perform testing such as soil classification, soil compaction tests, continuity tests, in-situ testing, and concrete slump, entrained air, temperature, yield, and strength tests.
2. Independent Certified Testing Laboratory to perform chemical analysis for items such as hazardous/non-hazardous soil classification.
3. Independent AWS Certified Welding Inspectors (CWI) to perform welding inspections.
4. Special Inspections and testing services to be performed by a qualified Independent Certified Inspection and Testing Agency retained by PPPL. These are Special Inspections as specified in the Subcontract Documents and as required by the governing jurisdiction, including the local building officials/codes, and are not to be confused with other types of inspections and testing. The following is a sample and an incomplete list of items requiring Special Inspections and testing services:
a. Excavation & Backfilling
1) Proof Rolling of existing and improved subgrades.
2) Density tests for all embankments, subgrades, aggregate bases and subbases.
b. Asphalt
1) Density tests for all courses, core analysis and comparison in accordance with NJDOT.
2) Temperature monitoring
c. Structural
1) Epoxy installation of dowels, anchor bolts, or other inserts in drilled holes in existing concrete including such items as pull-out tests.
2) Concrete structures including sumps, pits, equipment utility vaults, structural underpinning, pole foundations and such, inspection the inspection of items such as form work and ties, and placement of concrete in accordance with applicable codes.
3) Masonry Concrete Walls construction including placement of reinforcing steel and grouting.
4) Inspection of reinforcing bar placement to verify compliance with Subcontract Drawings and Specifications.

5) Steel fabrication and erection inspection for compliance with codes.
6) Structural steel welding and non-destructive testing.
7) Metal deck installation and welding inspection.
8) High strength bolt installation, inspection, and testing.
9) Nelson stud shear connector installation, inspection, and testing.
10) Compressive strength testing for concrete and grout.
11) Coring and testing of concrete samples to determine compressive strengths of existing concrete.

D. Where the required inspections and tests vary from the approved inspections and testing schedule, the Independent Inspection and Testing Agencies/Laboratories, in conjunction with the Subcontractor, shall provide a minimum of 5 Working days advance notice to PPPL prior to the date of the inspection and testing so that PPPL may witness the inspections and the tests.

E. Within 5 Working days after the completion of the inspections or tests performed, the Independent Inspection and Testing Agencies/Laboratories shall submit to PPPL and the Subcontractor, the results of the inspections and tests, indicating observations and the results of tests and indicating compliance or non-compliance with the Subcontract. With respect to the Special/continuous inspections and tests of Article 1.8 herein, the Independent Inspection and Testing Agencies/Laboratories shall also submit directly to the Permitting Agency two copies, or as required, of inspection and test reports.

F. Re-testing or re-inspecting required (including re-inspecting by building officials) because of non-conformance to specified requirements shall be performed by the same independent agency/laboratory on instructions by PPPL. All expenses related to re-testing and re-inspecting shall be borne solely by the Subcontractor.

G. Subcontractor is responsible to coordinate all required inspections, tests, including re-tests, by all independent inspection and testing laboratories/agencies so as to avoid unnecessary delays to the construction schedule.

H. The SQCC is required to complete and issue a Subcontractor Non-Conformance Report (SNCR) for each occurrence of non-conforming work or material. PPPL shall be furnished a copy of all SNCRs within 1 Working day upon initial generation and upon “closeout”. Subcontractor Non-Conformance Reports (SNCR) with REPAIR or USE AS-IS dispositions require review and acceptance by PPPL prior to the performance of repairs or subsequent incorporation of the non-conforming work into the Subcontract. Additionally, subsequent work may not progress on work that is the subject of a SNCR until conditions adverse to quality are corrected as documented by the SQCC via the SNCR and accepted by PPPL. SNCRs dispositioned as REJECT or REWORK, do not require PPPL’s approval. A copy of the updated SNCR Log shall be provided weekly to PPPL or as otherwise approved by PPPL.
    1. The Subcontractor shall investigate the root cause of the non-conformance, and take appropriate corrective actions to prevent recurrence. Failure to properly identify and prevent recurrence may be cause for removal of Key Personnel, as determined by PPPL, from the Subcontract. Responses with preventive measures from Suppliers, lower-tier subcontractors, and others shall be recorded on or attached to the SNCR.
    2. Any member of the Subcontractor’s staff can initiate a SNCR. Each initiated SNCR shall be logged and validated by the SQCC. SNCRs determined to be

invalid shall be marked “VOID”, with an explanation and VOID status clearly indicated on the SNCR Log.

1.11 COMPLETION INSPECTION

A. SQC Preliminary Punch-Out Inspection: Near the end of the Work, or any increment of the Work established by a time stated in the Subcontract for Time Allowance for Completion of Work, the SQCC shall conduct an inspection of the Work. A preliminary Punch Out of items that do not conform to the Subcontract shall be prepared and included with the quality records. The list of deficiencies or non-conforming items shall include the estimated date by which the items will be corrected. The SQCC shall make a second inspection to verify that these items have been corrected. Once this is accomplished, the SQCC shall notify PPPL that the facility or final product is ready for PPPL A/E Punch List Inspection.

B. A/E Punch List Inspection: The A/E design team will perform a Punch List Inspection to verify that the facility or final product is complete and ready for use, and as result, will develop a PPPL Punch List that will be provided to the SQCC. The SQCC shall ensure that deficiencies or non-conforming items identified on this list have been corrected within the time slated for the completion the Subcontract or any increment of work if the Subcontract is divided into increments by separation completion dates. Once this is accomplished, the SQCC shall notify PPPL so that a Final Inspection with Operations and Maintenance and other Departments can be scheduled.

C. Final Acceptance Inspection: The SQCC, other Subcontractor management staff, A/E design representative, and PPPL shall be in attendance at the Final Acceptance Inspection. Additional PPPL personnel including, but not limited to, those from Operations and Maintenance or other Departments may also be in attendance. PPPL, based on the results of PPPL Punch List Inspection, will schedule the Final Acceptance Inspection. The SQCC shall give PPPL an advance notice of 14 Days prior to the Final Acceptance Inspection and shall assure that all work is in compliance with the Subcontract by the date scheduled for the Final Acceptance Inspection. Failure of the Subcontractor to have all Subcontract work acceptably complete for this inspection will not allow PPPL to accept the project or products until all items are completed and verified.

1.12 DOCUMENTATION

A. The Subcontractor shall maintain current quality records to provide factual evidence that all activities, including quality control activities such as inspections and tests have been performed, accepted, and comply with the Subcontract Documents and the QAP. These quality records shall include the work of the Subcontractor, lower-tier subcontractors, Suppliers inspection services and testing laboratories, and shall be on an acceptable form.

B. Quality records shall include but not limited to Certificates of Compliance, checklists, QAP, lower-tier subcontractor and Supplier Quality Programs, inspection reports, receiving inspection reports, calibration test reports, material test reports, SNCRs, personnel information (qualifications, certifications, resumes and approvals), minutes of the Preparatory Phase, Initial Phase, and Follow up Phase meetings, procedures, Submittals and RFI logs, Punch Lists, and training records.

C. Subcontractor Inspector (SI) Daily Inspection Report shall include, as a minimum, the following information provided by the Subcontractor’s Inspectors (SI):

1. Subcontractor/lower-tier Subcontractor and their area of responsibility.
2. Description of the weather conditions encountered and the work performed

each day, giving location, description, and by whom. Identify each phase of work performed each day by CPM schedule activity or Work Breakdown Structure (WBS), as applicable.

3. Test and control activities performed with results and references to specifications/drawings requirements. The control phase shall be identified (Preparatory, Initial, Follow-up). List of deficiencies noted, along with the required corrective actions.
4. Material and equipment received at the site with statement as to acceptability (including verification/validation of required material identification, certifications, and quantities), storage and maintenance thereof, as required by the QAP.
5. Surveillance and audit activities, both on- and off-site, performed and documented as required by the QAP.
6. Documentation generated from the Preparatory Phase, Initial Phase, and Follow up Phase work.
7. Each Subcontractor/Inspector's report shall clearly indicate that the work performed and material and or equipment incorporated into the Subcontract, is acceptable or unacceptable. Unacceptable work that is considered by the SQCC to be in-process, which will be brought into conformance with the Subcontract Documents; both within one (1) Working Day, or at a time deemed to be reasonable by the SQCC, and subject to approval of PPPL, shall be clearly indicated on the Daily Inspector Report as requiring follow-up acceptance inspection. Work that is found to be non- conforming, that can't be brought into conformance with the Subcontract Documents, shall be documented and processed as required by the QAP with the Subcontractor's Non-Conformance Report number indicated on the Daily Inspection Report.
8. Daily Inspection Reports and completed Quality Control inspection checklists or forms as described in the QAP, and related laboratory test report results for work performed for the activities described within the Report shall be attached to the Daily Inspection Report. Reports need not be submitted for days on which no work is performed. As a minimum, one report shall be prepared and submitted for every 7 days of no work and on the last day of a no work period. All Days shall be accounted for throughout the life of the Subcontract.

1.13 NOTIFICATION OF NON-COMPLIANCE

A. In the event, for whatever reason, that the Subcontractor refuses to generate a Subcontractor Non-Conformance Report, PPPL may issue a Non- Compliance Notice to the Subcontractor for any detected non-compliance in the work or portion thereof that has not been performed in accordance with the Subcontract Documents. Such Notice, when delivered to the Subcontractor at the work site, shall be deemed sufficient for the purpose of notification.

B. The Subcontractor's Project Manager and SQCC shall take immediate corrective action after receipt of such Notice. The Subcontractor shall provide a written Response to the Notice within 5 Working Days after receipt of the Notice. The Subcontractor's response shall (a) explain why the Subcontractor believes that the work was performed in accordance with the Subcontract Documents or (b) the Subcontractor will generate a SNCR, provide the SNCR number to PPPL, and process the SNCR as required by the QAP.

C. If the Subcontractor disputes issuance of the Notice, PPPL will respond within 5 Working

Days after receipt of dispute by either (a) withdrawing the Non-Compliance Notice or (b) directing the Subcontractor to correct the work. Such determination from PPPL shall be final and conclusive of the matter. If PPPL directs the Subcontractor to correct the work, the Subcontractor shall do so within 5 Working Days after receipt of such direction from PPPL, or such other time as may be agreed to with PPPL.

D. If the Subcontractor fails or refuses to comply promptly, PPPL may issue an order stopping all or part of the work until satisfactory corrective action has been taken to prevent installation of non-conforming work. The Subcontractor shall make no part of the time lost due to such stop orders the subject of a claim for extension of time or for excess costs or damages.

PART 2 PRODUCTS

2.1 MATERIALS

A. Only new first quality materials will be supplied and used in this project.

B. Materials shall be delivered to the site in unopened, original containers, bearing the manufacturers printed label stating the quality, brand or trade name, batch number, date of manufacture and directions for use. Delivery of small day-to-day supplies will not be permitted.

C. PPPL & DOE have active programs to prevent use of Suspect & Counterfeit Items (SCI). Material(s) and/or product(s), including those components, parts, and materials that are permanently installed into assemblies furnished under this purchase order/subcontract shall be new. The Subcontractor shall verify conformance of purchased items to drawing and specification requirements and inspect items to detect signs of use or indications that raise questions of authenticity. Subcontractor shall actively prevent introduction of SCI to PPPL by inspection of all items brought on site to detect:

   a. Remanufactured, rebuilt, or used parts represented as new.

   b. Counterfeit parts (fraudulently labeled or marked with another manufacturer's name).

   c. Misrepresented parts.

   d. The Subcontractor shall assure that fasteners do not have head marks listed on the DOE Head mark List (See ATTACHMENT 1).

   e. Regardless of application, high strength bolts (tensile strength equal to or greater than 100 ksi) in accordance with the Fastener Quality Act, and must have the original manufacturer's Certified Material Test Reports (CMTR), and must be traceable to the CMTR(s) using lot numbers or other unique identification. Certified Material Test Reports (CMTR), showing actual material composition and physical properties and traceable to the actual fasteners, are required for each lot supplied. Results must be on the original letterhead of the entity performing the tests and not transferred to alternate letterhead. Fasteners shall exhibit grade marks and the manufacturer's identification symbol (headstamp) as shown in Attachment 1. Fasteners having a headmark that is displayed on the suspect fastener list will not be accepted. Where high strength fasteners are not required, it is recommended that they not be used.

PART 3 EXECUTION – Not Used

END OF SECTION 014100

ATTACHMENT 1

DOE Headmark List

ANY BOLT ON THIS LIST SHOULD BE TREATED AS DEFECTIVE WITHOUT FURTHER TESTING.



ALL GRADE 5 AND GRADE 8 FASTENERS OF FOREIGN ORIGIN WHICH DO NOT BEAR ANY MANUFACTURERS' HEADMARKS:

 **GRADE 5**  **GRADE 8**

GRADE 5 FASTENERS WITH THE FOLLOWING MANUFACTURERS' HEADMARKS:

MARK	MANUFACTURER	MARK	MANUFACTURER
J	Jinn Her (TW)	KS	Kosaka Kogyo (JP)

GRADE 8 FASTENERS WITH THE FOLLOWING MANUFACTURES' HEADMARKS:

MARK	MANUFACTURER	MARK	MANUFACTURER
A	Asahi Mfg (JP)	KS	Kosaka Kogyo (JP)
NF	Nippon Fasteners (JP)	RT	Takai Ltd (JP)
H	Hinomoto Metal (JP)	FM	Fastener Co of Japan (JP)
M	Minamida Sleybo (JP)	KY	Kyoei Mfg (JP)
MS	Minato Kogyo (JP)	J	Jinn Her (TW)
Hollow Triangle	Infasca (CA TW JP YU) (Greater than 1/2 inch dia.)		
E	Dalai (JP)	UNV	Unytite (JP)

GRADE 8.2 FASTENERS WITH THE FOLLOWING HEADMARKS:

MARK	MANUFACTURER
KS	Kosaka Kogyo (JP)

GRADE A325 FASTENERS (BENNETT DENVER TARGET ONLY) WITH THE FOLLOWING HEADMARKS:

	MARK	MANUFACTURER
Type 1	A325 KS	Kosaka Kogyo (JP)
Type 2		
Type 3		

Key: CA-Canada, JP-Japan, TW-Taiwan, YU-Yugoslavia

SECTION 015000 – TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. The following applies to this section:
   1. PPPL General Provisions for Non-Commercial Subcontracts Part G - Provisions & Conditions for Fixed Price Construction Subcontracts
   2. Related requirements in other Subcontract Documents including, but not necessarily limited to, the Subcontract Drawings and Specifications.

B. Related Sections:
   1. SECTION 017700 – CLOSEOUT PROCEDURES
   2. SECTION 017419 – CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

C. Related requirements in other Subcontract Documents listed in the Agreement.

D. Review these documents for coordination with additional requirements and information that apply to work under this Section.

E. All temporary facilities, utilities, equipment, field office, traffic control and protection, electrical work, etc. are subject to PPPL's review and approval before any work is commenced.

F. Relocation of any temporary facilities or controls for the convenience of the Work shall be at no additional cost to PPPL.

1.2 SUMMARY

A. Section Includes:
   1. Temporary electricity
   2. Telephone service
   3. Temporary water service
   4. Temporary sanitary facilities
   5. Portable hand washing facilities
   6. Temporary ventilation
   7. Barriers
   8. Traffic control
   9. Environmental controls
   10. Protection of installed work
   11. Protection of existing property
   12. Site security
   13. Progress cleaning and waste removal
   14. Field offices and sheds

1.3 REFERENCES

A. The following documents form a part of these specifications to the extent stated herein:
    1. American Conference of Governmental Industrial Hygienists (ACGIH):
        a. ACGIH Industrial Ventilation Manual
    2. Federal Highway Administration (FHWA):
        a. Manual on Uniform Traffic Control Devices (MUTCD)

1.4 SUBMITTALS

A. Provide detailed submittals describing plans for complying with each of the requirements for temporary facilities and controls as presented in this section.

B. Submittals shall include drawings showing plans and details of proposed facilities, written descriptions of proposed facilities and procedures, vendors' literature, and all other information necessary to fully describe the proposed temporary facilities and controls for each construction work area.

C. The Subcontractor shall prepare a site Logistics Plan detailing the impacts and plans to mitigate conflicts with existing conditions and operations. The Logistics Plan shall be updated by the Subcontractor as needed to inform the owner and subcontractors of any change in conditions impacting the site. The Logistics plan shall be submitted for PPPL review and approval 10 days following the Notice to Proceed.

1.5 TEMPORARY ELECTRICITY

A. See Terms and Conditions of the Subcontract. As soon as possible after award of the subcontract, notify PPPL of Power Requirements. Installation shall be done by the Subcontractor and shall conform to electrical codes. Subcontractor shall submit drawings for temporary power to PPPL for review and approval.

B. Temporary utilities, equipment and field office installations, traffic controls, and other temporary facilities are subject to review by PPPL before purchase of any items or commencement of work.

C. Provide temporary power drawing to PPPL to review and approve. The Subcontractor will then perform arc flash calculations and provide arc flash labels. The process of the Subcontractor performing calculations and providing arc flash labels is 5-7 days and no energization can be commenced without these labels.

D. Conditions:
    1. Subcontractor shall be responsible for connections/tie-ins to electricity from PPPL on-site sources.
    2. Exercise reasonable care to conserve PPPL -furnished power and take the following minimal energy conservation measures:
        a. Turn off all unnecessary construction lights and equipment at the close of each workday.
        b. Set thermostats in the construction office and any other facilities under Subcontractor's control at the following: 78°F for cooling and 65°F for heating.

c. Install a 7-day time clock on construction offices and any other facilities under Subcontractor's control that will provide for off-hour as well as weekend electrical power shutdown.

3. PPPL does not guarantee amounts available nor will PPPL be responsible for interruptions in service.
4. Temporary service installed by and for Subcontractor shall be removed and utilities restored to their initial condition by the Subcontractor at the completion of the subcontract.
5. Provide and maintain the electrical power distribution system downstream of PPPL -furnished electrical service.
6. Temporary power connection shall be per NFPA 70 National Electrical Code, applicable sections including Article 590, and PPPL requirements. Subcontractor to submit drawings showing routing of temporary power and include all back-up information including, but not limited to power survey and analysis.
7. All Temporary Facilities & Controls to comply with OSHA Illumination Standard 29 CFR 1926.56.
8. Temporary generator power, if needed, shall be provided by the Subcontractor.

1.6 TEMPORARY LIGHTING

A. Construction Lighting: Work conducted at night or under conditions of deficient daylight and at all times underground shall be suitably lighted to insure proper work performance and to afford adequate work facilities for inspection and safe working conditions. The lighting shall include appropriate lighting of yard and parking space areas, as well as PPPL's office space, and all Subcontractor facilities required for construction and support activities at each work area.

B. Lighting shall be directed to ensure safe illuminated working areas, and shall also be shielded to minimize nuisance to surrounding users. Glare from temporary lighting shall be mitigated as required and by applicable permits and regulations.

C. Temporary Lighting: The Subcontractor shall provide a general, weatherproof, grounded temporary lighting system in every area of construction to provide sufficient illumination for safe working and traffic conditions. Run circuit wiring generally overhead, and rise vertically in locations where it will be least exposed to possible damage from construction operations or other areas of possible damage or abuse. Protect wiring from construction hazards and mark cords accordingly.

1.7 TELEPHONE SERVICE

A. Voice and data services shall be provided by PPPL.

B. The subcontractor shall coordinate installation of voice/data with PPPL PTR and IT Department.

1.8 PARKING AND SUBCONTRACTOR'S STAGING/STORAGE AREAS

A. The Subcontractor's staging/storage areas shall be as shown or as determined during the pre-construction meeting and illustrated on the site Logistics Plan. PPPL may

designate and arrange for the Subcontractor's use, a portion of the property for its exclusive use during the term of the subcontract as a storage and shop area for its construction operations on the Work. At completion of Work, the Subcontractor shall return all staging/storage areas to their original condition, including grading and landscaping, as applicable.

B. The Subcontractor's use of staging and storage areas shall be coordinated with the work of adjacent construction subcontracts as specified.

C. The Subcontractor shall be responsible for making arrangements for any off-Site storage or shop areas that may be needed to carry out the work.

D. The Subcontractor shall construct and use separate containment areas at each site for storing hazardous materials used in constructing the Work.
    1. For the purpose of this paragraph, hazardous materials to be stored in the separate area are products labeled with any of the following terms: Warning, Caution, Poisonous, Toxic, Flammable, Corrosive, Reactive, or Explosive. In addition, whether or not so labeled, the following materials shall be stored in the separate area: diesel fuel, gasoline, new and used motor oil, hydraulic fluid, cement, paints and paint thinners, 2 part epoxy coatings, sealants, asphaltic products, glues, solvents, wood preservatives, sandblast materials, and spill absorbent.
    2. Hazardous materials shall be stored in groupings according to the Safety Data Sheets.
    3. The Subcontractor shall develop and submit to PPPL a plan for storing the materials above.
    4. The separate storage area shall meet the requirements of authorities having jurisdiction over the storage of hazardous materials.
    5. The separate storage area shall be inspected by PPPL prior to construction of the area, upon completion of construction of the area, and upon cleanup and removal of the area.
    6. Hazardous materials that are delivered in containers shall be stored in the original containers until use. Hazardous materials delivered in bulk shall be stored in containers which meet the requirements of authorities having jurisdiction.

E. Maintain traffic and parking areas in a sound condition, free of excavated material, construction equipment, mud, and construction materials. The Subcontractor shall repair breaks, potholes, low areas which collect standing water, and other deficiencies.
    1. Subcontractor's vehicles and equipment shall not be permitted to block fire lanes, building entrances or employee parking areas, or access routes required by adjacent construction subcontracts and workers.
    2. Provide sufficient parking for construction workers..
    3. Vehicle idling is limited to 60 seconds, unless the engine is needed to power other equipment. This policy applies to all vehicles at PPPL: personal, commercial, government, heavy equipment, etc.

1.9 TEMPORARY WATER SERVICE

A. Water: Reasonable amounts of temporary water service will be provided to the Subcontractor from the existing campus water service. The subcontractor shall install and maintain all necessary temporary connections and distributions lines and shall provide backflow-preventers as required.

1.10 TEMPORARY SANITARY FACILITIES

A. Toilet Facilities: Toilet facilities are not available. Provide and maintain portable chemical toilets for use at locations PPPL has approved and remove them from the site upon completion of the subcontract. Provide adequate facilities and clean and maintain your toilet facilities as required.

1.11 PORTABLE HAND WASHING FACILITIES

A. Provide and maintain portable facilities for washing hands at locations PPPL has approved and remove them from the site upon completion of the subcontract.

1.12 TEMPORARY HEATING AND COOLING

A. The Subcontractor shall provide adequate fixed temporary heating and cooling of enclosed construction Work, construction offices, fabrication shops and similar temporary facilities requiring climate control.

B. The maintenance of proper heat, ventilation and adequate drying out of the work are the responsibilities of the Subcontractor and any work damaged by dampness or insufficient or abnormal heat, shall be replaced to the satisfaction of PPPL and at the expense of the Subcontractor.

C. The Subcontractor may employ the permanent heating and cooling system as specified for this project for temporary purposes. The permanent heating and cooling system shall be thoroughly checked, reconditioned and repaired as required to bring it to the standard required by the specification, at no cost to PPPL.

D. Temporary electric power for construction purposes shall not be used as fuel for electrical resistance heaters except as approved by PPPL

E. All temporary installations required for temporary heating and cooling shall be removed by the Subcontractor, when no longer required, at no cost to PPPL.

1.13 TEMPORARY FIRE PROTECTION

A. Temporary fire protection shall be the responsibility of the subcontractor. Hose connections and hose, water casks, chemical equipment, or other sufficient means shall be provided for fighting fires in the temporary structures and other portions of the Work, and responsible persons shall be designated and instructed in the operation of such fire apparatus so as to prevent or minimize the hazard of fire. The Subcontractor's fire protection program shall conform to the requirements of NFPA 241.

1.14 TEMPORARY VENTILATION

A. Ventilate enclosed areas to achieve curing of materials, to dissipate humidity, and to prevent inhalation exposure and potential contamination caused by the generation of dust, fumes, vapors, or gases. Use ventilation to maintain the concentrations of air contaminants below standards applicable at PPPL. Provide ventilation in accordance with ACGIH "Industrial Ventilation Manual."

1.15 BARRIERS

A. Provide and maintain suitable temporary barricades, fences, or other structures as required for the protection of public traffic and employees; provide walks around any obstructions; and maintain on or near the construction sufficient light to protect all personnel from injury.

B. Provide and maintain temporary dust-proof partitions between construction areas (new or existing building) and areas occupied by PPPL. Partitions shall be constructed of metal studs, sheathing, and insulation and shall be sealed top and bottom in order to obtain a dust-tight seal. A fire rated temporary partition installation may be required depending on the specific area/condition encountered.

C. Provide a PPE free entrance and path to the main field office for visitor sign-in.

D. Provide and maintain signage at all site entrances with company name, project name, and name and contact info for subcontractor PM, superintendent, PPPL PTR and CM. Signs shall include: Danger: Construction Area; Do Not Enter. Visitors shall obtain release from PPPL PTR/CM, sign in at the field office, know the current hazards and conditions, and be escorted. Additional info on that sign or a separate sign shall be a list of required PPE: hard hat, safety glasses, reflective vest, safety-toed work boots, and long pants.

E. Provide and maintain protective closure facilities, such as roofing, canopies, and/or seals at existing buildings where connections or modifications are being made, to prevent the entry of rain and other weather elements so that equipment, facilities and structure are protected and retained in operating condition.

F. Protect vehicular traffic, PPPL parked vehicles, stored materials, site, and structures from damage.

1.16 TRAFFIC CONTROL

A. Give at least 48 hour notice to PPPL PTR/CM whenever large shipments or deliveries are expected at the jobsite.

B. Provide full-time, trained flaggers and adequate traffic control whenever heavy equipment or trucks are crossing or entering onto all PPPL roads including entrances and exits to PPPL property.

C. All temporary Traffic Control to comply with NJ DOT. Submit Traffic Control Plan to be reviewed and approved by PPPL.

D. PPPL speed limit in the construction zone is 10 mph.

E. Temporary road closure plans to be reviewed and approved by PPPL.

1.17 ENVIRONMENTAL CONTROLS

A. Dust Control: Perform dust control as required for the alleviation and prevention of any dust nuisance at, or in the vicinity of, the construction site as it pertains to its work. "Dust nuisance" is defined as airborne soil in sufficient quantity to be visible at any building or location adjacent to the construction site. Methods of dust control shall include:
   1. Spraying water, without causing runoff, on loose soil that may become airborne.
   2. Covering all stockpiled excavated material containing soil to preclude wind and water erosion and dispersal during storage.

B. Erosion Control: Exposed ground areas that are planned to be reworked at dates greater than two weeks after initial grading shall be sown with a fast germinating native grass seed and watered until vegetation is established and/or stabilized using approved soil binders, jute netting, or other methods approved in advance by PPPL.

C. Water Control: Grade site to drain. Maintain excavations free of water. Provide, operate, and maintain pumping equipment as needed to maintain water-free conditions at no additional cost to PPPL.

D. Provide erosion and sedimentation control plan for all construction-based soil disturbances greater than 5,000 square feet as required by the NJ Soil Erosion and Sediment Control Act (N.J.S.A. 4:24-39). The Soil Erosion and Sediment Control Plan shall contain a site map(s) which shows the construction site perimeter, existing and proposed buildings, lots, roadways, storm water collection and discharge points, general topography both before and after construction, and drainage patterns across the project. The plan shall apply Best Management Practices (BMPs) included in the “Standards for Soil Erosion and Sediment Control in NJ,” latest edition.

1.18 PROTECTION OF INSTALLED WORK

A. Protect installed work and provide special protection where specified in individual specification sections.

B. Provide temporary and removable protection for installed products. Control activity in the immediate work area to prevent damage.

1.19 PROTECTION OF EXISTING PROPERTY

A. Protect any and all existing infrastructure, equipment and property within and en route to the project limits to prevent damage during demolition and construction activities.

B. Protect existing structures and all trees and shrubbery to remain against damage. Repair or replace structures, streets, curbs, utilities, and landscaping damaged due to Subcontractor's construction activities at no additional cost to PPPL. Replace each tree removed or damaged with two boxed specimens, 6 inch minimum trunk diameter,

of like kind at locations directed by PPPL. Provide for temporary watering of existing trees and ground cover where existing irrigation is disrupted by construction. Replace damaged or removed irrigation at completion of work.

C. The Subcontractor shall provide all temporary means of protection to adjacent areas impacted by the project, including but not limited to asphalt paving, floor protection in hallway/stairs and any interior space, and ground covering for exterior material movements laydown, etc. Existing grass areas utilized for laydown shall require filter fabric with medium aggregate to be removed before restoration of area.

D. The Subcontractor shall be responsible for providing signed and sealed engineering calculations for any loading or rigging of equipment or materials interfacing with PPPL equipment or structure.

1.20 SITE SECURITY

A. Provide and maintain suitable temporary barricades, fences, or other structures as required for the protection of public, traffic, and personnel about the work site, provide walks around and obstruction made in public places; and maintain sufficient light on or near the construction to protect workmen, travelers, and other personnel from injury during all hours of darkness. As needed the Subcontractor shall provide traffic control personnel for work impacting traffic on any PPPL roads.

1.21 PROGRESS CLEANING AND NON-HAZARDOUS WASTE REMOVAL

A. Keep the construction areas clean at all times and remove accumulated debris, non-hazardous waste materials, and rubbish each day in accordance with the Subcontractor's Waste Management Plan. Assign required subcontractor and lower-tier subcontractor manpower to perform clean up and provide dumpsters for rubbish, debris, and nonhazardous waste materials. If, in the opinion of PPPL PM, PTR, or CM the jobsite has not been kept clean and orderly and presents a potential safety or fire hazard, Subcontractor may be required to stop work in the affected area and immediately correct the defects at no additional cost to PPPL, including lost time for clean-up effort. Refer to SECTION 017700 CLOSEOUT PROCEDURES for final cleaning. See SECTION 017419 – CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL for Non-Hazardous Waste Disposal .

B. All hazardous wastes generated from this project must be disposed of properly. See SECTION 017419 – CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL for Hazardous Waste Disposal.

1.22 FIELD OFFICES AND SHEDS

A. Provide all temporary storage and office space that may be required at the site for the safe and proper storage of tools, materials, and like items. Locate field offices and sheds where directed by PPPL’s PM, PTR or CM. Remove field offices and sheds at completion of work and restore areas to original condition. This includes complete removal of stone/aggregate and re-seeding areas accordingly.

B. Prior to installation, submit field office and fencing plans to PPPL demonstrating compliance with all of the items below.

1. Traffic Safety: Provide a traffic Safety Plan. (Plan shall include vehicle barriers and signage as per the Manual for Uniform Traffic Control Devices (MUTCD) standards. Project parking plan shall be included.
2. Fire Department Access: Maintain unobstructed Fire Department Access to and around the field office. The field office shall not obstruct existing fire access. Plan shall address egress, entrances and fire protection equipment for surrounding areas or buildings.
3. Electrical Safety: Electrical must comply with N.E.C. article 590. Submit engineered temporary power plan to PPPL for approval. Obtain PPPL inspections prior to energizing the field office.
4. Environmental Impact: All field office sites must comply with project-specific and PPPL site-specific Soil Erosion and Sediment Control Plans, BMP, approved drainage, fencing and provide for safe pedestrian routes in affected area(s).
5. Fire Exposure to Existing Buildings or Buildings Under Construction: Place construction field offices a minimum of thirty feet from existing structures or structures under construction, in accordance with NFPA 241-2009, Section 4.2 requirements. Field offices over 20 feet in length will require greater separation distances per the NFPA table. The primary mitigation if a construction field office must be placed closer to the work area is to require a field office of noncombustible construction.
6. Building Identification: Obtain a building address for the field office via the PPPL PTR or CM. Post address on the field office as directed by PPPL to allow identification of the structure by responders during an emergency.
7. Jobsite fence shall be configured such that all personnel must enter the job field office prior to entering the construction site unless otherwise approved by PPPL.
8. Field office electrical panels shall have all ARC FLASH labels as required by PPPL.

1.23 MAINTENANCE OF THE WORK AREA

A. The Subcontractor shall:
   1. Maintain the work areas in a safe condition. Remove all accumulations of rubbish and surplus materials at the end of each working day. Restore the work areas to a condition equal to that which existed prior to the start of work, and leave them at completion of the subcontract in a clean, orderly fashion.
   2. Control accumulation of waste materials and rubbish. Collect waste from construction areas and elsewhere daily. Comply with requirements of NFP A 241 for removal of combustible waste material and debris. Enforce requirements strictly. This task includes but is not limited to the following:
      a. Clean interior spaces prior to the start of finish work and maintain areas free of dust and other contaminants during finishing operations. Utilize dust sweeping compound when cleaning interior spaces.
      b. Handle hazardous, dangerous, or unsanitary waste materials separately from other waste by containerizing properly and disposing of material off-site in a lawful manner.

c. Sweep all public streets, roads, and pedestrian walkways and dispose of all collected debris on a daily basis.
d. Provide dumpsters at each site or other suitable means of disposing of all construction debris and trash generated by the Subcontractor's forces during the course of construction. See Section 017419 – CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.
e. Construct, operate, and maintain truck wheel wash facilities at each worksite as required to prevent tracking of mud onto public thoroughfares.

1.24 DUST CONTROL

A. The Subcontractor shall practice care to minimize dust generation. The Subcontractor shall furnish all labor, equipment, and means required to carry out effective measures whenever and as often as necessary to prevent its operation from producing dust in amounts damaging to property or cultivated vegetation, or causing a nuisance to local residents. The Subcontractor shall be responsible for any damage resulting from dust originating from its operations.

1.25 REMOVAL OF UTILITIES, FACILITIES, AND CONTROLS

A. Remove temporary utilities, equipment, facilities, and materials prior to final inspection and as directed by the Construction Coordinator.

B. Remove temporary underground installations to the minimum depth required or as indicated on the subcontract documents. Grade site as indicated or restore to original condition.

C. Clean and repair damage caused by work or use of temporary work.
1. Restore existing and pertinent facilities used during construction to original condition.

PART 2 - PRODUCTS - NOT USED

PART 3 - EXECUTION - NOT USED

END OF SECTION 015000

SECTION 017419 - CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. The following applies to this section:
   1. The PPPL Subcontract General Provisions and Special Terms and Conditions for Construction Subcontracts
   2. Related requirements in other Subcontract Documents including, but not necessarily limited to, the Subcontract Drawings and Specifications.

B. Related Sections:
   1. SECTION 015000 - TEMPORARY FACILITIES AND CONTROLS
   2. SECTION 017700 – CLOSEOUT PROCEDURES
   3. SECTION 018113 - SUSTAINABLE DESIGN REQUIREMENTS

C. Related requirements in other Subcontract Documents listed in the Agreement.

D. Review these documents for coordination with additional requirements and information that applies to work under this Section.

1.2 SUMMARY

A. Section Includes:
   1. Construction waste management plan.
   2. Construction waste recycling.

1.3 PLAN REQUIREMENTS

A. Develop, submit, and implement a Construction Waste Management plan to recycle or salvage at least 50 percent of the non-hazardous construction, demolition and land clearing materials, excluding soil, for compliance with DOE Guiding Principle.

B. The Subcontractor shall dispose of all construction debris in accordance with Federal, State, and Local requirements. In an effort to reduce waste to landfill per DOE E.O. 13834, the Subcontractor shall perform an inventory of waste to be generated and put in process means to recycle and reuse materials generated. Inventory of materials generated for the purpose of diverting waste from landfill shall be provided to the PTR. Additionally, the subcontractor shall provide the PTR data for all items recycled by type. Data shall be reported in tons and provided monthly.

1.4 SUBMITTALS

A. Construction Plan: Submit construction waste management plan describing methods and procedures for implementation and monitoring compliance including the following:
   1. Transportation company hauling construction waste to waste processing facilities.
   2. Recycling and adaptive reuse processing facilities and waste type each facility will accept.

3. Construction waste materials anticipated for recycling and adaptive reuse.
4. On site sorting and site storage methods.

B. Submit documentation prior to Substantial Completion, substantiating construction waste management plan was maintained and goals were achieved.
   1. Trash: Quantity by weight deposited in landfills. Provide receipt tickets from the disposal facility throughout the demolition/removal process to document weight.
   2. Salvaged Material: Quantity by weight with destination for each type of material salvaged for resale, recycling, or adaptive reuse.

1.5 CONSTRUCTION WASTE MANAGEMENT PLAN

A. Construction Waste Landfill Diversion: A goal of 80 percent by weight of construction waste materials for the duration of the Project through resale, recycling, or adaptive reuse.

B. Implement construction waste management plan at start of construction.

C. Review construction waste management plan at pre-construction meeting and progress meetings

D. Distribute approved construction waste management plan to subcontractors and others affected by Plan Requirements.

E. Oversee plan implementation, instruct construction personnel for plan compliance, and document plan results.

1.6 CONSTRUCTION WASTE RECYCLING

A. The Subcontractor shall have the option of sorting the materials off-site or on-site.
   1. Off-Site: If the Subcontractor chooses to sort the material off-site, he shall provide the name of the vendor, and provide procedures on the documentation of the recyclable waste.
   5. On-Site: The Subcontractor shall provide separate collection containers for a minimum of the following materials:
      a) Untreated lumber.
      b) Gypsum wallboard.
      c) Paper, paper products, and cardboard.
      d) Plastics.
      e) Metals.
      f) Glass.
      g) Other salvageable materials.

1.7 CONSTRUCTION WASTE ADAPTIVE RE-USE

A. Arrange with processing facility for salvage of construction material and processing for reuse. Do not reuse construction materials on site except as accepted by PPPL.

1.8 QUALITY ASSURANCE

A. Regulatory Requirements: Comply with applicable local ordinances and regulations.

B. Disposal Sites, Recyclers, and Waste Materials Processors: Use only facilities properly permitted by the State of New Jersey, and by local authorities where applicable.

C. Prior to beginning work at the site, schedule and conduct a conference to review the Construction Waste Management Plan and discuss procedures, schedules and specific requirements for waste materials recycling and disposal. Discuss coordination and interface between the Subcontractor and other construction activities. Identify and resolve problems with compliance with requirements. Record minutes of the meeting, identifying all conclusions reached and matters requiring further resolution.
   1. Attendees: The Subcontractor and related Subcontractor personnel associated with the work of this section, including personnel to be in charge of the waste management program; the PPPL Sustainability representative, Construction Quality Manager; and such additional personnel as PPPL deems appropriate.
   6. Plan Revision: Make any revisions to the Construction Waste Management Plan agreed upon during the meeting and incorporate resolutions agreed to be made subsequent to the meeting. Submit the revised plan to PPPL for approval.

D. Implementation: Designate an on-site person responsible for instructing workers and implementing the Construction Waste Management Plan. Distribute copies of the Construction Waste Management Plan to the job site foreman and each subcontractor. Post Waste Management Plan on the job site bulletin board. Include waste management and recycling in worker orientation and review periodically. Provide on-site instruction on appropriate separation, handling, recycling, and salvaging methods to be used by all parties at the appropriate stages of the work at the site. Include waste management and recycling discussion in pre-fabrication meetings with subcontractors and fabricators. Also include discussion of waste management and recycling in regular job meetings and job safety meetings conducted during the course of work at the site.

PART 2 PRODUCTS – NOT USED

PART 3 EXECUTION

2.1 CONSTRUCTION WASTE COLLECTION

A. Collect construction waste materials in marked bins or containers and arrange for transportation to recycling centers or adaptive salvage and reuse processing facilities. Remove all indicated recyclable materials from the work location to approved containers daily. Failure to remove waste materials will be considered cause for withholding payment and termination of Subcontract.

B. Containers: Change out loaded containers for empty ones as demand requires, but not less than weekly. See Section 015000 for temporary protections for containers.

C. Handling: Deposit all indicated recyclable materials in the containers in a clean (no mud, adhesives, solvents, petroleum contamination), debris-free condition. Do not deposit contaminated materials into the containers until such time as such materials have been cleaned.

D. If the contamination chemically combines with the material so that it cannot be cleaned, do not deposit into the recycle containers. In such a case, request resolution by the Construction Quality Manager as to disposal of the contaminated material. Directions from the Construction Quality Manager do not relieve the Subcontractor from compliance with all legal and regulatory requirements for disposal, nor shall such directions cause a request for modification of the Subcontract

E. Maintain recycling and adaptive reuse storage and collection area in orderly arrangement with materials separated to eliminate co-mingling of materials required to be delivered separately to the waste processing facility.

F. Store construction waste materials to prevent environmental pollution, fire hazards, hazards to persons and property, and contamination of stored materials.

G. Cover construction waste materials subject to disintegration, evaporation, settling, or runoff to prevent polluting air, water, and soil.

2.2 CONSTRUCTION WASTE DISPOSAL

A. Deliver construction waste to waste processing facilities. Obtain receipt for deliveries.

B. Dispose construction waste not capable of being recycled or adaptively reused by delivery to landfill, incinerator, or other legal disposal facility. Obtain receipt for deliveries.

END OF SECTION 017419

SECTION 017700 – CLOSEOUT PROCEDURES

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. General:
   1. The following applies to this section:
      a. PPPL General Provisions for Non-Commercial Subcontracts Part G - Provisions & Conditions for Fixed Price Construction Subcontracts
      b. Related requirements in other Subcontract Documents including, but not necessarily limited to, the Drawings and the Specification Sections in the Project Manual.
   2. Review these documents for coordination with additional requirements and information that apply to work under this Section.

1.2 SUMMARY

A. Section includes:
   1. Closeout procedures
   2. Final cleaning
   3. Adjusting
   4. Spare parts and maintenance products
   5. Maintenance service
   6. Training
   7. Warranties
   8. Commissioning

B. Related Sections:
   1. SECTION 010010 GENERAL REQUIREMENTS
   2. SECTION 015000 TEMPORARY FACILITIES AND CONTROLS
   3. SECTION 017823 OPERATION AND MAINTENANCE DATA
   4. SECTION 017836 WARRANTIES
   5. SECTION 017900 DEMONSTRATION AND TRAINING
   6. SECTION 019113 GENERAL COMMISSIONING REQUIREMENTS
   7. PPPL Subcontract General Provisions and Special Terms and Conditions for Construction Subcontracts

1.3 CLOSEOUT PROCEDURES

A. Submit written certification that subcontract documents have been reviewed, work has been inspected, is complete in accordance with subcontract documents, and is ready for PPPL review and inspection.

B. Provide all submittals to PPPL (or the AE) required by the subcontract documents in accordance with SECTION 013300 - SUBMITTALS.

C. Submit final application for payment identifying total adjusted subcontract sum, previous payments, and sum remaining due.

1.4 FINAL CLEANING

A. Execute final cleaning prior to final project assessment.

B. Clean debris from drainage systems.

C. Clean site; remove SWPPP BMPs, sweep paved areas, rake clean landscaped surfaces.

D. Remove non-hazardous waste, recycling and surplus materials, rubbish, and construction facilities from the site in accordance with SECTION 017419 CONSTRUCTION WASTE MANAGEMENT.

1.5 ADJUSTING

A. Adjust operating products and equipment in accordance with manufacturer's specifications and recommendations, to ensure smooth and unhindered operation.

1.6 SPARE PARTS AND MAINTENANCE PRODUCTS

A. Provide spare parts, maintenance, and extra products in quantities as specified in individual specification sections.

B. Deliver to project site and place in location as directed by the PTR or CM. Obtain signed receipt acknowledging delivery and include copy with request for final payment.

1.7 MAINTENANCE SERVICE

A. Furnish service and maintenance of all components indicated in individual specification sections for the duration of warranty from date of final acceptance.

B. Examine system components at a frequency consistent with reliable operation. Clean, adjust, and lubricate as required.

C. Prior to final acceptance include documentation of systematic examination, adjustment, and lubrication of components. Repair or replace parts whenever required. Use parts produced by the manufacturer of the original component.

D. Maintenance service shall not be assigned or transferred to any agent or lower-tier subcontractor without prior written consent of PPPL.

1.8 FINAL COMPLETION

A. Submit a written request for final inspection for acceptance. On receipt of request, PPPL will either proceed with inspection or notify the Subcontractor of unfulfilled requirements. PPPL will prepare a final Certificate for Payment after inspection or will notify the Subcontractor of construction that must be completed or corrected before the certificate will be issued.

1. Re-Inspection: Request re-inspection when the work identified in previous inspections as incomplete or unacceptable is completed or corrected.

B. Final Payment, before submitting application for final payment and/or release of retention, complete and submit the following:
   1. Final/ completed punch lists, and all items required for substantial completion if not already delivered.
   2. Certified copy of Architect's Substantial Completion inspection list of items to be completed or corrected (punch list), endorsed and dated by Architect. The certified copy of the list shall state that each item has been completed or otherwise resolved for acceptance.
   3. Final written notification of Commissioning Authority (CA) acceptance.
   4. Completion of the work, including as-builts and other submittals if and as required.
   5. Final Inspection Report (signature by Subcontractor).
   6. Certified Payroll Log and final Certified Payrolls.
   7. Return of Identification (ID) badges, Dosimeter Badges (if any) and keys (if any).
   8. Final Invoice.
   9. Release Upon Final Payment.

1.9 OWNER OCCUPANCY REQUIREMENTS

A. Partial Owner Occupancy: PPPL reserves the right to occupy, place and install equipment in completed areas of the building before Substantial Completion, provided such occupancy does not interfere with completion of the Work. Such use of building shall not constitute acceptance of the total Work.
   1. The Architect will prepare a Certificate of Substantial Completion for each specific portion of the Work to be occupied prior to partial Owner occupancy.
   2. PPPL will obtain a Certificate of Occupancy (CO) or Temporary Certificate of Occupancy (TCO) from the authorities having jurisdiction prior to Owner occupancy.
   3. Prior to partial Owner occupancy, mechanical and electrical systems shall be fully operational and required tests and inspection shall be successfully completed.
   4. Upon partial occupancy, maintenance and operation of mechanical and electrical systems serving occupied areas of the building shall become the Owner's responsibility.
   5. Upon partial occupancy, all maintenance and custodial service shall become the PPPL's responsibility.

1.10 OPERATION AND MAINTENANCE MANUALS

A. Refer to Section 017823 – Operation and Maintenance Data, for requirements.

1.11 WARRANTIES

A. Refer to Section 017836 – Warranties for warranty requirements

1.12 DEMONSTRATION AND TRAINING

A. Refer to Section 017900 – Demonstration and Training, for requirements.

B. Subcontract closeout will require completion of all demonstration and training activities (including factory training sessions).

1.13 COMMISSIONING

A. Commissioning is the final acceptance activity of the completed construction. Commissioning must be completed prior to subcontract closeout. Refer to Section 019113 – General Commissioning Requirements.

B. Commissioning may identify any system performance concerns that may delay subcontract closeout, application for final payment, or payment of retention.

C. The Subcontractor will be responsible for the resolution of system performance concerns. If these concerns are not the direct or indirect responsibility of the Subcontractor, then a letter indicating the scope and extent of the situation shall be submitted for PPPL review and approval.

PART 2 PRODUCTS – Not Used

PART 3 EXECUTION – Not Used

END OF SECTION 017700

SECTION 017823 - OPERATION AND MAINTENANCE DATA

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. General:
   1. The following applies to this section:
      a. PPPL General Provisions for Non-Commercial Subcontracts, Part G - Provisions & Conditions for Fixed Price Construction Subcontracts
      b. The Subcontract Drawings and Specifications.
      c. Related requirements in other Subcontract Documents listed in the Agreement.
   2. Review these documents for coordination with additional requirements and information that apply to work under this Section.

1.2 SUMMARY

A. Section includes:
   1. Format and content of manuals
   2. Instruction of PPPL's personnel
   3. Schedule of submittals

1.3 RELATED SECTIONS

A. SECTION 013300 –SUBMITTALS

B. SECTION 017700 – CLOSEOUT PROCEDURES

C. SECTION 017836 – WARRANTIES

D. SECTION 017900 – DEMONSTRATION AND TRAINING

E. SECTION 019113 – GENERAL COMMISSIONING REQUIREMENTS

F. Individual specifications sections for specific requirements for operation and maintenance data

1.4 GENERAL REQUIREMENTS

A. Compile operating and maintenance data in the form of manuals appropriate for care and maintenance of products provided under the subcontract and specific information requested in various technical sections of these specifications.

1.5 FORMAT

A. Prepare data in the form of an instructional manual.

B. Binders: Commercial quality, 8-1/2 x 11 inches, three side-ring type "D" binders with durable plastic covers. When multiple binders are used, correlate data into related consistent groupings.

C. Cover: Identify each binder with typed or printed title OPERATION AND MAINTENANCE MANUALS. Identify title of project, identify building, structure, equipment, system, and identify subject matter of contents.

D. Dividers: Provide tabbed dividers for each separate product and system with typed description of product and major component parts of equipment.

E. Text: Manufacturer's printed data or typewritten data on 20 pound bond paper.

F. Drawings: Provide with reinforced punched binder tab. Bind in with text; fold larger drawings to the size of text pages.

G. Content: Prepare a table of contents for each volume with each equipment or system identified in three parts as follows:
   1. Part 1: Directory, listing names, addresses, and telephone numbers of, Subcontractor, lower-tier subcontractors, and major equipment suppliers.
   2. Part 2: Operation and maintenance instructions arranged by equipment and or system and subdivided by specification section. For each category, identify names, addresses, and telephone numbers of subcontractors and suppliers. Identify the following:
      a. Significant design criteria
      b. Equipment list and description
      c. Parts list for each component, including recommended spare parts
      d. Operating instructions
      e. Maintenance instructions for equipment and systems
      f. Maintenance instructions for special finishes including recommended cleaning methods and materials and special precautions identifying detrimental agents
   3. Part 3: Project documents and certificates including the following:
      a. Shop drawings and product data
      b. Certificates
      c. Photocopies of warranties

H. Electronic copy shall be in PDF format with bookmarks and searchable text.

1.6 CONTENT OF EACH VOLUME

A. Table of Contents: Provide title of project; names, addresses, and telephone numbers of Project Engineer and Subcontractor with name of responsible parties; schedule of products and systems indexed to content of the volume and indexed by reference to the subcontract documents.

B. For Each Product or System: List names, addresses, and telephone numbers of lower-tier subcontractors and suppliers including local source of supplies and replacement parts.

C. Product Data: Mark each sheet to clearly identify specific products and component parts and data applicable to installation. Delete inapplicable information.

D. Drawings: Supplement product data to illustrate relations of component parts of equipment and systems to show control and flow diagrams. Coordinate drawings with information in the project as-built documents.

E. Typed Text: As required to supplement product data. Provide logical sequence of instructions for each procedure incorporating manufacturer's instructions.

F. Warranties: As specified in SECTION 017836 WARRANTIES.

1.7 MANUAL FOR MATERIALS AND FINISHES

A. Building Products, Applied Materials, and Finishes: Include manufacturer and (name, address, phone number) product data, with catalog number, size, composition, and color and texture designations. Provide information for reordering custom manufactured products.

B. Instructions for Care and Maintenance: Include manufacturer's recommendations for cleaning agents and methods, precautions against detrimental agents and methods, and recommended schedule for cleaning and maintenance.

C. Moisture Protection and Weather Exposed Products: Include product data listing applicable reference standards, chemical composition, and details of installation. Provide recommendations for inspections, maintenance, and repair.

D. Additional Requirements: As specified in individual product specification sections.

1.8 MANUAL FOR EQUIPMENT AND SYSTEMS

A. Each Item of Equipment and Each System: Include description of unit or system and component parts. Identify function, normal operating characteristics, and limiting conditions. Include performance curves with engineering data, tests, and complete nomenclature and model number of replaceable parts. Performance data shall be provided in sufficient detail to provide a clear indication of energy usage (i.e., power demand and consumption, and fuel consumption.

B. Panel board Circuit Directories: Provide electrical service characteristics, controls, and communications; typed.

C. Include color coded wiring diagrams as installed.

D. Operating Procedures: Include start-up, break-in, and routine normal operating instructions and sequences. Include regulation, control, stopping, shut-down, and

emergency instructions. Include summer, winter, and any special operating instructions.

E. Maintenance Requirements: Include routine procedures and guide for preventative maintenance and troubleshooting; disassembly, repair, and reassembly instructions; and alignment, adjusting, balancing, and checking instructions.

F. Provide servicing and lubrication schedule and list of lubricants required.

G. Include manufacturer's printed operation and maintenance instructions.

H. Provide original manufacturer's parts list, illustrations, assembly drawings, and diagrams required for maintenance.

I. Provide list of original manufacturer's spare parts, current prices, and recommended quantities to be maintained in storage.

J. Additional Requirements: As specified in individual product specification sections.

K. Provide a listing in table of contents for design data, with tabbed dividers and space for insertion of data.

1.9 INSTRUCTION OF PPPL PERSONNEL

A. Refer to Section 017900 for detailed requirements for Training.

B. Before final inspection, instruct PPPL's designated personnel in operation, adjustment, and maintenance of products, equipment, and systems at agreed upon times. Length of instructions shall be based upon times indicated in the individual specification sections.

C. Use operation and maintenance manuals as basis for instruction. Review contents of manual with personnel in detail to explain all aspects of operation and maintenance.

D. Prepare and insert additional data in the operation and maintenance manual when need for such data becomes apparent during instruction.

1.10 SUBMITTAL SCHEDULE

A. Prepare and insert additional data, including visual recording of instruction, in operation and maintenance manual when need for such data becomes apparent during instruction.

B. For equipment, or component parts of equipment put into service during construction and operated by PPPL, submit documents within ten days after acceptance of referenced equipment.

C. Submit one electronic copy of completed volumes of the manuals 14 calendar days prior to final inspection. One copy will be reviewed and returned after final inspection

with PPPL (and/or AE) comments. Revise content of all document sets as required prior to final submission.

D. Submit three sets of revised final volumes and one electronic copy in final form within 10 days after receipt of PPPL (and/or AE) comments.

PART 2 PRODUCTS – Not Used

PART 3 EXECUTION – Not Used

END OF SECTION 017823

SECTION 017836 – WARRANTIES

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. General:
1. The following applies to this section:
a. PPPL General Provisions for Non-Commercial Subcontracts Part G - Provisions & Conditions for Fixed Price Construction Subcontracts
b. Related requirements in other Subcontract Documents including, but not necessarily limited to, the Subcontract Drawings and Specifications.
2. Review these documents for coordination with additional requirements and information that apply to work under this Section.

1.2 SUMMARY

A. Section Includes:
1. General Guarantee of Work
2. Preparation and submittal of warranties.
3. Time and Schedule of submittals.

B. Related Sections:
1. SECTION 017700 – CLOSEOUT PROCEDURES
2. Individual specifications sections for warranties, guarantees, and service and maintenance contracts required for specific products or work.

1.3 GUARANTEE

A. The General Construction Subcontractor is required to provide the warranty for work performed, including rework and installation, labor and material, where applicable. Warranty shall be for period of one year from the date of final acceptance of the work, in accordance with Federal Acquisition Regulation (FAR) 52.246-21 and PPPL Subcontract General Provisions and Special Terms and Conditions for Construction Subcontracts

B. Warranty for equipment, subsystems, and manufactured components shall be for a period of 12 months from the date of being put into use as documented by Certificate of Occupancy. The warranty obligation includes field labor and material to remedy the problem, as well as replacement of defective parts, components, any whole equipment items as may be required. All manufacturer normal warranties that go beyond the warranty as described above shall remain in force and turned over to

PPPL.

C. Whenever a warranty or guarantee provision is included in the specifications, Subcontractor's responsibility for defects after the stated performance period in the warranty or guarantee shall be at a minimum as provided in the General Provisions Clause A21.

D. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by the Subcontract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve Subcontractor of obligations under requirements of the Subcontract Documents.

E. All materials and equipment shall meet all manufacturer specifications to obtain a minimum system warranty established by the manufacturer.

1.4 FORM OF SUBMITTALS

A. Bind in commercial quality 8-1/2 x 11 inches, three side-ring type "D" binders with durable plastic cover.

B. Cover: Identify each binder with typed or printed title WARRANTIES. Identify with title of project; name, address, and telephone number of Subcontractor and equipment supplier; and name of responsible company principal.

C. Table of Contents: Neatly typed, in the sequence of the table of contents of the project manual with each item identified with the number and title of the specification section in which specified and the name of product or work item.

D. Separate each warranty with index tab sheets keyed to the table of contents listing. Provide full information using separate typed sheets as necessary. List Subcontractor, supplier, and manufacturer with name, address, and telephone number of responsible company principal.

E. In addition to the hard copy submittals described above, submit an electronic copy in accordance with Section 013300 Submittals.

1.5 PREPARATION OF SUBMITTALS

A. Obtain warranties, guarantees, bonds, and service and maintenance subcontracts executed in quadruplicate by responsible Subcontractors, suppliers, and manufacturers, within 14 calendar days after acceptance of the applicable item of work. Except for items put into use with PPPL’s permission, leave date of beginning of time of warranty until the date of substantial completion is determined.

B. Provide a copy of each warranty/guarantee and service subcontract issued. Include an information sheet for PPPL personnel giving:
   1. Proper procedures in the event of failure
   2. Required maintenance to maintain subcontracts
   3. Instances which might affect the validity of subcontracts

C. Verify that documents are in proper form, comply with subcontract documents, contain full information, and are notarized.

D. Co-execute submittals when required.

E. Retain warranties until time specified for submittal.

F. Submit three original, signed copies, each, and an electronic copy.

1.6 TIME OF SUBMITTALS

A. For equipment or component parts of equipment put into service during construction with PPPL's permission, submit documents within 14 calendar days after acceptance of said equipment by PPPL.

B. Make other submittals within 14 calendar days after date of substantial completion, prior to final application for payment.

C. For items of work for which acceptance is delayed beyond the date of substantial completion, submit within 14 calendar days after acceptance, listing the date of acceptance as the beginning of the warranty period.

1.7 SPECIFIC WARRANTIES

A. Specific warranties of construction items, called out in individual sections of divisions 2 through 33 shall take precedence over the warranties stated above.

PART 2 PRODUCTS – Not Used

PART 3 EXECUTION – Not Used

END OF SECTION 017836

SECTION 017900 – DEMONSTRATION AND TRAINING

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. It is the Subcontractor's responsibility to refer to all related sections and documents for additional conditions and requirements pertaining to this section, including but not limited to those listed below. In the event of conflict between specific requirements of the various documents, the more restrictive, more extensive requirement shall govern.

1. PPPL General Provisions for Non-Commercial Subcontracts, Part G - Provisions & Conditions for Fixed Price Construction Subcontracts
2. The Subcontract Drawings and Specifications.
3. Related requirements in other Subcontract Documents listed in the Agreement.

1.2 SUMMARY

A. This section includes administrative and procedural requirements for instructing PPPL's personnel, including the following:
1. Demonstration of operation of systems, subsystems and equipment.
2. Training in operation and maintenance of systems, subsystems and equipment.

B. Training is required in three (3) general categories.
1. Classroom overview and general system architecture training.
2. On the-job-training with the equipment as applied to the system.
3. Factory training.

C. Demonstration and training video recordings

1.3 TRAINING AND DOCUMENTATION

A. Instruction Program: Submit two (2) copies of the outline of the training program for demonstration and operational and maintenance training. Include proposed dates, times, length of instruction time, and instructors' names for each training module.

B. Attendance Record: For each training session, submit summary of subject matter, list of participants and length of instruction time.

C. Demonstration and Training Video Recordings: Submit two copies within seven days of the end of each training module.

1. Identification: On each copy, provide an applied label with the following

information:

a. Name of Project

b. Name and address of photographer

c. Name of Architect

d. Name of Subcontractor

e. Date videotape was recorded

f. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction

2. Transcript: Prepared and bound in format matching operation and maintenance manuals. Mark appropriate identification on the front and spine of each binder. Include a cover sheet with same label information as the corresponding video recording. Include name of Project and date of video recording on each page.

1.4 QUALITY ASSURANCE

A. Facilitator Qualifications: An individual familiar with the project and the training requirements.

B. Instructor Qualifications: An individual experienced in training or educating operating and maintenance personnel for the systems and equipment indicated for this project. A factory-authorized service representative experienced in operation and maintenance procedures and training is preferred.

1.5 COORDINATION

A. Coordinate instruction schedule with PPPL's operators. Adjust schedule as required to obtain maximum participation as well as minimize disrupting Owner's operations.

B. Providing notification of dates, times, length of instruction time, and course content to PPPL, Commissioning Agent, and appropriate subcontractors. See Section 019113 GENERAL COMMISSIONING REQUIREMENTS for additional information and details.

C. Coordinate content of training with content of approved operation, and maintenance manuals. Do not submit an instruction program until operation and maintenance data has been reviewed and approved by PPPL.

1.6 INSTRUCTION PROGRAM

A. Training is required for all systems that require operation and maintenance. Develop an instruction program that includes training modules for each system and equipment as required by individual specification sections. Samples are:

1. Conveying systems, including elevators, lifts and hoists
2. HVAC systems, including air-handling equipment, air distribution systems and terminal equipment and devices.
3. Heat generation, including boilers feed water equipment, pumps, steam distribution piping and water distribution piping.
4. Refrigeration systems, including chillers, cooling towers, condensers, pumps and distribution piping.
5. Leak detection systems (exterior or interior)
6. HVAC instrumentation and controls.
7. Electrical distribution, including transformers, switchboards, panel boards.
8. Uninterruptible power supplies.
9. Motor controls.
10. Emergency power systems.
11. Lighting equipment and controls.
12. Communication and Audio VDO systems, including intercommunication, surveillance, clocks and programming, voice and data and television equipment.
13. Motorized doors, including overhead coiling doors, overhead coiling grilles, and automatic entrance doors.
14. Equipment, including projection screens, loading dock equipment, trash compactors, food-service equipment, appliances, etc.
15. Fire-protection systems, including fire pumps and fire-extinguishing systems and fire shutters
16. Fire alarm.
17. Security systems including automated doors and windows
18. Transportation systems, including elevators, wheelchair lifts and escalators.
19. Sump pump systems

B. Training sessions: Develop a teaching outline for each module. Include a description of specific skills and knowledge that participant is expected to master. For each module, include instruction for the following:
    1. Basis of System Design, Operational Requirements, and Maintenance Criteria: Include the following:
        a. System, subsystem and equipment descriptions
        b. Performance and design criteria
        c. Operating standards and limitations
        d. Equipment functions
        e. Operating characteristics
        f. Performance parameters
    2. Documentation: Review the following items in detail:
        a. Emergency manuals
        b. Operations manuals
        c. Maintenance manuals
        d. Warranties and bonds
        e. Maintenance service agreements and similar continuing commitments
    3. Operations: Address the following, as applicable:

a. Start/stop procedures
b. Break-in procedures
c. Routine and normal operations
d. Control sequences
e. Safety procedures
f. Operating procedures for system, subsystem or equipment failure
g. Seasonal and weekend operating instructions
h. Special operating instructions and procedures

4. Adjustments: Address the following as applicable:
   a. Alignments
   b. Adjustments
   c. Calibration
   d. Loop tuning
5. Troubleshooting: Include the following:
   a. Diagnostic instructions
   b. Test and inspection procedures
6. Maintenance: Include the following:
   a. Inspection procedures
   b. Cleaning agents and procedures for routine cleaning
   c. Preventive maintenance
   d. Use of special tools
7. Emergencies: Include the following, as applicable:
   a. Instructions on meaning of warnings, trouble indications, and error messages.
   b. Emergency shutdown instructions for each type of emergency.
   c. Operating instructions for conditions outside of normal operating limits.
   d. Special operating instructions and procedures.
8. Repairs: Include the following:
   a. Diagnosis.
   b. Repair instructions.
   c. Disassembly; component removal, repair, and replacement; and reassembly instructions.
   d. Instructions for identifying parts and components.
   e. Review of spare parts needed for operation and maintenance.

1.7 PREPARATION

A. Review and finalize instruction schedule.

B. Prepare lesson plan.

C. Prepare instructional aides.

D. Arrange for and inspect classroom facilities.

E. Arrange for on-the-job training location and equipment/system availability for instructional demonstrations.

F. Schedule instruction for mutually satisfactory times in accordance with the overall project schedule. Schedule training with Commissioning Authority (CA) through PPPL, with at least seven (7) days' advance notice.

G. For equipment that requires seasonal operation, provide instruction at the start of each season.

H. For instruction that must occur outside, review forecasted weather conditions and procedures to ensure conditions are favorable.

1.8 INSTRUCTION

A. The instructor must present all aspects of the training program in a sequence that will be conducive to the overall training and education of the students.

B. PPPL will identify and furnish the Subcontractor with names and positions of participants.

1.9 DEMONSTRATION AND TRAINING VIDEOS

A. General: Engage a qualified commercial photographer to record demonstration and training in digital media format. Record each training module separately. Include classroom instructions and demonstrations, board diagrams, and other visual aids, but not student practice.

1. At the beginning of each training module, record each chart containing the learning objective and lesson outline.

B. Digital Video Recordings: Provide high-resolution, digital video in MPEG format, produced by a digital camera with minimum sensor resolution of 12 megapixels and capable of recording in full HD mode with vibration reduction technology.

1. Submit video recordings by uploading to a web-based Project software site.

2. File Hierarchy: Organize folder structure and file locations according to Project Manual table of contents. Provide a complete screen-based menu.

3. File Names: Utilize file names based on the name of equipment generally described in the video segment, as identified in Project specifications.

4. Subcontractor and Installer Contact File: Using appropriate software, create a file for inclusion on the equipment demonstration and training recording that describes the following for each Subcontractor involved on the Project, arranged according to Project Manual table of contents:

a. Name of Subcontractor/Installer.

b. Business address.

c. Business phone number.

d. Point of contact.

e. Email address.

C. Recording: Mount camera on tripod before starting recording, unless otherwise necessary to show area of demonstration and training. Display continuous running time.

1. Film training session(s) in segments not to exceed 15 minutes.

a. Produce segments to present a single significant piece of equipment per segment.

b. Organize segments with multiple pieces of equipment to follow the order of the Project Manual table of contents.

c. Where a training session on a particular piece of equipment exceeds 15 minutes, stop filming and pause training session. Begin training session again upon commencement of new filming segment.

D. Light Levels: Verify light levels are adequate to properly light equipment. Verify equipment markings are clearly visible prior to recording.

1. Furnish additional portable lighting as required.

E. Narration: Describe scenes on digital media either by audio narration by microphone while or dubbing audio narration off-site after digital media is recorded. Include description of items being viewed. Describe vantage point, indicating location, direction (by compass point), and elevation or story of construction.

F. Preproduced Video Recordings: Provide video recordings used as a component of training modules in the same format as recordings of live training.

PART 2 PRODUCTS – Not Used

PART 3 EXECUTION – Not Used

END OF SECTION 017900

SECTION 018113 – SUSTAINABLE DESIGN REQUIREMENTS

PART 1 GENERAL

1.A SUMMARY

A. Section Includes:
1. The Work contributes to the High Performance Sustainable Building (HPSB) requirements for the Building. Work includes environmental, sustainable, and green building practice requirements related to energy conservation and efficiency, indoor air quality, and resource efficiency.
2. HPSB applicability: All HPSB Guiding Principles (GP) shall be addressed as YES or NOT APPLICABLE through design, construction methods or other means. Specific documentation may be required to be submitted to PPPL to show compliance with the DOE Requirement as described in Section 1.2.A.1. A copy of the DOE HPSB Compliance checklist for New Construction & Major Renovations is appended to this section for reference (Attachment A).
3. Projects which are not New Construction & Major Renovations shall follow instructions provided by the project design documents or PPPL statement of work.

B. Related Sections:
1. Divisions 01 through 33 Sections for HPSB requirements specific to the work of each of these Sections by reference to the submittals that are required. Should sustainable design requirements be applicable, but not listed in the specific section, the requirements of this section shall govern.
2. Section 017419 Construction Waste Management and Disposal.
3. Section 018119 Indoor Air Quality During Construction.
4. Section 019113 General Commissioning Requirements.

1.B HIGH PERFORMANCE SUSTAINABLE BUILDING REQUIREMENTS

A. General requirements:
1. New Construction & Major Renovations projects shall follow the 2016 Guiding Principles for Sustainable Federal Buildings published in Executive Order 13693, on February 26, 2016 by the White House Council on Environmental Quality. A checklist that summarizes the specific project requirements shall be provided by the AE design consultant. An example of the checklist is shown in Attachment A.
2. Provide all submittals and backup information to demonstrate compliance with the Guiding Principles.
    a. All submittals shall conform to Section 013300 Submittals.

1.C QUALITY ASSURANCE

A. HPSB Project Management and Coordination: Prior to the pre-construction meeting, the Subcontractor shall identify and assign one person on Subcontractor's staff to be

the HPSB Project Manager who shall be responsible for environmental issues compliance and coordination:

1. Experience: Similar responsibilities and successful performance for previous sustainable building construction project(s).
2. Responsibilities: Review and Identify Subcontract Documents for environmental issues, products, coordination and oversee Project Environmental Goals.
   a. Retain approved Submittals associated with HPSB, tabulation charts and other records to document progress toward meeting environmental requirements.
   b. Identify applicable records (tagged for HPSB record) available for review by A/E or PPPL.
   c. Provide Action Plans, Progress Reports and final documentation as needed and requested by PPPL.
3. HPSB Project Manager duties include the following:
   a. Commissioning coordination.
   b. Obtain calculations (where applicable).
   c. Complete assigned Templates (where applicable).
   d. Other tasks as necessary for compliance with HPSB guidelines.
   e. Meetings: Lead discussion of Environmental Goals at following meetings:
      1) Pre-construction meetings.
      2) Waste management conference.
      3) Pre-installation meetings.
      4) Regularly scheduled job-site meetings.
      5) Special sustainability issues meetings.

PART 2 PRODUCTS

2.1 MATERIAL SELECTION

A. PPPL's prime contract requires the purchase of certain types of environmentally preferable products such as, but not limited to, EPEAT-registered electronic devices, ENERGYSTAR and FEMP-listed energy-consuming equipment, Water-Sense listed products, bio-based materials, EPA-designated recycled content products, non-ozone-depleting chemicals and non-toxic or less toxic alternatives when practical. Where such products are available and meet the technical requirements of the work, they should be specified to the maximum extent practical. Additional information on environmentally preferable products is available online at: https://pppl.princeton.edu/PPPL_Environmentally_Preferable_Purchasing

PART 3 EXECUTION – Not Used

END OF SECTION 018113

ATTACHMENT A

2016 Guiding Principles Checklist for New Construction and Modernization v2.0

Per Executive Order 13693, the 2016 Guiding Principles were issued on February 26, 2016 by the White House Council on Environmental Quality, Office of Sustainability, in two documents: "Determining Compliance with the Guiding Principles for Sustainable Federal Buildings" (GP Compliance Document), which outlines evaluation criteria on which the Guiding Principles will be scored, and the "Guiding Principles for Sustainable Federal Buildings and Associated Instructions," which provides instructions, guidance and recommended practices. This Checklist was developed by the U.S. Department of Energy, Federal Energy Management Program, as a voluntary tool for evaluating and tracking a building's progress towards meeting the evaluation criteria defined in the GP Compliance Document. Agencies are invited to use this Checklist as is, or to reformat it to suit agency needs. Some cells in this worksheet are protected. For the password, contact the FEMP sustainability program at the link below.

For more information about the Guiding Principles and FEMP Support: http://energy.gov/eere/femp/guiding-principles-sustainable-federal-buildings

Agency:	Department/Region:	Federal Real Property ID:
Building Name:	Address:	Checklist Manager:
Federal Campus:	City & State:	Date:

Instructions: 1 or new construction, 20 out of 21 metrics are required, and for modernization, 19 out of 21 metrics are required. 1 or building evaluation purposes, "not applicable" may be used when the building's inherent function, mission, safety, or designation prevents compliance with a specific Guiding Principle, element, or sub-element. "Not applicable" is equivalent to compliance with that Guiding Principle, element, or sub-element. Metrics marked as "N/A" should be justified and documented.

Select Guiding Principle Metrics Type:	New Construction Metrics	20	In Compliance
		(click menu above)	

Guiding Principle	Required or Additional	Metric	Suggested Evidence of Compliance (may need one or more)	Compliance (dropdown box) Yes/No/In-Process/Not Applicable	Notes/Comments
I. Employ Integrated Assessment, Operation, and Management Principles					
Integrated Assessment, Operation, and Management	REQ	1. Consider the environmental impact of siting decisions and use an integrated project team to: establish energy and other environmental performance goals in the design process; follow sustainable landscape design principles; evaluate electric vehicle charging needs; consider design choices that improve environmental performance, support health and wellness of building occupants and consider climate risks including wildfire; and consider all stages of the building's life cycle.	☐ Integrated Team roster or equivalent ☐ Baseline Guiding Principles assessment ☐ Site assessment ☐ Sustainable design charrette ☐ Other	Yes	[Sample] Project uses Integrated Project Team: Energy goals and environmental goals established via LEED Gold; Sustainable design charrette was held on May 16, 2016 with the project team. Meeting results are available.
Commissioning	REQ	2. Commission and recommission at least every 4 years to optimize building performance using commissioning agents who are independent of the design and construction or operating team. Commissioning should be consistent with the Energy Independence and Security Act (EISA) section 432[1] and Federal Energy Management Program (FEMP) commissioning guidance.[2]	☐ Commissioning report ☐ Other	Yes	
II. Optimize Energy Performance					
Energy Efficiency	REQ	3. A. For new construction, ensure energy efficiency is 30% better than the current American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) 90.1 standard, OR	☐ Document reduction calculation method and results ☐ ENERGY STAR Portfolio Manager documentation	Yes	
		B. For modernization, ensure: 1) Energy use is 20% below the fiscal year (FY) 2015 energy use baseline, OR 2) Energy use is 30% below the FY 2003 energy use baseline, OR 3) The building has an ENERGY STAR® rating of 75 or higher, OR 4) For building types not in ENERGY STAR Portfolio Manager, where adequate benchmarking data exists, the building is in the top quartile of energy performance for its building type, AND	☐ ENERGY STAR Portfolio Manager documentation ☐ Document reduction calculation method and results		
Renewable and Clean Energy		C. Use energy efficient products, as required by statute	☐ Agency or site purchasing policy ☐ Purchasing contracts ☐ Purchase orders ☐ Affirmative procurement reports ☐ Other		
	ADD	4. Evaluate and implement, where appropriate, life cycle cost effective renewable energy projects on site; consider long term off site renewable sources and Renewable Energy Certificates (RECs); and utilize clean and alternative energy where possible.	☐ Renewable energy contract ☐ Utility records ☐ Photos of on site renewable ☐ Photos of alternative energy ☐ Proof of purchase of RECs ☐ Other	Yes	
Metering	ADD	5. Install building level meters for electricity, natural gas, and steam; install advanced or standard meters as appropriate.	☐ Building-level utility bills ☐ Design specifications for advanced meters ☐ Photos of building-level meters ☐ Other	Yes	
Benchmarking	ADD	6. Benchmark building performance at least annually, preferably using ENERGY STAR Portfolio Manager; regularly monitor building energy performance against historic performance data and peer buildings.4	☐ ENERGY STAR Portfolio Manager documentation ☐ Other	Yes	

Guiding Principle	Required or Additional	Metric	Suggested Evidence of Compliance (may need one or more)	Compliance (dropdown box) Yes/No/In-Process/Not Applicable	Notes/Comments
III. Protect and Conserve Water					
Indoor Water Use	REQ	7. A. Build to ASHRAE standard 189.1-2014 sections 6.3.2, 6.4.2, and 6.4.3, or current comparable ASHRAE standards, AND B. Use water efficient products: install building level water meters; optimize cooling tower operations; and eliminate single pass cooling.	☐ Photos of building level meters ☐ Building-level utility bills ☐ ENERGY STAR Portfolio Manager documentation ☐ Agency or site purchasing policy ☐ Document efficient products (photos, lists, product specs, etc.) ☐ Other	Yes	
Outdoor Water Use	ADD	8. A. Separately meter water for irrigation systems greater than 25,000 square feet, AND B. Use water efficient landscapes, AND C. Limit potable water use for irrigation to 50% or more below conventional practices using methodologies from (but not the numeric requirements contained in) ASHRAE standard 189.1 2014 section 6.5.1, or current comparable ASHRAE standards, to calculate water use of conventional practices.	☐ Landscape plan and/or narrative of landscape ☐ Building-level utility bills (showing metered use for irrigation) ☐ Document when < 25,000 sf2 of landscape is irrigated ☐ Document when no water is used to irrigate ☐ Photos ☐ Other	Yes	[Sample] Drip irrigation to be used along side native plants for 15% of the landscape area. The remainder will have no irrigation.
Alternative Water	ADD	9.Consider alternative sources of water where cost-effective and permitted by local laws and regulations.	☐ Document consideration of alternative sources ☐ Document use of alternative water (photos, narrative, etc.) ☐ Other	Yes	
Stormwater Management	ADD	10. For new construction meet or exceed EISA section 438 stormwater management requirements.	☐ Stormwater plan ☐ Stormwater permit ☐ Document stormwater strategies (photos, narrative, etc.) ☐ Other	Yes	
IV. Enhance Indoor Environmental Quality					
Ventilation and Thermal Comfort	REQ	11. Meet the current ASHRAE 55 and either 62.1or 62.2 standards for ventilation and thermal comfort.	☐ ENERGY STAR Certification ☐ Documentation from licensed architect, engineer or qualified building professional ☐ Other	Yes	
Daylighting and Lighting Controls	ADD	12. Maximize opportunities for daylighting in regularly occupied space, automatic dimming controls or accessible manual controls, task lighting, and shade and glare control.	☐ Schematic of floor layout ☐ Document (photos, list, narrative, design specifications, etc.) ☐ Other	Yes	
Indoor Air Quality	ADD	13. Develop and implement an indoor air quality policy that considers the following: moisture control, use of low emitting materials and products with low pollutant emissions, necessary protocols to protect indoor air quality during construction and in the finished building, prohibition of smoking in any form inside and within 25 feet of all building entrances, operable windows, and building ventilation intakes, and use of integrated pest management techniques.	☐ Indoor air quality policy ☐ Compilation of individual policies ☐ Other	Yes	
Occupant Health and Wellness	ADD	14. Promote opportunities for voluntary increased physical movement of building occupants such as making stairwells an option for circulation, active workstations, fitness centers, and bicycle commuter facilities; and support convenient access to healthy dining options, potable water, daylight, plants, and exterior views.	☐ Building or site-wide health and fitness programs ☐ Fitness centers ☐ Healthy dining options ☐ Active/ergonomic work stations ☐ Other	Yes	

Guiding Principle	Required or Additional	Metric	Suggested Evidence of Compliance (may need one or more)	Compliance (dropdown box) Yes/No/In-Process/Not Applicable	Notes/Comments
V. Reduce the Environmental Impact of Materials					
Material Content and Performance	RQD	**15.** Procure products that meet the following requirements where applicable: A. RCRA section 6002, AND B. FSRIA section 9002, AND C. Federally Recommended Specifications, Standards and Ecolabels or are on the Federal Green Procurement Compilation for other green products, as appropriate, AND	☐ Agency or site purchasing policy ☐ Other	Yes	
		D. Avoid ozone depleting compounds and high GWP chemicals.	☐ Document no ODC and GWP chemicals ☐ Plan to avoid ODC and GWP chemicals ☐ Other		
Waste Diversion	RQD	**16.** Where markets exist, provide reuse and recycling services for building occupants and divert at least 50% of non-hazardous non-construction related materials from landfills.	☐ Recycling contract ☐ Waste stream report/s ☐ Other	Yes	
Materials Management	ADD	**17.** Where markets exist, divert at least 50% of construction and demolition materials from landfills.	☐ Construction and Demolition (C&D) contract ☐ Construction and demolition policy ☐ Other	Yes	
VI. Assess and Consider Climate Change Risks					
Mission Criticality	RQD	**18.** Determine long-term mission criticality of the physical asset and operations to be housed in the facility to inform the design of new construction and modernization to increase climate resilience.	☐ Climate resilience plan ☐ Other	Yes	
Floodplain Considerations	RQD - NC	**19.** For new construction, avoid, to the extent possible, the long- and short-term adverse impacts associated with the occupancy and modification of floodplains and avoid floodplain development whenever there is a practicable alternative.	☐ Site plan ☐ Other	Yes	
Facility Design	RQD - NC	**20.** For new construction, balance options to address predicted climate change impacts against mission criticality, cost, and security to determine design parameters; at a minimum, include low and no cost resilience measures to address predicted climate conditions.	☐ Document building design features Climate resilience plan ☐ Climate resilience plan ☐ Other	Yes	
Facility Adaptation	RQD - M	**21.** For modernization, take action to mitigate identified risks, considering mission criticality, climate impacts, cost, and phased adaptation over time.	☐ Document building design features ☐ Climate resilience plan ☐ Other	No	

SECTION 018119 - INDOOR AIR QUALITY DURING CONSTRUCTION

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Subcontract, including 010010 General Conditions and other Division 1 Specification sections apply to this Section.

1.2 SUMMARY

A. Section Includes:
   1. Construction indoor air quality (IAQ) management plan.
   2. HVAC air filters.
   3. Building flush-out.

B. This section shall apply for all New Construction & Major Renovations and as indicated on the project documents.

C. Related Sections
   1. SECTION 018113, SUSTAINABLE DESIGN REQUIREMENTS

1.3 PLAN REQUIREMENTS

A. Develop, submit and implement construction IAQ management plan in accordance with SMACNA IAQ “Guidelines For Occupied Building Under Construction”. The plan shall include building flushout after construction and prior to occupancy.

1.4 SUBMITTALS

A. Product Data: Submit description and performance data for filters, including MERV ratings.

B. Construction Plan: Submit construction IAQ management plan describing methods and procedures for implementation and monitoring compliance.

1.5 CLOSEOUT SUBMITTALS

A. Project Record Documents: Submit construction photographs showing compliance with construction IAQ management plan.

B. Flushout documentation: Submit signed letter documenting that the building flushout is successfully accomplished in accordance with the approved IAQ management plan.

1.6 CONSTRUCTION IAQ MANAGEMENT PLAN

A. HVAC PROTECTION
   1. Seal off all supply louvers, return air grilles, and air intake/discharge points to prevent construction dust and debris from entering.
   2. Seal off all ductwork openings and air outlets with plastic sheeting to protect the duct system from dust and debris. Do not re-open until the end of activities that produce dust or pollution, such as drywall sanding, concrete cutting, masonry work, wood sawing, and so forth.
   3. If for some unforeseen reason, there should arise a circumstance wherein the return air system is required to be used during the construction phase, install temporary filters (as determined by ASHRAE Standard 52.2-1999 and of type as used in the final installation) at each return air opening and supply air intakes, and provide frequent inspection and maintenance. If inspections by AE or Commissioning Agent reveal that the ductwork has become contaminated due to inadequate protection, the ductwork shall be cleaned professionally prior to the first phase of occupancy, using procedures established in ACR 2013 published by the National Air Duct Cleaners Association. Replace these filters prior to building flushout.
   4. Under no circumstances shall air be returned from a construction area and then recirculated through the permanent supply ductwork, unless and until the level of construction in the relevant area involves final finishes and trim and the construction has reached a point of complete building dry-in with no sanding and is free from dust, debris, and contaminants.
   5. Do not use fan rooms to store construction or waste materials, and keep them clean and neat.

B. SOURCE CONTROL
   1. Limit construction traffic and motor idling in the vicinity of air intake louvers when the HVAC systems are activated. Restrict motor vehicles to areas well-removed from air intakes, preventing emissions from being drawn into the building.
   2. Use electric or natural gas alternatives for gasoline and diesel equipment where possible and practical.
   3. Cycle equipment off when not being used or needed.
   4. Avoid the use of materials and products with high VOC and/or particulate levels. Use products and installation methods with low VOCs such as paints, sealers, insulation, adhesives, caulking and cleaners. Comply with the requirements in other Specification Sections.
   5. Keep containers of wet products closed as much as possible. Cover and seal waste materials which can release odor or dust.
   6. Protect all materials, especially absorbent materials such as insulated ductwork, against moisture during delivery to and storage at the job site. Store materials inside the structure in a dry and clean environment pending installation

C. PATHWAY INTERRUPTION
   1. Use dust curtains or temporary enclosures to prevent dust from migrating to other areas when applicable. During construction, isolate areas of work to prevent contamination of clean or occupied areas.
   2. Keep pollutant sources as far away as possible from ductwork and areas occupied by workers when feasible.

3. Isolate work areas and/or create pressure differentials to prevent the migration of contaminants.
4. Use portable fan systems to exhaust contaminated air directly to the outside of the building, and discharge the air in a means to prevent it from re-entering.

D. HOUSEKEEPING
1. General housekeeping, and dust suppression programs shall include the use of wetting agents or sweeping compounds. Use efficient and effective dust collecting methods such as damp cloths, wet mops, and vacuums with particulate filters, or wet scrubbers. Institute cleaning activities of building areas on a daily basis, and of HVAC equipment as required.
2. Keep all coils, air filters, dampers, fans, and ductwork clean during installation, and clean them as required prior to performing the testing, adjusting and balancing of the systems.
3. Avoid accumulations of water inside the building, and promptly remove any that may occur. Especially protect porous materials such as insulation and ceiling tiles from exposure to moisture
4. Provide photographs of the above activities during construction to document compliance

E. SCHEDULING
1. Sequence construction activities carefully over the duration of the project to minimize the impact on IAQ. Plan adequate time to conduct flushout before occupancy.
2. Install new filters, of types required and specified for the final installation, at the central fan system, immediately prior to occupancy for each respective system.

PART 2 PRODUCTS

1.1 HVAC AIR FILTERS

A. Return Filters: Filtration media rated for minimum efficiency reporting value (MERV) when tested in accordance with ASHRAE 52.2.
1. Construction Return Filters: MERV of 8, minimum.
2. Flush-Out Return Filters: MERV of 13, minimum.
3. Permanent Filters: As specified in Mechanical Section or Equipment Schedule

PART 3 EXECUTION

2.1 IAQ PLAN

A. Implement Construction IAQ Management Plan at start of construction.

B. Review Construction IAQ management plan at pre-construction meeting and progress meetings

C. Distribute approved Construction IAQ Management Plan to subcontractors and others affected by Plan Requirements.

D. Oversee plan implementation, instruct construction personnel for plan compliance, and document plan results.

2.2 FILTER INSTALLATION AND REPLACEMENT

A. Install construction return filter at each return grille before operating permanent air handlers during construction.

B. Replace filters after completing construction and before conducting building flush-out.
   1. Replace construction return filters with flush-out return filters.
   2. Replace supply filters.

C. Replace filters after conducting building flush-out and before occupancy.
   1. Replace flush-out return filters with permanent filters.
   3. Replace supply filters.

2.3 BUILDING FLUSH-OUT

A. Conduct building flush-out after construction ends and before occupancy by installing new filtration media and supplying a total air volume of 14,000 cubic feet of outdoor air per square foot of gross floor area while maintaining an internal temperature of at least 60°F and no higher than 80°F and relative humidity no higher than 60%.

2.4 CONSTRUCTION PHOTOGRAPHS

A. Photograph construction operations to show compliance with SMACNA IAQ and construction IAQ management plan.
   1. Take minimum six photographs on minimum three different occasions during construction to show consistent adherence with specified requirements.

END OF SECTION 018119

SECTION 019113 – GENERAL COMMISSIONING REQUIREMENTS

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. General:
   1. The following applies to this section:
      a. The terms of the Subcontract, including General Provisions, and General Requirements.
      b. Related requirements in other Subcontract Documents including, but not necessarily limited to, the Drawings and the Specification Sections in the Project Manual & Statement of Work.
   2. Review these documents for coordination with additional requirements and information that apply to work under this Section.
   3. Commissioning shall be in accordance with the commissioning plan and ASHRAE Guidelines or PPPL approved alternate standard at minimum.
   4. Fire Protection systems shall be commissioned in compliance with NFPA 3.

1.2 SUMMARY

A. Section includes:
   1. General requirements that apply to implementation of commissioning without regard to specific systems, assemblies, or components.

B. Commissioning Agent(s) will be procured by PPPL.

C. Commissioning during construction and acceptance phases is intended to achieve following objectives:
   1. Verify that all applicable building systems and the building's energy related systems including the utility systems (Mechanical, Electrical, Fire Life Safety, Security, HVAC, Controls, AV System) are installed, calibrated and perform according to the Systems Requirement Document (SRD) and the construction documents.
   2. Verify and document equipment and systems receive complete installation and operational checkout by installing lower-tier subcontractors.
   3. Verify and document that equipment and systems are functionally tested and the performance of these systems meets the SRD.
   4. Meet the requirements of High Performance Sustainable Buildings Guiding Principle for "Commissioning" as outlined in Section 018113 "Sustainable Design Requirements".

D. Commissioning procedures will be developed and coordinated by the Commissioning Agent (CxA). The Subcontractor shall be responsible for participation in the commissioning process as outlined below, and in references and attachments throughout the subcontract documents.

E. The Commissioning (Cx) process does not relieve or reduce responsibility of system designers or installing lower-tier subcontractors to provide finished and fully functioning product.

F. Related Sections
   1. SECTION 017900, DEMONSTRATION AND TRAINING
   2. DIVISION 23 – HEATING, VENTILATING AND AIR CONDITIONING REQUIREMENTS (HVAC)
   3. DIVISION 26 – ELECTRICAL

1.3 COORDINATION

A. Commissioning Team
   1. Commissioning Agent
   2. PPPL Technical Staff
   3. Subcontractor
   4. Lower-Tier Subcontractors – Mechanical, Electrical, Fire Life Safety System, HVAC and Controls, Security System, TAB, etc.
   5. Manufacturer's Representatives/Vendors
   6. Architect/Engineer

B. Management
   1. Commissioning Agent: Manages and directs the commissioning process with assistance of the Commissioning Team.

C. Scheduling: Commissioning Agent to work with PPPL Technical Staff, and Subcontractor to schedule commissioning activities.
   1. Subcontractor to integrate commissioning activities into master construction schedule.
   2. Involved parties to address scheduling issues in a timely manner to expedite commissioning process.

1.4 DEFINITIONS

A. Acceptance Phase: The phase of construction after startup and initial checkout when functional and overall system performance tests, O&M documentation review and training occurs.

B. Architect/Engineer: The prime consultant (architect) and sub-consultants (engineers) who comprise the design team.

C. Areas of Conflict: Where commissioning requirements conflict with technical design provisions or other requirements of the Subcontract Documents, prompting the Subcontractor to issue a request for clarification.

D. Basis of Design: A document that records concepts, calculations, decisions, and product selections used to meet the project requirements and to satisfy applicable regulatory requirements, standards, and guidelines. The document includes both narrative descriptions and lists of individual items that support the design process.

E. Commissioning Agent (CxA): An independent entity (or agent employed on behalf of PPPL) not associated with the work of the A/E design team members or Subcontractor. This entity leads, plans, schedules, coordinates the Cx team and implements the Commissioning Plan.

F. Commissioning (Cx): Documented confirmation that building systems function in compliance with criteria set forth in the Subcontract Documents including Subcontract document, Design document, and PPPL's operational criteria.

G. Commissioning Plan: A document prepared by the CxA that outlines the Cx team organization, communication paths, schedule, allocation of resources, matrix of responsibilities and documentation requirements (guidelines) of the commissioning process. The plan shall be implemented by the CxA at the start of the construction. The plan shall be updated by the CxA as the construction process evolves including Functional performance test checklist (FPTC) and functional test procedure (FTP) from examples.

H. Subcontract Documents: The documents binding on parties involved in the construction of this project (drawings, specifications, change orders, amendments, subcontracts, Cx Plan, etc.).

I. Subcontractor: The term "subcontractor" as used throughout these specifications refers to the general contractor and the term "lower-tier subcontractor" refers to the general contractor's subcontractors.

J. Lower-tier subcontractor's Pre-Commissioning Checklists: Includes installation and start-up items as specified to be completed by the appropriate subcontractors independent of the CxA Commissioning Plan.

K. Building Automation System (BAS): The central building energy management control system. Building and central utility operation, monitoring and controls. Local and remote (Mechanical, electrical, fire life safety, security etc.)

L. Deferred Testing: Tests that are performed after substantial completion due to partial occupancy, equipment, seasonal requirements, design or other site conditions that disallow the test from being performed. Deferred tests must be approved by the CxA.

M. Deficiency: A condition in the installation or function of a component, piece of equipment or system that is not in compliance with the subcontract documents.

N. Design Intent: A dynamic document that provides the explanation of the ideas, concepts and criteria that are considered to be very important to PPPL. It is initially the outcome of the programming and conceptual design phases as represented in this document.

O. Final Commissioning Report: Document prepared by the CxA, which details the actual Cx procedures performed, inspection and testing results, and the final version of the deficiencies list indicating that all issues discovered through the Cx process have been verified as resolved or accepted. The report also includes key items for maintenance staff, such as fan & pump curves for equipment furnished. A copy of the final report is furnished to PPPL as a menu-driven CD for future reference.

P. Functional Performance Test Checklists and Procedures (FPTC and FTP) The Functional Performance Test checklists are prepared by the CxA and used to test the dynamic function and operation of systems/equipment by using manual (direct observation) or monitoring methods. The CxA develops the written functional test procedures form in a sequential format and then coordinates, oversees and documents the actual testing which is usually performed by the installing Subcontractor or vendor. Functional performance testing takes place after pre-functional (installation and start-up) tests have been completed.

Q. Issues Log: A list of systems/equipment or procedural deficiencies and issues that have been noted. The list is prepared by the CxA and includes the current disposition of each issue and the date of final resolution. Deficiencies include products/material, installation, service, or systems/equipment performance that do not comply with the Subcontract Documents and/or Commissioning Plan.

R. Monitoring: Recording parameters (i.e. flow, current, status, pressure, etc.) of equipment operation using data loggers or the trending capabilities of BASs.

S. O&M Manuals: Manuals used for the training, operations and maintenance of systems & equipment. Subcontractor shall reference additional requirements in the individual systems/equipment specification sections.

T. PPPL's Representatives: Individuals with the authority to make technical decisions on behalf of PPPL. The Commissioning Plan should be kept up to date with the names and contact information for each of the PPPL's representatives.

U. PPPL's Project Requirements: The System Requirements Document (SRD) document developed under the leadership of, and primarily by, the A/E based on discussions with PPPL. It states the concepts, and performance criteria (including functional requirements) to which the completed project must conform. It includes expectations of how the building will be used and operated. This document is the basis for building commissioning and includes items such as project goals, measurable performance criteria, cost considerations, benchmarks, success criteria, and supporting information. It is used by the engineers to develop their basis of design/design intent documentation.

V. Pre-functional Test Checklists: The Pre-functional Test checklists are prepared by the CxA and used to verify systems/equipment that installation and start-up activities comply with the Subcontract Documents and equipment submittals. CxA verification can include inspections, tests and procedures to prepare equipment for initial start-up and proper/continuous operation (such as cleanliness, belt tension, oil levels, labels affixed, gages in place, sensors calibrated, initial settings, etc.). Operationally, the

checklists shall be used to verify system set points, operating strategies, required component testing, correct rotation, and damper positions prior to functional performance tests. They shall incorporate manufacturers' start-up instructions. The checklists require final sign-off by the Subcontractor and CxA prior to continuing the commissioning process.

W. Project Manager (PM): PPPL Project Manager.

X. Seasonal Performance Tests: Functional performance tests that are deferred until system design conditions can be replicated or simulated.

Y. Specifications: The construction specifications are a part of the subcontract documents and supplement the design drawings. Specifications can also include procedures implied in the Commissioning Plan.

Z. Start-up: The initial start-up or activation of dynamic equipment.

AA. Testing and Balancing (TAB): Lower-tier contracting specialist who performs air and hydronic measurements on the HVAC systems and adjusts the flows as required to achieve optimum performance of the building environmental equipment.

BB. Training: Instructions and hands-on demonstration of operating and maintaining systems and equipment.

CC. Trending: Monitoring of equipment and sequences using the building control system (BAS) and/or data loggers.

DD. Vendors/Manufacturer's Representatives: Supplier of equipment of service.

1.5 COMMISSIONING PROCESS

A. Following activities describe commissioning tasks coordinated by Commissioning Agent and Subcontractor during construction in order of occurrence.
   1. Kick-off, Scoping Meeting: Members of the construction team involved in the commissioning process are to meet and agree on responsibilities of work, tasks, schedules, deliverables, and implementation of Commissioning Plan.
   2. Commissioning Plan: Provides guidance in execution of commissioning processes.
   3. Project Manual (Specifications & Plans) takes precedence over Commissioning Plan.
   4. Submittals: Equipment documentation submitted to Commissioning Agent during normal submittals, including detailed start-up procedures.
   5. Start-Up/Pre-Functional Checklists: Commissioning Agent works with Subcontractor to develop startup plans and documentation formats, including providing pre-functional checklists to be completed during the startup process.
   6. Issues Log: Commissioning Agent maintains a list of noted deficiencies discovered as a result of the Cx process. This list also includes the

current disposition of issues, and the date(s) of final resolution as confirmed by the Commissioning Agent. Deficiencies are defined as those issues where products, execution or performance do not satisfy the specifications and/or the design intent. Subcontractor is to review the list and clarify, rectify, repair, and resolve the issue to the Commissioning Agent.

7. Functional Performance Testing: Commissioning Agent develops specific equipment and system functional performance test procedures.
    a. Subcontractor reviews the test procedures and provides written comments in a timely manner.
    b. Testing procedures to be executed by Subcontractor, under direction of, and documented by Commissioning Agent.
    c. Subcontractor to set-up and record performance tests and trends under direction of, and documented by Commissioning Agent.
    d. Deficiencies and Resolution: Commissioning Agent documents items of non-compliance in materials, installation or operation.
    e. Items are corrected at Subcontractor's expense and equipment or systems are retested.
8. Deferred Testing: Deferred or seasonal testing is conducted, as required.

1.6 COMMISSIONING RESPONSIBILITIES

A. Commissioning Agent:
1. Primary Role: Develop and coordinate execution of Commissioning Plan to verify and document systems are functioning in accordance with PPPL's project requirements, SRD and Construction Documents.
2. May assist with problem-solving deficiencies, but ultimately that responsibility resides with the Subcontractor.
3. Construction and Acceptance Phase:
    a. Reviews the project requirements and SRD.
    b. Develops and updates the commissioning plan.
    c. Coordinates and directs commissioning activities.
    d. Coordinates with Subcontractor to verify commissioning activities are scheduled.
    e. Maintains a current Commissioning Plan.
    f. Facilitates scoping commissioning meetings and contributes to various start-up and technical meetings.
    g. Requests and reviews additional information required to perform commissioning tasks, including O&M materials, subcontractor start-up and checkout procedures, and sequences of operation.
    h. Reviews Subcontractor submittals applicable to commissioned systems, concurrent with Architect and Engineer reviews.
    i. Develops and distributes pre-functional checklists, Function Performances Test Checklist (FPTC), Functional Test Procedure (FTP), and forms.
    j. Performs site visits to observe component and system installations.
    k. Attends selected construction job-site meetings to monitor construction and commissioning progress.

l. Reviews completed pre-functional checklist and start-up reports.
m. Assists with coordination of start-up requirements with TAB requirements.
n. Develops functional performance test procedures for equipment and systems.
o. Coordinates with subcontractor and customer, witnesses, and documents functional performance tests completed by lower-tier subcontractors.
p. Coordinates retesting as necessary until satisfactory performance is verified including system periodic and seasonal operation and performance of the system and equipment.
q. Maintains master deficiency and resolution record and test results with recommended actions.
r. Prepares Final Commissioning Report.

4. Warranty Period:
   a. Coordinates and supervises seasonal or deferred testing and deficiency corrections.

B. Architect/Engineer:
   1. Construction and Acceptance Phase:
      a. Attends commissioning scoping meetings, BAS integration meeting and additional meetings, as necessary.
      b. Provides design intent and sequence of operation documentation as required by Commissioning Agent.
      c. Provides updates to the Systems Requirements Document.
      d. Assists in resolution of system deficiencies identified during commissioning.
   2. Warranty Period:
      a. Assists in resolution of system deficiencies identified during warranty period commissioning.

C. PPPL Technical Staff:
   1. Construction and Acceptance Phase:
      a. Provides input to project requirements.
      b. Plans/Attends commissioning scoping meetings, start-up meetings and additional meetings, as necessary.
      c. Arranges for facility operating and maintenance personnel to participate in commissioning activities and training sessions.
      d. Reviews final acceptance for commissioning work.
   2. Warranty Period:
      a. Ensure seasonal or deferred testing and deficiency issues are addressed.

D. Subcontractor:
   1. Construction and Acceptance Phase:
      a. Facilitate coordination of commissioning work by Commissioning Agent.
      b. Attend commissioning scoping meetings, start-up meetings, and additional meetings, as necessary.

c. Furnish copies of construction documents, addenda, change orders and approved submittals and design drawings related to commissioned equipment to Commissioning Agent.
d. Ensure lower-tier subcontractors execute their commissioning responsibilities according to Project Manual, Construction Documents, and Commissioning Plan.
e. Develop start-up and checkout plan for commissioned equipment based on manufacturer's recommendations and pre-functional checklists from Commissioning Agent.
    1) Submit to Commissioning Agent for review and approval prior to start-up.

2. Warranty Period:
    a. Execute required seasonal or deferred functional performance testing.
    b. Correct deficiencies as a result of seasonal or deferred functional performance testing.
3. Lower-tier Subcontractors (Mechanical, Electrical, Fire Alarm, HVAC and Controls, Security Systems, and TAB) Construction and Acceptance Phases:
    a. Attend commissioning scoping meeting, BAS integration meeting and additional meetings, as necessary.
    b. Provide additional requested documentation, prior to normal O&M manual submittals, to the Commissioning Agent for development of start-up and functional testing procedures.
    c. Assist in clarification of operation and control of commissioned equipment where Project Manual, control drawings or equipment documentation is not sufficient for writing detailed testing procedures.
    d. During startup and checkout process, execute pre-functional checklists for commissioned equipment.
        1) Perform and document completed startup and system operational checkout procedures.
    e. Resolve punch list items before functional testing.
        1) Air and water TAB to be completed with discrepancies and problems resolved before functional testing.
    f. Perform functional performance testing, under direction of Commissioning Agent, for commissioned equipment.
    g. TAB lower-tier subcontractor shall demonstrate onsite to the Commissioning Agent that air and hydronic systems are properly adjusted, calibrated, and balanced.
    h. Resolve equipment or system deficiencies and retest as required to verify modified performance.
    i. Provide training for the systems specified. Submit the training plan and agenda, at least two weeks in advance of the scheduled training, to the Commissioning Agent for approval. Training shall not be performed until the Cx process is 100% complete and the training plan and agenda have been approved by the Commissioning Agent.
4. Warranty Period

a. Execute seasonal or deferred functional performance testing

E. Lower-tier subcontractor (Temperature Control). Commissioning responsibilities of Temperature Control lower-tier subcontractor, during construction and acceptance phases in addition to those listed elsewhere are:
   1. Prepare Sequences of Operation Submittals. Temperature controls submittals to include complete and detailed sequences of operation for each piece of equipment, regardless of completeness and clarity of sequences in Project Manual and include:
      a. Narrative description of system, describing its purpose, components and function.
      b. Interactions and interlocks with other systems.
      c. Delineation of control interactions between packaged controls and building automation system (BAS), including listing of monitored points, controlled points, and adjustable points.
      d. Written sequences of control for packaged controlled equipment.
      e. Sequences of control for modes of operation.
      f. Capacity control sequences and equipment staging.
      g. Temperature and pressure control sequences (setbacks, resets, etc.).
      h. Sequences for control strategies (economizer control, optimum start/stop, optimization, demand limiting, etc.).
      i. Effects of power or equipment failure with standby component functions.
      j. Sequences for alarms and emergency shutdowns.
      k. Seasonal operational requirements.
      l. Provide training for the systems specified. Submit the training plan and agenda, at least two weeks in advance of the scheduled training, to the Commissioning Agent for approval. Training shall not be performed until the Cx process is 100% complete and the training plan and agenda have been approved by the Commissioning Agent.
   2. Control Drawings Submittal
      a. Control drawings to have key to abbreviations.
      b. Control Drawings: Provide graphic schematic depictions of systems and each component – PPPL Standard graphics and Input/Output description and control set point including alarm set point.
      c. Schematics to include system and component layout of equipment that BAS monitors, enables or controls, even if equipment is primarily controlled by packaged or integral controls – systems/equipment SOP.
   3. Attend commissioning scoping meetings, controls integration meetings, and additional meetings as necessary.
   4. Setup and execute trending of various Building Automation System (BAS) points to verify correct performance of systems and equipment.

5. Prepare a written plan indicating step-by-step manner, and procedures that will be followed to test, checkout and adjust BAS prior to functional performance testing.
6. Provide Commissioning Agent up-to-date controls sequences, set points and control algorithms prior to functional performance testing.
7. Provide a signed and dated certification to Commissioning Agent upon completion of BAS checkout.

F. Manufacturer's Representatives/Vendors
1. Provide requested submittal data, including detailed start-up procedures and specific responsibilities of PPPL to keep warranties in effect.
2. Provide information requested by Commissioning Agent regarding equipment sequence of operation and testing procedures.

1.7 Commissioning Scope: All building systems may be commissioned. This includes, but is not limited to mechanical, electrical, plumbing, fire sprinkler, fire alarm, AV, IT/Telecom, Security, and telecommunications systems, and as listed below. Commissioning shall be comprehensive to investigate systems from the component level to system and intersystem levels. This list may be subject to change during the fit-up design phase. Systems and equipment is subject to change during the fit-up design phase.

1.8 TEST EQUIPMENT

A. Standard testing equipment required to perform startup and initial checkout and required functional performance testing to be provided by the relevant lower-tier subcontractor for equipment being tested.

B. Testing equipment to be of sufficient quality and accuracy to test or measure system performance as required by Project Manual.

C. Data logging equipment or software required to test equipment to be provided by Subcontractor but shall not become property of PPPL.

1.9 MEETINGS

A. Scoping Meeting:
1. Commissioning Agent will schedule, plan and conduct Commissioning Scoping Meeting and Controls Integration Meetings with entire commissioning team in attendance. The Subcontractor shall attend these meetings.
2. Provide minutes to document significant interactions and coordination meetings concerning the commissioning aspects of the project.

B. Miscellaneous Meetings. Other meetings to be planned and conducted by Commissioning Agent as construction progresses.
1. Meetings to cover coordination, deficiency resolution and planning issues.

1.10 REPORTING

A. Final Commissioning Report: Commissioning Agent compiles a summary Commissioning Report which summarizes tasks, findings, and documentation of commissioning process.
    1. Report addresses actual performance of building systems in reference to project requirements, SRD and subcontract documents.
    2. Report includes:
        a. Executive Summary
        b. Commissioning Plan
        c. Commissioning site visit reports
        d. Submittal reviews
        e. O&M data reviews
        f. Training record & evaluations
        g. Pre-functional test and functional performance test procedures
        h. System/equipment data sheets
        i. Subcontractor pre-commissioning checklists with start-up reports
        j. Pre-functional test and functional performance test checklists with test results
        k. Inspection reports
        l. Copies of related testing and inspection reports (by others) related to the commissioning process.
        m. Issues Log, including both cleared and open items
        n. Recommendations

1.11 SUBMITTALS

A. Commissioning Agent reviews submittals related to commissioned equipment for conformance to Construction Documents as it relates to commissioning process.

B. Commissioning Agent reviews intended primarily to aid in development of functional performance test procedures.

C. Commissioning Agent may submit written requests for additional information from all lower-tier subcontractors to facilitate the commissioning process.

1.12 START-UP, PRE-FUNCTIONAL TEST CHECKLISTS

A. Following procedures apply to equipment commissioned, according to Article 1.7, Commissioning Scope.

B. General: Pre-functional Test Checklists are developed and completed for major equipment and systems being commissioned.
    1. Checklist documents equipment nameplate and characteristics data and confirms as-built status of equipment or system.
    2. Checklists ensure systems are installed correctly, complete and operational.

C. Completion of Pre-functional Test Checklists and Startup.

1. Subcontractor to complete pre-functional checklists.
2. Subcontractor to schedule startup and checkout with PPPL, lower-tier subcontractor and Commissioning Agent.
3. Commissioning Agent may observe start-up for each piece of primary equipment.
4. Only individuals who have completed or witnessed line item task to make initials or checks on forms.

D. Deficiencies, Non-Conformance and Approval in Checklists and Startup.
   1. Lower-tier subcontractors to clearly list notes for any items of start-up and pre-functional procedures not successfully completed at bottom of form or on an attached sheet.
      a. Procedures form and any outstanding deficiencies are provided to the Commissioning Agent within two days of test completion.
   2. Commissioning Agent reviews and verifies pre-functional test checklists and start-up reports and recommends approval to PPPL.
   3. Commissioning Agent assists the Subcontractor in correcting and retesting any deficiencies or uncompleted items, involving other members of the Commissioning Team as necessary.

1.13 FUNCTIONAL PERFORMANCE TESTING

A. Following procedures apply to all equipment commissioned, according to Article 1.7, Commissioning Scope and to all commissioning functional performance testing.

B. Objectives and Scope:
   1. Demonstrate each system is operating according to documented project requirements, SRD and Construction Documents.
   2. Functional performance testing comprises tests to verify components, equipment, systems, and interfaces between systems operate correctly and includes operating modes, interlocks, control sequences, and responses to emergency conditions.
   3. Verification procedures are directed, witnessed, and documented by Commissioning Agent and PPPL representative.

C. Development of Test Procedures:
   1. Commissioning Agent develops specific test procedures and forms to verify and document proper operation of each piece of equipment and system.
   2. Commissioning Agent provides test procedures to Subcontractor and Lower-tier Subcontractors who review the tests for feasibility, safety, equipment and warranty protection prior to execution.
   3. Commissioning Agent shall submit tests to PPPL for review.
   4. Test procedure forms, developed by Commissioning Agent, may include the following information:
      a. Project name
      b. System and equipment or component name(s)
      c. Equipment location and ID number
      d. Date of test

e. Participating parties
f. Instructions for setting up test, including special cautions, alarm limits, etc.
g. Specific step-by-step procedures to execute test
h. Acceptance criteria of proper performance, Pass-Fail status, and data trending requirements.
i. Section for comments or notes

D. Execution of Functional Performance Tests:
1. Test Methods: Functional performance testing and verification may be achieved by direct manipulation of system inputs (i.e. heating or cooling sensors), manipulation of system inputs with building automation system (i.e. software override of sensor inputs), trend logs of system inputs and outputs using building automation system, or short-term monitoring of system inputs and outputs using stand alone data loggers.
a. Subcontractor shall perform the functional performance tests witnessed by the Commissioning Agent.
b. Combination of methods may be required to test the complete sequence of operations.
c. Commissioning Agent determines which method, or combination, is most appropriate.
2. Setup: Each test procedure is performed under conditions that simulate normal operating conditions as closely as possible.
a. Lower-tier subcontractor executing tests provides necessary system modifications to produce specified conditions (flows, pressures, temperatures, etc) necessary to execute tests and, at completion of test, returns affected building equipment and systems to their pre-test conditions.
3. Test values: After two attempts at testing, failures are still present, additional testing shall be at the Subcontractor's expense.

E. Coordination and Scheduling: Subcontractor provides sufficient notice to Commissioning Agent regarding completion schedule for pre-functional checklists and startup of equipment and systems.
1. Functional Performance Testing is scheduled with the Subcontractor, Lower-tier subcontractors and the Commissioning Agent provided the following sequential priorities are adhered to:
a. Equipment pre-start checklist items and manufacturer's pre-start procedures are completed and moisture, dust and other environmental and building integrity issues have been addressed.
b. Functional performance testing begins after pre-functional testing is complete, start-up and TAB is completed for a given system.
c. Controls system and equipment controls have been functionally tested and all points have been calibrated and pre-functional checklists are completed. Temperature Control Lower-tier subcontractor has submitted the point-to-point system checklist.

1.14 DOCUMENTATION, NON-CONFORMANCE AND APPROVAL OF TESTS

A. Documentation:
   1. Commissioning Agent witnesses and documents results of functional performance tests using forms developed for that purpose.
      a. Prior to testing, forms are provided to the Subcontractor and PPPL for review and approval.

B. Non-Conformance:
   1. Commissioning Agent records results of functional test on procedure or test form.
      a. Deficiencies identified during verification testing are documented on standard form and reported to PPPL and Subcontractor.
      b. Deficiency Report: Includes details of components or systems found to be non-compliant with parameters of test plans.
      c. Report details adjustments or alterations required to correct system operation, and identifies the responsible party.
   2. Corrections of minor deficiencies identified may be made during tests at discretion of Commissioning Agent and deficiency and resolution documented on procedure form.
   3. Expedite testing process and minimize unnecessary delays, while not compromising integrity of procedures.
   4. Identified deficiency resolution:
      a. Undisputed:
         1) Commissioning Agent documents deficiency and adjustments or alterations required to correct it.
         2) Subcontractor corrects the deficiency and notifies the Commissioning Agent that equipment is ready to be retested.
         3) Subcontractor and Commissioning Agent reschedule test and test is repeated.
      b. Disputed deficiency:
         1) Deficiency is documented on non-compliance form and copy given to PPPL and Subcontractor.
         2) Resolutions are made at the lowest management level possible and additional parties are brought into discussions as needed.
         3) Final interpretive authority is with the PPPL team.
         4) Final acceptance authority is with PPPL.
         5) Commissioning Agent documents resolution process.
         6) Once interpretation and resolution have been decided, the appropriate party corrects deficiency and notifies Commissioning Agent that equipment is ready to be retested.
         7) Subcontractor and Commissioning Agent reschedules test and test is repeated until satisfactory performance is achieved.

C. Approval:

1. Commissioning Agent makes formal approval of functional performance tests after review.
2. The Commissioning Agent recommends acceptance of each test to PPPL.
3. PPPL gives final approval on each test.

1.15 DEFERRED TESTING

A. Unforeseen Deferred Tests: Functional testing that cannot be completed due to building structure, required occupancy condition, or other deficiency may be delayed upon approval of PPPL.

B. Tests are conducted in the same manner as seasonal tests as soon as possible.

C. Seasonal Testing: Seasonal variation in operations or control strategies may require additional testing during the opposite season to verify performance of HVAC systems and controls.
   1. During warranty period: Seasonal testing and other deferred testing is completed as required to fully test sequences of operation.
      a. Commissioning Agent coordinates these activities.
   2. Tests are executed and documented; with deficiencies corrected by appropriate sub-contactor.

1.16 TRAINING OF PPPL OPERATIONS AND MAINTENANCE STAFF

A. Refer to SECTION 017900, DEMONSTRATION AND TRAINING, for requirements.

Construction Commissioning Responsibility Table

Activity		Commissioning Agent	Subcontractor Lower-tier subcontractors including Manufacturer Reps/Vendors	PPPL	Design Professionals Architect/Engineer
1.	Project Requirements and SRD	Reviews and verifies.	Ensures the project meets requirements.	Provides input and approves project requirements	Develops & updates to SRD
2.	Cx Plan	Writes and manages the Cx plan	Reviews plan for understanding and scope issues	Reviews and accepts	Reviews and comments
3.	Commissioning Meetings	Plan, conduct and take and distribute minutes for regular	Attend commissioning meetings.	Attend commissioning meetings.	Attend selected commissioning meetings as required by PPPL

		commissioning meetings.			
4.	Construction Submittals of commissioned equipment and assemblies	Reviews and utilizes in the development pre-functional checklists & functional testing procedures	Provide in compliance with specifications	Reviews and approves.	Reviews
5.	Pre-functional Checklists, Tests and Startup forms	Develops and provides Checklists and Start-Up forms.	Manages the completion of Checklists and Start-Up forms.	Reviews and approves.	None
6.	Equipment Start-up	Witness selected systems. Reviews completed forms.	Coordinates tasks and schedules with Subcontractors.	Observe selected systems.	May witness selected systems.
7.	Construction Observation	Limited observation of equipment & systems.	Observe to ensure Subcontractors are executing accepted installation procedures	Site Walks & Observations.	As required by PPPL
8.	Management of Issues Log	Manages (develops and keeps updated).	Resolves issues	Reviews and provides input	Assists in resolving issues
9.	Functional Performance Test Procedure Development	Writes procedures for all tests.	Reviews for scope, technical correctness safety, and warranty.	Reviews and approves.	May review
10	Scheduling of Functional Performance Tests	Assists in the Scheduling of Testing.	Schedules Functional Performance Tests	Coordinates and Reviews.	None
11	TAB of Air and Water Systems	Spot checks TAB	TAB to assist in spot testing specific areas.	Reviews	Review T&B Report and help resolve issues
12	Functional Performance Test Execution	Oversees and witnesses all tests and	Performs testing	Participate in selected tests	None

		documents testing.			
13	Commissioning Report	Develops and submits to PPPL	Reviews	Receives, Reviews, and accepts	None
14	Seasonal and Deferred Testing	Oversees and witnesses all tests and documents testing.	Executes testing	Participate in selected tests	None
15	Training	Develops Training agenda, Records and assures training complete	Provides training to PPPL Facilities personnel	Observes training	None

PART 2 PRODUCTS – Not Used

PART 3 EXECUTION – Not Used

END OF SECTION 019113

SECTION 024119

SELECTIVE DEMOLITION AND ALTERATION WORK

PART 1 GENERAL

1.1 GENERAL REQUIREMENTS

A. Work of this Section, as shown or specified, shall be in accordance with the requirements of the Contract Documents.

1.2 SECTION INCLUDES

A. Work of this Section includes all labor, materials, equipment and services necessary to complete the selective demolition and alteration work as shown on the drawings and/or specified herein, including but not limited to the following:

1. Alterations, selective demolition and removals as noted on drawings and as required to accommodate new construction.
2. Removal of debris.
3. Protection of existing building and spaces to remain, and shoring of the structure as required for structural integrity and personal safety.
4. Protection of existing curbs and sidewalks.
5. Temporary coverage passageways.
6. Patching and refinishing of existing surfaces damaged as a result of this work.
7. Substrate preparation and refinishing.
8. Protection.

1.3 QUALITY ASSURANCE

A. The Contractor shall comply with the requirements of all applicable Federal, State and local safety and health regulations regarding the demolition of structures including ANSI/NFPD 241-Building Construction and Demolition Operations.

B. The Contractor shall be responsible for any damage to any adjacent structures or buildings to remain.

C. Qualifications: Qualifications of Contractor for work of this Section shall not be less than ten (10) years of field experience in work of this nature.

D. Professional Engineering: The Contractor shall retain the services of a Professional Engineer licensed in the State of New Jersey.

E. All work related to the demolition of Architectural, Mechanical, Electrical, Plumbing, Fire, and Structural and related elements are outlined in the contract documents. Due to physical access inability, visual disruption and other site elements, some items may not be shown in the drawings but are still to be included in the scope of demolition work by the Contractor. The Contractor is to perform a complete site survey prior to demolition to ensure the scope of work is understood and unknowns noted. Should any items not shown in the contract drawings be present in field and require further information on demolition, the Contractor is to notify the Design Team and Owner immediately for confirmation of removal. Unknown elements that arise during demolition not previously surveyed by the Contractor or noted by the design team, the cost of work is to be charged to the allowance set forth by the Owner.

1.4 RELATED SECTIONS

A. Alteration and removal requirements for mechanical and electrical work - Mechanical and Electrical Sections.

1.5 SUBMITTALS

A. Schedule of Demolition Operations: Submit demolition procedures and operational sequence for Owner's review prior to start of work. Submit a written request to Owner well in advance of executing any cutting or alteration which affects:

1. The work of tying in or connecting to operational systems of the building, including electrical, mechanical and security systems.

2. The work of the Owner or any separate Contractor.

3. The structural value or integrity of any element of the project or of adjacent structures.

4. The integrity or effectiveness of weather-exposed and moisture-resistant elements or systems.

5. The efficiency, operational life, maintenance, or safety of operational elements or systems.

B. Notice of Differing Conditions: Submit a written notification if, during the work of demolition and cutting, conditions are discovered which significantly vary from those shown on the drawings. Do not commence work until approval of Architect.

C. Shop Drawings: Submit the following prior to starting work:

1. Submit for Architect's information shop drawings indicating location and typical construction details of temporary dustproof and weatherproof partitions.

2. Submit drawings of temporary structural shoring, bracing, framing or support, for the information of the Owner. Such drawings will be reviewed

by the Structural Engineer for the effects of such temporary members on the structural elements to remain. These drawings shall include the reason for such temporary members, the location, the direction and magnitude of design reaction forces on existing structure, and details showing how these reaction forces will be applied to the existing structure.

a. Shop drawings shall be submitted with the Seal of the P.E. engaged by Contractor; P.E. must be licensed in the State of New Jersey.
b. The Architect will receive acknowledgment for concepts shown. Such acknowledgments shall be of the concept only and not of actual capacities or structural design and shall not in any way diminish or limit the Contractor's responsibility for the quality and performance of the work and for protecting existing structures and facilities.

1.6 SPECIAL PRECAUTION

A. If Hazardous materials are encountered during demolition operations notify Owner for remediation.

1.7 JOB CONDITIONS

A. Condition of Structure

1. The Contractor for the work of this Section shall be held to have visited the site, examined the premises, determined for himself the existing conditions, character of equipment and facilities needed for the performance of the work, and all matters which may in any way affect the work before submitting a bid.

a. Information regarding existing construction or conditions is based on available record drawings which may or may not truly reflect existing conditions. Such information is included on the assumption that it may be of interest to the Contractor, but the Architect, Owner and their consultants do not assume responsibility for its accuracy or completeness.
b. Notify the Architect if, during the course of demolition, conditions are discovered which significantly vary from those shown on the drawings. Do not proceed until authorized by Architect.

2. The Contractor shall accept the condition of the site and structures as found. The Architect and Owner assume no responsibility for condition of site or structures nor the continuation of the condition existing at time of bidding or thereafter.

B. Areas of building to be demolished or altered will be vacated and discontinued in use prior to the start of the work.

1. Surrounding areas of the building shall remain operational by the Owner.

C. Partial Removal

1. Items of savable value to the Contractor may be removed from the structure as the work progresses. Salvaged items must be transported from the site as they are removed.

2. Storage or sale of removed items on the site will not be permitted.

D. Explosives: The use of explosives will not be permitted.

E. Traffic

1. Conduct demolition operations and the removal of debris to ensure minimum interference with roads, streets, walks and other adjacent occupied or used facilities.

2. Do not close or obstruct streets, walks or other occupied or used facilities without permission from authorities having jurisdiction. Provide alternate routes around closed or obstructed traffic ways if required by governing regulations.

F. Utilities

1. Refer to Division 22 and 26 of the specifications for special requirements concerning utilities and services.

2. Maintain any existing utilities required to remain; keep in service and protect against damage during demolition operations.

3. Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by authorities having jurisdiction. Provide temporary services during interruptions to existing utilities, as acceptable to the governing authorities.

4. Disconnect and seal any abandoned utilities before starting demolition operations. Coordinate all work with local utility companies having jurisdiction.

1.8 SCHEDULING

A. Before commencing any alteration or demolition work, submit for review by the Architect, and approval of the Owner, a schedule showing the commencement, the order, and the completion dates for the various parts of this work.

B. Before starting any work relating to existing utilities (electrical, sewer, water, heat, gas, fire lines, etc.) that will temporarily discontinue or disrupt service to the structures to remain, notify the Architect and the Owner 7 days in advance and obtain the Owner's approval in writing before proceeding with this phase of the work.

PART 2 PRODUCTS

Refer to Part 3 - Execution, for Product Requirements

PART 3 EXECUTION

3.1 PROTECTION

A. Take full precautions to protect workmen, passersby or any other persons from falling debris and other hazards of demolition operations.

B. Execute demolition work to ensure protection of existing portions of building to remain against damages which might occur from falling debris or other cause. Do not interfere with use of adjacent occupied buildings and areas. Maintain free, safe passage to and from occupied adjacent buildings.

C. Materials Placement: Do not load structure with weight that will endanger, overload or cause excessive deflection of the existing structure, or that will damage finished surfaces adjacent to and/or supported by the existing structure, except portions being removed.

D. Construction Operations: Do not employ any construction operation, equipment or vehicles that will endanger, overload or cause excessive deflection of the existing structure, or that will damage finished surfaces adjacent to and/or supported by the existing structure, except portions being removed.

E. Take precautions to guard against movement, settlement, damage, or collapse of any part of building, sidewalks, adjacent property or street passages; be liable for any such movement, settlement or collapse. If such damage does accidentally occur, Contractor shall repair promptly at no cost to Owner.

F. Provide the necessary safeguards to prevent accidents, to avoid all necessary hazards and protect the public, the work and property at all times, including Saturdays, Sundays, and holidays.

G. Be responsible for any and all damages which may arise or occur to any party whatsoever by reason of the neglect in providing proper lights, guards, barriers, or any other safeguards to prevent damage to property, life and limb.

H. Make such explorations and probes as are necessary to ascertain any required protective measures before proceeding with demolition and removal. Give particular attention to shoring and bracing requirements so as to prevent any damage to existing construction.

1. Provide interior and exterior shoring, bracing, or support to prevent movement or settlement or collapse of structures to be demolished and adjacent facilities to remain. The Contractor's Professional Engineer shall advise on bracing, shoring, underpinning, or other structural requirements. The Contractor shall bear all responsibility for prevention of movement or other structural fault.

2. The Contractor shall restore, by repair or otherwise, the portions of structure or their contents altered by the Contractor in furtherance of his underpinning and support operations. Restoration shall be completed to the conditions which existed prior to the start of the work. Any damage

caused by inadequate support shall also be restored by the Contractor at no cost to the Owner.

I. Provide, erect and maintain catch platforms, lights, barriers, weather protection, warning signs, and other items as required for proper protection of the workmen engaged in demolition and alteration operations, occupants of the building, public and adjacent property. Any damage caused by the Contractor's operations shall be promptly repaired by the Contractor at no cost to the Owner.

J. Provide and maintain temporary protection of the existing structure designated to remain where demolition, removal, and new work are being done, connections made, materials handled, or equipment moved.

K. Take necessary precautions to prevent dust and dirt from rising. Protect unaltered portions of the existing building affected by the operations under this Section by dustproof partitions and other adequate means.

L. Provide adequate fire protection in accordance with local Fire Department requirements.

M. Do not close or obstruct walkways, passageways, or stairways. Do not store or place materials in passageways, stairs, or other means of egress. Conduct operations with minimum traffic interference.

N. Be responsible for any damage to the existing structure or contents by reason of the insufficiency of protection provided.

O. Promptly repair damages caused to adjacent facilities by demolition operations at no cost to the Owner.

P. Provide and maintain weather protection at exterior openings so as to fully protect the interior premises against damage from the elements until such openings are closed by new construction.

3.2 INSPECTION

A. Verify that areas of demolition work are protected and temporary dustproof partitions have been installed.

B. Verify that construction to be removed is not load bearing or has been properly braced, framed or supported.

C. Inspect existing conditions of the project, including elements subject to damage or to movement during demolition and cutting.

D. After uncovering work, inspect the conditions affecting the installation or performance of the work.

1. Report differing or questionable conditions to the Architect in writing; do not proceed with the work until the Architect has provided further instructions.

3.3 PREPARATION

A. Provide adequate temporary support as necessary to ensure the structural value or integrity of the affected portion of the work.

B. Provide devices and methods to protect other portions of the project from damage.

C. Pollution Controls

1. Use temporary enclosures, and other suitable methods to limit the amount of dust and dirt rising and scattering in the air to the lowest practical level. Comply with governing regulations pertaining to environmental protection.

3.4 DEMOLITION AND CUTTING

A. Selectively demolish existing construction in conformance with the drawings and these specifications.

1. Execute cutting and demolition by methods which will prevent damage to other work and will provide proper surface to receive installation of work by others and patching of finish surfaces.

2. Do all cutting or removal so as to leave neat, true, plumb and square edges, at edges to remain. Use carborundum or diamond saw equipment for cutting masonry, concrete and stone work, where edges or surfaces are to remain.

3. Do not cut or remove construction which might weaken or impair the structural integrity or strength of the structural framing or support systems which are to remain.

4. Demolish and remove materials as shown on the drawings without damage to the remaining parts of the structure or mechanical/electrical/utility systems.

5. Remove materials so as to not impose excessive loads in supporting walls, floors or framing and so as not to damage remaining undemolished portions of the structure.

6. Where portions of structures are to be removed, remaining portions shall be protected from damage and prepared to fit new construction. Damage to portions of structures to remain shall be repaired.

7. Reinforcing steel in existing structures shall be left in place, cleaned and aligned to provide tie with new work.

8. Existing waterproofing systems and flashings shall be carefully exposed and protected to maintain workable conditions of fitting new work with existing construction.

9. Proceed with demolition in a systematic manner.

10. Demolish concrete and masonry in small sections.

11. Remove structural framing members and lower to ground by means of hoists, derricks, or other suitable methods.

B. Shoring

1. Design, provide, erect and maintain necessary temporary shoring, bracing, framing, or support where load bearing structural or supporting members are removed or weakened by cuts or openings or are subject to damage from demolition operations, and otherwise as required for safety or to protect finish surfaces from damage.

2. Construction and adequacy of the shoring shall be the entire responsibility of the Contractor. Any damage caused by the inadequacy of the shoring or other support shall be the responsibility of the Contractor to remedy at no additional expense to the Owner.

3. Shoring and bracing shall remain until new structural framing and/or supports are installed. Coordinate operations fully with other trades.

4. Be ready at any time to promptly provide, add to, or strengthen temporary shoring, bracing, or support for existing work, in case existing construction begins to show signs of structural stress.

3.5 WORKMANSHIP STANDARDS FOR ALTERATION AND REMOVAL WORK

A. Cut, remove, alter, temporarily remove and replace, or relocate existing work as required for performance of the work. Perform such work required with due care, including shoring and bracing.

B. Coordinate patching involving the various trades whether or not specifically mentioned in the respective specification Sections.

C. Materials or items demolished and not designated to become the property of the Owner or to be reinstalled shall become the property of the Contractor and shall be removed from the Owner's property.

D. Execute the work in a careful and orderly manner, with the least possible disturbance to the public and to the occupants of the adjacent buildings.

E. In general, demolish masonry in small sections. Where necessary to prevent collapse of any construction, install temporary shores, struts, or bracing.

F. Materials to be removed by existing elevators shall be put in enclosed containers.

G. Where existing equipment and/or fixtures are indicated to be reused, repair such equipment and/or fixtures and refinish to put in perfect working order. Refinish as directed.

H. Cut out embedded anchorage and attachment items as required to properly provide for patching and repair of the respective finishes.

I. Confine cutting of existing roof areas designated to remain to the limits required for the proper installation of the new work. Cut and fold back existing roofing. Cut and remove insulation and related items. Provide temporary weathertight protection as required until new roofing and flashings are installed. Consult the Owner to ascertain if existing guarantee bonds are in force and execute the work so as not to invalidate such bonds.

J. Where utilities are removed, relocated or abandoned, cap, valve, plug, or by-pass to make complete and working installation.

K. Restore existing pipe and duct coverings damaged by work under this Contract to original undamaged condition.

L. Immediately restore to service and repair any damage caused by Contractor's workmen to existing pipe and conduits, wires, cables, etc., of utility services or of fire protection systems and communications systems which are not scheduled for discontinuance or abandonment.

M. Upon completion of contract, deliver work complete. Damage that may be caused by Contractor or Contractor's workmen to existing structures designated to remain, grounds, and utilities shall be repaired by Contractor and left in as good condition as existed prior to damaging.

N. Restore finish work of floors, walls, and ceilings remaining in place but damaged or defaced because of demolition or alteration work to condition equal that which existed at beginning of work under this Contract.

O. Where alteration or removals expose damaged or unfinished surfaces or materials, refinish such surfaces or materials, or remove them and provide new or salvaged materials to make continuous surfaces uniform.

P. Perform new work and restore and refinish existing work in conformance with applicable requirements of the specifications, except as follows:

1. Materials for use in repair of existing surfaces, but not otherwise specified, shall conform to the highest standards of the trade involved, and be in accordance with approved industry standards, and shall be as required to match existing surfaces.

2. Workmanship for repair of existing materials shall, unless otherwise specified, be equal to similar workmanship existing in or adjacent to the space where the work is being done.

3. Installation of salvaged items where no similar items exist shall be done in accordance with the highest standards of the trade involved and in accordance with approved shop drawings.

Q. Materials or items designated to become the property of the Owner shall be as shown on the drawings. Remove such items with care and store them in a location at the site to be designated by the Owner.

R. Materials or items designated to be reinstalled shall be as shown on the drawings. Remove such items with care under the supervision of the trade responsible for reinstallation; protect and store until required. Replace materials or items damaged in their removal with similar new material.

S. The existing building shall not be used to store materials or for layout. Neither shall the furnishings or equipment in any room be used as work benches. Should any damage occur during the progress of the work to any furniture, fixtures, equipment, or appurtenances therein, such damage shall be repaired, replaced or made good by the Contractor without extra cost to the Owner.

T. Where removing existing floor finish and base, remove all adhesive and leave floors and walls smooth and flush, ready to receive new finish.

U. Finish new and adjacent existing surfaces as specified for new work. Clean existing surfaces of dirt, grease and loose paint before refinishing.

3.6 DISPOSAL OF DEMOLISHED MATERIALS

A. General

1. Remove from the site debris, rubbish and other materials resulting from work of this Section.

2. Burning of removed materials from demolished structures will not be permitted on the site.

B. Removal: Transport materials removed from demolished structures and legally dispose of off site. Pay any and all fees associated with disposal work. Leave the site in an orderly condition to the approval of the Architect.

3.7 CLEANING UP

A. Remove debris as the work progresses. Maintain existing premises in a neat and clean condition.

END OF SECTION

SECTION 035114 – CEMENTITIOUS ROOF DECK

PART 1 - GENERAL

1.01 SUMMARY

A. Section Includes:

1. Cementitious Wood Fiber Acoustic Plank Roof Deck System known as
Tectum I Acoustic Roof Deck System.

B. Related Sections:

1. Division 5 Section: Steel Framing.
2. Division 6 Sections: Wood Framing.
3. Division 7 Sections: Roofing.
4. Division 9 Sections: Acoustical Materials

1.02 REFERENCES

A. ASTM International:

1. ASTM C177 Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded-Hot-Plate Apparatus.
2. ASTM C518 Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus.
3. ASTM E96 Standard Test Methods for Water Vapor Transmission of Materials.

B. Underwriters Laboratories, Inc. (UL):

1. UL 580 Standard for Safety for Tests for Uplift Resistance of Roof Assemblies.

1.03 SYSTEM DESCRIPTION

A. Design Requirements: Provide roof deck assembly designed and tested according to the following:

1. Underwriters Laboratories UL 580 (UL Class 90 Design): [Design No. NM504] [Design No. NM511] [Design No. NM512] [Design No. NM517] [Design No. NM533] [Design No. 474] [Design No. 451]. (Select One, only required if there are atypical wind unload requirements. Default is none).

2. Structural Performance Requirements: Provide a roof deck system that has been manufactured, fabricated, and installed to provide deflection of [Less than l/240 at design load (Default)] [Specify required maximum deflection.].
3. Acoustic Performance: NRC Value 0.60 Minimum generated from UL Classified face material of 2" to 3" thickness.

1.04 SUBMITTALS:

A. General: Submit listed submittals in accordance with Conditions of the Contract and Division 1 Submittals Procedures Sections.

B. Product Data: Submit manufacturer's product data and installation instructions.

C. Shop Drawings: Provide drawings indicating locations and spacing of structural supports and penetrations including panel layout, attachment details and termination details.

D. Samples:

1. 6 inch square sample of each wood fiber composite deck unit required indicating exposed texture expected in completed work.
2. Labeled set of required fasteners and accessories for a complete installation.

E. Quality Assurance/Control Submittals:

1. Test Reports: Upon request, submit certified test reports from recognized test laboratories.

2. Sustainability Reports:

a. Third Party Verified Environmental Product Declaration (EPD).
b. Third Party Verified Health Product Declaration (HPD).
c. Living Product Imperative Certification for Cementitious Wood Fiber Plank.

F. Closeout Submittals

1. Standard Manufacturer's Warranty Document.

1.05 QUALITY ASSURANCE:

A. Installer Qualification:

1. Utilize an installer having documented experience on projects of a similar size and complexity.
2. Letter of certification for the manufacturer stating that the installer

has appropriate experience and is certified to install system.

B. Regulatory Requirements and Approvals: [Comply with requirements below.] [Specify applicable requirements of regulatory agencies. (Default is ICC-ES ESR-1112)].

1. International Code Council (ICC):
   a. ICC-ES Evaluation Report ESR-1112.

2. State of California:
   a. DSA Number PA-008.

C. Certifications: [Specify requirement for certifications.].

Specifier Note: Retain paragraph below if reinstallation meeting is required.

D. Preinstallation Meetings: [Specify requirements for meeting.].

1.06 DELIVERY, STORAGE AND HANDLING

A. General: Comply with applicable Division 1 Sections.

B. Delivery: Deliver materials in manufacturer's original, unopened, undamaged containers with identification labels intact.

1. Provide labels indicating brand name, deck type, panel size, and panel thickness.

C. Storage and Protection: Store materials protected from exposure to harmful environmental condition and at temperature and humidity conditions recommended by the manufacturer.

1. Prevent soiling, physical damage or wetting.

Specifier Note: Retain article below when requiring compliance with thermal performance warranty.

Coordinate article with Conditions of the Contract and with Division 1 Closeout Submittals (Warranty) Section. Use this article to require special or extended warranty or bond covering the work of this Section.

1.07 WARRANTY

A. Project Warranty: Refer to Conditions of the Contract for project warranty provisions.

B. Manufacturer's Warranty: Submit manufacturer's standard 15 year thermal performance warranty.

PART 2 – PRODUCTS

2.01 ROOF DECK AND FORM SYSTEMS.

A. Manufacturer: Tectum Inc.
1. Armstrong World Industries
- Contact: Matt Septak **MRSeptak@armstrongceilings.com** 640.250.2506

B. Proprietary Systems. Cementitious deck form board systems, including the following configurations:
1. Tectum Roof Deck Plank.
2. Tectum Roof Deck Tile.
3. Tectum Roof Deck Long Span.

2.02 MANUFACTURED UNITS:

A. Tectum I Roof Deck Panel consisting of Tectum Cementitious Wood Fiber Board.
1. Total Panel Thickness and R-Value: [Select one.].

Total Thickness	R-Value
3”	6.43

2. Panel Width: 32.5”
3. Panel Length(s): 83.5”

2.03 COMPONENTS:

A. Cementitious Wood Fiber Board, 3 inch/NRC 0.80 thick Tectum consisting of FSC Certified aspen wood fibers bonded with inorganic hydraulic cement.

2.04 ACCESSORIES:

A. Fasteners:
1. DEKFAST 14 gage steel. Length to penetrate structural member minimum of 1-1/2 inch.

B. Washers:
1. 2” Washers

C. Adhesive:
1. SFA-66

D. Grout:
1. Tectum Grout (For use with Tectum Tile). Gypsum cement grout, ready for mixing with potable water.

E. Steel channel
1. 16 gauge steel channel (For use with Tectum Long span). Applied to standard plank to allow for spans up to six feet.

PART 3 - EXECUTION

3.01 EXAMINATION:

A. Site Verification of Conditions:

1. Verify that site conditions are acceptable for installation of roof deck system.
2. Do not proceed with installation of roof deck system until unacceptable conditions are corrected.
3. Do not proceed with installation if precipitation or freezing temperatures are forecast during installation.

3.02 INSTALLATION: Comply with Manufacturer's published instructions and recommended installation procedures.

A. Place panel on joists with square cut ends butted tightly together.

B. Stagger end joints. Seal joints larger than 1/4 inch with adhesive and larger joints with foam strips or expanding foam.

C. Support panels with bent plates (steel or other support material) at roof transitions. Including, but not limited to ridge, valley, perimeter, and panel direction change conditions.

D. Panels require a minimum 1-inch bearing on structural members. Must be glued and screwed at transitions.

E. Panel ends are required to terminate over structural members or supports with a minimum of 1 inch bearing on structural members.

F. Cut panels neatly to abut to parapets around openings and penetrations. Use manufacturer recommended saw and techniques for field cutting.

G. Apply adhesive to support members and on top of plank tongue in accordance with manufacturer's recommendations.

H. Use manufacturer's recommended slide hammer or other tools to assure a tight joint at panel-to-panel joints. Hold panels in position until screws are installed.

I. Install screws at each structural support in conformance with approved

Shop Drawings and manufacturer's recommended spacing and quantity.

J. Field Installed Overlay: If indicated on Shop Drawings, install continuous 7/16-inch OSB overlay staggering panel joints in both directions for a minimum joint coverage of 1'-0". Install overlay with adhesive and 1-inch staples as indicated on approved Shop Drawings.

3.03 CLEANING:

A. Clean exposed surfaces of panel installation.

B. Remove visible adhesive from exposed surfaces.

C. Remove and replace work that cannot be successfully cleaned or repaired, or which indicates structural damage.

3.04 PROTECTION:

A. Protect installed work from damage due to weather related moisture.

B. Protect installed work from damage due to subsequent construction activity.

C. Provide temporary protection as necessary to protect installed material from exposure to excessive moisture prior to installation of roofing material.

END OF SECTION 035114

SECTION 042000

UNIT MASONRY

PART 1 GENERAL

1.1 GENERAL REQUIREMENTS

A. Work of this Section, as shown or specified, shall be in accordance with the requirements of the Contract Documents.

1.2 SECTION INCLUDES

A. Work of this Section includes all labor, materials, equipment and services necessary to complete the unit masonry work as shown on the drawings and/or specified herein, including but not necessarily limited to the following:

1. Concrete block walls and partitions.
2. Metal joint reinforcing, anchors, ties, closures and related accessories for masonry.
3. Chases, recesses, pockets and openings in masonry as required for installation of work by others.
4. Building in of items furnished by others into masonry, including access doors, door frames, anchors, sleeves and inserts, and other similar items to be embedded in masonry.
5. Grouting in of metal items built into masonry work.
6. Protection, pointing and cleaning of masonry.

1.3 SUBMITTALS

A. Submit technical and installation information for the following:

1. Mortar materials, each material and mortar type.
2. Certification of mortar mix.
3. Concrete block, joint reinforcing, anchors, ties and joint filler; submit manufacturer's technical and descriptive literature.
4. Block manufacturer shall submit certifications of compliance with ASTM C 90, C 331 and UL 618 prior to any job site delivery. Field sampling of concrete block may be tested by an Independent Testing Laboratory retained by the Owner according to the requirements of ASTM C 140.

B. Construction Procedures (Submit the following)

1. Procedures and materials for cleaning masonry work; including certification that cleaner will not adversely affect stone, gaskets, sealants, etc.

1.4 QUALITY ASSURANCE

A. Conform to the following non-cumulative tolerances (any masonry work not meeting these standards shall be re-built as directed by the Architect).

1. Variation from the plumb:

a. In lines and surfaces of columns, walls and arrises:
   1). In 10 feet 1/8"
   2). In any story of 25 feet maximum 1/4"
   3). In 40 feet or more 1/4"
b. For external corners, expansion joints and other conspicuous lines:
   1). In any story of 25 feet maximum 1/4"
   2). In 40 feet or more 3/8"

2. Variation from the level or the grades indicated on the drawings; for exposed lintels, sills, parapets, horizontal grooves and other conspicuous lines:

a. In any bay or 20 feet maximum 1/4"
b. In 40 feet or more 1/4"

3. Variation of the linear building lines from established position in plan related portion of columns and partitions:

a. In any bay or 20 feet maximum 1/4"
b. In 40 feet or more 1/2"

4. Variation in cross-sectional dimensions of columns and in thickness of walls:

a. Minus 1/8"
b. Plus 1/8"

5. Variation in dimensions of masonry openings:

a. Horizontal dimension -0" + 1/16"
b. Vertical dimension +0" - 1/16"

B. Work of this Section shall conform to the requirements of the following (unless otherwise superseded by prevailing Building Code):

1. 2008 ACI 530/ASCE 5/TMS 402 Building Code Requirements for Masonry Structures.

2. 2008 ACI 530-1/ASCE 6/TMS 602 Specifications for Masonry Structures.

3. Brick Industry Association (BIA) "Technical Notes on Brick Construction."

C. Pre-Construction Conference: Prior to installation of masonry and associated work, Contractor shall arrange a meeting with Masonry Subcontractor, installers of related work, and other entities concerned with masonry wall performance, including the Architect and Owner. Contractor shall record discussions and agreements and furnish copy to each participant. Provide at least seventy-two (72) hours' advance notice to participants prior to convening conference. Review methods and procedures related to masonry work, including, but not limited to, the following:

1. Review masonry requirements (drawings, specifications and other Contract Documents).

2. Review required submittals, both completed and yet to be completed.

3. Review and finalize construction schedule related to masonry work and verify availability of materials, installer's personnel, equipment and facilities needed to make progress and avoid delays.

4. Review required inspection, testing, certifying and material usage accounting procedures.

1.5 PRODUCT HANDLING

A. General: Deliver, store, handle and protect all materials from damage, moisture, dirt and intrusion of foreign matter. Store all masonry units and mortar materials on raised platforms and under ventilated and waterproof cover. Store packaged materials in manufacturer's unopened containers, marked with manufacturer's name and product brand name. Immediately reseal containers after partial use. Remove and replace damaged materials.

B. Masonry Units: Pack, deliver and store to prevent breakage, cracking, chipping, spalling or other damage. Store, protect and ventilate units at project site.

C. Aggregate: Store with provisions for good drainage.

D. Reinforcement and Anchors: Store and protect so that when placed, joint reinforcement and anchors will be free of soil, dirt, ice, loose rust, scale, or other coatings which would destroy or reduce bond with mortar, and will not be disfigured or bent out of shape.

1.6 JOB CONDITIONS

A. In cold weather, when the outside temperature is below forty (40) degrees F., conform to the requirements of "Cold Weather Masonry Construction and Protection Recommendations" publication by Brick Industry Association (BIA). No anti-freeze admixtures are permitted.

1. In addition, conform to the following:

a. Masonry materials must be warmed as required.
b. Work must be protected a minimum of 24 hours after installation so as to maintain enough heat for hydration of the cement in the mortar.

B. Hot-Weather Requirements: Protect unit masonry work when temperature and humidity conditions produce excessive evaporation of water from mortar and grout. Provide artificial shade and wind breaks and use cooled materials as required. Do not apply mortar to substrates with temperatures of 100 deg. F. and above. In addition, conform to the following:

1. Masonry materials must be cool.

2. Mortar must be used within 2 hours of initial mixing.

C. Protection of Masonry: During erection, cover tops of walls, projections, and sills with waterproof sheeting at end of each day's work. Cover partially completed masonry when construction is not in progress.

1. Extend cover a minimum of 24" down both sides and hold cover securely in place.

2. Where one wythe of multi-wythe masonry walls is completed in advance of other wythes, secure cover a minimum of 24" down face next to unconstructed wythe and hold cover in place.

D. Stain Prevention: Prevent grout, mortar, and soil from staining the face of masonry to be left exposed or painted. Immediately remove grout, mortar, and soil that come in contact with such masonry.

1. Protect base of walls from mortar splatter by coverings spread on floor and over wall surface.

2. Protect sills, ledges, and projections from mortar droppings.

3. Protect surfaces of window and door frames, as well as similar products with painted and integral finishes, from mortar droppings.

PART 2 PRODUCTS

2.1 MATERIALS

A. Standard Concrete Block

1. Portland cement, ASTM C 150, Type 1, low alkali (less than 65) one source.

2. Aggregates, ASTM C 331, lightweight expanded shale, clay or slate aggregates, manufactured by the rotary kiln process equal to "Solite," "Norlite," or "Haydite."

a. Block scheduled to receive painted finish shall contain normal weight aggregate meeting ASTM C-33 in addition to light weight aggregate in order to receive a smooth, uniform finish.

3. Concrete Masonry Units: Load bearing lightweight aggregate concrete masonry units conforming to the requirements of ASTM C 90.

a. Block for rated walls shall be 75% solid units.
b. All other block may be hollow units.

4. The producer of the concrete masonry units shall furnish certification from an independent testing laboratory confirming that all 8" or larger masonry units meet all of the UL 618 requirements for two (2) hours or better (as required), referencing full scale fire test reports (ASTM E 119). All 4" and 6" units shall conform to "National Bureau of Standards" and "National Research Council" full scale fire tests.

5. Sizes and Shapes: Nominal face size 8" x 16" by thickness as indicated on drawings, with stretcher units, jamb units, header units, square corner units (at ends and corners of exposed or painted work), sash units (at control joints within masonry wall), lintel units and other special shapes and sizes required to complete the work.

6. Finish: For exposed or painted block surfaces, in addition to ASTM requirements, block shall have uniformly dense, flat, fine grain texture, with no cracks, chips, spalls, or other defects which would impair appearance. For concealed CMU, surfaces shall be free from deleterious materials that would stain plaster or corrode metal.

7. Curing: All concrete block shall be steam cured, and air dried for not less than thirty (30) days before delivery.

8. Density of concrete block shall not exceed one hundred and five (105) lbs. per cubic foot.

9. Shrinkage: Shrinkage of concrete blocks shall not exceed 0.065% when tested in accordance with ASTM C 426-99.

10. Water Content

    a. At the time of delivery to the job site, concrete masonry units shall have a value, in weight of contained water, of not more than thirty (30) percent of the fully saturated content for the unit tested.
    b. Ship all units from the factory, and store at the job site, with all necessary protection to prevent increase of water content from rain and other sources.

B. Joint Reinforcing for Masonry Walls

1. Super heavy duty reinforcing fabricated of 3/16" dia. side and cross rods, truss or ladder design, ties, spaced every block course. Provide prefabricated pieces at corners and intersections of walls or partitions.

    a. Reinforcing assembly shall be hot dip galvanized steel finish conforming to ASTM A 153 with zinc coating of 1.5 oz. of zinc per sq. ft., after fabrication.

2. Wire used in assemblies noted above shall be cold drawn steel wire conforming to ASTM A 82.

3. Approved Joint Reinforcing Manufacturers

    a. Hohmann & Barnard
    b. Wire-Bond
    c. Heckmann Building Products
    d. National Wire Products Industries, Inc.

C. Anchors and Ties

1. Wire Mesh: Hot-dip galvanized sixteen (16) gage steel wire, square mesh, width 3" by length to suit condition; No. 268 by Heckmann Building Products, or approved equal by manufacturer noted above in Para. B.3.

2. Provide strap anchors to accommodate connecting new to existing construction.

D. Reinforcing Bars and Rods: ASTM A 615, Grade 60. See Drawings for size.

2.2 MORTAR MATERIALS

A. Portland Cement: ASTM C 150, Type 1, standard color, one source.

B. Hydrated Lime: ASTM C 207, Type S, as manufactured by Corsons, or approved equal.

C. Aggregate: Clean, washed, buff colored sand, graded per ASTM C 144.

D. Aggregate for Grout: ASTM C 404.

E. Water: Clean, fresh and suitable for drinking.

2.3 MORTAR MIX

A. Interior Masonry Construction: Provide Portland cement/lime mortar conforming to ASTM C 270, Type N, for load bearing conditions, mortar shall conform to ASTM C 270, Type M.

B. Reinforced Concrete Block: Provide Portland cement/lime mortar conforming to ASTM C 270, Type S.

C. Mortar for Cement Cants: One (1) part Portland cement and four (4) parts sand, by volume.

D. Grout for Unit Masonry: Comply with ASTM C 476 for grout for use in construction of unit masonry. Use grout of consistency (fine or coarse) at time of placement which will completely fill all spaces intended to receive grout. Grout shall have a minimum compressive strength of 3000 psi when tested in accordance with ASTM C 1019.

E. Mixing

1. General: Add cement just before mixing and mix dry. Use sufficient amount of water as necessary to produce workable mix. Mix in small batches to make plastic mass.

2. Mixing: Machine mix all mortars in approved type mixer with device to accurately and uniformly control water. Add hydrated lime dry. Mix dry materials not less than two (2) minutes. Add water, then mix not less than three (3) minutes, not to exceed five (5) minutes. Mix only amount of mortar that can be used before initial set. Do not use mortar which has reached its initial set or two (2) hours after initial mixing, whichever comes earlier. Mortar may not be re-tempered. Clean mixer for each batch, whenever mortar type is changed, and at end of each day's work.

3. Acceleration or other admixtures not permitted.

4. Mortar shall have a flow after suction of not less than seventy-five (75) percent of that immediately after mixing as determined by ASTM C 91.

F. Admixtures

1. No air-entraining admixtures or cementitious materials containing air-entraining admixtures shall be used in the mortar.

2. Calcium chloride or admixtures containing calcium chloride shall not be used in mortar.

PART 3 EXECUTION

3.1 SURFACE CONDITIONS

A. Inspection

1. Prior to all work of this Section, carefully inspect the installed work of all other trades and verify that all such work is complete to the point where this installation may properly commence.

2. Verify that masonry may be completed in accordance with all pertinent codes and regulations, the referenced standards, and the original design.

3. Do not start any work until mock-ups are approved by the Architect.

B. Discrepancies

1. In the event of discrepancy, immediately notify the Architect in writing.

2. Do not proceed with installation in areas of discrepancy until all such discrepancies have been fully resolved.

3. Starting of work by the Contractor means acceptance by the Contractor of the substrate.

3.2 COORDINATION

A. Carefully coordinate with all other trades to ensure proper and adequate interface of the work of other trades with the work of this Section.

3.3 PREPARATION

A. Concrete Block: Do not wet concrete block units.

3.4 INSTALLATION

A. General

1. Build walls to the full thickness shown. Build single wythe walls to the actual thickness of the masonry units, using units of nominal thickness shown.

2. Build chases and recesses as shown or required for the work of other trades.

3. Leave openings for equipment to be installed before completion of masonry work. After installation of equipment, complete masonry work to match work immediately adjacent to the opening.

4. Lay out walls in advance for accurate spacing of surface bond patterns with uniform joint widths and to properly locate openings, movement type joints, returns and off-sets. Avoid the use of less than half size units at corners, jambs and wherever possible.

5. Lay up walls plumb and true with courses level, accurately spaced and coordinated with other work.

6. Pattern Bond: Lay concealed concrete block with all units in a wythe bonded by lapping not less than two (2) inches. Bond and interlock each course of each wythe at corners. Do not use units of less than four (4) inches horizontal face dimensions at corners or jambs.

7. Where possible, masonry walls and partitions shall be built after all overhead ducts, pipes and conduits are in place and tested. Masonry shall be neatly built around the items above. Walls and partitions shall be plumb, true to line and free from defects such as open cells, voids, dry joints and other similar defects.

8. Mortar, ties and reinforcement must not extend into or bridge any expansion joints.

9. New CMU is to align with the existing. Tool joints to match existing joints.

10. Lay CMU to blend and to match the pattern of the existing CMU as much as possible.

B. Mortar Bedding and Jointing

1. Lay concrete masonry units with full mortar coverage on horizontal and vertical face shells. Bed webs in mortar in starting course in all courses of piers, columns and pilasters, where solid CMU is used and where adjacent to cells or cavities to be reinforced or filled with concrete or grout.

2. Lay masonry walls with 3/8" joints unless otherwise shown on drawings.

3. Tool exposed joints slightly concave to match existing after the mortar joint is “thumbprint” hard. Concealed joints shall be struck flush, including at any CMU scheduled to receive a waterproofing or air barrier membrane.

4. Remove masonry units disturbed after laying; clean and reset in fresh mortar. Do not pound corners at jambs to fit stretcher units which have been set in position. If adjustments are required, remove units, clean off mortar and reset in fresh mortar.

5. On both sides of the doors and openings; fill the cells solid with grout, min. 8" each side.

C. Stopping and Resuming Work: Rake back 1/2 unit length in each course; do not tooth. Clean exposed surfaces of set masonry and remove loose masonry units and mortar prior to laying fresh masonry.

D. Built-In Work

1. As the work progresses, build in items specified under this and other Sections of these specifications. Fill in solidly with masonry around built-in items.

2. Mortar in door frames, access doors, louvers and other metal items embedded or built into masonry work solidly with mortar as the masonry units are laid up.

3. Grout under lintels, bearing plates, and steel bearing on masonry with solid bed grout.

4. Sleeves, pipes, ducts and all other items which pass through masonry walls shall be caulked with interior grade sealant meeting requirements of Section 079200, so as to be airtight and prevent air leakage. Refer to Section 078413 for packing of voids in rated masonry walls.

5. Fill vertical cells of masonry units solid with grout which have anchoring, reinforcing rods, supporting or hanging devices embedded in the cell including stone anchors and window or curtain wall anchors and at hollow metal door frames.

6. Fill vertical cells of masonry units solid with mortar on each side of door frames to sixteen (16) inches beyond.

7. Unless otherwise noted, fill vertical cells of masonry units solid with grout which are below steel bearing plates, steel beams, and ends of lintels, to eight (8) inches beyond bearing and from floor to bearing.

8. Place wire mesh in horizontal joint below masonry unit cells to be filled with mortar, to prevent mortar from dropping into unfilled cells below.

9. Masonry indicated as being reinforced shall have all voids filled solid with grout. Grout shall be consolidated in place by vibration or other methods which insure complete filling of cells. When the least clear dimension of the grouted cell is less than two (2) inches, the maximum height of grout

pour shall not exceed twelve (12) inches. When the least clear dimension is two (2) inches or more, maximum height of grout pour shall not exceed forty-eight (48) inches. When grouting is stopped for one (1) hour or longer, the grout pour shall be stopped 1-1/2" below the top of a masonry unit. Vertical bar reinforcing shall be accurately placed and held in position while being grouted, and shall be in place before grouting starts. All such reinforcing shall have a minimum clear cover of 5/8". Lap all bars a minimum of forty (40) bar diameters and provide steel spacer ties (not to exceed 192 bar diameter) to secure and position all vertical steel and prevent displacement during grouting. Provide continuous horizontal reinforcement embedded in mortar joints every second course.

E. Cutting and Patching

1. All exposed masonry which requires cutting or fitting shall be cut accurately to size with motorized carborundum or diamond saw, producing cut edges.

2. All masonry which requires patching in exposed work, if approved by Architect, shall be patched neatly with mortar to match appearance of masonry as closely as possible and to the Architect's satisfaction. Rake back joints and use pointing mortar to match as required.

F. Solid Wall Construction

1. Fill the vertical longitudinal joint between wythes solidly with mortar by parging the in-place wythe and shoving units into the parging.

2. Tie wythes with continuous horizontal reinforcement embedded in mortar joints sixteen (16) inches o.c. vertically.

G. Ties and Anchors for Masonry Construction

1. Provide ties and anchors as shown or specified, but not less than one metal tie, spaced not to exceed sixteen (16) inches o.c. horizontally and/or vertically. Provide additional ties within 1'-0" of all openings and adjacent to expansion joints and spaced not more than 16" apart around perimeter of openings.

2. Anchor masonry to structure complying with the following:

a. Provide an open space not less than 1/2" in width between masonry and structural member, unless otherwise shown. Keep open space free of mortar or other rigid materials.

3.5 CLEANING, PROTECTION, ADJUSTMENT

A. Protection

1. The Contractor shall take adequate precautions for the protection of all surfaces against mortar spatter, and shall immediately remove any such spatter should it inadvertently occur, leaving no stain or discoloration.

2. Excess mortar shall be wiped off the masonry surfaces as the work progresses.

3. Wood coverings shall be placed over all such masonry surfaces as are likely to be damaged during the progress of the entire project.

4. Protective measures shall be performed in a manner satisfactory to the Architect.

5. Damaged masonry units shall be replaced to satisfaction of the Architect.

B. Clean-Up

1. Upon completion, all exposed masonry shall be thoroughly cleaned following recommendations of the BIA Technical Note No. 20. Before applying any cleaning agent to the entire wall, it shall be applied to a sample wall area of approximately 4' x 4' in a location approved by the Architect. No further cleaning work may proceed until the sample area has been approved by the Architect, after which time the same cleaning materials and method shall be used on the remaining wall area. If stiff brushes and water do not suffice, the surface shall be thoroughly saturated with clear water and then scrubbed with a solution of an approved detergent masonry cleaner, equal to "Vana Trol" made by ProSoCo Inc. or equal made by Diedrich or approved equal, mixed and applied as per manufacturer's directions, followed immediately by a thorough rinsing with clear water. All adjacent non-masonry surfaces shall be thoroughly protected during cleaning.

   a. Unless otherwise required by cleaning agent manufacturer use only low pressure device (30 to 50 psi) for application of cleaning agent and water rinsing.

C. Pointing: Point any defective joint with mortar identical with that specified for that joint.

END OF SECTION

SECTION 055000

MISCELLANEOUS METALS

PART 1 GENERAL

1.1 GENERAL REQUIREMENTS

A. Work of this Section, as shown or specified, shall be in accordance with the requirements of the Contract Documents.

1.2 SECTION INCLUDES

A. Work of this Section includes all labor, materials, equipment, and services necessary to complete the miscellaneous metal work as indicated on the drawings and/or specified herein, including, but not limited to, the following:

1. Rough hardware.
2. Exterior cage ladder.
3. Exterior Ship's ladders.
4. Steel pipe handrails and railings.
5. Light steel framing and supports not included as part of work of other trades.
6. Steel framing, bracing, supports, anchors, bolts, shims, fastenings, and all other supplementary parts indicated on drawings or as required to complete each item of work of this Section.
7. Prime painting, touch-up painting, galvanizing and separation of dissimilar metals for work of this Section.
8. Cutting, fitting, drilling and tapping work of this Section to accommodate work of other Sections and of concrete, masonry or other materials as required for attaching and installing work of this Section.

1.3 QUALITY ASSURANCE

A. Field Measurements: Take field measurements prior to preparation of shop drawings and fabrication, where possible. Do not delay job progress; allow for trimming and fitting where taking field measurements before fabrication might delay work.

B. Shop Assembly: Pre-assemble items in shop to greatest extent possible to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for re-assembly and coordinated installation.

C. Reference Standards: The work is subject to requirements of applicable portions of the following standards:

1. "Manual of Steel Construction," American Institute of Steel Construction.

2. AWS D1-1 "Structural Welding Code," American Welding Society.

3. SSPC SP-3 "Surface Preparation Specification No. 3, Power Tool Cleaning," Steel Structures Painting Council.

4. SSPC PA-1 "Painting Application Specification," Steel Structures Painting Council.

5. "Handbook on Bolt, Nut and Rivet Standards," Industrial Fasteners Institute.

D. Steel Materials: For steel to be hot dip-galvanized, provide steel chemically suitable for metal coatings complying with the following requirements: carbon below 0.25 percent, silicon below 0.24 percent, phosphorous below 0.05 percent, and manganese below 1.35 percent. Notify galvanizer if steel does not comply with these requirements to determine suitability for processing.

E. Engage the services of a galvanizer who has demonstrated a minimum of five (5) years' experience in the successful performance of the processes outlined in this specification in the facility where the work is to be done and who will apply the galvanizing and coatings within the same facility as outlined herein. The Architect has the right to inspect and approve or reject the galvanizer/galvanizing facility.

F. The galvanizer/galvanizing facility must have an ongoing Quality Control/Quality Assurance program which has been in effect for a minimum of five years and shall provide the Architect with process and final inspection documentation. The galvanizer/galvanizing facility must have an on-premise testing facility capable of measuring the chemical and metallurgical composition of the galvanizing bath and pickling tanks.

G. Inspection and testing of hot-dip galvanized coating shall be done under the guidelines provided in the American Hot-Dip Galvanizers Association (AGA) publication "Inspection of Products Hot-Dip Galvanized After Fabrication."

1.4 PERFORMANCE STANDARDS

A. Stairs and railings shall be constructed to conform to the following performance standards:

1. Stairs and platforms shall support a live load of one hundred (100) psf and a concentrated live load of three hundred (300) lbs. and shall have a live load deflection limited to 1/360 of the span. Loads shall not apply simultaneously.

2. Railings shall be designed to resist loads per Building Code.

1.5 SUBMITTALS

A. Manufacturer's Literature: Submit manufacturer's specifications, load tables, dimension diagrams, anchor details and installation instructions for products to be used in the fabrication of miscellaneous metal work, including paint products.

B. Shop Drawings: Shop drawings for the fabrication and erection of all assemblies of miscellaneous iron work which are not completely shown by manufacturer's data sheets. Include plans and elevations at not less than 1" to 1'-0" scale, and include details of sections and connections at not less than 3" to 1'-0" scale. Show anchorage and accessory items.

C. Engineering Data

1. Before any stairs, ladders or railings are fabricated, submit engineering data drawings to the Architect for review indicating how performance standards specified here shall be met. The Contractor is responsible for the structural design and supports for these systems and must show his proposed systems on these drawings.

2. These drawings must show all load conditions and design calculations relative to connections, fastening devices and anchorage, as well as size and gauge of members. Calculations and drawings must be prepared by a Structural Engineer licensed in the State of New Jersey and shall be signed and sealed by this Engineer.

D. Welding shall be indicated on shop drawings using AWS symbols and showing length, size and spacing (if not continuous). Auxiliary views shall be shown to clarify all welding. Notes such as 1/4" weld, weld and tack weld are not acceptable.

E. Certification: For items to be hot-dip galvanized, identify each item galvanized and to show compliance of application. The Certificate shall be signed by the galvanizer and shall contain a detailed description of the material processed and the ASTM standard used for the coating and, the weight of the coating. In addition, and as attachment to Certification, submit reports of testing and inspections indicating compliance with the provisions of this Section.

PART 2 PRODUCTS

2.1 MATERIALS

A. Metals

1. Metal Surfaces, General: For fabrication of miscellaneous metal work which will be exposed to view, use only materials which are smooth and free of surface blemishes including pitting, seam marks, roller marks, rolled trade names and roughness.

2. Steel Plates, Shapes and Bars: ASTM A 36.

3. Steel Bar Grating: ASTM A 1011 or ASTM A 36.

4. Steel Tubing: Cold formed, ASTM A 500; or hot rolled, ASTM A 501.

5. Structural Steel Sheet: Hot rolled, ASTM A 1011; or cold rolled, ASTM A 1008, Class 1; of grade required for design loading.

6. Galvanized Structural Steel Sheet: ASTM A 924, of grade required for design loading. Coating designation G90.

7. Steel Pipe: ASTM A 53, type and grade as selected by fabricator and as required for design loading; black finish unless galvanizing is indicated; standard weight (Schedule 40), unless otherwise indicated.

8. Gray Iron Castings: ASTM A 48, Class 30, unless another class is indicated or required by structural loads.

9. Malleable Iron Castings: ASTM A 47, grade as selected by fabricator.

10. Brackets, Flanges and Anchors: Cast or formed metal of the same type material and finish as supported rails, unless otherwise indicated.

11. Concrete Inserts: Threaded or wedge type; galvanized ferrous castings, either malleable iron, ASTM A 47, or cast steel, ASTM A 27. Provide bolts, washers and shims as required, hot-dip galvanized, ASTM A 153.

B. Grout: Non-shrink, non-metallic grout conforming to the requirements of Section 033000.

C. Fasteners

1. General: Provide zinc-coated fasteners for exterior use or where built into exterior walls. Select fasteners for the type, grade and class required.

2. Bolts and Nuts: Regular hexagon head type, ASTM A 307, Grade A.

3. Anchor Bolts: ASTM F 1554, Grade 36.

4. Lag Bolts: ASME B18.2.1.

5. Machine Screws: ASME B18.6.3.

6. Plain Washers: Round, carbon steel, ASME B18.22.1.

7. Masonry Anchorage Devices: Expansion shields, FS FF-S-325.

8. Toggle Bolts: Tumble-wing type, FS FF-B-588, type, class and style as required.

9. Lock Washers: Helical spring type carbon steel, ASME B18.21.1.

D. Shop Paint: Shop prime all non-galvanized miscellaneous metal items using Series 88 Azeron Primer made by Tnemec, ICI Devoe "Rust Guard" quick dry

alkyd shop coat No. 41403, or "Interlac 393" by International Protection Coatings.

1. If steel is to receive high performance coating as noted in Section 099000, shop prime using primer noted in Section 099000.

E. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D 1187.

F. Galvanizing Repair Coating: For touching up galvanized surfaces after erection, provide repair coating that is V.O.C. compliant, equal to "Silver Galv" made by Z.R.C. Worldwide or approved equal. Apply to a dry film thickness of 1.5 to 3.0 mils.

2.2 PRIME PAINTING

A. Scope: All ferrous metal (except galvanized steel) shall be cleaned and shop painted with one coat of specified ferrous metal primer. No shop prime paint required on galvanized steel or aluminum work.

B. Cleaning: Conform to Steel Structures Painting Council Surface Preparation Specification SP 3 (latest edition) "Power Tool Cleaning" for cleaning of ferrous metals which are to receive shop prime coat.

1. Steel to get high performance coating as noted in Section 099000 shall be cleaned as per SSPC SP.6 "Commercial Blast Cleaning."

C. Application

1. Apply shop prime coat immediately after cleaning metal. Apply paint in dry weather or under cover. Metal surfaces shall be free from frost or moisture when painted. Paint all metal surfaces including edges, joints, holes, corners, etc.

2. Paint surfaces which will be concealed after shop assembly prior to such assembly. Apply paint in accordance with approved paint manufacturer's printed instructions, and the use of any thinners, adulterants or admixtures shall be only as stated in said instructions.

3. Paint shall uniformly and completely cover the metal surfaces, 2.0 mils minimum dry film thickness. No work shall be shipped until the shop prime coat thereon has dried.

D. Touch-Up: In the shop, after assembly and in the field, after installation of work of this Section, touch-up damaged or abraded portions of shop prime paint with specified ferrous metal primer.

E. Apply one shop coat to fabricated metal items, except apply two (2) coats of paint to surfaces inaccessible after assembly or erection. Change color of second coat to distinguish it from the first.

2.3 GALVANIZING

A. Scope: All ferrous metal exposed to the weather, and all ferrous metals indicated on drawings or in specifications to be galvanized, shall be cleaned and then hot-dipped galvanized after fabrication as provided by Duncan Galvanizing or approved equal.

B. Avoid fabrication techniques that could cause distortion or embrittlement of steel items to be hot-dip galvanized. Fabricator shall consult with hot-dip galvanizer regarding potential warpage problems or handling problems during the galvanizing process that may require adjustment of fabrication techniques or design before finalizing shop drawings and beginning of fabrication.

C. Cleaning: Thoroughly clean metal surfaces of all mill scale, rust, dirt, grease, oil, moisture and other contaminants prior to galvanizing.

D. Application: Hot-dip galvanizing shall conform to the following:

1. ASTM A 143: Safeguarding Against Embrittlement of Hot-Dip Galvanized Structural Steel.

2. ASTM A 123: Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products.

3. ASTM A 153: Galvanized Coating on Iron and Steel Hardware - Table 1.

4. ASTM A 384: Practice for Safeguarding Against Warpage and Distortion During Hot-Dip Galvanizing of Steel Assemblies.

5. ASTM A 385: Practice for Providing High Quality Zinc Coatings.

6. ASTM A 924: Galvanized Coating on Steel Sheets.

7. Minimum weight of galvanized coating shall be two (2) oz. per square foot of surface.

E. Fabricate joints which will be exposed to weather in a manner to exclude water or provide weep holes where water may accumulate.

F. All galvanized materials must be inspected for compliance with these specifications and marked with a stamp indicating the name of the galvanizer, the weight of the coating, and the appropriate ASTM number.

G. To minimize surface imperfection (e.g.: flux inclusions), material to be galvanized shall be dipped into a solution of Zinc Ammonium Chloride (pre-flux) immediately prior to galvanizing. The type of galvanizing process utilizing a flux blanket overlaying the molten zinc will not be permitted.

H. After galvanizing all materials not exposed to view must be chromated by dipping material in a 0.2% chromic acid solution.

I. Galvanized surfaces, where exposed to view, must have a smooth, level surface finish. Where this does not occur, piece shall be rejected and replaced to the acceptance of the Architect.

2.4 PROTECTIVE COATINGS

A. Whenever dissimilar metals will be in contact, separate contact surfaces by coating each contact surface prior to assembly or installation with one coat of specified bituminous paint, which shall be in addition to the specified shop prime paint. Mask off those surfaces not required to receive protective coating.

2.5 WORKMANSHIP

A. General

1. Miscellaneous metal work shall be fabricated by an experienced fabricator or manufacturer and installed by an experienced tradesman.

2. Materials, methods of fabrication, fitting, assembly, bracing, supporting, fastening, operating devices, and erection shall be in accordance with drawings and specifications, approved shop drawings, and best practices of the industry, using new and clean materials as specified, having structural properties sufficient to safely sustain or withstand stresses and strains to which materials and assembled work will be subjected.

3. All work shall be accurately and neatly fabricated, assembled and erected.

B. Shop Assembly: Insofar as practicable, fitting and assembly of work shall be done in shop. Shop assemble work in largest practical sizes to minimize field work. It is the responsibility of the miscellaneous metal subcontractor to ensure that the shop-fabricated miscellaneous metal items will properly fit the field condition. In the event that shop-fabricated miscellaneous metal items do not fit the field condition, the item shall be returned to the shop for correction.

C. Cutting: Cut metal by sawing, shearing, or blanking. Flame cutting will be permitted only if cut edges are ground back to clean, smooth edges. Make cuts accurate, clean, sharp and free of burrs, without deforming adjacent surfaces or metals.

D. Holes: Drill or cleanly punch holes; do not burn.

E. Connections: Make connections with tight joints, capable of developing full strength of member, flush unless indicated otherwise, formed to exclude water where exposed to weather. Locate joints where least conspicuous. Unless indicated otherwise, weld or bolt shop connections; bolt or screw field connections. Provide expansion and contraction joints to allow for thermal movement of metal at locations and by methods approved by Architect.

1. Welding

a. Shall be in accordance with AWS D1.1 Structural Welding Code of the American Welding Society and shall be done with electrodes and/or

methods recommended by the manufacturer of the metals being welded.

b. Welds shall be continuous, except where spot welding is specifically permitted. Welds exposed to view shall be ground flush and dressed smooth with and to match finish of adjoining surfaces; undercut metal edges where welds are required to be flush.

c. All welds on or behind surfaces which will be exposed to view shall be done so as to prevent distortion of finished surface. Remove weld spatter and welding oxides from all welded surfaces.

2. Bolts and Screws: Make threaded connections tight with threads entirely concealed. Use lock nuts. Bolts and screw heads exposed to view shall be flat and countersunk. Cut off projecting ends of exposed bolts and screws flush with nuts or adjacent metal.

F. Operating Mechanism: Operating devices (i.e. pivots, hinges, etc.) mechanism and hardware used in connection with this work shall be fabricated, assembled, installed and adjusted after installation so that they will operate smoothly, freely, noiselessly and without excessive friction.

G. Built-In Work: Furnish anchor bolts, inserts, plates and any other anchorage devices, and all other items specified under this Section of the Specifications to be built into concrete, masonry or work of other trades, with necessary templates and instructions, and in ample time to facilitate proper placing and installation.

H. Supplementary Parts: Provide as necessary to complete each item of work, even though such supplementary parts are not shown or specified.

I. Coordination: Accurately cut, fit, drill and tap work of this Section to accommodate and fit work of other trades. Furnish or obtain, as applicable, templates and drawings to or from applicable trades for proper coordination of this work.

J. Exposed Work

1. In addition to requirements specified herein and shown on drawings, all surfaces exposed to view shall be clean and free from dirt, stains, grease, scratches, distortions, waves, dents, buckles, tool marks, burrs, and other defects which mar appearance of finished work.

2. Metal work exposed to view shall be straight and true to line or curve, smooth arrises and angles as sharp as practicable, miters formed in true alignment, profiles accurately intersecting, and with joints carefully matched to produce continuity of line and design.

3. Exposed fastenings, where permitted, shall be of the same material, color and finish as the metal to which applied, unless otherwise indicated, and shall be of the smallest practicable size.

K. Preparation for Hot-Dip Galvanizing: Fabricator shall correctly prepare assemblies for galvanizing in consultation with galvanizer and in accordance with applicable Reference Standards and applicable AGA publications for the "Design of Products to be Hot-Dip galvanized After Fabrication." Preparation shall include but not be limited to the following:

1. Remove welding flux.

2. Drill appropriate vent holes and provide for drainage in inconspicuous locations of hollow sections and semi-enclosed elements. After galvanizing, plug vent holes with shaped lead and grind smooth.

2.6 CAGE LADDER

A. Cage Ladder with Roofover Rail Extension.

1. Model 522 as manufactured by O'Keeffe's Inc. or approved equal.

2. Materials:

a. Cage Ladder with Roofover Rail Extension.
b. Model 522 as manufactured by O'Keeffe's Inc.

3. Fabrication:

a. Rungs: Not less than 1-1/4 inches (32 mm) in section and 18–3/8 inches (467mm) long, formed from tubular aluminum extrusions. Squared and deeply serrated on all sides.
1). Rungs shall withstand a 1,500 pound (454 kg) load without deformation or failure.
b. Channel Side Rails: Not less than 1/8 inch (3 mm) wall thickness by 3 inches (76 mm) wide.
c. Heavy Duty Tubular Side Rails: Assembled from two interlocking aluminum extrusions no less than 1/8 inch (3 mm) wall thickness by 3 inches (76 mm) wide. Construction shall be self-locking stainless steel fasteners, full penetration TIG welds and clean, smooth and burr-free surfaces.
d. Walk-Through Rail and Roof Rail Extension: Not less than 3 feet 6 inches (1067 mm) above the landing and shall be fitted with deeply serrated, square, tubular grab rails.
e. Ladder Safety Post: Retractable hand hold and tie off.
f. Safety Cages:
1). Fabricate ladder safety cages to comply with authority having jurisdiction. Assemble by welding. Spacing of primary hoops, secondary hoops and vertical bars shall not exceed that required by code.
2). Safety cage hoops and vertical bars: 3/16 inch (5 mm) by 2 inches (51 mm) aluminum bar.

4. Finish: Clear Anodic Finish: AA-M10C22A41 Mechanical finish as fabricated. Architectural Class I, clear coating 0.018 mm or thicker.

2.7 ALUMINUM SHIPS LADDER

A. Aluminum Ships Ladder and Components: Ladder, mounting brackets and handrails on both sides.

1. Model: Model 522 Aluminum Ships Ladder as manufactured by O'Keefe's Ladders, Inc., or approved equal.

2. Capacity: Unit shall support a 500 lb (227 kg) total load without failure.

3. Ladder Stringer: 5 inch by 2 inch by 3/16 inch (127 mm by 51 mm by 5 mm) extruded 6005-T5 aluminum channel. Pitch: 60 degrees.

4. Ladder Mounting Brackets:

   a. Floor Bracket: 2 inch by 3 inch by 1/4 inch (51 mm by 76 mm by 6 mm) aluminum angle.
   b. Top Bracket: 4-3/4 inch by 5 inch by 1/4 inch (121 mm by 127 mm by 6 mm) aluminum angle.

5. Handrails: 1-1/4 inches (32 mm) Schedule 40, 6005-T5 aluminum pipe provided with internal aluminum fittings.

6. Width: 2'-6" minimum, unless noted otherwise on the drawings.

B. Completely fabricate ladder ready for installation before shipment to the site.

C. Completely fabricate handrail components ready for field assembly to ladder before shipment to site.

D. Finish: Clear Anodic Finish: AA-M10C22A41 Mechanical finish as fabricated. Architectural Class I, clear coating 0.018 mm or thicker.

2.8 MISCELLANEOUS METALS ITEMS

A. Rough Hardware

1. Furnish bent or otherwise custom fabricated bolts, plates, anchors, hangers, dowels and other miscellaneous steel and iron shapes as required for framing and supporting woodwork, and for anchoring or securing woodwork to concrete or other structures. Straight bolts and other stock rough hardware items are specified in Division 6 Sections.

2. Fabricate items to sizes, shapes and dimensions required. Furnish malleable iron washers for heads and nuts which bear on wood connections; elsewhere, furnish steel washers.

B. Steel Pipe Handrails

1. Steel pipe of size shown on Drawings, Schedule 40. Fittings shall be flush type, malleable of cast iron. Brackets shall be malleable iron, design as selected by the Architect.

2. Construction: Form direction changes in rails using solid bar stock or elbows. Connections shall be shop welded and ground smooth and flush, except where field connections and expansion joints are required. Field connections may be welded, internal sleeve and plug weld, or internal sleeve and set screw.

3. Secure handrails to walls with wall brackets. Provide brackets of malleable iron castings, with not more than three (3) inches clearance from inside face of handrail to wall surface. Neatly drill wall plate portion of the bracket into concrete or masonry to receive bolts for concealed anchorage. For installation at drywall, Drywall trades shall provide plate to receive wall plate portion of bracket and anchor or bolt wall plate through drywall to supporting steel plate. Locate brackets at not more than 5'-0" o.c. unless otherwise shown.

4. Provide wall return fittings of cast iron, flush type, with the same projection as that specified for wall brackets.

5. Longitudinal members shall be parallel with each other and with floor surface or shape of stair to a tolerance of 1/8" in 10'-0" linear feet. Center line of members within each run of railing shall be in the plane.

6. For steel pipe posts where indicated, anchor posts in concrete by means of pipe sleeves set and anchored into concrete. Provide sleeves of galvanized steel pipe, not less than six (6) inches long and having an inside diameter not less than 1/2" greater than outside diameter of the inserted pipe. Provide steel plate closure secure to bottom of sleeve and of width and length not less than one (1) inch greater than outside diameter of sleeve. After posts have been inserted into sleeves, fill annular space between post and sleeve solid with non-shrink, non-ferrous grout. Cover anchorage joint with a round steel flange welded to post. Posts shall be set plumb within 1/8" vertical tolerance.

7. Steel pipe handrails shall be capable of resisting a two hundred (200) lb. force applied to rail from any direction and a uniformly distributed load of fifty (50) lbs. per linear foot applied downward or horizontally, loads not to act simultaneously.

C. Miscellaneous Light Steel Framing

1. Light steel framing, bracing, supports, framing, clip angles, shelf angles, plates, etc., shall be of such shapes and sizes as indicated on the drawings and details or as required to suit the condition and shall be provided with all necessary supports and reinforcing such as hangers, braces, struts, clip angles, anchors, bolts, nuts, welds, etc., as required to properly support and rigidly fasten and anchor same in place and to steel, concrete, masonry and all other connecting and adjoining work.

2. All light steel framing steel shall be furnished and erected in accordance with the applicable requirements of the "Specifications for the Design,

Fabrication and Erection of Structural Steel for Buildings" by the American Institute of Steel Construction and as specified herein.

D. Miscellaneous Steel Trim: Provide shapes and sizes for profiles shown. Except as otherwise indicated, fabricate units from structural steel shapes and plates and steel bars, with continuously welded joints and smooth exposed edges. Use concealed field splices wherever possible. Provide cutouts, fittings and anchorages as required for coordination of assembly and installation with other work.

PART 3 EXECUTION

3.1 INSPECTION

A. Examine the areas and conditions where miscellaneous metal is to be installed and correct any conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions are corrected to permit proper installation of the work.

3.2 ERECTION

A. Fastening to In-Place Construction: Provide anchorage devices and fasteners where necessary for securing miscellaneous metal fabrications to in-place construction; including threaded fasteners for concrete and masonry inserts, toggle bolts, through-bolts, lag bolts, wood screws, and other connectors as required.

B. Cutting, Fitting and Placement: Perform cutting, drilling and fitting required for installation of miscellaneous metal fabrications. Set work accurately in location, alignment and elevation, plumb, level, true and free of rack, measured from established lines and levels. Provide temporary bracing or anchors in formwork for items which are to be built into concrete, masonry, or similar construction.

C. Fitting Connections: Fit exposed connections accurately together to form tight hairline joints. Weld connections which are not to be left as exposed joints but cannot be shop welded because of shipping size limitations. Grind exposed joints smooth and touch up shop paint coat. Do not weld, cut or abrade the surfaces of exterior units which have been hot dip galvanized after fabrication, and are intended for bolted or screwed field connections.

D. Field Welding: Comply with AWS Code for procedures of manual shielded metal-arc welding, appearance, and quality of welds made, and methods used in correcting welding work.

E. Touch-Up Painting: Immediately after erection, clean field welds, bolted connections, and abraded areas of shop paint, and paint exposed areas with same material as used for shop painting. Apply by brush or spray to provide a minimum dry film thickness of 2.0 mils.

F. Field Touch-Up of Galvanized Surfaces: Touch-up shop applied galvanized coatings damaged during handling and installation. Use galvanizing repair coating specified herein for galvanized surfaces.

END OF SECTION

SECTION 062000

CARPENTRY

PART 1 GENERAL

1.1 GENERAL REQUIREMENTS

A. Work of this Section, as shown or specified, shall be in accordance with the requirements of the Contract Documents.

1.2 SECTION INCLUDES

A. Work of this Section includes all labor, materials, equipment, and services necessary to complete the carpentry work as shown on the drawings and/or specified herein, including, but not limited to, the following:

1. Blocking and miscellaneous wood.
2. Rough hardware.
3. Installation only of finish hardware.
4. Installation only of doors and hollow metal frames.
5. Construction of field built mechanical equipment curbs.

1.3 RELATED SECTIONS

A. Roofing - Division 7.

B. Steel doors and frames - Section 081113.

C. Finish hardware - Section 087100.

1.4 QUALITY ASSURANCE

A. Lumber Standard: Comply with PS 20.

B. Plywood Standard: Comply with PS 1 and American Plywood Assoc. (APA).

C. Shop fabricate carpentry work to the extent feasible and where shop fabrication will result in better workmanship than feasible for on-site fabrication.

D. Grade Marks: Identify lumber and plywood by official grade mark.

1. Lumber: Grade stamp to contain symbol of grading agency certified by Board of Review, American Lumber Standards Committee, mill number or name, grade of lumber, species grouping or combination designation, rules under which graded where applicable, and condition of seasoning at time of manufacture.

a. S-Dry: Maximum nineteen (19) percent moisture content as per ASTM D 2016.

E. Installation of doors, frames and hardware shall conform to the minimum standards of "Installation Guides for Doors and Hardware" of the Door and Hardware Institute.

1.5 SUBMITTALS

A. Pressure Treatment: Include certification by treating plant stating chemicals and process used, net amount of salts retained and conformance with applicable standards.

B. Fire-Retardant Treatment: Include certification by treating plant that treatment material complies with governing ordinances and that treatment will not bleed through finished surfaces.

1.6 PRODUCT HANDLING

A. Deliver carpentry materials to the site ready to use with each piece of lumber clearly marked as to grade, type and mill, and place in an area protected from the elements.

B. Deliver rough hardware in sealed containers which shall bear labels as to type and kind.

C. Pile lumber for rough usage, when delivered to the site in stacks to ensure drainage and with a minimum clearance of six (6) inches above grade. Cover stacks with tarpaulins or other watertight coverings. Store grounds and similar small sized lumber inside the building as soon as possible after delivery.

D. Do not store seasoned lumber in wet or damp portions of the building.

E. Protect fire retardant treated materials against high humidity and moisture during storage and erection.

F. Remove delivered materials which do not conform to specified grading rules or are otherwise not suitable for installation from the job site and replace with acceptable materials.

G. All items specified in Section 087100 of this specification entitled "Finish Hardware" shall be received, accounted for, stored and applied under this Section.

H. Hardware shall be sorted and stored in space assigned by Contractor and shall be kept at all times under lock and key. The safety and preservation of all items delivered will be the responsibility of the Contractor.

1.7 JOB CONDITIONS

A. Installer must examine the substrates and supporting structure and the conditions under which the carpentry work is to be installed, and notify the

Contractor in writing of conditions detrimental to the work. Do not proceed with the installation until unsatisfactory conditions have been corrected in a manner acceptable to the Installer and the Architect.

B. Coordination: Fit carpentry work to other work; scribe and cope as required for accurate fit. Correlate location of furring, nailers, blocking, grounds and similar supports to allow proper attachment of other work.

PART 2 PRODUCTS

2.1 WOOD MATERIAL

A. General

1. All wood shall be sound, flat, straight, well seasoned, thoroughly dry and free from all defects. Warped or twisted wood shall not be used.

2. For miscellaneous wood blocking, grounds, furring as required, use Utility Grade Coastal Douglas Fir or Southern Pine, free from knots, shakes, rot or other defects, straight, square edges and straight grain, air seasoned with maximum moisture content of nineteen (19) percent. Wood shall be S4S, S-Dry, complying with PS-20.

B. Wood Treatment

1. For exterior blocking, roofing and sheet metal, pressure treat wood with copper azole, Type B (CA-B); ammoniacal copper quat (ACQ) or similar preservative product that contains no arsenic or chromium. Preservative shall comply with AWPA Standard U1, (.25 lbs./cubic foot of chemical in wood).

a. After treatment, kiln dry to a maximum moisture content of fifteen (15) percent. Treatment shall be equal to "Wolmanized Natural Select" made by Arch Wood Protection Inc. or approved equal.

2. Treated wood which is cut or otherwise damaged shall be further treated in accordance with the AWPA Standard M-4.

2.2 HARDWARE

A. Rough Hardware for Treated Woods and Exterior Use: Hot-dipped galvanized or Type 304 stainless steel.

B. Nails: Common steel wire, untreated for interior work as per ASTM F 1667.

C. Bolts: Standard mild steel, square head machine bolts with square nuts and malleable iron or steel plate washers or carriage bolts with square nuts and cut washers conforming to the following:

1. Bolts: ASTM A 307, Grade A.

2. Nuts: ASTM A 563.

3. Lag Screws and Bolts: ASME B 18.2.1.

D. Expansion Anchors: Anchor bolt and sleeve assembly of material indicated below with capability to sustain, without failure, a load equal to 6 times the load imposed when installed in unit masonry assemblies and equal to 4 times the load imposed when installed in concrete as determined by testing per ASTM E 488 conducted by a qualified independent testing and inspecting agency.

1. Material: Carbon-steel components, zinc plated to comply with ASTM B 633, Class Fe/Zn 5.

2. Material: Stainless steel with bolts and nuts complying with ASTM F 593 and ASTM F 594, Alloy Group 1 or 2; use stainless steel for treated woods and exterior use.

E. Wood Screws: ASME B 18.6.1.

F. Concrete and Masonry Anchors: Standard expansion-shield self-drilling type concrete anchors where so shown or noted on the drawings, or where approved by the Architect.

PART 3 EXECUTION

3.1 INSPECTION

A. Examine the areas and conditions where carpentry is to be installed and correct any conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions are corrected to permit proper installation of the work.

3.2 INSTALLATION OF FINISH HARDWARE

A. Hardware shall be carefully fitted and securely attached, in accordance with these specifications and the instructions of the various manufacturers.

B. Unless otherwise noted, mount hardware units at heights established in Section 081113.

C. Install each hardware item in compliance with the manufacturer's instructions and recommendations. Wherever cutting and fitting is required to install hardware onto or into surfaces which are later to be painted or finished in another way, install each item completely and then remove and store in a secure place during the finish application. After completion of the finishes, re-install each item. Do not install surface-mounted items until finishes have been completed on the substrate.

D. Set units level, plumb and true to line and location. Adjust and reinforce the attachment substrate as necessary for proper installation and operation.

E. Drill and countersink units which are not factory prepared for anchorage fasteners. Space fasteners and anchors in accordance with industry standards.

F. Cut and fit threshold and floor covers to profile of door frames, with mitered corners and hair-line joints. Join units with concealed welds or concealed mechanical joints. Cut smooth openings for spindles, bolts and similar items, if any.

G. All keys used shall be construction keys which are to be tagged with fiber discs as approved, clearly labeled with identifying inscriptions and then neatly arranged in a temporary cabinet. All construction keys shall be returned to the Owner.

H. Adjusting and Cleaning

1. Adjust and check each operating item of hardware and each door, to ensure proper operation and function of every unit. Lubricate moving parts with type lubrication recommended by manufacturer (graphite type if no other recommended). Replace units which cannot be adjusted and lubricated to operate freely and smoothly as intended for the application made.

2. Final Adjustment: Wherever hardware installation is made more than one month prior to acceptance or occupancy of a space or area, return to the work during the week prior to acceptance or occupancy, and make a final check and adjustment of all hardware items in such space or area. Clean and re-lubricate operating items as necessary to restore proper function and finish of hardware and doors. Adjust door control devices to compensate for final operation of heating and ventilating equipment.

3.3 INSTALLATION OF DOORS AND FRAMES

A. Preparation

1. Remove welded-in shipping spreaders installed at factory.

2. Prior to installation and with installation spreaders in place, adjust and securely brace standard steel door frames for squareness, alignment, twist, and plumb to the following tolerances:

a. Squareness: Plus or minus 1/16 inch, measured at door rabbet on a line 90 degrees from jamb perpendicular to frame head.
b. Alignment: Plus or minus 1/16 inch, measured at jambs on a horizontal line parallel to plane of wall.
c. Twist: Plus or minus 1/16 inch, measured at opposite face corners of jambs on parallel lines, and perpendicular to plane of wall.
d. Plumbness: Plus or minus 1/16 inch, measured at jambs on a perpendicular line from head to floor.

3. Drill and tap doors and frames to receive non-templated mortised and surface-mounted door hardware.

B. Installation

1. General: Provide doors and frames of sizes, thicknesses, and designs indicated. Install steel doors and frames plumb, rigid, properly aligned, and securely fastened in place; comply with Drawings and manufacturer's written instructions.

2. Set frames accurately in position; plumbed, aligned, and braced securely until permanent anchors are set. After wall construction is complete, remove temporary braces, leaving surfaces smooth and undamaged.

    a. Install frames in accordance with ANSI/SDI A250.11, Recommended Erection Instructions for Steel Frames, unless more stringent requirements are specified herein.
    b. At fire-protection-rated openings, install frames according to NFPA 80.
    c. Where frames are fabricated in sections due to shipping or handling limitations, field splice at approved locations by welding face joint continuously; grind, fill, dress, and make splice smooth, flush, and invisible on exposed faces.
    d. Frames set in masonry walls shall have door silencers installed in frames before grouting.
    e. Remove temporary braces necessary for installation only after frames have been properly set and secured.
    f. Check plumb, squareness, and twist of frames as walls are constructed. Shim as necessary to comply with installation tolerances.

3. Floor Anchors: Provide floor anchors for each jamb and mullion that extends to floor and secure with post-installed expansion anchors.

    a. Floor anchors may be set with powder-actuated fasteners instead of post-installed expansion anchors if so indicated and approved on Shop Drawings.

4. Metal-Stud Partitions: Solidly pack mineral-fiber insulation behind frames conforming to the requirements of Section 072100 "Thermal Insulation."

5. Masonry Walls: Coordinate installation of frames to allow for solidly filling space between frames and masonry with mortar; refer to Section 042000 "Unit Masonry" for installation of frames in masonry walls.

6. In-Place Concrete or Masonry Construction: Secure frames in place with post-installed expansion anchors. Countersink anchors, and fill and make smooth, flush, and invisible on exposed faces.

7. In-Place Gypsum Board Partitions: Secure frames in place with post-installed expansion anchors through floor anchors at each jamb. Countersink anchors, and fill and make smooth, flush, and invisible on exposed faces.

8. Ceiling Struts: Extend struts vertically from top of frame at each jamb to supporting construction above, unless frame is anchored to masonry or to other structural support at each jamb. Bend top of struts to provide flush

contact for securing to supporting construction above. Provide adjustable wedged or bolted anchorage to frame jamb members.

9. Installation Tolerances: Adjust steel door frames for squareness, alignment, twist, and plumb to the tolerance given in HMMA 841 of ANSI/NAAMM, current edition.

10. Steel Doors: Fit hollow metal doors accurately in frames to the tolerances given in HMMA 841 of ANSI/NAAMM, current edition.

    a. Fire-Rated Doors: Install doors with clearances according to NFPA 80.

C. Adjustments: Check and readjust operating finish hardware items just prior to final inspection. Leave work in complete and proper operating condition. Remove and replace defective work, including doors or frames which are warped, bowed or otherwise unacceptable.

3.4 BLOCKING AND MISCELLANEOUS WOOD

A. General

1. Erect rough carpentry true to line, levels and dimensions required; squared, aligned, plumbed, and securely fastened in place.

2. Shim where required to true up furring, blocking and the like. Use wood or metal shims only.

3. Do all cutting, fitting, drilling and tapping of other work as required to secure work in place and to perform the work included herein. Do all the cutting and fitting of carpentry work, for the work of other trades as required.

B. Blocking and Miscellaneous Wood

1. Furnish and install all wood grounds, furring, blocking, curbs, bucks, nailers, etc., that may be necessary and required in connection with the carpentry and with the work described for any other trades and including required carpentry for electrical fixtures. All blocking and nailers shall be continuous wherever required, whether or not so indicated.

2. Blocking shall be as required for the proper installation of the finished work and for items in mechanical sections as required. Blocking, edgings, stops, nailing strips, etc., shall be continuous, unless distinctly noted otherwise. Provide blocking as required to install all equipment. Provide blocking and nailers where shown or required to fasten interior sheet metal work.

3. Fastening for wood grounds, furring and blocking shall be of metal and of type and spacing as best suited to conditions. Hardened steel nails, expansion screws, toggle bolts, self-clinching nails, metal plugs, inserts or similar fastenings shall be used, of suitable type and size to draw the members into place and securely hold same.

C. Rough Lumber for Roofing and Sheet Metal

1. Furnish and install all wood nailing strips and wood blocking required in connection with respective types of roofing, fans, flashings, and sheet metal work, using preservative treated wood as herein before specified.

2. Wood blocking shall be of sizes and shapes as indicated on the drawings and/or designed for the reception of curb flashings for roof ventilators and similar items.

3. All nailing strips and blocking shall be carried out in accordance with the printed installation instructions, and/or recommendations of the accepted manufacturer of the roofing materials, and in coordination and cooperation with the sheet metal work trades.

4. All blocking and nailing strips shall be firmly secured in place using counter bored bolt and nut fastenings, or secured by any other proposed flush surfaced fastenings.

5. Wood nailing strips or blocking required to be embedded in concrete work shall be furnished in time due for placing, prior to start of concrete operations. Locations and spacings of nailing strips or blocking shall be performed in coordination with the concrete trades, as required for respective installations.

3.5 ROUGH HARDWARE

A. Securely fasten rough carpentry together. Nail, spike, lag screw or bolt as required by conditions encountered in the field and the Contract Documents.

B. Provide rough or framing hardware, such as nails, screws, bolts, anchors, hangers, clips, inserts, miscellaneous fastenings, and similar items of the best quality and of the proper size and kind to adequately secure the work together and in place, in a rigid and substantial manner.

C. Secure rough carpentry to masonry with countersunk bolts in expansion sleeves or other acceptable manner, with fastenings not more than sixteen (16) inches apart. Secure woodwork to hollow masonry with toggle bolts spaced not more than sixteen (16) inches apart.

D. Countersink bolts in nailers and other rough woodwork and include washers and nuts. Cut bolts off flush with surfaces and peen as may be required to receive finished work.

E. Furnish to the mason for building into the work, or attaching the work which is to be built in, anchors, bolts, wall plates bolted to masonry, corrugated wall plugs, nailing blocks, etc., which are required for the proper fastening and installation for the work or other items as called for in this Section.

F. Detailed instructions with sketches of necessary requirements, shall be given to the masonry trade showing the location and other details of such nailing devices.

3.6 CLEANING UP

A. General: Keep the premises in a neat, safe and orderly condition at all times during execution of this portion of the work, free from accumulation of sawdust, cut-ends and debris.

B. Sweeping

1. At the end of each working day, or more often if necessary, thoroughly sweep all surfaces where refuse from this portion of the work has settled.

2. Remove the refuse to the area of the job site set aside for its storage.

3. Upon completion of this portion of the work, thoroughly broom clean all surfaces.

END OF SECTION

SECTION 075323

EPDM MEMBRANE ROOFING AND ROOF INSULATION

PART 1 GENERAL

1.1 GENERAL REQUIREMENTS

A. Work of this Section, as shown or specified, shall be in accordance with the requirements of the Contract Documents.

1.2 SECTION INCLUDES

A. Work of this Section includes all labor, materials, equipment, and services necessary to complete the membrane roofing and roof insulation as shown on the drawings and/or specified herein, including, but not limited to, the following:

1. EPDM sheet membrane roofing, fully adhered.
2. Rigid roof insulation below roof membrane.
3. Sheet flashing.
4. Substrate board.
5. Walkway pads.

1.3 RELATED SECTIONS

A. Carpentry - Section 062000.

B. Sheet Metal Flashing - Section 076200.

C. Plumbing - Division 22, for drains and vents.

1.4 DESCRIPTION OF THE SYSTEM

A. The membrane roofing system specified herein shall consist of factory fabricated large sections of sheet membrane fully adhered over the rigid roof insulation. Provide flashing at roof penetrations and vertical surfaces.

1.5 QUALITY ASSURANCE

A. Qualifications

1. The membrane roofing system specified herein shall be the product of a manufacturer who can furnish supporting evidence of experience in the manufacture of the membrane roofing system and of having been regularly engaged in this business for not less than five (5) years. Such experience shall be in projects similar to the requirements and scope for this project.
2. The details and specifications are based on a particular manufacturer. It is not the intention of this specification to restrict competition. If a

manufacturer other than the one specified is selected, it shall be his obligation and responsibility to modify and adjust his materials to suit the encountered conditions and to consult and coordinate his work with other trade Contractors to assure that the installation will be watertight and function for use intended and that the guarantee will be issued to the Owner.

3. Acceptable Manufacturers

   a. Carlisle Syntec Incorporated Fleeceback, with factory applied tape seam technology.

B. Installer: A firm with not less than 5 years of successful experience in installation of roofing systems similar to those required for this project and which is acceptable to or licensed by the manufacturer of the primary roofing materials.

C. Roofing manufacturer's technical representative shall visit the site periodically and provide reports within 72 hours of visits.

D. UL Listing: Provide system which has been tested and listed by UL for application indicated and which has a "Class A" rating.

E. FM Global Listing: Roofing, base flashings, and component materials shall comply with requirements in FM Global 4450 or FM Global 4470 as part of a roofing system, and shall be listed in FM Global's "RoofNav" for Class 1 or noncombustible construction, as applicable. Identify materials with FM Global markings.

1. Fire/Windstorm Classification: Class 1A-90 TO BE CONFIRMED.

1.6 SUBMITTALS

A. The samples and certificates listed below are required to be submitted by the Contractor to the Architect, for review. An omission of an item or items does not relieve the Contractor from this responsibility and for compliance with the Contract Documents, of which this is a part.

1. Samples

Item No.	Size	Description
a. S1	6" x 6"	Membrane w/splice
b. S2	6" x 6"	Rigid insulation
c. S3	6" x 6"	Flashing materials
d. S4	6" x 6"	Walkway material

2. Notarized Certificates of Compliance

Item No.	Description	Standard
a. C1	Sheet membrane	As specified
b. C2	Submit manufacturer's published specifications completely describing the preparation of surfaces and application of	

roofing systems.

c. C3 Submit a letter from membrane manufacturer issuing sample warranty and approving the applicator, prior to pre-application conference.

B. Submit complete project specific shop drawings showing details, dimensions, fabrication and fastening elements for each condition encountered, insulation layout, layout of each sheet noting seam locations, perimeter and penetration flashing, and other details where roofing abuts other materials and/or conditions.

C. Submit copies of pre-roofing conference records.

D. Submit a letter signed by the manufacturer and Contractor acknowledging that the submitted roofing system complies with ASCE-7 and FM 1-90 for wind speed code requirements based on height and geographic location of project.

1.7 PRODUCT DELIVERY, STORAGE AND HANDLING

A. Deliver materials to the site ready for use in the manufacturer's original and unopened containers and packaging, bearing labels as to type and brand. Delivered materials shall match approved samples. Fire classification labels shall be intact and visible.

B. Store materials under cover in a dry and clean location, off the ground, and remove materials which are damaged, torn, or otherwise not suitable for installation and replace with acceptable materials.

C. Keep insulation and membrane dry before and during installation. Remove wet materials from project site.

D. Store roofing materials on platforms or pallets, above ground, on roof level and cover with tarpaulins or on other suitable watertight covering. Store membrane and handle, in such a way as to prevent damage to edges or ends.

1.8 PREROOFING CONFERENCE

A. Prior to ordering of materials, and only after approval of submittals, a pre-roofing conference will be held to discuss the specified roofing system and its proper application. Conference shall include installer, roofing manufacturer, installers of related work, Architect, and representatives of the Owner. Record discussions and agreements and furnish copy to each participant. Provide at least seven (7) days' advance notice to participants prior to convening conference.

B. Coordinate application of the roofing system in such a manner that the complete installation is weather-tight and in accordance with warranty requirements.

1.9 ENVIRONMENTAL REQUIREMENTS

A. Work shall not be installed when the roof deck is damp, wet or spotted with frost or if the ambient temperature is 35 deg. F. and falling or if there is a forecast for inclement weather that may adversely affect the proper installation of the roofing system.

1.10 WARRANTY

A. Provide manufacturer's 30-year Total System Warranty covering both labor and material with no dollar limitation. The maximum wind speed coverage shall be peak gusts of 72 mph measured at 10 meters above ground level. Certification is required with bid submittal indicating the manufacturer has reviewed and agreed to such wind coverage.

B. Pro-rated System Warranties shall not be accepted.

C. Evidence of the manufacturer's warranty reserve shall be included as part of the project submittals for the specifier's approval.

PART 2 PRODUCTS

2.1 MATERIALS

A. Materials: The materials provided shall be part of a roofing system developed by the approved manufacturer and shall in every respect be compatible with each other and with the substrates and conditions encountered in the field.

B. Membrane Sheets: Furnish white EPDM fleece backed 145-mil membrane. The membrane shall conform to the minimum physical properties of ASTM D4637-96, Type III (Fabric-backed membrane). Membrane sheets are 10' wide and 50' or 100' long and incorporate factory-applied splice tape (FAT) along the length of the membrane. Smaller 5' x 40' rolls are also available in the 100-mil and 115-mil thickness.

1. Dynamic Puncture Resistance (ASTM D5635-04a) of 25 joules for 145-mil.

2. Static Puncture Resistance (ASTM D120) of 22 lbf for 145-mil.

C. Membrane Flashing: 0.060" thick uncured EPDM; or as recommended by roofing manufacturer.

D. Bonding Adhesives, Mastics and Splicing Cement: Compatible with the materials with which they will come in contact.

E. Lap Sealant: For sealing the exposed edge of the splices and as otherwise required shall be of a consistency recommended by the manufacturer.

F. Prefabricated Pipe Seal Assemblies: Provide assemblies to accommodate vents, pipe penetrations and other similar roof penetrations. Provide prefabricated split pipe seals, square tubing wraps and curb wrap corners.

G. Sealers: Provide sealers and other similar accessory materials as recommended by the manufacturer.

H. Cant Strips, Tapered Edge Strips, and Flashing Accessories: Types recommended by membrane manufacturer, including adhesive tapes, flashing cements, and sealants.

I. Membrane Adhesive: As recommended by membrane manufacturer for particular substrate and project conditions, formulated to withstand ASCE 7-02 wind uplift force requirements of the geographic area of the building.

1. Provide adhesives that comply with local requirements limiting amounts of volatile organic compounds.

J. Substrate Board: ASTM C 1177, glass-mat, water-resistant gypsum substrate, 5/8" thick; provide "Dens-Deck Prime" by Georgia-Pacific Corporation, or equal.

K. Vapor Retarder: VapAir Seal 725TR

L. Roof Insulation: Minimum 2" thick boards, flat with top layer tapered (1/4" per foot), polyisocyanurate board roof insulation conforming to ASTM C 1289, faced with proper facing to allow membrane to be adhered to it without delamination. Roof insulation must have an LTTR R-Value of 6.0/inch at 75 deg. F. when tested in accordance with ASTM C 1303, and achieve a total value (average) of R=30.

1. Manufacturer of roofing system must approve use of insulation in writing in advance.

M. Walkways: Factory-formed, nonporous, heavy-duty, solid-rubber, slip-resisting, surface-textured walkway pads, approximately 3/16 inch thick, and acceptable to membrane roofing system manufacturer.

1. Weight: 22 lb./sq. ft. unless greater is required to resist wind uplift pressures indicated for the Project.

2. Adhere walkway products to substrate with compatible adhesive according to roofing system manufacturer's written instructions, and to resist wind uplift pressures indicated.

PART 3 EXECUTION

3.1 INSPECTION

A. Examine the areas and conditions where roofing is to be installed and correct any conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions are corrected to permit proper installation of the work.

3.2 INSTALLATION

A. Provide for independent inspection of the roofing at each stage of the installation.

B. Roof Deck Board: Fasten substrate board to top flanges of concrete and cementitious fiber deck according to recommendations in FM Global's "RoofNav" and FM Global Loss Prevention Data Sheet 1-29 for specified Windstorm Resistance Classification.

C. Nailers

1. Continuous pressure treated (See Section 062000) nailers shall be firmly anchored to resist a force of 75 pounds per lineal foot in any direction. The thickness of the nailer shall be such that the top of the nailer is flush with the surface to which the membrane is attached at the horizontal plane.

2. Nailers shall be installed continuous at perimeters and around all roof penetrations unless otherwise noted.

3. Wood Nailer Securement: Wood nailers shall be attached with No. 10 galvanized screws in conjunction with galvanized steel washers (minimum 5/8 inch outside diameter) using a staggered fastening pattern in two rows at 24 inches apart. Within eight feet of outside corners, the staggered fastening pattern shall be increased to a maximum 12 inches on center in each row. When additional wood nailers are required, they must be attached with galvanized screws and washers that penetrate into the bottom wood nailer at 1-1/4 inches using a staggered fastening pattern in two rows at 24 inches on center in addition to the enhanced fastening pattern within eight feet of outside corners. Contractor shall examine existing wood nailers to remain to determine if existing wood nailers are attached in compliance with the above criteria. If not, existing wood nailers shall be refastened in accordance with the fastening requirements noted above.

D. Install substrate board with long joints in continuous straight lines, perpendicular to roof slopes with end joints staggered between rows. Tightly butt substrate boards together.

E. Insulation

1. Clean the deck prior to installation of insulation. Mechanically fasten insulation to deck in pattern to meet wind uplift requirements specified herein, including greater requirements for corners and perimeters as required. For tapered insulation, follow pattern of taper to ensure correct pitch.

2. Moderately butt end joints; stagger joints in adjacent boards. Do not install more insulation in any one day than can be covered by the membrane roof sheets.

3. Where two layers of insulation are required, stagger joints 24" in both directions.

4. Neatly cut around all projections encountered and at abutting vertical surfaces. Where large gaps occur fill with a urethane foam pack.

5. Conform to applicable specified FM class wind uplift resistance test.

F. Sheet Membrane Application (Fully Adhered): Where required by manufacturer, install membrane by unrolling over prepared substrate, lapping adjoining sheets. Apply adhesive to surfaces to be bonded and roll into place when adhesive has properly cured. Treat seams with cleaner and prime finish with 4" seam tape and apply sealant to exposed sheet edges, tapering application as recommended by manufacturer. Install mechanical fasteners, flashings and counterflashings, and accessories at locations indicated and as recommended by manufacturer.

G. Splicing

1. Fold the top sheet back about 12" and clean both mating surfaces at the splice area using clean rags with membrane manufacturer's recommended cleaner.

2. Apply the inseam tape primer with a synthetic scrub pad at a rate of 375 lineal feet of 5" splice per gallon. Allow tape primer to dry to the touch.

3. Form all EPDM seams with 6” seam tape.

4. Roll the top sheet toward the splice area until the cemented area is nearly touching the cement on the bottom sheet along the entire length of the splice. Allow the top sheet to fall freely into place avoiding stretching and wrinkling. Roll the splice with a 2" wide steel roller, using positive pressure, toward the outer edge of the splice.

5. Solvent clean the splice edge, extending at least 1" onto the top and bottom membranes. Apply a bead of lap sealant completely covering the splice edge, feathering the lap sealant with a preformed putty knife or trowel.

6. Lap sealant application shall be completed on all splices by the end of each working day.

H. Membrane Flashing

1. Perimeter flashing and flashing around vents and other roof penetrations shall be preformed using the recommended flashing, compatible with the approved roofing system and utilizing the longest pieces practicable.

2. The splice between the flashing and the main roof sheet should be completed before bonding the flashing to the vertical surface. Seal this splice at least 3" beyond the fasteners which attach the membrane to the horizontal nailer.

3. Bonding adhesive shall be applied to both the flashing and the surface to which it is being bonded. After the adhesive has dried to the point where it does not string or stick to a dry finger touch, roll the flashing into the adhesive. Take care to assure that the flashing is not bridging where there is any change of direction of the flashing (e.g., where the parapet meets the roof deck).

4. Nail the flashing at the top every 12" on center maximum under metal counterflashing or cap. Metal counter is specified under Section 076200.

I. Pipe Flashing

1. Flashing for pipes, conduits and other similar items which are scheduled to penetrate (pass through) the membrane shall be provided with factory prefabricated elements when such use is possible. When prefabricated devices are not possible, field fabricated seals shall be used.

2. Bases of the pipe seals shall be spliced to the membrane roofing sheet as specified above for sheet laps and the top portion shall be secured to the pipe with a stainless steel clamping ring and continuously sealed with sealant in a watertight manner.

3. Field fabricated pipe seals shall be fabricated with base and cap membrane flashing which shall be spliced to the membrane and to itself and continuously sealed with sealant in a watertight manner.

J. Drains: At drain locations, where the insulation is tapered to form a smooth transition from roof surface to membrane, the membrane sheet shall be accurately cut-out so as to fit the encountered clamping ring, and shall be secured to the ring with the addition of the approved mastic in a secure, neat and watertight manner.

K. Curbs, Corners

1. Field fabricated outside corners shall consist of approved membrane flashing which shall have not less than 6" horizontal legs which shall be spliced to the roof membrane, and vertical legs as required which shall be nailed at 12" o.c. maximum. Corners shall be lapped a minimum of 3" and be secured by splicing to each flashing section

2. Field fabricated inside corners shall consist of approved membrane flashing with 6" horizontal legs which shall be spliced to the roof membrane, and vertical legs as required which shall be nailed at 12" o.c. maximum. Corners shall be lapped a minimum 6" and secured by splicing to each flashing section.

3. Install lap type sealant along all seams to insure a watertight installation.

L. Daily Seal: Care should be exercised to ensure that the water does not flow beneath any completed sections of roof. Temporarily seal loose edge of membrane with sealant when weather is threatening.

1. Mix the two components thoroughly according to the instructions on the label.

2. Apply the sealant at a rate of 100 lineal feet per gallon, on smooth surface, 12" back from edge of sheet onto exposed substrate surface. If necessary, use a trowel to spread material in order to achieve complete seal.

3. After embedding membrane in sealant, check for continuous contact. Then weight the edge, providing continuous pressure over the length of the cutoff. The recommended weight for the continuous pressure is a 10 foot length of 2-1/2" tubing filled with dry sand.

4. When work is resumed, pull sheet free before continuing installation.

M. Walkways: Install walkway products in locations indicated. Adhere walkway products to substrate with compatible adhesive according to roofing system manufacturer's written instructions.

3.3 CLEANING AND PROTECTION

A. From time to time during the progress of the work and at the completion of the work, remove all rubbish, debris, dirt, equipment and unused materials from the site. Clean adjoining surfaces which may have been soiled by roofing work.

B. Protect installed roofing from damage and abuse by other trades. Repair damages to watertight conditions at no additional cost to the Owner.

C. Exercise care to protect installed work. Work which does become damaged in any way or is not watertight, shall be repaired and/or replaced as directed to the satisfaction of Architect and/or the Owner at no additional cost or time.

END OF SECTION

SECTION 076200

SHEET METAL WORK

PART 1 GENERAL

1.1 GENERAL REQUIREMENTS

A. Work of this Section, as shown or specified, shall be in accordance with the requirements of the Contract Documents.

1.2 SECTION INCLUDES

A. Work of this Section includes all labor, materials, equipment, and services necessary to complete the sheet metal work as indicated on the drawings and/or specified herein, including, but not limited to, the following:

1. Stainless steel cap and counter flashings.
2. Field fabricating (including bending, cutting, soldering, etc.), if required, of stainless steel flashing.
3. Stainless steel flashing elsewhere, where metal flashing is indicated on drawings.
4. Separation of contacting surfaces of dissimilar metals.

1.3 RELATED SECTIONS

A. Roofing - Division 7.

1.4 SUBMITTALS

A. Shop Drawings: Submit, showing all materials, finishes, fastenings, joint details, fabrication, construction and relation to adjoining construction.

B. Samples: Submit 12" x 12" samples of flashing materials and finishes.

1.5 PRODUCT HANDLING

A. Protection: Use all means necessary to protect the materials of this Section before, during, and after installation, and to protect the installed work and materials of all other trades.

B. Replacements: In the event of damage, immediately make all repairs and replacements necessary at no additional cost to the Owner.

1.6 WARRANTY

A. The Contractor shall warrant that all Metal Flashing Work executed under this Section will be free from defects in materials and workmanship for a period of

ten (10) years from date of acceptance of the Project, and he shall remedy any defects in the Metal Flashing Work.

PART 2 PRODUCTS

2.1 MATERIALS

A. Stainless Steel Flashing Materials

1. Stainless Steel Flashing: ASTM A 240, Type 304, stainless steel, with 2D finish, dead soft temper, fully annealed, as manufactured by International Nickel Co., Republic Steel Corp., United States Steel, or Washington Steel Corp. Thickness of stainless steel shall be as listed below.

   a. Concealed Flashings: 0.012" thick, thirty (30) gauge (U.S. Standard).
   b. Exposed Flashings: 0.015" thick, twenty-eight (28) gauge (U.S. Standard).
   c. Edge Strips: 0.025" thick, twenty-four (24) gauge (U.S. Standard).

2. Accessories and Fastenings: AISI, Types 302 and 304 stainless steel.

3. Solder: Composed of sixty (60) percent block tin and forty (40) percent pig lead, except that solder at seams exposed to public view shall be eighty (80) percent tin and twenty (20) percent lead.

4. Flux: An acid type flux manufactured specifically for soldering stainless steel, as approved.

B. Bituminous Coating: Cold-applied asphalt mastic, SSPC-Paint 12, compounded for 15-mil dry film thickness per coat. Provide inert-type non-corrosive compound free of asbestos fibers, sulfur components, and other deleterious impurities.

PART 3 EXECUTION

3.1 INSPECTION

A. Examine the areas and conditions where sheet metal work is to be installed and correct any conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions are corrected to permit proper installation of the work.

3.2 METAL FLASHING INSTALLATION

A. Reference Standard: Conform to the requirements of 7th Edition of the Sheet Metal and Air Conditioning Contractors Association (SMACNA) Architectural Sheet Metal Manual.

B. General: Fabricate and install metal flashing work in accordance with details and specifications of above Reference Standard, with manufacturer's instructions, and as herein specified, to provide a watertight installation. Apply

metal flashing to smooth, even, sound, clean, dry surfaces free from defects. Make provisions to allow for expansion and contraction of metal flashing work. Wherever practicable, shop form all metal flashing work and deliver ready for installation. Form metal flashing work accurately to required profiles, with flat surfaces, straight edges and corners, free from defects. Fold exposed metal edges back not less than 1/2" and form drip.

C. Nailing: Confine to sheets twelve (12) inches or less in width. Confine nailing to one edge only, locate nails where concealed. Use No. 12 x 1" long flat headed, annular threaded, Type 302 stainless steel nails for nailing to wood blocking; use one (1) inch long masonry nails for nailing to concrete. Space nails four (4) inches o.c. maximum.

D. Cleating: Use cleats where sheets are more than twelve (12) inches in width. Space cleats approximately twelve (12) inches o.c. Cleats two (2) inches wide by three (3) inches long, of the same material and weight as the metal flashing being installed. Secure one end of the cleat with two (2) nails and fold edge back over the nail heads. Lock other end into seam or into folded edge of metal flashing sheets. Pre-tin cleats for soldered seams.

E. Joining: Join metal flashings with one (1) inch locked and soldered seams except at slip joints. Mallet seams flat and solder full length of seam as specified below.

F. Soldering: Clean and pre-tin edges of metal flashing to be soldered before soldering is begun with solder on both sides for a width of not less than 1-1/2". Solder slowly with well heated metal surfaces. Use ample solder. Show not less than one full inch of evenly flowed solder on seam. Seams shall have a liberal amount of flux brushed in before soldering is commenced. Where soldering paste or killed acid is employed as a flux, soldering shall follow immediately after application of the flux. Upon completion of soldering, clean surfaces of all flux.

G. Slip Joints: Locate slip joints not more than twenty-four (24) feet apart and not more than eight (8) feet from corners. Form slip joints as three (3) inch wide joints with cover piece behind flashing and fill locked ends neatly with sealant.

H. Miscellaneous Flashing: Provide all other miscellaneous metal flashing not specifically mentioned herein but indicated on drawings and/or required to provide a watertight installation.

I. Separation of Dissimilar Materials: Back paint surfaces of metal flashing in contact with dissimilar metals or with concrete or masonry with bituminous paint.

END OF SECTION

SECTION 077100

ROOF SPECIALTIES AND ACCESSORIES

PART 1 GENERAL

1.1 GENERAL REQUIREMENTS

A. Work of this Section, as shown or specified, shall be in accordance with the requirements of the Contract Documents.

1.2 SECTION INCLUDES

A. The Work of this Section includes all labor, materials, equipment and services necessary to complete the roof specialties and accessories as shown on the drawings and/or specified herein, including but not necessarily limited to the following:

1. Aluminum gravel stops and copings.

2. Roof hatches.

3. Roof smoke vents.

1.3 SUBMITTALS

A. Before any roof specialties and accessories are delivered to the job site, submit shop drawings showing profiles and anchoring devices.

B. Product Certificates: For each type of roof specialty copings and roof-edge flashings that is ANSI/SPRI/FM 4435/ES-1 tested and FM Approvals approved for use based on porject requirements.

C. Product Test Reports: For copings and roof-edge flashings, for tests performed by a qualified testing agency.

D. Research Reports: For copings and roof-edge flashings, from an agency acceptable to authorities having jurisdiction showing compliance with ANSI/SPRI/FM 4435/ES-1.

E. Qualification Statements: For manufacturer.

F. Sample warranties.

1.4 PERFORMANCE REQUIREMENTS

A. General Performance: Roof specialties to withstand exposure to weather and resist thermally induced movement without failure, rattling, leaking, or fastener disengagement due to defective manufacture, fabrication, installation, or other defects in construction.

B. FM Approvals' Listing: Manufacture and install copings, roof-edge specialties that are listed in FM Approvals' "Approval Guide" and approved for windstorm classification, Class as required per roofing specificaton section. Identify materials with FM Approvals' markings.

C. SPRI Wind Design Standard: Manufacture and install copings, roof-edge specialties tested in accordance with ANSI/SPRI/FM 4435/ES-1 and capable of resisting the following design pressures:

1. Design Pressure: As indicated on Drawings, unless ASCE-7 based on location and height is greater.

D. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes to prevent buckling, opening of joints, hole elongation, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Provide clips that resist rotation and avoid shear stress as a result of thermal movements. Base calculations on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

1. Temperature Change: 120 deg F (67 deg C), ambient; 180 deg F (100 deg C), material surfaces.

1.5 PRODUCT HANDLING

A. Protection: Use all means necessary to protect the materials of this Section before, during and after installation and to protect the installed work and materials of all other trades.

B. Replacements: In the event of damage, immediately make all repairs and replacements necessary.

PART 2 PRODUCTS

2.1 ALUMINUM GRAVEL STOPS AND COPINGS

A. Fabricate of .063" thick aluminum alloy 5005-H154, smooth, no pattern.

B. Provide concealed splice plates 12'-0" o.c. fabricated of .050" thick aluminum to match exposed aluminum; finished to match exposed aluminum.

C. Provide pre-fabricated mitered and welded corner units.

D. For gravel stops, provide concealed anchors and hold down clips 24" o.c.

E. For copings, provide galvanized steel anchor plates, anchors spaced 6'-0" o.c. and snap-lock coping design; all anchors concealed.

F. High-Performance Organic Finish: AA-C12C42R1x (Chemical Finish: Cleaned with inhibited chemicals; Chemical Finish: Acid-chromate-fluoride-phosphate conversion coating; Organic Coating: As specified below). Prepare, pretreat,

and apply coating to exposed metal surfaces to comply with coating and resin manufacturer's written instructions.

1. Fluoropolymer Two-Coat System: Manufacturer's standard two-coat, thermo-cured system consisting of specially formulated inhibitive primer and fluoropolymer color topcoat containing not less than 70 percent polyvinylidene fluoride resin by weight; complying with AAMA 2605-02.

2. Custom color and gloss as selected by the Architect.

G. Provide units manufactured by OMG Roofing, or equal made by Cheney, Johns Manville or approved equal.

2.2 ROOF HATCH

A. Provide thermally broken shop-primed, galvanized steel roof hatch units of sizes shown on drawings, with 1" rigid insulation at curbs and door and standard self-lifting mechanism. Provide manufacturer's standard hardware, including hold-open device, hinges, latch and operating handles for inside operation. Construct units for 40 lbs. per sq. ft. live load.

B. Safety Railing System: Manufacturer's standard complete system including rails, clamps, fasteners, safety barrier at railing opening, and all accessories required for a complete installation.

1. Height: 42 inches above finished roof deck.

2. Test load per code requirements.

3. Provide self-latching gate fabricated of same materials as safety railing system.

C. Provide units manufactured by Bilco, Babcock-Davis, Milcor or approved equal.

2.3 SMOKE VENTS

A. Provide shop primed galvanized steel heat and smoke vent units of sizes shown on drawings, with 1" rigid insulation at curbs and door, standard lifting mechanism and automatic heat and smoke sensitive release devices. Provide manufacturer's standard hardware including hold-open device, hinges, latch and operating handles for inside and outside operation.

B. Provide units manufactured by Bilco, Babcock-Davis, Milcor or approved equal.

PART 3 EXECUTION

3.1 INSPECTION

A. Examine the areas and conditions where roof specialties and accessories are to be installed and correct any conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions are corrected to permit proper installation of the work.

3.2 INSTALLATION

A. General: Comply with manufacturer's instructions and recommendations. Coordinate with installation of roof deck and other substrates to receive accessory units, and with roof insulation, roofing and flashing; as required to ensure that each element of the work performs properly, and that combined elements are waterproof and weathertight. Anchor units securely to supporting structural substrates, adequate to withstand lateral and thermal stresses as well as inward and outward loading pressures.

B. Isolation: Where metal surfaces of units are to be installed in contact with non-compatible metal or corrosive substrates, including wood, apply bituminous coating on concealed metal surfaces, or provide other permanent separation.

C. Cap Flashing: Where cap flashing is required as component of accessory, install to provide adequate waterproof overlap with roofing or roof flashing (as counter flashing). Seal with thick bead of mastic sealant, except where overlap is indicated to be left open for ventilation.

D. Operational Units: Test operational units with operable components. Clean and lubricate joints and hardware. Adjust for proper operation.

3.3 CLEANING AND PROTECTION

A. Clean exposed metal surfaces in accordance with manufacturer's instructions. Touch up damaged metal coatings.

END OF SECTION

SECTION 07 72 60 – Rooftop Fall Protection and Accessories

PART 1 - GENERAL

1.1 SECTION INCLUDES

A. Free-standing rooftop fall protection systems. (KEE GUARD)

B. Free-standing rooftop fall protection systems. (KEE GUARD CONTRACTOR)

1.2 RELATED SECTIONS

A. Section 05500 - Metal Fabrications: Associated metal supports.

B. Section 07400 - Membrane Roofing: Coordination of roof edge protection installation.

1.3 REFERENCES

A. Americans with Disabilities Act Accessibility Guidelines (ADA).

B. American Society of Civil Engineer (ASCE):
1. ASCE 7-16 - Minimum Loads and Associated Criteria for Buildings and Other Structures.

C. ASTM International (ASTM):
1. ASTM A47 - Standard Specification for Ferritic Malleable Iron Castings.
2. ASTM A53 - Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless.
3. ASTM A123 - Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
4. ASTM A500 - Standard Specification for cold-formed welded and seamless carbon steel structural tubing.
5. ASTM B221 - Standard Specification for Aluminum and Aluminum Alloy extruded bars, rods, wires, profiles, and tubes.

D. Occupational Safety and Health Administration (OSHA):
1. OSHA 1910.29 - Fall Protection systems and falling object protection.
2. OSHA 1926.502 - Fall Protection systems criteria and practices.

E. Underwriters Laboratories (UL): UL 94 - Tests for Flammability of Plastic Materials for Parts in Devices and Appliances.

1.4 SUBMITTALS

A. Submit under provisions of Section 01300.

B. Product Data: Manufacturer's data sheets on each product to be used, including:
1. Preparation instructions and recommendations.
2. Storage and handling requirements and recommendations.
3. Installation methods.

1.5 QUALITY ASSURANCE

A. Railings Structural Requirements:
   1. Handrail, wall rail and guardrail assemblies and attachments shall withstand a minimum concentrated load of 200 lbs (90,719 g) applied horizontally or vertically down at any point on the top rail.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Deliver, Store and handle materials and products in strict compliance with manufacturer's instructions and recommendations and industry standards. Store materials within absolute limits for temperature and humidity recommended by the manufacturer.
   1. Materials to be delivered to the job site in good condition and adequately protected against damage as handrails are a finished product.

PART 2 – PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturer: Kee Safety, Inc., which is located at: 100 Stradtman St.; Buffalo, NY 14206; Toll Free Tel: 800-851-5181; Tel: 716-896-4949; Fax: 716-896-5696; Email: info@keesafety.com; Web: http://keesafety.com

B. Substitutions: Not permitted.

C. Requests for substitutions will be considered in accordance with the provisions of Section 01600.

2.2 SAFETY RAILINGS PERFORMANCEAND DESIGN REQUIREMENTS

A. Performance and Design Requirements: Pipe and tube railing and guardrail design.

   1. Comply with the following:

      a. OSHA Standard Pipe Railing: 1910.29 Fall Protection systems and falling object protection.

   2. Structural Performance: Railings and Attachments: Withstand effects of gravity loads and the following loads as specified.

      a. Recommended Maximum Post Spacing: 8 feet

      b. Minimum Height: 42 inches (1067 mm).

      c. Intermediate Rail Height: 21 inches (533 mm).

      d. Handrails and Top Rail of Railing Systems:

         1) Uniform Load: 50 lbf per ft. (0.73 kN per m) in any direction.

         2) Concentrated Load: 200 lbf (0.89 kN) in any direction.

         3) Uniform and concentrated loads need not be assumed to act concurrently.

   3. Allow expansion and contraction due to thermal movements caused by temperature

changes.

2.3 FREE-STANDING ROOFTOP FALL PROTECTION SYSTEMS (KEE GUARD)

A. Free-Standing Rooftop Edge Fall Protection Systems: Kee Guard as manufactured by Kee Safety Inc. Does not penetrate the roofing system. Systems do not require physical fixing into the roof's structure/membrane. The complete system's design, manufacture, testing and installation has been externally assessed.

1. Permanent, modular, free-standing, roof edge fall protection railing system for areas where regular access for maintenance and inspection is required.

2. Components include but are not limited to pipe railings, uprights, bases, counterweights, fittings, and accessories as indicated or required to match design indicated on Drawings and to provide complete installation.

3. Compliance:

a. Testing and Certification:

1) OSHA Standard 29 CFR 1910.29 (B) (1); (B) (3).

a) 42 inch (1067 mm) minimum height providing a pedestrian egress barrier on the roof , withstanding a load of 200 lb. (90719 g).

2) OSHA Standard 29 CFR 1926.501 (b) (1); (b) (2) (ii).

3) OSHA Standard 29 CFR 1926.502 (B) (1) - (B) (14).

4) Canadian National Building Code 4.1.5.14

5) Ontario Building Code Section 4.1.5.14

6) California Building Code 1710A.3.1; 24 hour 400 lbs (181 kg) point load.

7) California Code of Regulations Title 8 Sections 3209 and 3294.

8) NF E85-003.

9) EN ISO 14122: PT3.

10) EN 13374 Class A.

11) EN 1991-1-4.

12) BS 13700 Permanent counterweighted guardrail systems - Specification.

13) BS 6399; Part 2 Code of Practice for Wind Code.

b. Work at Height Regulations:

1) HSG-33 Health and Safety in Roof Work.

2) HSE Sheet 21 "Working on Flat Roofs Protection Against Falls."

3) EU Directives and CDM Regulations.

c. Wind Loading:

1) Installations are to be wind speed calculated to ASCE 7 - The American Society of Civil Engineers design standard: “Minimum Design Loads for Buildings and Other Structures.”

d. BGV A1: 2000, BGG 928 & BGR 184.

e. Health, Safety and Welfare Regulation 13 "Falls or Falling Objects."

f. TUV Tested.

4. Materials:

a. Tubing: Gatorshield.

1) Tubing Cut On Site: Zinc rich paint is applied to the cut end of the tube.

b. Base Metal: ASTM A500 Carbon Steel.

c. Zinc Spelter - ASTM B6 High Grade Special High Grade Zinc.

d. Chemical Composition per ASTM A-500.

e. Material Testing – ASTM 500 and ASTM E8.

f. Top Guardrails and Intermediate Rails: Steel. 1.9 inch (48.3 mm) external diameter.

1) Wall Thickness: 0.109 inches (2.77 mm).

2) Supplied Length: 10 ft 6 inches (3.2 m).

g. Vertical Support Legs: Steel. 1.9 inch (48.3 mm) external diameter.

1) Wall Thickness: 0.109 inches (2.77 mm).

2) Leg Type: Vertical.

h. Cantilever Tubes: Steel. 1.66 inch (42.2 mm) external diameter

1) Wall Thickness: 0.109 inches (02.77 mm).

2) Supplied Length: 42.32 inches (1075 mm).

i. Cast Clamps Used to Join the Guardrail: Galvanized malleable cast iron produced to ASTM A47 and galvanized to ASTM A123.

1) Tapped Holes: Protected with Threadkoat; applied to all tapped holes.

2) Grub Screws: Carbon steel and have KEE KOAT protection applied to ensure minimal maintenance.

3) Fittings:

a) Base Foot.

b) 90 degree Elbow.

c) Saddle Clamp.

d) Straight Coupling.

e) Wall / Ladder Clamp.

1) Anchor Bolts. Stainless steel.

f) Collar.

4) Finish: Galvanized.

j. Counter Weights: Recycled PVC. Weight: 29 lbs 5 oz (13.3 kg).

k. Access Gates: Kee Gate Standard Self-Closing Safety Gates. OSHA complaint. Gate Width Range: 16.5 to 50 inches (419 to 1270 mm). Spring loaded to close automatically. Supplied with U-bolts and fastener packs for secure, quick assembly to existing structures and posts.

1) Safety tested to 50,000 actuations.

2) Gate Height: 19.69 inch (500 mm).

a) Material: Galvanized steel.

l. Metal components in contact with the roof membrane are covered with 0.118 inch (3 mm) rubber.

m. Fasteners: Galvanized.

5. Layout:

a. Guardrail Height: 42 inches (1067 mm)/

b. Vertical Support Spacing. A maximum 8 ft (2.438 m) Center-to-center depending on the system.

c. Recycled PVC Counter Weights: Attached to each vertical leg set at no more than 8 ft (2.438 m) center-to-center.

d. Stop Ends: Are appropriately counter weighted according to roof pitch and membrane or supported by way of a wall/ladder clamp.

6. System Design: Designed for applications with flat or low slope roofs up to 5 degrees.

a. Roof Type: TPO - Single Ply Membrane and EPDM - Single Ply Membrane.

1) Counterbalances: With one fixing collar per counterbalance.

a) For slopes up to 5 degrees:

1) End Counter Balance: CB7 - Seven Weight Counterbalance.

2) End Bay Length: 8 ft (2.438 m).

3) Second Vertical Leg Counter Balance: CB1 - One Weight Counterbalance.

4) Subsequent Counter Balances: CB1 - One Weight Counterbalance.

5) Subsequent Bay Length: 8 ft (2.438 m).

2.4 FREE-STANDING ROOFTOP FALL PROTECTION SYSTEMS (KEE GUARD CONTRACTOR)

A. Free-Standing Rooftop Fall Protection Systems: KeeGuard Contractor as manufactured by Kee Safety Inc.

1. Description: Portable, modular, free-standing, roof edge railing system that does not penetrate the roofing system.

2. Provide components including but not limited to pipe railings, uprights, bases, weights, fittings, and accessories as indicated or required to match design indicated on Drawings and to provide complete installation.

3. Compliance:

a. OSHA Standard 29 CFR 1910.29 (B) (1); (B) (3).

1) 42 inch (1067 mm) minimum height to provide a pedestrian egress barrier on the roof to withstand a minimum load of 200 lbs (90719 g) in any direction to the top rail.

b. OSHA Standard 29 CFR 1926.501 (b) (1); (b) (2) (ii).

c. OSHA Standard 29 CFR 1926.502 (B) (1) – (B) (14).

d. Canadian National Building Code 4.1.5.14

e. Ontario Building Code Section 4.1.5.14

f. Wind Loading: All installations are wind speed calculated to ASCE 7 - The

American Society of Civil Engineers design standard: “Minimum Design Loads for Buildings and Other Structures.”

4. System Design: Designed for applications with flat or low slope roofs up to 5 degrees.
    a. Bases: Produced in two halves.
        1) Materials: 100 percent recycled PVC.
        2) Weights: As indicated on Drawings.
5. System Configuration: As indicated on Drawings.
6. Height of Guardrail: 42 inches (1067 mm).
7. Vertical Supports: Set at maximum 10 ft centers.
8. Each vertical support is inserted into the PVC Base. The leg is secured in place by tightening the collar inserted within the mold. All stop ends have returns or triple counter weighted, using standard KeeGuard components or supported by way of a wall/ladder clamp.
9. Gatorshield Tubing:
10. Base Metal: ASTM A500 Carbon Steel.
11. Zinc Spelter - ASTM B6 High Grade Special High Grade Zinc.
12. Chemical Composition - ASTM A500.
13. Material Testing - ASTM-A500 and ASTM E8.
14. Guardrail top and intermediate rails: 1.9 inch (48.3 mm) external diameter.
    a. Wall Thickness: 0.109 inches (2.77 mm).
15. Vertical Support Legs: 1.9 inch (48.3 mm) external diameter.
    a. Wall Thickness: 0.109 inches (2.77 mm).
16. Bases: Recycled PVC. Produced in two halves.
17. Cast Clamps: Galvanized malleable cast iron produced to ASTM A47.
18. Threadkoat applied to all tapped holes.
19. Grub screws are carbon steel and have Keekoat protection applied to ensure minimal maintenance.
20. Safety Gates: Spring Loaded, self-closing safety gate for North America. Each model can be adjusted on site without need for welding. Complete with fixing pack.

a. Material : Galvanized steel to ASTM A53.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install in accordance with manufacturer's instructions including the following:

1. Fit exposed connections accurately together to form tight joints. For all connections with Kee Klamp fittings, each set screw is to be tightened to 29 ft-lbs (39.3 N-m) of torque.

2. Perform cutting, drilling, and fitting required for installation of handrails. Set handrails and accurately in location, alignment, and elevation, measured from established lines and levels.

3. Set posts plumb within a tolerance of 1/8 inch (3 mm).

3.2 PROTECTION

A. Protect installed products until completion of project.

B. Touch-up, repair or replace damaged products before Substantial Completion.

END OF SECTION 07 72 60

SECTION 079200

JOINT SEALERS

PART 1 GENERAL

1.1 GENERAL REQUIREMENTS

A. Work of this Section, as shown or specified, shall be in accordance with the requirements of the Contract Documents.

1.2 SECTION INCLUDES

A. Work of this Section includes all labor, materials, equipment and services necessary to complete the joint sealers work as shown on the drawings and/or specified herein, including but not necessarily limited to the following:

1. Flashing reglets and retainers.
2. Coping joints.
3. Exterior wall joints not specified to be sealed in other Sections of work.
4. Interior wall joints not specified to be sealed in other Sections of work, including caulking to fill between architectural woodwork and any wall, floor and/or ceiling imperfections.
5. Control and expansion joints in walls.
6. Joints at wall penetrations.
7. Joints between items of equipment and other construction.
8. All other joints required to be sealed to provide a positive barrier against penetration of air and moisture.

1.3 QUALITY ASSURANCE

A. Qualification of Installers: Use only personnel who are thoroughly familiar, skilled and specially trained in the techniques of sealant work, and who are completely familiar with the published recommendations of the sealant manufacturer.

B. Pre-Construction Field Adhesion Testing: Before installing elastomeric sealants, field test their adhesion to project joint substrates according to the method in ASTM C 794 and C 1521 that is appropriate for the types of Project joints.

C. Perform testing per ASTM C 1248 on interior and exterior sealants to determine if sealants or primers will stain adjacent surfaces. No sealant work shall start

until results of these tests have been submitted and the Architect has given written approval to proceed with the work.

1.4 SUBMITTALS

A. Shop Drawings: Submit shop drawings showing all joint conditions, indicating relation of adjacent materials, all sealant materials (sealant, bond breakers, backing, primers, etc.), and method of installation.

1. Submit joint sizing calculations certifying that movement capability of sealant is not being exceeded.

B. Samples: Submit the following:

1. Color samples of sealants, submit physical samples (not color chart).

2. Sealant bond breaker and joint backing.

C. Product Data: Submit manufacturer's technical information and installation instructions for:

1. Sealant materials, indicating that material meets standards specified herein.

2. Backing rods.

D. Submit manufacturer's certification as required by Article 1.6 herein.

E. Submit results of testing required in Article 1.4 herein.

1.5 MANUFACTURER'S RESPONSIBILITY AND CERTIFICATION

A. Contractor shall require sealant manufacturer to review the Project joint conditions and details for this Section of the work. Contractor shall submit to the Architect written certification from the sealant manufacturer that joints are of the proper size and design, that the materials supplied are compatible with adjacent materials and backing, that the materials will properly perform to provide permanent watertight, airtight or vaportight seals (as applicable), and that materials supplied meet specified performance requirements.

1.6 ENVIRONMENTAL CONDITIONS

A. Temperature: Install all work of this Section when air temperature is above forty (40) degrees F. and below eighty (80) degrees F., unless manufacturer submits written instructions permitting sealant use outside of this temperature range.

B. Moisture: Do not apply work of this Section on surfaces which are wet, damp, or have frost.

1.7 PRODUCT HANDLING

A. Protection: Use all means necessary to protect the materials of this Section, before, during and after installation and to protect the installed work and materials of all other trades.

B. Replacements: In the event of damage, immediately make all repairs and replacements necessary.

C. Storage

1. Store sealant materials and equipment under conditions recommended by their manufacturer.

2. Do not use materials stored for a period of time exceeding the maximum recommended shelf life of the material.

3. Material shall be stored in unopened containers with manufacturers' name, batch number and date when shelf life expires.

1.8 GUARANTEE

A. Provide a written, notarized guarantee from the manufacturer stating that the applied sealants shall show no material failure for a period of twenty (20) years.

B. Contractor to provide a written, notarized, guarantee stating that the applied sealants shall show no failure due to improper installation for a period of five (5) years.

C. Guarantee shall be in a form acceptable to the Owner and executed by an authorized individual.

D. Include in guarantee provision, agreement to repair and/or replace, at Contractor's expense, sealant defects which develop during guarantee period, because of faulty labor and/or materials.

PART 2 PRODUCTS

2.1 SEALANT MATERIALS

A. Exterior Wall Sealant: Provide one (1) part non-sag sealant equal to "DOWSIL 790" or "DOWSIL 795" made by The Dow Chemical Company, "Silpruf SCS 2000" or "LM SCS 2700" made by G.E., "Spectrem 1" or "Spectrem 3" made by Tremco conforming to the minimum standards of ASTM C 920, Type S, Grade NS, Class 50.

B. Interior Sealant: Provide a one (1) part acrylic based sealant conforming to ASTM C 834, equal to "AC-20+ Silicone" made by Pecora, Masterseal NP 520 by BASF or equal made by Tremco.

C. Colors: Colors selected from manufacturer's standard selection.

2.2 MISCELLANEOUS MATERIALS

A. Back-Up Materials: Provide back-up materials and preformed joint fillers, non-staining, non-absorbent, compatible with sealant and primer, and of a resilient nature, equal to "HBR" made by Nomaco Inc. or approved equal, twenty-five (25) percent wider than joint width. Materials impregnated with oil, bitumen or similar materials shall not be used. Provide back-up materials only as recommended by sealant manufacturer in writing.

B. Provide bond breakers, where required, of polyethylene tape as recommended by manufacturer of sealant.

C. Provide primers recommended by the sealant manufacturer for each material to receive sealant. Note that each exterior joint must be primed prior to sealing.

D. Provide solvent, cleaning agents and other accessory materials as recommended by the sealant manufacturer.

E. Materials shall be delivered to the job in sealed containers with manufacturer's original labels attached. Materials shall be used per manufacturer's printed instructions.

PART 3 EXECUTION

3.1 INSPECTION

A. Examine the areas and conditions where joint sealers are to be installed and correct any conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions are corrected to permit proper installation of the work.

3.2 INSTALLATION

A. Sealant Installation Standard: Comply with instructions and recommendations of the manufacturer and in accordance with ASTM C 1193 for use of joint sealants as applicable to materials, applications and conditions required by this Project where more stringent installation requirements are specified herein, such requirements shall apply.

B. Sample Section of Sealant

1. During sealant installation work in exterior wall, the manufacturer of sealant shall send their representative to the site, under whose supervision a section of the wall (used as "control section") shall be completed for purposes of determining performance characteristics of sealant in joints. Architect shall be informed of time and place of such installation of control section.

2. Control section shall be installed according to specification given herein and shall not be considered as acceptable until written acceptance is provided by the Architect.

3. Accepted control section shall be standard to which all other sealant work must conform.

C. Supervision: The Contractor shall submit to the Architect written certification from the sealant manufacturer that the applicators have been instructed in the proper application of their materials. The Contractor shall use only skilled and experienced workmen for installation of sealant.

D. Apply sealant under pressure with a hand or power actuated gun or other appropriate means. Gun shall have nozzle of proper size and provide sufficient pressure to completely fill joints as detailed. Neatly point or tool joint to provide the contour as indicated on the drawings.

E. Preparation and Application

1. Thoroughly clean all joints, removing all foreign matter such as dust, oil, grease, water, surface dirt and frost. Sealant must be applied to the base surface. Previously applied film must be entirely removed.

2. Stone, masonry and concrete surfaces to receive sealant shall be cleaned where necessary by grinding, water blast cleaning, mechanical abrading, or combination of these methods as required to provide a clean, sound base surface for sealant adhesion.

   a. Do not use any acid or other material which might stain surfaces.
   b. Remove laitance by grinding or mechanical abrading.
   c. Remove loose particles present or resulting from grinding, abrading, or blast cleaning by blowing out joints with compressed air, oil and water free, or vacuuming joints prior to application of primer or sealant.

3. Clean non-porous surfaces such as metal and glass chemically. Remove protective coatings on metallic surfaces by solvent that leaves no residue and is compatible with sealant. Use solvent and wipe dry with clean, dry lint free paper towels. Do not allow solvent to air dry without wiping. Clean joint areas protected with masking tape or strippable films as above after removal of tape film.

4. Do not seal joints until they are in compliance with drawings, or meet with the control section standard.

5. Joint Size and Sealant Size: Joints to receive sealant shall be at least 1/4" wide. In joint 1/4" to 3/8" wide, sealant shall be 1/4" deep. In joints wider than 3/8" and up to 1" wide, sealant depth shall be one half the joint width. For joints wider than 1", sealant depth shall be as recommended by the sealant manufacturer. Depth of joint is defined as distance from outside face of joint to closest point of the filler.

6. Primer: Thoroughly clean joints and apply primer to all surfaces that will receive sealant. Apply primer on clean, dry surfaces, and prior to installation of joint backing. Completely wet both inner faces of the joint

with primer. Mask adjacent surfaces of joint with non-staining masking tape prior to priming. Apply primer with clean brush and only when temperature is above 45 deg. F.

7. Joint Backing: In joints where depth of joint exceeds required depth of sealant, install joint backing (after primer is dry) in joints to provide backing and proper joint shape for sealant. Proper shape for sealant is a very slight "hourglass" shape, with back and front face having slight concave curvature. Use special blunt T-shaped tool or roller to install joint backing to the proper and uniform depth required for the sealant. Joint backing shall be installed with approximately twenty-five (25) percent compressions. Do not stretch, twist, braid, puncture, or tear joint backing. Butt joint backing at intersections.

8. Bond Breaker: Install bond breaker smoothly over joint backing so that sealant adheres only to the sides of the joint and not backing.

9. Sealant Application: Apply sealant in accordance with the manufacturer's application manual and manufacturer's instructions, using hand guns or pressure equipment, on clean, dry, properly prepared substrates, completely filling joints to eliminate air pockets and voids. Mask adjacent surfaces of joint with non-staining masking tape. Force sealant into joint in front of the tip of the "caulking gun" (not pulled after it) and force sealant against sides to make uniform contact with sides of joint and to prevent entrapped air or pulling of sealant off of sides. Fill sealant space solid with sealant.

10. Tooling: Tool exposed joints to form smooth and uniform beds, with slightly concave surface conforming to joint configuration per Figure 5A in ASTM C 1193. Finished joints shall be straight, uniform, smooth and neatly finished. Remove masking tape immediately after tooling of sealant and before sealant face starts to "skin" over. Neatly remove any excess sealant from adjacent surfaces of joint, leaving the work in a neat, clean condition.

11. Replace sealant which is damaged during construction process.

3.3 FIELD QUALITY CONTROL

A. Field-Adhesion Testing: Field test joint-sealant adhesion to joint substrates as follows:

1. Extent of Testing: Test completed and cured sealant joints as follows:

   a. Perform 5 tests for the first 1000 feet of joint length for each kind of sealant and joint substrate.
   b. Perform one test for each 5,000 feet of joint length thereafter or one test per each floor per elevation.

2. Test Method: Test joint sealants according to Method A, Field-Applied Sealant Joint Hand Pull Tab, in Appendix X1 in ASTM C1193 or Method A, Tail Procedure, in ASTM C1521.

a. For joints with dissimilar substrates, verify adhesion to each substrate separately; extend cut along one side, verifying adhesion to opposite side. Repeat procedure for opposite side.

3. Inspect tested joints and report on the following:

   a. Whether sealants filled joint cavities and are free of voids.
   b. Whether sealant dimensions and configurations comply with specified requirements.
   c. Whether sealants in joints connected to pulled-out portion failed to adhere to joint substrates or tore cohesively. Include data on pull distance used to test each kind of product and joint substrate. Compare these results to determine if adhesion complies with sealant manufacturer's field-adhesion hand-pull test criteria.

4. Record test results in a field-adhesion-test log. Include dates when sealants were installed, names of persons who installed sealants, test dates, test locations, whether joints were primed, adhesion results and percent elongations, sealant material, sealant configuration, and sealant dimensions.

5. Repair sealants pulled from test area by applying new sealants following same procedures used originally to seal joints. Ensure that original sealant surfaces are clean and that new sealant contacts original sealant.

B. Evaluation of Field-Adhesion-Test Results: Sealants not evidencing adhesive failure from testing or noncompliance with other indicated requirements will be considered satisfactory. Remove sealants that fail to adhere to joint substrates during testing or to comply with other requirements. Retest failed applications until test results prove sealants comply with indicated requirements.

END OF SECTION

SECTION 080671 – DOOR HARDWARE SCHEDULE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section references specification sections relating to commercial door hardware for the following:

1. Swinging doors.
2. Sliding Doors.
3. Other doors to the extent indicated.

B. Commercial door hardware includes, but is not necessarily limited to, the following:

1. Mechanical door hardware.
2. Electromechanical and access control door hardware.
3. Electromechanical and access control door hardware power supplies, back-ups and surge protection.
4. Automatic operators.
5. Cylinders specified for doors in other sections.

C. Related Sections:

1. Division 08 Section “Door Hardware”.

D. Codes and References: Comply with the version year adopted by the Authority Having Jurisdiction.

1. ANSI A117.1 - Accessible and Usable Buildings and Facilities.
2. ICC/IBC - International Building Code.
3. NFPA 70 - National Electrical Code.
4. NFPA 80 - Fire Doors and Windows.
5. NFPA 101 - Life Safety Code.
6. NFPA 105 - Installation of Smoke Door Assemblies.
7. State Building Codes, Local Amendments.

E. Standards: Reference Related Sections for requirements regarding compliance with applicable industry standards.

1.3 SUBMITTALS

A. Product Data: Manufacturer's product data sheets including installation details, material descriptions, dimensions of individual components and profiles, operational descriptions and finishes.

B. Door Hardware Schedule: Prepared by or under the supervision of supplier, detailing fabrication and assembly of door hardware, as well as procedures and diagrams. Coordinate the final Door Hardware Schedule with doors, frames, and related work to ensure proper size, thickness, hand, function, and finish of door hardware.

1. Format: Comply with scheduling sequence and vertical format in DHI's "Sequence and Format for the Hardware Schedule."

2. Organization: Organize the Door Hardware Schedule into door hardware sets indicating complete designations of every item required for each door or opening. Organize door hardware sets in same order as in the Door Hardware Sets at the end of Part 3. Submittals that do not follow the same format and order as the Door Hardware Sets will be rejected and subject to resubmission.

3. Content: Include the following information:

a. Type, style, function, size, label, hand, and finish of each door hardware item.
b. Manufacturer of each item.
c. Fastenings and other pertinent information.
d. Location of door hardware set, cross-referenced to Drawings, both on floor plans and in door and frame schedule.
e. Explanation of abbreviations, symbols, and codes contained in schedule.
f. Mounting locations for door hardware.
g. Door and frame sizes and materials.

4. Submittal Sequence: Submit the final Door Hardware Schedule at earliest possible date, particularly where approval of the Door Hardware Schedule must precede fabrication of other work that is critical in the Project construction schedule. Include Product Data, Samples, Shop Drawings of other work affected by door hardware, and other information essential to the coordinated review of the Door Hardware Schedule.

C. Keying Schedule: Prepared under the supervision of the Owner, separate schedule detailing final keying instructions for locksets and cylinders in writing. Include keying system explanation, door numbers, key set symbols, hardware set numbers and special instructions. Owner to approve submitted keying schedule prior to the ordering of permanent cylinders.

D. Product Test Reports: Indicating compliance with cycle testing requirements, based on evaluation of comprehensive tests performed by manufacturer and witnessed by a qualified independent testing agency.

E. Operating and Maintenance Manuals: Provide manufacturers operating and maintenance manuals for each item comprising the complete door hardware installation in quantity as required in Division 01, Closeout Submittals. The manual to include the name, address, and contact information of the manufacturers providing the hardware and their nearest service

representatives. The final copies delivered after completion of the installation test to include "as built" modifications made during installation, checkout, and acceptance.

F. Warranties and Maintenance: Special warranties and maintenance agreements specified in the Related Sections.

1.4 DELIVERY, STORAGE, AND HANDLING

A. Inventory door hardware on receipt and provide secure lock-up and shelving for door hardware delivered to Project site. Do not store electronic access control hardware, software or accessories at Project site without prior authorization.

B. Tag each item or package separately with identification related to the final Door Hardware Schedule, and include basic installation instructions with each item or package.

C. Deliver, as applicable, permanent keys, cylinders, cores, access control credentials, software and related accessories directly to Owner via registered mail or overnight package service. Instructions for delivery to the Owner shall be established at the "Keying Conference".

1.5 WARRANTY

A. General Warranty: Reference Division 01, General Requirements. Special warranties specified in this Article shall not deprive Owner of other rights Owner may have under other provisions of the Contract Documents and shall be in addition to, and run concurrent with, other warranties made by Contractor under requirements of the Contract Documents.

1.6 MAINTENANCE SERVICE

A. Maintenance Tools and Instructions: Furnish a complete set of specialized tools and maintenance instructions as needed for Owner's continued adjustment, maintenance, and removal and replacement of door hardware.

PART 2 - PRODUCTS

2.1 SCHEDULED DOOR HARDWARE

A. Refer to “PART 3 – EXECUTION” for required specification sections.

PART 3 - EXECUTION

3.1 DOOR HARDWARE SETS

A. The door hardware sets represent the design intent and direction of the owner and architect. They are a guideline only and should not be considered a detailed hardware schedule. Discrepancies, conflicting hardware and missing items should be brought to the attention of the architect with corrections made prior to the bidding process. Omitted items not included in a

hardware set should be scheduled with the appropriate additional hardware required for proper application and functionality.

1. Quantities listed are for each pair of doors, or for each single door.

2. The supplier is responsible for handing and sizing all products.

3. Where multiple options for a piece of hardware are given in a single line item, the supplier shall provide the appropriate application for the opening.

4. At existing openings with new hardware the supplier shall field inspect existing conditions prior to the submittal stage to verify the specified hardware will work as required. Provide alternate solutions and proposals as needed.

B. Products listed in the hardware sets shall be supplied by and in accordance with the requirements described in the specification section as noted for each item.

1. Section 08 71 00 – Door Hardware.

C. Manufacturer's Abbreviations:

1. MK - McKinney
2. RO - Rockwood
3. SA - SARGENT
4. RF - Rixson
5. NO - Norton
6. PE - Pemko

Hardware Sets

Set: 1.0

Doors: C40-4A

2	Hinge, Full Mortise, Hvy Wt	T4A3386 NRP 4-1/2" x 4-1/2"	US32D	MK	087100
1	Store Door Lock	8226 LNNJ	US32D	SA	087100
2	Keyed Core	By Owner	US26D		087100
1	Conc Overhead Stop	1-X36	630	RF	087100
1	Kick Plate	K1050 10" H. x CSK BEV	US32D	RO	087100
1	Gasketing	303AV TKSP x Head & Jambs		PE	087100
1	Door Sweep	3452CNB TKSP x Width of Door		PE	087100
1	Threshold	181A MSES25SS x Width of Opening		PE	087100
1	Door Position Switch	By Security Vendor			087100

Set: 2.0

Doors: C40-4B, C40-4D

4	Hinge, Full Mortise, Hvy Wt	T4A3386 NRP 4-1/2" x 4-1/2"	US32D	MK	087100
1	Store Door Lock	8226 LNNJ	US32D	SA	087100
2	Keyed Core	By Owner	US26D		087100
1	Conc Overhead Stop	1-X36	630	RF	087100
1	Surface Closer	7500	689	NO	087100
1	Kick Plate	K1050 10" H. x CSK BEV	US32D	RO	087100
1	Gasketing	303AV TKSP x Head & Jambs		PE	087100
1	Door Sweep	3452CNB TKSP x Width of Door		PE	087100
1	Threshold	181A MSES25SS x Width of Opening		PE	087100
1	Door Position Switch	By Security Vendor			087100

Set: 3.0

Doors: C40-4E

8	Hinge, Full Mortise, Hvy Wt	T4A3386 NRP 4-1/2" x 4-1/2"	US32D	MK	087100
1	Flush Bolt	555-24	US26D	RO	087100
1	Flush Bolt	555-12	US26D	RO	087100
1	Dust Proof Strike	570	US26D	RO	087100
1	Store Door Lock	8226 LNNJ	US32D	SA	087100
2	Keyed Core	By Owner	US26D		087100
2	Conc Overhead Stop	1-X36	630	RF	087100
2	Surface Closer	7500	689	NO	087100
2	Kick Plate	K1050 10" H. x CSK BEV	US32D	RO	087100
1	Gasketing	303AV TKSP x Head & Jambs		PE	087100
2	Door Sweep	3452CNB TKSP x Width of Door		PE	087100
1	Threshold	171A MSES25SS x Width of Opening		PE	087100
2	Door Position Switch	By Security Vendor			087100

Notes: Metal astragal to be provided with the hollow metal doors.

END OF SECTION 080671

SECTION 081113 - HOLLOW METAL DOORS AND FRAMES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Standard and custom hollow metal doors and frames.
2. Steel sidelight, borrowed lite and transom frames.
3. Louvers installed in hollow metal doors.
4. Light frames and glazing installed in hollow metal doors.

B. Related Sections:

1. Division 01 Section "General Conditions".
2. Division 04 Section "Unit Masonry" for embedding anchors for hollow metal work into masonry construction.
3. Division 08 Section "Glazing" for glass view panels in hollow metal doors.
4. Division 08 Section "Door Hardware".
5. Division 09 Sections "Exterior Painting" and "Interior Painting" for field painting hollow metal doors and frames.

C. Codes and References: Comply with the version year adopted by the Authority Having Jurisdiction.

1. ANSI/SDI A250.8 - Recommended Specifications for Standard Steel Doors and Frames.
2. ANSI/SDI A250.4 - Test Procedure and Acceptance Criteria for Physical Endurance for Steel Doors, Frames, Frames Anchors and Hardware Reinforcing.
3. ANSI/SDI A250.6 - Recommended Practice for Hardware Reinforcing on Standard Steel Doors and Frames.
4. ANSI/SDI A250.10 - Test Procedure and Acceptance Criteria for Prime Painted Steel Surfaces for Steel Doors and Frames.
5. ANSI/SDI A250.11 - Recommended Erection Instructions for Steel Frames.
6. ASTM A1008 - Standard Specification for Steel Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability.
7. ASTM A653 - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.
8. ASTM A924 - Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process.
9. 10. SDI-113 Standard Practice for Determining the Steady-State Thermal Transmittance of Steel Door & Frame Assemblies.

10. ASTM C 1363 - Standard Test Method for Thermal Performance of Building Assemblies by Means of a Hot Box Apparatus.
11. ASTM C1199 - Standard Test Method for Measuring the Steady-State Thermal Transmittance of Fenestration Systems Using Hot Box Methods
12. ASTM E1423 - Practice for Determining Steady State Thermal Transmittance of Fenestration Systems.
13. ANSI/BHMA A156.115 - Hardware Preparation in Steel Doors and Frames.
14. ANSI/SDI 122 - Installation and Troubleshooting Guide for Standard Steel Doors and Frames.
15. ANSI/NFPA 80 - Standard for Fire Doors and Fire Windows; National Fire Protection Association.
16. ANSI/NFPA 105: Standard for the Installation of Smoke Door Assemblies.
17. NFPA 252 - Standard Methods of Fire Tests of Door Assemblies; National Fire Protection Association.
18. UL 10C - Positive Pressure Fire Tests of Door Assemblies.
19. UL 1784 - Standard for Air Leakage Tests of Door Assemblies.

1.3 SUBMITTALS

A. Product Data: For each type of product indicated. Include construction details, material descriptions, core descriptions, hardware reinforcements, profiles, anchors, fire-resistance rating, and finishes.

B. Door hardware supplier is to furnish templates, template reference number and/or physical hardware to the steel door and frame supplier in order to prepare the doors and frames to receive the finish hardware items.

C. Shop Drawings: Include the following:

1. Elevations of each door design.
2. Details of doors, including vertical and horizontal edge details and metal thicknesses.
3. Frame details for each frame type, including dimensioned profiles and metal thicknesses.
4. Locations of reinforcement and preparations for hardware.
5. Details of anchorages, joints, field splices, and connections.
6. Details of accessories.
7. Details of moldings, removable stops, and glazing.
8. Details of conduit and preparations for power, signal, and control systems.

D. Samples for Verification:

1. Samples are only required by request of the architect and for manufacturers that are not current members of the Steel Door Institute.

1.4 QUALITY ASSURANCE

A. Source Limitations: Obtain hollow metal doors and frames through one source from a single manufacturer wherever possible.

B. Quality Standard: In addition to requirements specified, furnish SDI-Certified manufacturer products that comply with ANSI/SDI A250.8, latest edition, "Recommended Specifications for Standard Steel Doors and Frames".

C. Fire-Rated Door Assemblies: Assemblies complying with NFPA 80 that are listed and labeled by a qualified testing agency, for fire-protection ratings indicated, based on testing at positive pressure according to UL10C (neutral pressure at 40” above sill) or UL 10C.

1. Oversize Fire-Rated Door Assemblies Construction: For units exceeding sizes of tested assemblies, attach construction label certifying doors are built to standard construction requirements for tested and labeled fire rated door assemblies except for size.
2. Temperature-Rise Limit: Where indicated and at vertical exit enclosures (stairwell openings) and exit passageways, provide doors that have a maximum transmitted temperature end point of not more than 450 deg F (250 deg C) above ambient after 30 minutes of standard fire-test exposure.
3. Smoke Control Door Assemblies: Comply with NFPA 105.

a. Smoke "S" Label: Doors to bear “S” label, and include smoke and draft control gasketing applied to frame and on meeting stiles of pair doors.

D. Fire-Rated, Borrowed-Light Frame Assemblies: Assemblies complying with NFPA 80 that are listed and labeled, by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire-protection ratings indicated, based on testing according to NFPA 257. Provide labeled glazing material.

E. Pre-Submittal Conference: Conduct conference in compliance with requirements in Division 01 Section "Project Meetings" with attendance by representatives of Supplier, Installer, and Contractor to review proper methods and procedures for installing hollow metal doors and frames and to verify installation of electrical knockout boxes and conduit at frames with electrified or access control hardware.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver hollow metal work palletized, wrapped, or crated to provide protection during transit and Project site storage. Do not use non-vented plastic.

B. Deliver welded frames with two removable spreader bars across bottom of frames, tack welded to jambs and mullions.

C. Store hollow metal work under cover at Project site. Place in stacks of five units maximum in a vertical position with heads up, spaced by blocking, on minimum 4-inch high wood blocking. Do not store in a manner that traps excess humidity.

1. Provide minimum 1/4-inch space between each stacked door to permit air circulation. Door and frames to be stacked in a vertical upright position.

1.6 PROJECT CONDITIONS

A. Field Measurements: Verify actual dimensions of openings by field measurements before fabrication.

1.7 COORDINATION

A. Coordinate installation of anchorages for hollow metal frames. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors. Deliver such items to Project site in time for installation.

B. Building Information Modeling (BIM) Support: Utilize designated BIM software tools and obtain training needed to successfully participate in the Project BIM processes. All technical disciplines are responsible for the product data integration and data reliability of their Work into the coordinated BIM applications.

1.8 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace doors that fail in materials or workmanship within specified warranty period.

B. Warranty includes installation and finishing that may be required due to repair or replacement of defective doors.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Manufacturers: Subject to compliance with requirements, provide steel doors and frames from a SDI Certified manufacturer:

1. CECO Door Products (C).
2. Curries Company (CU).

2.2 MATERIALS

A. Cold-Rolled Steel Sheet: ASTM A 1008/A 1008M, Commercial Steel (CS), Type B; suitable for exposed applications.

B. Metallic-Coated Steel Sheet: ASTM A 653/A 653M, Commercial Steel (CS), Type B; with minimum G60 (Z180) or A60 (ZF180) metallic coating.

C. Frame Anchors: ASTM A 653/A 653M, Commercial Steel (CS), Commercial Steel (CS), Type B; with minimum G60 (Z180) or A60 (ZF180) metallic coating.

2.3 HOLLOW METAL DOORS

A. General: Provide 1-3/4 inch doors of design indicated, not less than thickness indicated; fabricated with smooth surfaces, without visible joints or seams on exposed faces unless otherwise indicated. Comply with ANSI/SDI A250.8 and ANSI/NAAMM HMMA 867.

B. Exterior Doors: Face sheets fabricated of commercial quality hot-dipped zinc coated steel that complies with ASTM A 653/A 653M, Coating Designation A60. Provide doors complying with

requirements indicated below by referencing ANSI/SDI A250.8 for level and model and ANSI/SDI A250.4 for physical performance level:

1. Design: Flush panel.
2. Core Construction: Manufacturer's standard polystyrene. Where indicated, provide doors fabricated as thermal-rated assemblies with a minimum R-value of 2.8 or better.
3. Level/Model: Level 3 and Physical Performance Level A (Extra Heavy Duty), Minimum 16 gauge (0.053-inch - 1.3-mm) thick steel, Model 2.
4. Top and Bottom Edges: Reinforce tops and bottoms of doors with a continuous steel channel not less than 16 gauge, extending the full width of the door and welded to the face sheet. Doors with an inverted top channel to include a steel closure channel, screw attached, with the web of the channel flush with the face sheets of the door. Plastic or composite channel fillers are not acceptable.
5. Hinge Reinforcement: Minimum 7 gauge (3/16") plate 1-1/4" x 9" or minimum 14 gauge continuous channel with pierced holes, drilled and tapped.
6. Hardware Reinforcements: Fabricate according to ANSI/SDI A250.6 with reinforcing plates from same material as door face sheets.

C. Manufacturers Basis of Design:

1. Curries Company (CU) - Polystyrene Core - 707 Series.

2.4 HOLLOW METAL FRAMES

A. General: Comply with ANSI/SDI A250.8 and with details indicated for type and profile.

B. Exterior Frames: Fabricated of hot-dipped zinc coated steel that complies with ASTM A 653/A 653M, Coating Designation A60.

1. Fabricate frames with mitered or coped corners. Profile as indicated on drawings.
2. Frames: Minimum 14 gauge (0.067-inch -1.7-mm) thick steel sheet.
3. Manufacturers Basis of Design:

    a. Curries Company (CU) - M CM Series.

C. Fire rated frames: Fabricate frames in accordance with NFPA 80, listed and labeled by a qualified testing agency, for fire-protection ratings indicated.

D. Hardware Reinforcement: Fabricate according to ANSI/SDI A250.6 Table 4 with reinforcement plates from same material as frames.

2.5 FRAME ANCHORS

A. Jamb Anchors:

1. Masonry Type: Adjustable strap-and-stirrup or T-shaped anchors to suit frame size, formed from A60 metallic coated material, not less than 0.042 inch thick, with corrugated or perforated straps not less than 2 inches wide by 10 inches long; or wire anchors not less than 0.177 inch thick.
2. Stud Wall Type: Designed to engage stud and not less than 0.042 inch thick.

B. Floor Anchors: Floor anchors to be provided at each jamb, formed from A60 metallic coated material, not less than 0.042 inches thick.

C. Mortar Guards: Formed from same material as frames, not less than 0.016 inches thick.

2.6 LIGHT OPENINGS AND GLAZING

A. Stops and Moldings: Provide stops and moldings around glazed lites where indicated. Form corners of stops and moldings with butted or mitered hairline joints at fabricator's shop. Fixed and removable stops to allow multiple glazed lites each to be removed independently. Coordinate frame rabbet widths between fixed and removable stops with the type of glazing and installation indicated.

B. Moldings for Glazed Lites in Doors and Loose Stops for Glazed Lites in Frames: Minimum 20 gauge thick, fabricated from same material as door face sheet in which they are installed.

C. Fixed Frame Moldings: Formed integral with hollow metal frames, a minimum of 5/8 inch (16 mm) high unless otherwise indicated. Provide fixed frame moldings and stops on outside of exterior and on secure side of interior doors and frames.

D. Preformed Metal Frames for Light Openings: Manufacturer's standard frame formed of 0.048-inch-thick, cold rolled steel sheet; with baked enamel or powder coated finish; and approved for use in doors of fire protection rating indicated. Match pre-finished door paint color where applicable.

2.7 ACCESSORIES

A. Mullions and Transom Bars: Join to adjacent members by welding or rigid mechanical anchors.

B. Grout Guards: Formed from same material as frames, not less than 0.016 inches thick.

2.8 FABRICATION

A. Fabricate hollow metal work to be rigid and free of defects, warp, or buckle. Accurately form metal to required sizes and profiles, with minimum radius for thickness of metal. Where practical, fit and assemble units in manufacturer's plant. When shipping limitations so dictate, frames for large openings are to be fabricated in sections for splicing or splining in the field by others.

B. Tolerances: Fabricate hollow metal work to tolerances indicated in ANSI/SDI A250.8.

C. Hollow Metal Doors:

1. Exterior Doors: Provide optional weep-hole openings in bottom of exterior doors to permit moisture to escape where specified.
2. Glazed Lites: Factory cut openings in doors with applied trim or kits to fit. Factory install glazing where indicated.
3. Astragals: Provide overlapping astragals as noted in door hardware sets in Division 08 Section "Door Hardware" on one leaf of pairs of doors where required by NFPA 80 for fire-

performance rating or where indicated. Extend minimum 3/4 inch beyond edge of door on which astragal is mounted.

D. Hollow Metal Frames:

1. Shipping Limitations: Where frames are fabricated in sections due to shipping or handling limitations, provide alignment plates or angles at each joint, fabricated of same thickness metal as frames.
2. Welded Frames: Weld joints continuously through full throat width of frames, including rabbets, soffits, and stops; grind, fill, dress, and make smooth, flush, and invisible.

    a. Welded frames are to be provided with two steel spreaders temporarily attached to the bottom of both jambs to serve as a brace during shipping and handling. Spreader bars are for bracing only and are not to be used to size the frame opening.

3. High Frequency Hinge Reinforcement: Provide high frequency hinge reinforcements at door openings 48-inches and wider with mortise butt type hinges at top hinge locations.
4. Provide countersunk, flat- or oval-head exposed screws and bolts for exposed fasteners unless otherwise indicated for removable stops, provide security screws at exterior locations.
5. Mortar Guards: Provide guard boxes at back of hardware mortises in frames at all hinges and strike preps regardless of grouting requirements.
6. Floor Anchors: Weld anchors to bottom of jambs and mullions with at least four spot welds per anchor.
7. Jamb Anchors: Provide number and spacing of anchors as follows:

    a. Masonry Type: Locate anchors not more than 18 inches from top and bottom of frame. Space anchors not more than 32 inches on-center and as follows:

        1) Two anchors per jamb up to 60 inches high.
        2) Three anchors per jamb from 60 to 90 inches high.
        3) Four anchors per jamb from 90 to 120 inches high.
        4) Four anchors per jamb plus 1 additional anchor per jamb for each 24 inches or fraction thereof above 120 inches high.

    b. Stud Wall Type: Locate anchors not more than 18 inches from top and bottom of frame. Space anchors not more than 32 inches o.c. and as follows:

        1) Three anchors per jamb up to 60 inches high.
        2) Four anchors per jamb from 60 to 90 inches high.
        3) Five anchors per jamb from 90 to 96 inches high.
        4) Five anchors per jamb plus 1 additional anchor per jamb for each 24 inches or fraction thereof above 96 inches high.
        5) Two anchors per head for frames above 42 inches wide and mounted in metal stud partitions.
8. Door Silencers: Except on weatherstripped or gasketed doors, drill stops to receive door silencers. Silencers to be supplied by frame manufacturer regardless if specified in Division 08 Section "Door Hardware".
9. Bituminous Coating: Where frames are fully grouted with an approved Portland Cement based grout or mortar, coat inside of frame throat with a water based bituminous or asphaltic emulsion coating to a minimum thickness of 3 mils DFT, tested in accordance

with UL 10C and applied to the frame under a 3rd party independent follow-up service procedure.

E. Hardware Preparation: Factory prepare hollow metal work to receive template mortised hardware; include cutouts, reinforcement, mortising, drilling, and tapping according to the Door Hardware Schedule and templates furnished as specified in Division 08 Section "Door Hardware."

1. Locate hardware as indicated, or if not indicated, according to ANSI/SDI A250.8.
2. Reinforce doors and frames to receive non-template, mortised and surface mounted door hardware.
3. Comply with applicable requirements in ANSI/SDI A250.6 and ANSI/DHI A115 Series specifications for preparation of hollow metal work for hardware.
4. Coordinate locations of conduit and wiring boxes for electrical connections with Division 26 Sections.

2.9 STEEL FINISHES

A. Prime Finishes: Doors and frames to be cleaned, and chemically treated to insure maximum finish paint adhesion. Surfaces of the door and frame exposed to view to receive a factory applied coat of rust inhibiting shop primer.

1. Shop Primer: Manufacturer's standard, fast-curing, lead and chromate free primer complying with ANSI/SDI A250.10 acceptance criteria; recommended by primer manufacturer for substrate; and compatible with substrate and field-applied coatings.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. General Contractor to verify the accuracy of dimensions given to the steel door and frame manufacturer for existing openings or existing frames (strike height, hinge spacing, hinge back set, etc.).

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Remove welded in shipping spreaders installed at factory. Restore exposed finish by grinding, filling, and dressing, as required to make repaired area smooth, flush, and invisible on exposed faces.

B. Prior to installation, adjust and securely brace welded hollow metal frames for square, level, twist, and plumb condition.

C. Tolerances shall comply with SDI-117 "Manufacturing Tolerances Standard Steel Doors and Frames."

D. Drill and tap doors and frames to receive non-template, mortised, and surface-mounted door hardware.

E. Verify tolerances against manufacturers installations instructions for tornado and hurricane storm shelter openings.

3.3 INSTALLATION

A. General: Install hollow metal work plumb, rigid, properly aligned, and securely fastened in place; comply with Drawings and manufacturer's written instructions.

B. Hollow Metal Frames: Install hollow metal frames of size and profile indicated. Comply with ANSI/SDI A250.11 and NFPA 80 at fire rated openings.

1. Set frames accurately in position, plumbed, leveled, aligned, and braced securely until permanent anchors are set. After wall construction is complete and frames properly set and secured, remove temporary braces, leaving surfaces smooth and undamaged. Shim as necessary to comply with installation tolerances.
2. Floor Anchors: Provide floor anchors for each jamb and mullion that extends to floor, and secure with post-installed expansion anchors.
3. Masonry Walls: Coordinate installation of frames to allow for solidly filling space between frames and masonry with mortar.
4. Grout Requirements: Do not grout head of frames unless reinforcing has been installed in head of frame. Do not grout vertical or horizontal closed mullion members.

C. Hollow Metal Doors: Fit hollow metal doors accurately in frames, within clearances specified below. Shim as necessary.

1. Non-Fire-Rated Standard Steel Doors:

    a. Jambs and Head: 1/8 inch plus or minus 1/16 inch.
    b. Between Edges of Pairs of Doors: 1/8 inch plus or minus 1/16 inch.
    c. Between Bottom of Door and Top of Threshold: Maximum 3/8 inch.
    d. Between Bottom of Door and Top of Finish Floor (No Threshold): Maximum 3/4 inch.

2. Fire-Rated Doors: Install doors with clearances according to NFPA 80.

D. Field Glazing: Comply with installation requirements in Division 08 Section "Glazing" and with hollow metal manufacturer's written instructions.

3.4 ADJUSTING AND CLEANING

A. Final Adjustments: Check and readjust operating hardware items immediately before final inspection. Leave work in complete and proper operating condition. Remove and replace defective work, including hollow metal work that is warped, bowed, or otherwise unacceptable.

B. Remove grout and other bonding material from hollow metal work immediately after installation.

C. Prime-Coat and Painted Finish Touchup: Immediately after erection, sand smooth rusted or damaged areas of prime coat, or painted finishes, and apply touchup of compatible air drying, rust-inhibitive primer, zinc rich primer (exterior and galvanized openings) or finish paint.

3.5 FIELD QUALITY CONTROL

A. Field Inspection (Punch Report): Reference Division 01 Sections “Closeout Procedures”. Produce project punch report for each installed door opening indicating compliance with approved submittals and verification hardware is properly installed, operating and adjusted. Include list of items to be completed and corrected, indicating the reasons or deficiencies causing the Work to be incomplete or rejected.

1. Organization of List: Include separate Door Opening and Deficiencies and Corrective Action Lists organized by Mark, Opening Remarks and Comments, and related Opening Images and Video Recordings.

END OF SECTION 081113

SECTION 087100 - DOOR HARDWARE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes commercial door hardware for the following:

1. Swinging doors.

B. Door hardware includes, but is not necessarily limited to, the following:

1. Mechanical door hardware.

C. Related Sections:

1. Division 08 Section “Hollow Metal Doors and Frames”.

D. Codes and References: Comply with the version year adopted by the Authority Having Jurisdiction.

1. ANSI A117.1 - Accessible and Usable Buildings and Facilities.
2. ICC/IBC - International Building Code.
3. NFPA 70 - National Electrical Code.
4. NFPA 80 - Fire Doors and Windows.
5. NFPA 101 - Life Safety Code.
6. NFPA 105 - Installation of Smoke Door Assemblies.
7. State Building Codes, Local Amendments.

E. Standards: All hardware specified herein shall comply with the following industry standards as applicable. Any undated reference to a standard shall be interpreted as referring to the latest edition of that standard:

1. ANSI/BHMA Certified Product Standards - A156 Series.
2. UL10C - Positive Pressure Fire Tests of Door Assemblies.
3. ANSI/UL 294 - Access Control System Units.
4. UL 305 - Panic Hardware.

1.3 SUBMITTALS

A. Product Data: Manufacturer's product data sheets including installation details, material descriptions, dimensions of individual components and profiles, operational descriptions and finishes.

B. Door Hardware Schedule: Prepared by or under the supervision of supplier, detailing, fabrication and assembly of door hardware, as well as procedures and diagrams. Coordinate the final Door Hardware Schedule with doors, frames, and related work to ensure proper size, thickness, hand, function, and finish of door hardware.

1. Format: Comply with scheduling sequence and vertical format in DHI's "Sequence and Format for the Hardware Schedule."
2. Organization: Organize the Door Hardware Schedule into door hardware sets indicating complete designations of every item required for each door or opening. Organize door hardware sets in same order as in the Door Hardware Sets at the end of Part 3. Submittals that do not follow the same format and order as the Door Hardware Sets will be rejected and subject to resubmission.
3. Content: Include the following information:

    a. Type, style, function, size, label, hand, and finish of each door hardware item.
    b. Manufacturer of each item.
    c. Fastenings and other pertinent information.
    d. Location of door hardware set, cross-referenced to Drawings, both on floor plans and in door and frame schedule.
    e. Explanation of abbreviations, symbols, and codes contained in schedule.
    f. Mounting locations for door hardware.
    g. Door and frame sizes and materials.
    h. Warranty information for each product.

4. Submittal Sequence: Submit the final Door Hardware Schedule at earliest possible date, particularly where approval of the Door Hardware Schedule must precede fabrication of other work that is critical in the Project construction schedule. Include Product Data, Samples, Shop Drawings of other work affected by door hardware, and other information essential to the coordinated review of the Door Hardware Schedule.

C. Keying Schedule: After a keying meeting with the owner has taken place prepare a separate keying schedule detailing final instructions. Submit the keying schedule in electronic format. Include keying system explanation, door numbers, key set symbols, hardware set numbers and special instructions. Owner must approve submitted keying schedule prior to the ordering of permanent cylinders/cores.

D. Informational Submittals:

1. Product Test Reports: Indicating compliance with cycle testing requirements, based on evaluation of comprehensive tests performed by manufacturer and witnessed by a qualified independent testing agency.

1.4 CLOSEOUT SUBMITTALS

A. Operating and Maintenance Manuals: Provide manufacturers operating and maintenance manuals for each item comprising the complete door hardware installation in quantity as required in Division 01, Closeout Procedures.

B. Project Record Documents: Provide record documentation of as-built door hardware sets in digital format (.pdf, .docx, .xlsx, .csv) and as required in Division 01, Project Record Documents.

1.5 QUALITY ASSURANCE

A. Manufacturers Qualifications: Engage qualified manufacturers with a minimum 5 years of documented experience in producing hardware and equipment similar to that indicated for this Project and that have a proven record of successful in-service performance.

B. Certified Products: Where specified, products must maintain a current listing in the Builders Hardware Manufacturers Association (BHMA) Certified Products Directory (CPD).

C. Installer Qualifications: A minimum 3 years documented experience installing both standard and electrified door hardware similar in material, design, and extent to that indicated for this Project and whose work has resulted in construction with a record of successful in-service performance.

D. Door Hardware Supplier Qualifications: Experienced commercial door hardware distributors with a minimum 5 years documented experience supplying both mechanical and electromechanical hardware installations comparable in material, design, and extent to that indicated for this Project. Supplier recognized as a factory direct distributor by the manufacturers of the primary materials with a warehousing facility in Project's vicinity. Supplier to have on staff a certified Architectural Hardware Consultant (AHC) available during the course of the Work to consult with Contractor, Architect, and Owner concerning both standard and electromechanical door hardware and keying.

E. Source Limitations: Obtain each type and variety of door hardware specified in this section from a single source unless otherwise indicated.

F. Each unit to bear third party permanent label indicating compliance with the referenced testing standards.

G. Keying Conference: Conduct conference to comply with requirements in Division 01 Section "Project Meetings." Keying conference to incorporate the following criteria into the final keying schedule document:

1. Function of building, purpose of each area and degree of security required.
2. Plans for existing and future key system expansion.
3. Requirements for key control storage and software.
4. Installation of permanent keys, cylinder cores and software.
5. Address and requirements for delivery of keys.

1.6 DELIVERY, STORAGE AND HANDLING

A. Inventory door hardware on receipt and provide secure lock-up and shelving for door hardware delivered to Project site. Do not store electronic access control hardware, software or accessories at Project site without prior authorization.

B. Tag each item or package separately with identification related to the final Door Hardware Schedule, and include basic installation instructions with each item or package.

C. Deliver, as applicable, permanent keys, cylinders, cores, access control credentials, software and related accessories directly to Owner via registered mail or overnight package service. Instructions for delivery to the Owner shall be established at the "Keying Conference".

1.7 COORDINATION

A. Templates: Obtain and distribute to the parties involved templates for doors, frames, and other work specified to be factory prepared for installing standard and electrified hardware. Check Shop Drawings of other work to confirm that adequate provisions are made for locating and installing hardware to comply with indicated requirements.

B. Door and Frame Preparation: Doors and corresponding frames are to be prepared, reinforced and pre-wired (if applicable) to receive the installation of the specified electrified, monitoring, signaling and access control system hardware without additional in-field modifications.

1.8 WARRANTY

A. General Warranty: Reference Division 01, General Requirements. Special warranties specified in this Article shall not deprive Owner of other rights Owner may have under other provisions of the Contract Documents and shall be in addition to, and run concurrent with, other warranties made by Contractor under requirements of the Contract Documents.

B. Warranty Period: Written warranty, executed by manufacturer(s), agreeing to repair or replace components of standard and electrified door hardware that fails in materials or workmanship within specified warranty period after final acceptance by the Owner. Failures include, but are not limited to, the following:

1. Structural failures including excessive deflection, cracking, or breakage.
2. Faulty operation of the hardware.
3. Deterioration of metals, metal finishes, and other materials beyond normal weathering.
4. Electrical component defects and failures within the systems operation.

C. Warranty Period: Unless otherwise indicated, warranty shall be one year from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MATERIALS

2.2 BUTT HINGES

A. Hinges: ANSI/BHMA A156.1 butt hinges with number of hinge knuckles and other options as specified in the Door Hardware Sets.

1. Quantity: Provide the following hinge quantity:

a. Two Hinges: For doors with heights up to 60 inches.
b. Three Hinges: For doors with heights 61 to 90 inches.
c. Four Hinges: For doors with heights 91 to 120 inches.
d. For doors with heights more than 120 inches, provide 4 hinges, plus 1 hinge for every 30 inches of door height greater than 120 inches.

2. Hinge Size: Provide the following, unless otherwise indicated, with hinge widths sized for door thickness and clearances required:

a. Widths up to 3’0”: 4-1/2” standard or heavy weight as specified.
b. Sizes from 3’1” to 4’0”: 5” standard or heavy weight as specified.

3. Hinge Weight and Base Material: Unless otherwise indicated, provide the following:

a. Exterior Doors: Heavy weight, non-ferrous, ball bearing or oil impregnated bearing hinges unless Hardware Sets indicate standard weight.
b. Interior Doors: Standard weight, steel, ball bearing or oil impregnated bearing hinges unless Hardware Sets indicate heavy weight.

4. Hinge Options: Comply with the following:

a. Non-removable Pins: With the exception of electric through wire hinges, provide set screw in hinge barrel that, when tightened into a groove in hinge pin, prevents removal of pin while door is closed; for all out-swinging lockable doors.

5. Manufacturers:

a. McKinney (MK) - TA/T4A Series, 5-knuckle.

2.3 DOOR OPERATING TRIM

A. Flush Bolts and Surface Bolts: Provide products conforming to ANSI/BHMA A156.3 and A156.16, Grade 1.

1. Flush bolts to be furnished with top rod of sufficient length to allow bolt retraction device location approximately six feet from the floor.
2. Furnish dust proof strikes for bottom bolts.

3. Surface bolts to be minimum 8” in length and U.L. listed for labeled fire doors and U.L. listed for windstorm components where applicable.
4. Provide related accessories (mounting brackets, strikes, coordinators, etc.) as required for appropriate installation and operation.
5. Manufacturers:

    a. Rockwood (RO).

2.4 CYLINDERS AND KEYING

A. General: Cylinder manufacturer to have minimum (10) years experience designing secured master key systems and have on record a published security keying system policy.

B. Cylinder Types: Original manufacturer cylinders able to supply the following cylinder formats and types:

1. Threaded mortise cylinders with rings and cams to suit hardware application.
2. Rim cylinders with back plate, flat-type vertical or horizontal tailpiece, and raised trim ring.
3. Bored or cylindrical lock cylinders with tailpieces as required to suit locks.
4. Tubular deadlocks and other auxiliary locks.
5. Mortise and rim cylinder collars to be solid and recessed to allow the cylinder face to be flush and be free spinning with matching finishes.
6. Keyway: Match Facility Standard.

C. Keying System: Each type of lock and cylinders to be factory keyed.

1. Supplier shall conduct a "Keying Conference" to define and document keying system instructions and requirements.
2. Furnish factory cut, nickel-silver large bow permanently inscribed with a visual key control number as directed by Owner.
3. Existing System: Field verify and key cylinders to match Owner's existing system.

D. Key Quantity: Provide the following minimum number of keys:

1. Change Keys per Cylinder: Two (2)
2. Master Keys (per Master Key Level/Group): Five (5).
3. Construction Keys (where required): Ten (10).

E. Construction Keying: Provide construction master keyed cylinders.

F. Key Registration List (Bitting List):

1. Provide keying transcript list to Owner's representative in the proper format for importing into key control software.
2. Provide transcript list in writing or electronic file as directed by the Owner.
3. Furnish a list of opening numbers with locking devices, showing cylinder types and quantities required when cylinders or cores are to be owner furnished.

2.5 MORTISE LOCKS AND LATCHING DEVICES

A. Mortise Locksets, Grade 1 (Heavy Duty): Provide ANSI/BHMA A156.13, Series 1000, Operational Grade 1 Certified Products Directory (CPD) listed mortise locksets. Listed manufacturers shall meet all functions and features as specified herein.

1. Manufacturers:

a. Sargent Manufacturing (SA) - 8200 Series.

2.6 LOCK AND LATCH STRIKES

A. Strikes: Provide manufacturer's standard strike with strike box for each latch or lock bolt, with curved lip extended to protect frame, finished to match door hardware set, unless otherwise indicated, and as follows:

1. Flat-Lip Strikes: For locks with three-piece antifriction latchbolts, as recommended by manufacturer.
2. Extra-Long-Lip Strikes: For locks used on frames with applied wood casing trim.
3. Aluminum-Frame Strike Box: Provide manufacturer's special strike box fabricated for aluminum framing.
4. Double-lipped strikes: For locks at double acting doors. Furnish with retractable stop for rescue hardware applications.

B. Standards: Comply with the following:

1. Strikes for Mortise Locks and Latches: BHMA A156.13.
2. Strikes for Bored Locks and Latches: BHMA A156.2.
3. Strikes for Auxiliary Deadlocks: BHMA A156.36.
4. Dustproof Strikes: BHMA A156.16.

2.7 SURFACE DOOR CLOSERS

A. All door closers specified herein shall meet or exceed the following criteria:

1. General: Door closers to be from one manufacturer, matching in design and style, with the same type door preparations and templates regardless of application or spring size. Closers to be non-handed with full sized covers.
2. Standards: Closers to comply with UL-10C for Positive Pressure Fire Test and be U.L. listed for use of fire rated doors.
3. Size of Units: Comply with manufacturer's written recommendations for sizing of door closers depending on size of door, exposure to weather, and anticipated frequency of use. Where closers are indicated for doors required to be accessible to the Americans with Disabilities Act, provide units complying with ANSI ICC/A117.1.
4. Closer Arms: Provide heavy duty, forged steel closer arms unless otherwise indicated in Hardware Sets.
5. Closers shall not be installed on exterior or corridor side of doors; where possible install closers on door for optimum aesthetics.

6. Closer Accessories: Provide door closer accessories including custom templates, special mounting brackets, spacers and drop plates as required for proper installation. Provide through-bolt and security type fasteners as specified in the hardware sets.

B. Door Closers, Surface Mounted (Heavy Duty): ANSI/BHMA A156.4, Grade 1 Certified Products Directory (CPD) listed surface mounted, heavy duty door closers with complete spring power adjustment, sizes 1 thru 6; and fully operational adjustable according to door size, frequency of use, and opening force. Closers to be rack and pinion type, one piece cast iron or aluminum alloy body construction, with adjustable backcheck and separate non-critical valves for closing sweep and latch speed control. Provide non-handed units standard.

1. Heavy duty surface mounted door closers shall have a 30-year warranty.

2. Manufacturers:

a. Norton Rixson (NO) - 7500 Series.

2.8 ARCHITECTURAL TRIM

A. Door Protective Trim

1. General: Door protective trim units to be of type and design as specified below or in the Hardware Sets.
2. Size: Fabricate protection plates (kick, armor, or mop) not more than 2" less than door width (LDW) on stop side of single doors and 1” LDW on stop side of pairs of doors, and not more than 1" less than door width on pull side. Coordinate and provide proper width and height as required where conflicting hardware dictates. Height to be as specified in the Hardware Sets.
3. Where plates are applied to fire rated doors with the top of the plate more than 16” above the bottom of the door, provide plates complying with NFPA 80. Consult manufacturer’s catalog and template book for specific requirements for size and applications.
4. Protection Plates: ANSI/BHMA A156.6 protection plates (kick, armor, or mop), fabricated from the following:

a. Stainless Steel: 300 grade, 050-inch thick.

5. Options and fasteners: Provide manufacturer's designated fastener type as specified in the Hardware Sets. Provide countersunk screw holes.
6. Manufacturers:

a. Rockwood (RO).

2.9 DOOR STOPS AND HOLDERS

A. General: Door stops and holders to be of type and design as specified below or in the Hardware Sets.

B. Overhead Door Stops and Holders: ANSI/BHMA A156.8, Grade 1 Certified Products Directory (CPD) listed overhead stops and holders to be surface or concealed types as indicated in

Hardware Sets. Track, slide, arm and jamb bracket to be constructed of extruded bronze and shock absorber spring of heavy tempered steel. Provide non-handed design with mounting brackets as required for proper operation and function.

1. Manufacturers:

    a. Norton Rixson (RF).

2.10 ARCHITECTURAL SEALS

A. General: Thresholds, weatherstripping, and gasket seals to be of type and design as specified below or in the Hardware Sets. Provide continuous weatherstrip gasketing on exterior doors and provide smoke, light, or sound gasketing on interior doors where indicated. At exterior applications provide non-corrosive fasteners and elsewhere where indicated.

B. Smoke Labeled Gasketing: Assemblies complying with NFPA 105 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for smoke control ratings indicated, based on testing according to UL 1784.

1. Provide smoke labeled perimeter gasketing at all smoke labeled openings.

C. Fire Labeled Gasketing: Assemblies complying with NFPA 80 that are listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction, for fire ratings indicated, based on testing according to UL-10C.

1. Provide intumescent seals as indicated to meet UL10C Standard for Positive Pressure Fire Tests of Door Assemblies, and NFPA 252, Standard Methods of Fire Tests of Door Assemblies.

D. Replaceable Seal Strips: Provide only those units where resilient or flexible seal strips are easily replaceable and readily available from stocks maintained by manufacturer.

E. Manufacturers:

1. Pemko (PE).

2.11 FABRICATION

A. Fasteners: Provide door hardware manufactured to comply with published templates generally prepared for machine, wood, and sheet metal screws. Provide screws according to manufacturers recognized installation standards for application intended.

2.12 FINISHES

A. Standard: Designations used in the Hardware Sets and elsewhere indicate hardware finishes complying with ANSI/BHMA A156.18, including coordination with traditional U.S. finishes indicated by certain manufacturers for their products.

B. Provide quality of finish, including thickness of plating or coating (if any), composition, hardness, and other qualities complying with manufacturer's standards, but in no case less than specified by referenced standards for the applicable units of hardware

C. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine scheduled openings, with Installer present, for compliance with requirements for installation tolerances, labeled fire door assembly construction, wall and floor construction, and other conditions affecting performance.

B. Notify architect of any discrepancies or conflicts between the door schedule, door types, drawings and scheduled hardware. Proceed only after such discrepancies or conflicts have been resolved in writing.

3.2 PREPARATION

A. Hollow Metal Doors and Frames: Comply with ANSI/DHI A115 series.

B. Wood Doors: Comply with ANSI/DHI A115-W series.

3.3 INSTALLATION

A. Install each item of mechanical and electromechanical hardware and access control equipment to comply with manufacturer's written instructions and according to specifications.

1. Installers are to be trained and certified by the manufacturer on the proper installation and adjustment of fire, life safety, and security products including: hanging devices; locking devices; closing devices; and seals.

B. Mounting Heights: Mount door hardware units at heights indicated in following applicable publications, unless specifically indicated or required to comply with governing regulations:

1. Standard Steel Doors and Frames: DHI's "Recommended Locations for Architectural Hardware for Standard Steel Doors and Frames."
2. DHI TDH-007-20: Installation Guide for Doors and Hardware.
3. Where indicated to comply with accessibility requirements, comply with ANSI A117.1 "Accessibility Guidelines for Buildings and Facilities."
4. Provide blocking in drywall partitions where wall stops or other wall mounted hardware is located.

C. Retrofitting: Install door hardware to comply with manufacturer's published templates and written instructions. Where cutting and fitting are required to install door hardware onto or into surfaces that are later to be painted or finished in another way, coordinate removal, storage, and

reinstallation of surface protective trim units with finishing work specified in Division 9 Sections. Do not install surface-mounted items until finishes have been completed on substrates involved.

D. Thresholds: Set thresholds for exterior and acoustical doors in full bed of sealant complying with requirements specified in Division 7 Section "Joint Sealants."

E. Storage: Provide a secure lock up for hardware delivered to the project but not yet installed. Control the handling and installation of hardware items so that the completion of the work will not be delayed by hardware losses before and after installation.

3.4 FIELD QUALITY CONTROL

A. Field Inspection (Punch Report): Reference Division 01 Sections “Closeout Procedures”. Produce project punch report for each installed door opening indicating compliance with approved submittals and verification hardware is properly installed, operating and adjusted. Include list of items to be completed and corrected, indicating the reasons or deficiencies causing the Work to be incomplete or rejected.

1. Organization of List: Include separate Door Opening and Deficiencies and Corrective Action Lists organized by Mark, Opening Remarks and Comments, and related Opening Images and Video Recordings.

3.5 ADJUSTING

A. Initial Adjustment: Adjust and check each operating item of door hardware and each door to ensure proper operation or function of every unit. Replace units that cannot be adjusted to operate as intended. Adjust door control devices to compensate for final operation of heating and ventilating equipment and to comply with referenced accessibility requirements.

3.6 CLEANING AND PROTECTION

A. Protect all hardware stored on construction site in a covered and dry place. Protect exposed hardware installed on doors during the construction phase. Install any and all hardware at the latest possible time frame.

B. Clean adjacent surfaces soiled by door hardware installation.

C. Clean operating items as necessary to restore proper finish. Provide final protection and maintain conditions that ensure door hardware is without damage or deterioration at time of owner occupancy.

3.7 DEMONSTRATION

A. Instruct Owner's maintenance personnel to adjust, operate, and maintain mechanical and electromechanical door hardware.

3.8 DOOR HARDWARE SETS

A. The hardware sets represent the design intent and direction of the owner and architect. They are a guideline only and should not be considered a detailed hardware schedule. Discrepancies, conflicting hardware and missing items should be brought to the attention of the architect with corrections made prior to the bidding process. Omitted items not included in a hardware set should be scheduled with the appropriate additional hardware required for proper application and functionality.

1. Quantities listed are for each pair of doors, or for each single door.
2. The supplier is responsible for handing and sizing all products.
3. Where multiple options for a piece of hardware are given in a single line item, the supplier shall provide the appropriate application for the opening.
4. At existing openings with new hardware the supplier shall field inspect existing conditions prior to the submittal stage to verify the specified hardware will work as required. Provide alternate solutions and proposals as needed.

B. Refer to Section 080671, Door Hardware Sets, for hardware sets.

END OF SECTION 087100

SECTION 099000

PAINTING AND FINISHING

PART 1 GENERAL

1.1 GENERAL REQUIREMENTS

A. Work of this Section, as shown or specified, shall be in accordance with the requirements of the Contract Documents.

1.2 SECTION INCLUDES

A. Work of this Section includes all labor, materials, equipment, and services necessary to complete the high performance paint as shown on the drawings and/or specified herein, including, but not limited to, the following:

1. Prime painting unprimed surfaces to be painted under this Section.

2. Painting all items furnished with a prime coat of paint, including touching up of or repairing of abraded, damaged or rusted prime coats applied by others.

3. Painting all ferrous metal (except stainless steel) exposed to view.

4. Painting all galvanized ferrous metals exposed to view, where painting of the galvanized substrate is specifically called fro in the Architectural Drawings.

1.3 QUALITY ASSURANCE

A. Qualification of Painters: Use only qualified journeyman painters for the mixing and application of paint on exposed surfaces.

B. Paint Coordination: Provide finish coats which are compatible with the prime paints used. Review other Sections of these specifications in which prime paints are to be provided to ensure compatibility of the total coatings system for the various substrates. Upon request from other subcontractors, furnish information on the characteristics of the finish materials proposed to be used, to ensure that compatible prime coats are used. Provide barrier coats over incompatible primers or remove and re-prime as required. Notify the Architect in writing of any anticipated problems using the coating systems as specified with substrates primed by others.

C. All paints must conform to the Volatile Organic Compounds (VOC) standards of prevailing codes and ordinances.

1.4 SUBMITTALS

A. Materials List

1. Before any paint materials are delivered to the job site, submit to the Architect a complete list of all materials proposed to be furnished and installed under this portion of the work.

2. This shall in no way be construed as permitting substitution of materials for those specified or accepted for this work by the Architect.

B. Samples

1. Accompanying the materials list, submit to the Architect copies of the full range of colors available in each of the proposed products.

2. Upon direction of the Architect, prepare and deliver to the Architect two (2) identical sets of Samples of each of the selected colors and glosses painted onto 8-1/2" x 11" x 1/4" thick material; whenever possible, the material for Samples shall be the same material as that on which the coating will be applied in the work.

C. Manufacturer's Recommendations: In each case where material proposed is not the material specified or specifically described as an acceptable alternate in this Section of these specifications, submit for the Architect's review the current recommended method of application published by the manufacturer of the proposed material.

D. All finishes shall comply with Smoke and flame spread index limitations of Table 803.1 of Building Code.

1.5 PRODUCT HANDLING

A. Deliver all paint materials to the job site in their original unopened containers with all labels intact and legible at time of use.

B. Protection

1. Store only the approved materials at the job site, and store only in a suitable and designated area restricted to the storage of paint materials and related equipment.

2. Use all means necessary to ensure the safe storage and use of paint materials and the prompt and safe disposal of waste.

3. Use all means necessary to protect paint materials before, during and after application and to protect the installed work and materials of all other trades.

C. Replacements: In the event of damage, immediately make all repairs and replacements necessary.

1.6 EXTRA STOCK

A. Upon completion of this portion of the Work, deliver to the Owner an extra stock of paint equaling approximately ten (10) percent of each color and gloss used

and each coating material used, with all such extra stock tightly sealed in clearly labeled containers.

1.7 JOB CONDITIONS

A. Apply water-based paints only when the temperature of surfaces to be painted and the surrounding air temperatures are between 50 degrees F. and 90 degrees F., unless otherwise permitted by the paint manufacturer's printed instructions.

B. Apply solvent-thinned paints only when the temperature of surfaces to be painted and the surrounding air temperatures are between 45 degrees F. and 95 degrees F. unless otherwise permitted by the paint manufacturer's printed instructions.

C. Do not apply paint in snow, rain, fog or mist; or when the relative humidity exceeds eighty-five (85) percent; or to damp or wet surfaces; unless otherwise permitted by the paint manufacturer's printed instructions.

D. Painting may be continued during inclement weather only if the areas and surfaces to be painted are enclosed and heated within the temperature limits specified by the paint manufacturer during application and drying periods.

PART 2 PRODUCTS

2.1 PAINT MANUFACTURERS

A. Provide Tnemec products listed below as the basis of design or approved equal by Carboline, Sherwin Williams, International Protective Coatings, Benjamin Moore or approved equal.

2.2 MATERIALS

A. Provide undercoat paint produced by the same manufacturer as the finish coats. Use only thinners approved by the paint manufacturer, and use only to recommended limits.

B. Colors and Glosses: All colors and glosses shall be as selected by the Architect. Certain colors will require paint manufacturer to prepare special factory mixes to match colors selected by the Architect. Color schedule (with gloss) shall be furnished by the Architect.

C. Coloring Pigment: Products of or furnished by the manufacturer of the paint or enamel approved for the work.

D. Linseed Oil: Raw or boiled, as required, of approved manufacture, per ASTM D 234 and D 260, respectively.

E. Turpentine: Pure distilled gum spirits of turpentine, per ASTM D 13.

F. Shellac: Pure gum shellac (white or orange) cut in pure denatured alcohol using not less than four (4) lbs. of gum per gallon of alcohol.

G. Driers, Putty, Spackling Compound, Patching Plaster, etc.: Best quality, of approved manufacture.

H. Heat Resistant Paint: Where required, use heat resistant paint when applying paint to heating lines and equipment.

2.3 GENERAL STANDARDS

A. The various surfaces shall be painted or finished as specified below in Article 2.4. However, the Architect reserves the right to change the finishes within the range of flat, semi-gloss or gloss, without additional cost to the Owner.

B. All paints, varnishes, enamels, lacquers, stains and similar materials must be delivered in the original containers with the seals unbroken and label intact and with the manufacturer's instructions printed thereon.

C. All painting materials shall bear identifying labels on the containers with the manufacturer's instructions printed thereon.

D. Paint shall not be badly settled, caked or thickened in the container, shall be readily dispersed with a paddle to a smooth consistency and shall have excellent application properties.

E. Paint shall arrive on the job color-mixed except for tinting of under-coats and possible thinning.

F. All thinning and tinting materials shall be as recommended by the manufacturer for the particular material thinned or tinted.

G. It shall be the responsibility of the Contractor to see that all mixed colors match the color selection made by the Architect prior to application of the coating.

2.4 SCHEDULE OF FINISHES

A. High Performance Coating on Galvanized Ferrous Metals

First Coat:	PPG- Amerlock 600 Epoxy Series; "Series 1224 " by Tnemec; "Epoxy Mastic Coating V160" by Benjamin Moore Corotech; or "Recoatable Epoxy Primer 867-45" by Sherwin Williams.
Second Coat:	PITT-Pitthane Ultra 95-812 (Gloss)" or "High Build 95-8800 (Semi-Gloss).; "Series 1094 (gloss) " or "Series 1095 (semi-gloss) " by Tnemec; "Aliphatic Acrylic Urethane V500 (Gloss)" or "V510 (Semi-Gloss)" by Benjamin Moore Corotech; or "Hi-Solids Urethane B65-300/350" by Sherwin Williams.

B. High Performance Coating on Non-Galvanized Ferrous Metals

Prime Coat:	PPG- Amercoat 68HS Epoxy Zinc-Rich Primer; "Series 94-H_2O Hydro-Zinc" by Tnemec; “Epoxy Mastic Coating V160” by Benjamin Moore; or "Zinc Clad II Plus Inorganic Zinc Rich Coating B69V212" by Sherwin Williams.
Second Coat:	PPG- Amerlock 600 Epoxy ; "Series 1224" by Tnemec; "Aliphatic Acrylic Urethane V500 (Gloss)" or "V510 (Semi-

Gloss)" by Benjamin Moore Corotech; or "Macropoxy 646 Fast Cure Epoxy B58-600" by Sherwin Williams.

Third Coat: PITT- Pitthane Ultra 95-812 (Gloss)" or "High Build 95-8800 (Semi-Gloss); "Series 1094 (gloss) " or "Series 1095 (semi-gloss) Fluoronar" by Tnemec; "Aliphatic Acrylic Urethane V500 (Gloss)" or "V510 (Semi-Gloss)" by Benjamin Moore Corotech; or "Hi-Solids Polyurethane B65-300/350" by Sherwin Williams.

PART 3 EXECUTION

3.1 INSPECTION

A. Examine the areas and conditions where painting and finishing are to be applied and correct any conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions are corrected to permit proper installation of the work.

3.2 GENERAL WORKMANSHIP REQUIREMENTS

A. Only skilled mechanics shall be employed. Application may be by brush or roller. Spray application only upon acceptance from the Architect in writing.

B. The Contractor shall furnish the Architect a schedule showing when he expects to have completed the respective coats of paint for the various areas and surfaces. This schedule shall be kept current as the job progresses.

C. The Contractor shall protect his work at all times, and shall protect all adjacent work and materials by suitable covering or other method during progress of his work. Upon completion of the work, he shall remove all paint and varnish spots from floors, glass and other surfaces. He shall remove from the premises all rubbish and accumulated materials of whatever nature not caused by others and shall leave his part of the work in clean, orderly and acceptable condition.

D. All materials shall be applied under adequate illumination, evenly spread and flowed on smoothly to avoid runs, sags, holidays, brush marks, air bubbles and excessive roller stipple.

E. Coverage and hide shall be complete. When color, stain, dirt or undercoats show through final coat of paint, the surface shall be covered by additional coats until the paint film is of uniform finish, color, appearance and coverage, at no additional cost to the Owner.

F. All coats shall be dry to manufacturer’s recommendations before applying succeeding coats.

3.3 PREPARATION OF SURFACES

A. General

1. The Contractor shall be held wholly responsible for the finished appearance and satisfactory completion of painting work. Properly prepare all surfaces to receive paint, which includes cleaning, sanding, and touching-up of all prime coats applied under other Sections of the work.

Broom clean all spaces before painting is started. All surfaces to be painted or finished shall be perfectly dry, clean and smooth.

2. Perform all preparation and cleaning procedures in strict accordance with the paint manufacturer's instructions and as herein specified, for each particular substrate condition.

3. Clean surfaces to be painted before applying paint or surface treatments. Remove oil and grease with clean cloths and cleaning solvents prior to mechanical cleaning. Program the cleaning and painting so that dust and other contaminants from the cleaning process will not fall in wet, newly painted surfaces.

B. Metal Surfaces

1. Weld Fluxes: Remove weld fluxes, splatters, and alkali contaminants from metal surfaces in an approved manner and leave surface ready to receive painting.

2. Bare Metal: Thoroughly clean off all foreign matter such as grease, rust, scale and dirt before priming coat is applied. Clean surfaces, where solder flux has been used, with benzene. Clean surfaces by flushing with mineral spirits. For aluminum surfaces, wipe down with an oil free solvent prior to application of any pre-treatment.

   a. Bare metal to receive high performance coating specified herein must be blast cleaned SSPC SP-6 prior to application if field applied primer; coordinate with steel trades furnishing ferrous metals to receive this coating to insure that this cleaning method is followed.

3. Shop Primed Metal: Clean off foreign matter as specified for "Bare Metal." Prime bare, rusted, abraded and marred surfaces with approved primer after proper cleaning of surfaces. Sandpaper all rough surfaces smooth.

4. Galvanized Metal: Prepare surface as per the requirements of ASTM D 6386.

5. Metal Filler: Fill dents, cracks, hollow places, open joints and other irregularities in metal work to be painted with an approved metal filler suitable for the purpose and meeting the requirements of the related Section of work; after setting, sand to a smooth, hard finish, flush with adjoining surface.

C. Testing for Moisture Content: Contractor shall test all plaster, masonry, and drywall surfaces for moisture content using a reliable electronic moisture meter. Contractor shall also test latex type fillers for moisture content before application of top coats of paint. Do not apply any paint or sealer to any surface or to latex type filler where the moisture content exceeds seven (7) percent as measured by the electronic moisture meter.

D. Touch-Up: Prime paint all patched portions in addition to all other specified coats.

3.4 MATERIALS PREPARATION

A. Mix and prepare painting materials in strict accordance with the manufacturer's directions.

B. Store materials not in actual use in tightly covered containers. Maintain containers used in storage, mixing, and application of paint in a clean condition, free of foreign materials and residue.

C. Stir all materials before application to produce a mixture of uniform density, and as required during the application of the materials. Do not stir any film which may form on the surface into the material. Remove the film and, if necessary, strain the material before using.

D. Tint each undercoat a lighter shade to facilitate identification of each coat where multiple coats of the same material are to be applied. Tint undercoats to match the color of the finish coat, but provide sufficient difference in shade of undercoats to distinguish each separate coat.

3.5 APPLICATION

A. General

1. Apply paint by brush or roller in accordance with the manufacturer's directions. Use brushes best suited for the type of material being applied. Use rollers of carpet, velvet back, or high pile sheep's wool as recommended by the paint manufacturer for material and texture required.

2. The number of coats and paint film thickness required is the same regardless of the application method. Do not apply succeeding coats until the previous coat has completely dried. Sand between each enamel or varnish coat application with fine sandpaper, or rub surfaces with pumice stone where required to produce an even, smooth surface in accordance with the coating manufacturer's directions.

3. Apply additional coats when undercoats, stains, or other conditions show through the final coat of paint, until the paint film is of uniform finish, color and appearance. Give special attention to insure that all surfaces, including edges, corners, crevices, welds, and exposed fasteners receive a film thickness equivalent to that of flat surfaces.

B. Scheduling Painting

1. Apply the first coat material to surfaces that have been cleaned, pre-treated or otherwise prepared for painting as soon as practicable after preparation and before subsequent surface deterioration.

2. Allow sufficient time between successive coatings to permit proper drying. Do not re-coat until paint has dried to where it feels firm, does not deform or feel sticky under moderate thumb pressure, and the application of another coat of paint does not cause lifting or loss of adhesion of the undercoat.

C. Pigmented (Opaque) Finishes: Completely cover to provide an opaque, smooth surface of uniform finish, color, appearance and coverage.

D. “Touching-Up” of Factory Finishes: Unless otherwise specified or shown, materials with a factory finish shall not be painted at the project site. To “touch-up,” the Contractor shall use the factory finished material manufacturer’s recommended paint materials to repair abraded, chipped, or otherwise defective surfaces.

3.6 PROTECTION

A. Protect work of other trades, whether to be painted or not, against damage by the painting and finishing work. Leave all such work undamaged. Correct any damages by cleaning, repairing or replacing, and repainting, as acceptable to the Architect.

B. Provide "Wet Paint" signs as required to protect newly painted finishes. Remove temporary protective wrappings provided by others for protection of their work after completion of painting operations.

3.7 CLEAN UP

A. During the progress of the work, remove from the site all discarded paint materials, rubbish, cans and rags at the end of each work day.

B. Upon completion of painting work, clean window glass and other paint spattered surfaces. Remove spattered paint by proper methods of washing and scraping, using care not to scratch or otherwise damage finished surfaces.

C. At the completion of work of other trades, touch-up and restore all damaged or defaced painted surfaces.

END OF SECTION

SECTION 22 00 00 - GENERAL PLUMBING REQUIREMENTS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawing and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specifications Sections, apply to this Section.

1.02 SECTION INCLUDES

A. General requirements for Plumbing work including:

1. Definitions.
2. Regulatory Requirements.
3. References.
4. Quality Assurance.
5. Scope of Work and Coordination.
6. Submittals.
7. Cutting and Patching.
8. Waste Management.
9. Project Closeout.

B. General Product Requirements.

1.03 DEFINITIONS

A. The term “furnished” shall mean, to obtain and deliver to the job site.

B. The term “install” shall mean to unpack, store, assemble, fix in position, and connect for use.

C. The term “provide” shall mean to furnish and install.

D. The term “contractor” shall mean the plumbing contractor or any plumbing sub-contractor.

E. Listed: Equipment is "listed" if of a kind mentioned in a list which:

1. Is published by a nationally recognized laboratory which makes periodic inspection of production of such equipment.
2. States that such equipment meets nationally recognized standards or has been tested and found safe for use in a specified manner.

F. Labeled: Equipment is "labeled" if:

1. It embodies a valid label, symbol, or other identifying mark of a nationally recognized testing laboratory such as Underwriters Laboratories, Inc.
2. Production is periodically inspected by a nationally recognized testing laboratory.
3. The labeling indicates compliance with nationally recognized standards or tests to determine safe use in a specified manner.

G. Certified: Equipment is "certified" if:

1. Equipment has been tested and found by a nationally recognized testing laboratory to meet nationally recognized standards or to be safe for use in a specified manner.

2. Production is periodically inspected by a nationally recognized testing laboratory.
3. It bears a label, tag, or other record of certification.

H. Nationally recognized Testing Laboratory: A testing laboratory which complies with OSHA regulations.

1.04 REGULATORY REQUIREMENTS:

A. Comply with applicable sections of the standards and codes in effect as of the date of the contract documents unless otherwise indicated.

1. National Standard Plumbing Code, NSPC
2. New Jersey Rehabilitation Subcode.
3. International Building Code, New Jersey Edition.
4. International Fuel Gas Code, IFGC
5. NSF 61 and NSF 372, Lead-Free Requirements.
6. Health Code
7. International Fire Code.
8. International Energy Conservation Code – ASHRAE 90.1.
9. National Electrical Code (NEC), including amendments by local Authority having jurisdiction.
10. Local Code Ordinances.
11. Authority Having Jurisdiction, AHJ.

1.05 REFERENCES

A. Unless the contract documents include more stringent requirements, applicable construction industry standards have the same force and effect as if copied directly into the contract documents to the extent referenced. Such standards are made part of the contract documents by reference.

B. Comply with standards and codes in effect as of the date of the contract documents unless otherwise indicated.

C. The contractor shall be familiar with industry standards applicable to the work. Where copies of standards are needed to perform a required construction activity, obtain copies directly from publication source.

D. The contractor shall comply with all laws, ordinances, statutes, and lawful orders issued by the authorities having jurisdiction and rules, conventions, and agreements within the construction industry that control the performance of the work.

1.06 QUALITY ASSURANCE:

A. Unless other standards and/or more stringent provisions are shown or specified, provide work meeting the following standards.

1. American National Standards Institute (ANSI).
2. American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE).
3. American Society of Mechanical Engineers (ASME).
4. American Society for Testing and Materials (ASTM).
5. American Welding Society (AWS).
6. Copper Development Association (CDA).
7. Factory Mutual System (FM).
8. Institute of Boiler and Radiation, Hydronics Institute (IBR).

9. Manufacturers Standardization Society of Valve and Fitting Industry (MSS).
10. National Electrical Manufacturers Association (NEMA).
11. National Fire Protection Association, NFPA
12. Underwriters Laboratories Inc. (UL): All material and equipment shall be listed, labeled or certified by Underwriters Laboratories, Inc., where such standards have been established. Equipment and material which are not covered by UL Standards are approved, provided equipment and materials are listed, labeled, certified or otherwise determined to meet safety requirements of a nationally recognized testing laboratory. Equipment of a class which no nationally recognized testing laboratory accepts, certifies, lists, labels, or determines to be safe, will be considered if inspected or tested in accordance with national industrial standards, such as NEMA, or ANSI. Evidence of compliance shall include certified test reports and definitive submittals.
13. Pressure vessels and safety devices shall be listed and certified in accordance with ASME Test Code.

1.07 SCOPE OF WORK AND COORDINATION

A. The work shall include, but not be limited to the following:

1. Provide new storm piping, roof drains and piping insulation as indicated.
2. Provide system testing as specified and as may be required by the local subcode official.
3. Provide the services of testing, equipment startup, manufacturer's training for setup, adjusting, and balancing of equipment and systems.

B. Existing building projects:

1. The contractor shall have limited use of the premises for construction operations as indicated on drawings by scope areas shown.
2. Maintain existing building in a weathertight condition throughout construction period. All work not specifically called for, but reasonably implied, including cutting, patching and repair of damage caused by construction operations shall be provided by the contractor.

C. The contractor shall ensure that all rules and regulations, including those which may be issued by the owner, are being observed. The contractor shall install signage, barriers, and other means to provide warnings and restrict access to construction areas.

D. In finished spaces, conceal work within walls or chases. At suspended ceilings, conceal work above ceiling unless indicated otherwise.

E. In finished spaces with or without ceilings, coordinate with the work of other trades.

F. In finished spaces without ceilings, and unless shown otherwise, install work in most direct manner, parallel with building lines.

G. Where physical interference cannot be readily resolved, prepare composite drawings at a scale of 1/4 inch = 1'-0" or larger. Show this work in relation to other work. Obtain written approval of proposed changes. Distribute drawings to other parties affected.

H. Subject to approval, provide modifications in layout as required to prevent conflict with existing conditions, other work, or for proper execution.

I. Do not install aluminum or copper products where they will be encased in concrete.

J. Do not install plastic piping in plenum ceilings or similar spaces.

K. Each contractor shall coordinate operations with the work of other trades to ensure the proper installation, clearance, connection and operation of systems.

L. Contractor shall be responsible to take field measurements as required to ensure the work will fit based on field conditions.

M. Contractor shall work with other trades to prepare project specific coordination drawings drawn accurately to scale. Do not base coordination drawings on reproductions of the contract documents or standard printed data. Indicate functional and spatial relationships of components of architectural, structural, civil, mechanical, and electrical systems. Indicate dimensions and make specific note of dimensions that appear to be in conflict with submitted equipment and minimum clearance requirements. Provide alternate sketches for resolution of such conflicts.

N. Contractor shall notify building owner in writing no fewer than 2 days prior to shutdown of any building system. Contractor shall include coordination with owner for any work during an extended shutdown that may be necessary to perform outside of normal business hours.

1.08 SUBMITTALS

A. Contractor shall prepare and submit electronic copy of submittals for work and items indicated. Provide submittals for the following: performance data and material specifications for all equipment listed in schedules, material specifications for items listed in products section, sequence of operations and material specifications for automatic temperature controls, and testing, adjusting, and balancing report.

B. Contractor shall allow for sufficient processing of submittals for review, prior to start dates of fabrication, purchasing, testing, and delivery necessary to meet project schedule. Allow 7 days for the review and transmission of each submittal.

C. All submittals shall include the following information:

1. Rock Brook’s Project name and location.
2. Date.
3. Name and address of Engineer.
4. Name and address of Contractor and/or Sub-contractor.
5. Name and address of Supplier.
6. Name of Manufacturer.
7. Submittal number.
8. Description of the work or item submitted, including relevant specification section number(s).

D. Identify all deviations from the Contract Documents on the submittals.

E. Indicate partial submittals as such.

F. Transmit submittals with a transmittal form listing all included submittals.

G. Resubmittals shall follow the same procedures as for submittals. Identify resubmittals as such and include the following:

1. Note date and content of previous submittal.
2. Note date and content of revision in label or title block and clearly indicate the extent of the revision(s).

    a. Resubmittals that do not clearly identify what has been changed, may be returned without review until changes are identified.

3. Resubmit submittals until they are marked “Approved” or “Approved as Noted”.
4. “Approved as Noted” submittals do not require resubmission for record.

H. Engineer shall review each submittal, make marks to indicate corrections or modifications, and stamp each submittal with an action stamp indicating action to be taken.

I. Approval of submittal shall neither relieve the contractor of the responsibility for compliance with the requirements of the plans and specifications, nor accuracy of measurements.

1.09 CUTTING AND PATCHING

A. The contractor shall remove, replace, patch, and repair materials and surfaces cut or damaged during installation by methods and with materials so as not to void existing warranties.

B. Cut in-place construction using methods least likely to damage elements to be retained, or adjoining construction. Cap, valve, or plug and seal remaining portion of pipe or conduit cut.

C. Patch construction by filling, repairing, refinishing, and similar operations following performance of the new work. Patch with durable seams that are as invisible as possible. Provide materials and comply with installation requirements specified by other sections.

D. Contractor shall repair or replace fireproofing and fire stopping disturbed during construction.

1.10 WASTE MANAGEMENT

A. The contractor shall follow the local ordinances for recycled materials.

B. The contractor shall dispose of all demolition and construction waste from the project site in a landfill or incinerator acceptable to authorities having jurisdiction.

1.11 PROJECT CLOSEOUT

A. The contractor shall complete or correct all work identified as not conforming to the contract documents.

B. The contractor shall prepare electronic record drawings in AutoCAD format to show actual as-built installation conditions.

C. The contractor shall prepare a complete set of operation and maintenance manuals indicating operation and maintenance requirements for each system, subsystem, and piece of equipment.

D. The contractor shall provide source information, manufacturer's maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list, maintenance service contracts, and warranty and bond information for each system, subsystem, and piece of equipment.

E. The contractor shall provide a complete set of emergency information indicating procedures for use by emergency personnel and by owner's operating personnel for types of emergencies indicated.

F. Unless otherwise specified in other Sections, the contractor shall develop an instruction program to train the owner's personnel in the operation, emergency procedures, adjustment, troubleshooting, maintenance, and repair of each system, subsystem, and piece of equipment.

PART 2 PRODUCTS

2.01 GENERAL PRODUCT REQUIREMENTS

A. Where two or more items of the same type of equipment are required, the product of one manufacturer shall be used.

B. The products shown in the contract documents are based on a specific manufacturer, make, and model. These products establish significant qualities related to type, functions, dimension in-service performance, physical properties, appearance, and other characteristics for purposes of evaluating products of other named manufacturers.

C. The Engineer shall consider requests for substitution when the following conditions are satisfied:

1. Requested substitution offers owner a substantial advantage in cost, time, or energy conservation, after deducting additional responsibilities the substitution may present to the owner, including compensation to the Engineer for redesign and evaluation services and increased construction costs or related work.
2. Requested substitution does not require extensive revisions to the contract documents.
3. Requested substitution is consistent with the construction documents and shall produce the indicated results.
4. Requested substitution is fully documented and properly submitted.
5. Requested substitution shall not adversely affect the contractor's construction schedule and has been coordinated with the work or related trades.

D. The contractor shall deliver, store, and handle products using means and methods that shall prevent damage, deterioration, and loss. Comply with manufacturer's written instructions. The contractor shall store materials in a manner that shall not endanger project structure.

E. The contractor shall provide the manufacturer's standard written product warranty to the Owner. Where special warranties are indicated, the contractor shall provide the written warranty for the given time period to the Owner. Warranties specified in other sections shall be in addition to, and run concurrent with, other warranties required by the contract documents. Manufacturer's disclaimers and limitations on product warranties do not relieve the contractor of obligations under the requirements of the contract documents.

PART 3 EXECUTION - Not Applicable.

END OF SECTION 22 00 00

SECTION 22 05 17 - SLEEVES AND SLEEVE SEALS FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Sleeves.
2. Stack-sleeve fittings.
3. Sleeve-seal systems.
4. Grout.
5. Silicone sealants.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

PART 2 - PRODUCTS

2.1 SLEEVES

A. Cast-Iron Pipe Sleeves: Cast or fabricated of cast or ductile iron and equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop collar.

B. Steel Pipe Sleeves: ASTM A53/A53M, Type E, Grade B, Schedule 40, galvanized, with plain ends and integral welded waterstop collar.

C. Galvanized-Steel Sheet Sleeves: 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint.

2.2 STACK-SLEEVE FITTINGS

A. Description: Manufactured, Duco-coated or Dura-coated cast iron sleeve with integral clamping flange for use in waterproof floors and roofs. Include clamping ring, bolts, and nuts for membrane flashing.

1. Underdeck Clamp: Clamping ring with setscrews.

2.3 SLEEVE-SEAL SYSTEMS

A. Description:

1. Modular sealing-element unit, designed for field assembly, for filling annular space between piping and sleeve.
2. Designed to form a hydrostatic seal of 20 psig minimum.
3. Sealing Elements: EPDM-rubber or Nitrile (Buna N) interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
4. Pressure Plates: Carbon steel or Composite plastic or Stainless steel.
5. Connecting Bolts and Nuts: Carbon steel, with corrosion-resistant coating, ASTM B633 or Stainless steel of length required to secure pressure plates to sealing elements.

2.4 GROUT

A. Description: Nonshrink, for interior and exterior sealing openings in non-fire-rated walls or floors.

B. Standard: ASTM C1107/C1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.

C. Design Mix: 5000-psi, 28-day compressive strength.

D. Packaging: Premixed and factory packaged.

2.5 SILICONE SEALANTS

A. Silicone, S, P, 25, T, NT: Single-component, pourable, plus 25 percent and minus 25 percent movement capability, traffic- and nontraffic-use, neutral-curing silicone joint sealant; ASTM C920, Type S, Grade P, Class 25, Uses T and NT. Grade P Pourable (self-leveling) formulation is for opening in floors and other horizontal surfaces that are not fire rated.

PART 3 - EXECUTION

3.1 SLEEVE INSTALLATION

A. Install sleeves for piping passing through penetrations in floors, partitions, roofs, and walls.

B. For sleeves that will have sleeve-seal system installed, select sleeves of size large enough to provide 1-inch annular clear space between piping and concrete slabs and walls.

1. Sleeves are not required for core-drilled holes.

C. Install sleeves in concrete floors, concrete roof slabs, and concrete walls as new slabs and walls are constructed.

1. Cut sleeves to length for mounting flush with both surfaces.

a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches above finished floor level.

2. Using grout or silicone sealant, seal the space outside of sleeves in slabs and walls without sleeve-seal system.

D. Install sleeves for pipes passing through interior partitions.

1. Cut sleeves to length for mounting flush with both surfaces.
2. Install sleeves that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation.
3. Seal annular space between sleeve and piping or piping insulation; use joint sealants appropriate for size, depth, and location of joint.

E. Fire-Resistance-Rated Penetrations, Horizontal Assembly Penetrations, and Smoke Barrier Penetrations: Maintain indicated fire or smoke rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with fire- and smoke-stop materials. Comply with requirements for firestopping and fill materials specified in Section 07.

3.2 STACK-SLEEVE-FITTING INSTALLATION

A. Install stack-sleeve fittings in new slabs as slabs are constructed.

1. Install fittings that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation.
2. Secure flashing between clamping flanges for pipes penetrating floors with membrane waterproofing. Comply with the requirements for flashing specified in Section 07.
3. Install section of cast-iron soil pipe to extend sleeve to 2 inches above finished floor level.
4. Extend cast-iron sleeve fittings below floor slab as required to secure clamping ring if ring is specified.
5. Use silicone sealant to seal the space around outside of stack-sleeve fittings.

B. Fire-Resistance-Rated Penetrations, Horizontal Assembly Penetrations, and Smoke Barrier Penetrations: Maintain indicated fire or smoke rating of floors at pipe penetrations. Seal pipe penetrations with fire- and smoke-stop materials. Comply with requirements for firestopping specified in Section 07.

3.3 SLEEVE-SEAL-SYSTEM INSTALLATION

A. Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at service piping entries into building.

B. Select type, size, and number of sealing elements required for piping material and size and for sleeve ID or hole size. Position piping in center of sleeve. Center piping in penetration, assemble sleeve-seal system components, and install in annular space between piping and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make a watertight seal.

3.4 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Leak Test: After allowing for a full cure, test sleeves and sleeve seals for leaks. Repair leaks and retest until no leaks exist.

B. Sleeves and sleeve seals will be considered defective if they do not pass tests and inspections.

C. Prepare test and inspection reports.

3.5 SLEEVE AND SLEEVE-SEAL SCHEDULE

A. Use sleeves and sleeve seals for the following piping-penetration applications:

1. Exterior Concrete Walls above Grade:

a. Piping Smaller Than NPS 6 : Steel pipe sleeves.
b. Piping NPS 6 and Larger: Steel pipe sleeves.

2. Exterior Concrete Walls below Grade:

a. Piping Smaller Than NPS 6: Steel pipe sleeves with sleeve-seal system

1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.

b. Piping NPS 6 and Larger: Steel pipe sleeves with sleeve-seal system

1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.

3. Concrete Slabs-on-Grade:

a. Piping Smaller Than NPS 6: Steel pipe sleeves with sleeve-seal system

1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.

b. Piping NPS 6 and Larger: Steel pipe sleeves with sleeve-seal system

1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.

4. Concrete Slabs above Grade:

a. Piping Smaller Than NPS 6: Stack-sleeve fittings.
b. Piping NPS 6 and Larger: Stack-sleeve fittings.

5. Interior Partitions:

    a. Piping Smaller Than NPS 6: Steel pipe sleeves.
    b. Piping NPS 6 and Larger: Galvanized-steel sheet sleeves.

END OF SECTION 22 05 17

SECTION 22 05 18 - ESCUTCHEONS FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Escutcheons.
2. Floor plates.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

PART 2 - PRODUCTS

2.1 ESCUTCHEONS

A. One-Piece, Cast-Brass Type: With polished, chrome-plated finish and setscrew fastener.

B. One-Piece, Deep-Pattern Type: Deep-drawn, box-shaped brass with polished, chrome-plated finish and spring-clip fasteners.

C. One-Piece, Stamped-Steel Type: With polished, chrome-plated finish and spring-clip fasteners.

D. Split-Plate, Stamped-Steel Type: With polished, chrome-plated finish; concealed hinge; and spring-clip fasteners.

2.2 FLOOR PLATES

A. Split Floor Plates: Cast brass with concealed hinge.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install escutcheons for piping penetrations of walls, ceilings, and finished floors.

B. Install escutcheons with ID to closely fit around pipe, tube, and insulation of insulated piping and with OD that completely covers opening.

1. Escutcheons for New Piping:

a. Piping with Fitting or Sleeve Protruding from Wall: One-piece, deep pattern.
b. Chrome-Plated Piping: One-piece cast brass with polished, chrome-plated finish.
c. Insulated Piping: One-piece stamped steel or split-plate, stamped steel with concealed hinge with polished, chrome-plated finish.
d. Bare Piping at Wall and Floor Penetrations in Finished Spaces: One-piece stamped steel or split-plate, stamped steel with concealed hinge with polished, chrome-plated finish.
e. Bare Piping at Ceiling Penetrations in Finished Spaces: One-piece stamped steel or split-plate, stamped steel with concealed hinge with polished, chrome-plated finish.
f. Bare Piping in Unfinished Service Spaces: One-piece stamped steel or split-plate, stamped steel with concealed hinge with polished, chrome-plated finish.
g. Bare Piping in Equipment Rooms: One-piece stamped steel or split-plate, stamped steel with concealed hinge with polished, chrome-plated finish.

C. Install floor plates for piping penetrations of equipment-room floors.

D. Install floor plates with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.

1. New Piping: One-piece, floor plate.

3.2 FIELD QUALITY CONTROL

A. Using new materials, replace broken and damaged escutcheons and floor plates.

END OF SECTION 22 05 18

SECTION 22 05 29 - HANGERS AND SUPPORTS FOR PLUMBING PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Metal pipe hangers and supports.
2. Thermal hanger-shield inserts.
3. Fastener systems.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.4 INFORMATIONAL SUBMITTALS

A. Welding certificates.

1.5 QUALITY ASSURANCE

A. Structural-Steel Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M.

B. Pipe Welding Qualifications: Qualify procedures and operators according to 2015 ASME Boiler and Pressure Vessel Code, Section IX.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer, as defined in Section 01 to design trapeze pipe hangers and equipment supports.

B. Structural Performance: Hangers and supports for plumbing piping and equipment shall withstand the effects of gravity loads and stresses within limits and under conditions indicated according to ASCE/SEI 7.

1. Design supports for multiple pipes, capable of supporting combined weight of supported systems, system contents, and test water.
2. Design equipment supports capable of supporting combined operating weight of supported equipment and connected systems and components.

2.2 METAL PIPE HANGERS AND SUPPORTS

A. Carbon-Steel Pipe Hangers and Supports:

1. Description: MSS SP-58, Types 1 through 58, factory-fabricated components.
2. Galvanized Metallic Coatings: Pregalvanized, hot-dip galvanized, or electro-galvanized.
3. Nonmetallic Coatings: Plastic coated or epoxy powder coated.
4. Padded Hangers: Hanger with fiberglass or other pipe insulation pad or cushion to support bearing surface of piping.
5. Hanger Rods: Continuous-thread rod, nuts, and washer made of carbon steel.

B. Stainless-Steel Pipe Hangers and Supports:

1. Description: MSS SP-58, Types 1 through 58, factory-fabricated components.
2. Padded Hangers: Hanger with fiberglass or other pipe insulation pad or cushion to support bearing surface of piping.
3. Hanger Rods: Continuous-thread rod, nuts, and washer made of stainless steel.

2.3 THERMAL HANGER-SHIELD INSERTS

A. Insulation-Insert Material for Cold Piping: ASTM C552, Type II cellular glass with 100-psig minimum compressive strength and vapor barrier.

B. Insulation-Insert Material for Hot Piping: ASTM C552, Type II cellular glass with 100-psig minimum compressive strength.

C. For Clamped Systems: Insert and shield shall cover entire circumference of pipe.

D. For Clevis or Band Hangers: Insert and shield shall cover lower 180 degrees of pipe.

E. Insert Length: Extend 2 inches beyond sheet metal shield for piping operating below ambient air temperature.

2.4 FASTENER SYSTEMS

A. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete, with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

B. Mechanical-Expansion Anchors: Insert-wedge-type anchors, for use in hardened portland cement concrete, with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

1. Indoor Applications: Zinc-coated or stainless steel.
2. Outdoor Applications: Stainless steel.

2.5 MATERIALS

A. Aluminum: ASTM B221.

B. Carbon Steel: ASTM A1011/A1011M.

C. Structural Steel: ASTM A36/A36M carbon-steel plates, shapes, and bars; black and galvanized.

D. Stainless Steel: ASTM A240/A240M.

E. Grout: ASTM C1107/C1107M, factory-mixed and -packaged, dry, hydraulic-cement, nonshrink and nonmetallic grout; suitable for interior and exterior applications.

1. Properties: Nonstaining, noncorrosive, and nongaseous.
2. Design Mix: 5000-psi, 28-day compressive strength.

PART 3 - EXECUTION

3.1 APPLICATION

A. Comply with requirements in Section 07 for firestopping materials and installation, for penetrations through fire-rated walls, ceilings, and assemblies.

B. Strength of Support Assemblies: Where not indicated, select sizes of components, so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb.

3.2 HANGER AND SUPPORT INSTALLATION

A. Metal Pipe-Hanger Installation: Comply with MSS SP-58. Install hangers, supports, clamps, and attachments as required to properly support piping from building structure.

B. Thermal Hanger-Shield Installation: Install in pipe hanger or shield for insulated piping.

C. Fastener System Installation:

1. Install powder-actuated fasteners for use in lightweight concrete or concrete slabs less than 4 inches thick in concrete, after concrete is placed and completely cured. Use operators that are licensed by powder-actuated tool manufacturer. Install fasteners according to powder-actuated tool manufacturer's operating manual.

2. Install mechanical-expansion anchors in concrete, after concrete is placed and completely cured. Install fasteners according to manufacturer's written instructions.

D. Install hangers and supports complete with necessary attachments, inserts, bolts, rods, nuts, washers, and other accessories.

E. Install hangers and supports to allow controlled thermal and seismic movement of piping systems, to permit freedom of movement between pipe anchors, and to facilitate action of expansion joints, expansion loops, expansion bends, and similar units.

F. Install lateral bracing with pipe hangers and supports to prevent swaying.

G. Install building attachments within concrete slabs or attach to structural steel. Install additional attachments at concentrated loads, including valves, flanges, and strainers, NPS 2-1/2 and larger and at changes in direction of piping. Install concrete inserts before concrete is placed; fasten inserts to forms, and install reinforcing bars through openings at top of inserts.

H. Load Distribution: Install hangers and supports, so that piping live and dead loads and stresses from movement will not be transmitted to connected equipment.

I. Pipe Slopes: Install hangers and supports to provide indicated pipe slopes and to not exceed maximum pipe deflections allowed by ASME B31.9 for building services piping.

J. Insulated Piping:

1. Attach clamps and spacers to piping.

    a. Piping Operating Above Ambient Air Temperature: Clamp may project through insulation.
    b. Piping Operating Below Ambient Air Temperature: Use thermal hanger-shield insert with clamp sized to match OD of insert.
    c. Do not exceed pipe stress limits allowed by ASME B31.9 for building services piping.

2. Install MSS SP-58, Type 39 protection saddles if insulation without vapor barrier is indicated. Fill interior voids with insulation that matches adjoining insulation.

    a. Option: Thermal hanger-shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.

3. Install MSS SP-58, Type 40 protective shields on cold piping with vapor barrier. Shields shall span an arc of 180 degrees.

    a. Option: Thermal hanger-shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.

4. Shield Dimensions for Pipe: Not less than the following:

    a. NPS 1/4 to NPS 3-1/2: 12 inches long and 0.048 inch thick.
    b. NPS 4: 12 inches long and 0.06 inch thick.
    c. NPS 5 and NPS 6: 18 inches long and 0.06 inch thick.
    d. NPS 8 to NPS 14: 24 inches long and 0.075 inch thick.

5. Pipes NPS 8 and Larger: Include wood insulation inserts of length at least as long as protective shield.
6. Thermal Hanger Shields: Install with insulation of same thickness as piping insulation.

3.3 METAL FABRICATIONS

A. Cut, drill, and fit miscellaneous metal fabrications for equipment supports.

B. Fit exposed connections together to form hairline joints. Field weld connections that cannot be shop welded because of shipping size limitations.

C. Field Welding: Comply with AWS D1.1/D1.1M procedures for shielded, metal arc welding; appearance and quality of welds; and methods used in correcting welding work; and with the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.
4. Finish welds at exposed connections, so no roughness shows after finishing and so contours of welded surfaces match adjacent contours.

3.4 ADJUSTING

A. Hanger Adjustments: Adjust hangers to distribute loads equally on attachments and to achieve indicated slope of pipe.

B. Trim excess length of continuous-thread hanger and support rods to 1-1/2 inches.

3.5 PAINTING

A. Touchup: Clean field welds and abraded, shop-painted areas. Paint exposed areas immediately after erecting hangers and supports. Use same materials as those used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.

1. Apply paint by brush or spray to provide a minimum dry film thickness of 2.0 mils.

B. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas, and apply galvanizing-repair paint to comply with ASTM A780/A780M.

3.6 HANGER AND SUPPORT SCHEDULE

A. Specific hanger and support requirements are in Sections specifying piping systems and equipment.

B. Comply with MSS SP-58 for pipe-hanger selections and applications that are not specified in piping system Sections.

C. Use hangers and supports with galvanized metallic coatings for piping and equipment that will not have field-applied finishes.

D. Use nonmetallic coatings on attachments for electrolytic protection where attachments are in direct contact with copper tubing.

E. Use carbon-steel pipe hangers and supports and attachments for general service applications.

F. Use stainless-steel pipe hangers and stainless-steel or corrosion-resistant attachments for hostile environment applications.

G. Use padded hangers for piping that is subject to scratching.

H. Use thermal hanger-shield inserts for insulated piping and tubing.

I. Horizontal-Piping Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Adjustable, Steel Clevis Hangers (MSS Type 1): For suspension of noninsulated or insulated, stationary pipes NPS 1/2 to NPS 30.
2. Yoke-Type Pipe Clamps (MSS Type 2): For suspension of up to 1050 deg F pipes NPS 4 to NPS 24, requiring up to 4 inches of insulation.
3. Carbon- or Alloy-Steel, Double-Bolt Pipe Clamps (MSS Type 3): For suspension of pipes NPS 3/4 to NPS 36, requiring clamp flexibility and up to 4 inches of insulation.
4. Steel Pipe Clamps (MSS Type 4): For suspension of cold and hot pipes NPS 1/2 to NPS 24 if little or no insulation is required.
5. Pipe Hangers (MSS Type 5): For suspension of pipes NPS 1/2 to NPS 4, to allow off-center closure for hanger installation before pipe erection.
6. Adjustable, Swivel Split- or Solid-Ring Hangers (MSS Type 6): For suspension of noninsulated, stationary pipes NPS 3/4 to NPS 8.
7. Adjustable, Steel Band Hangers (MSS Type 7): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
8. Adjustable Band Hangers (MSS Type 9): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
9. Adjustable, Swivel-Ring Band Hangers (MSS Type 10): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
10. Split Pipe Ring with or without Turnbuckle Hangers (MSS Type 11): For suspension of noninsulated, stationary pipes NPS 3/8 to NPS 8.
11. Extension Hinged or Two-Bolt Split Pipe Clamps (MSS Type 12): For suspension of noninsulated, stationary pipes NPS 3/8 to NPS 3.
12. U-Bolts (MSS Type 24): For support of heavy pipes NPS 1/2 to NPS 30.
13. Clips (MSS Type 26): For support of insulated pipes not subject to expansion or contraction.
14. Pipe Saddle Supports (MSS Type 36): For support of pipes NPS 4 to NPS 36, with steel-pipe base stanchion support and cast-iron floor flange or carbon-steel plate.
15. Pipe Stanchion Saddles (MSS Type 37): For support of pipes NPS 4 to NPS 36, with steel-pipe base stanchion support and cast-iron floor flange or carbon-steel plate, and with U-bolt to retain pipe.
16. Adjustable Pipe Saddle Supports (MSS Type 38): For stanchion-type support for pipes NPS 2-1/2 to NPS 36 if vertical adjustment is required, with steel-pipe base stanchion support and cast-iron floor flange.
17. Single-Pipe Rolls (MSS Type 41): For suspension of pipes NPS 1 to NPS 30, from two rods if longitudinal movement caused by expansion and contraction occurs.

18. Adjustable Roller Hangers (MSS Type 43): For suspension of pipes NPS 2-1/2 to NPS 24, from single rod if horizontal movement caused by expansion and contraction occurs.
19. Complete Pipe Rolls (MSS Type 44): For support of pipes NPS 2 to NPS 42 if longitudinal movement caused by expansion and contraction occurs but vertical adjustment is unnecessary.
20. Pipe Roll and Plate Units (MSS Type 45): For support of pipes NPS 2 to NPS 24 if small horizontal movement caused by expansion and contraction occurs and vertical adjustment is unnecessary.
21. Adjustable Pipe Roll and Base Units (MSS Type 46): For support of pipes NPS 2 to NPS 30 if vertical and lateral adjustment during installation, in addition to expansion and contraction, is required.

J. Vertical-Piping Clamps: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Extension Pipe or Riser Clamps (MSS Type 8): For support of pipe risers NPS 3/4 to NPS 24.
2. Carbon- or Alloy-Steel Riser Clamps (MSS Type 42): For support of pipe risers NPS 3/4 to NPS 24 if longer ends are required for riser clamps.

K. Hanger-Rod Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel Turnbuckles (MSS Type 13): For adjustment of up to 6 inches for heavy loads.
2. Steel Clevises (MSS Type 14): For 120 to 450 deg F piping installations.
3. Swivel Turnbuckles (MSS Type 15): For use with MSS Type 11 split pipe rings.
4. Malleable-Iron Sockets (MSS Type 16): For attaching hanger rods to various types of building attachments.
5. Steel Weldless Eye Nuts (MSS Type 17): For 120 to 450 deg F piping installations.

L. Building Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel or Malleable-Concrete Inserts (MSS Type 18): For upper attachment to suspend pipe hangers from concrete ceiling.
2. Top-Beam C-Clamps (MSS Type 19): For use under roof installations with bar-joist construction, to attach to top flange of structural shape.
3. Side-Beam or Channel Clamps (MSS Type 20): For attaching to bottom flange of beams, channels, or angles.
4. Center-Beam Clamps (MSS Type 21): For attaching to center of bottom flange of beams.
5. Welded Beam Attachments (MSS Type 22): For attaching to bottom of beams if loads are considerable and rod sizes are large.
6. C-Clamps (MSS Type 23): For structural shapes.
7. Top-Beam Clamps (MSS Type 25): For top of beams if hanger rod is required tangent to flange edge.
8. Side-Beam Clamps (MSS Type 27): For bottom of steel I-beams.
9. Steel-Beam Clamps with Eye Nuts (MSS Type 28): For attaching to bottom of steel I-beams for heavy loads.
10. Linked-Steel Clamps with Eye Nuts (MSS Type 29): For attaching to bottom of steel I-beams for heavy loads, with link extensions.

11. Malleable-Beam Clamps with Extension Pieces (MSS Type 30): For attaching to structural steel.
12. Welded-Steel Brackets: For support of pipes from below or for suspending from above by using clip and rod. Use one of the following for indicated loads:

    a. Light (MSS Type 31): 750 lb.
    b. Medium (MSS Type 32): 1500 lb.
    c. Heavy (MSS Type 33): 3000 lb.

13. Side-Beam Brackets (MSS Type 34): For sides of steel or wooden beams.
14. Plate Lugs (MSS Type 57): For attaching to steel beams if flexibility at beam is required.
15. Horizontal Travelers (MSS Type 58): For supporting piping systems subject to linear horizontal movement where headroom is limited.

M. Saddles and Shields: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel-Pipe-Covering Protection Saddles (MSS Type 39): To fill interior voids with insulation that matches adjoining insulation.
2. Protection Shields (MSS Type 40): Of length recommended in writing by manufacturer to prevent crushing insulation.
3. Thermal Hanger-Shield Inserts: For supporting insulated pipe.

N. Spring Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Restraint-Control Devices (MSS Type 47): Where indicated to control piping movement.
2. Spring Cushions (MSS Type 48): For light loads if vertical movement does not exceed 1-1/4 inches.
3. Spring-Cushion Roll Hangers (MSS Type 49): For equipping Type 41 roll hanger with springs.
4. Spring Sway Braces (MSS Type 50): To retard sway, shock, vibration, or thermal expansion in piping systems.
5. Variable-Spring Hangers (MSS Type 51): Preset to indicated load, and limit variability factor to 25 percent to allow expansion and contraction of piping system from hanger.
6. Variable-Spring Base Supports (MSS Type 52): Preset to indicated load, and limit variability factor to 25 percent to allow expansion and contraction of piping system from base support.
7. Constant Supports: For critical piping stress and if necessary to avoid transfer of stress from one support to another support, critical terminal, or connected equipment. Include auxiliary stops for erection, hydrostatic test, and load-adjustment capability. These supports include the following types:

    a. Horizontal (MSS Type 54): Mounted horizontally.
    b. Vertical (MSS Type 55): Mounted vertically.

O. Use powder-actuated fasteners or mechanical-expansion anchors instead of building attachments where required in concrete construction.

END OF SECTION 22 05 29

SECTION 22 05 53 - IDENTIFICATION FOR PLUMBING PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Warning signs and labels.
2. Pipe labels.
3. Warning tags.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

PART 2 - PRODUCTS

2.1 WARNING SIGNS AND LABELS

A. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/16 inch thick, and having predrilled holes for attachment hardware.

B. Letter Color: Black.

C. Background Color: Yellow.

D. Maximum Temperature: Able to withstand temperatures up to 160 deg F.

E. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

F. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-quarters the size of principal lettering.

G. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

H. Label Content: Include caution and warning information plus emergency notification instructions.

2.2 PIPE LABELS

A. General Requirements for Manufactured Pipe Labels: Preprinted, color-coded, with lettering indicating service, and showing flow direction.

B. Self-Adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.

C. Pipe Label Contents: Include identification of piping service using same designations or abbreviations as used on Drawings; also include pipe size and an arrow indicating flow direction.

1. Flow-Direction Arrows: Integral with piping-system service lettering to accommodate both directions or as separate unit on each pipe label to indicate flow direction.
2. Lettering Size: At least 1/2 inch for viewing distances up to 72 inches and proportionately larger lettering for greater viewing distances.

2.3 WARNING TAGS

A. Description: Preprinted or partially preprinted accident-prevention tags of plasticized card stock with matte finish suitable for writing.

1. Size: 3 by 5-1/4 inches minimum.
2. Fasteners: Reinforced grommet and wire or string.
3. Nomenclature: Large-size primary caption such as "DANGER," "CAUTION," or "DO NOT OPERATE."
4. Color: Safety yellow background with black lettering.

PART 3 - EXECUTION

3.1 PREPARATION

A. Clean piping and equipment surfaces of substances that could impair bond of identification devices, including dirt, oil, grease, release agents, and incompatible primers, paints, and encapsulants.

3.2 GENERAL INSTALLATION REQUIREMENTS

A. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.

B. Coordinate installation of identifying devices with locations of access panels and doors.

C. Install identifying devices before installing acoustical ceilings and similar concealment.

3.3 PIPE LABEL INSTALLATION

A. Piping Color Coding: Painting of piping is specified in Section 09.

B. Pipe Label Locations: Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:

1. Near each valve and control device.
2. Near each branch connection, excluding short takeoffs for fixtures and terminal units. Where flow pattern is not obvious, mark each pipe at branch.
3. Near penetrations through walls, floors, ceilings, and inaccessible enclosures.
4. At access doors, manholes, and similar access points that permit view of concealed piping.
5. Near major equipment items and other points of origination and termination.
6. Spaced at maximum intervals of 50 feet along each run. Reduce intervals to 25 feet in areas of congested piping and equipment.
7. On piping above removable acoustical ceilings. Omit intermediately spaced labels.

C. Directional Flow Arrows: Arrows shall be used to indicate direction of flow in pipes, including pipes where flow is allowed in both directions.

D. Pipe Label Color Schedule:

1. Storm Water Piping:

a. Background: Safety white.
b. Letter Color: Black.

3.4 WARNING-TAG INSTALLATION

A. Write required message on, and attach warning tags to, equipment and other items where required.

END OF SECTION 22 05 53

SECTION 22 07 19 - PLUMBING PIPING INSULATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes insulating the following plumbing piping services:

1. Storm-water piping.
2. Roof drains and rainwater leaders.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product indicated. Include thermal conductivity, water-vapor permeance thickness, and jackets (both factory- and field-applied, if any).

B. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.

1. Detail application of protective shields, saddles, and inserts at hangers for each type of insulation and hanger.
2. Detail insulation application at elbows, fittings, flanges, valves, and specialties for each type of insulation.
3. Detail application of field-applied jackets, where required.

1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified Installer.

B. Material Test Reports: From a qualified testing agency acceptable to authorities having jurisdiction indicating, interpreting, and certifying test results for compliance of insulation materials, sealers, attachments, cements, and jackets, with requirements indicated. Include dates of tests and test methods employed.

C. Field quality-control reports.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Skilled mechanics who have successfully completed an apprenticeship program or another craft training program certified by the Department of Labor, Bureau of Apprenticeship and Training.

B. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products according to ASTM E84 by a testing agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing agency.

1. Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.
2. Insulation Installed Outdoors: Flame-spread index of 75 or less, and smoke-developed index of 150 or less.

C. Comply with the following applicable standards and other requirements specified for miscellaneous components:

1. Supply and Drain Protective Shielding Guards: ICC A117.1.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Packaging: Insulation material containers shall be marked by manufacturer with appropriate ASTM standard designation, type and grade, and maximum use temperature.

1.7 COORDINATION

A. Coordinate sizes and locations of supports, hangers, and insulation shields specified in Section 220529 "Hangers and Supports for Plumbing Piping and Equipment."

B. Coordinate clearance requirements with piping Installer for piping insulation application. Before preparing piping Shop Drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance.

1.8 SCHEDULING

A. Schedule insulation application after pressure testing systems and, where required, after installing and testing heat tracing. Insulation application may begin on segments that have satisfactory test results.

B. Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

PART 2 - PRODUCTS

2.1 INSULATION MATERIALS

A. Comply with requirements in "Piping Insulation Schedule, General," "Indoor Piping Insulation Schedule" articles for where insulating materials shall be applied.

B. Products shall not contain asbestos, lead, mercury, or mercury compounds.

C. Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C871.

D. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C795.

E. Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.

F. Mineral-Fiber, Preformed Pipe Insulation:

1. Type I, 850 Deg F Materials: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C547, Type I, Grade A, with factory-applied ASJ-SSL. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.

2.2 INSULATING CEMENTS

A. Mineral-Fiber Insulating Cement: Comply with ASTM C195.

2.3 ADHESIVES

A. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated, unless otherwise indicated.

B. Mineral-Fiber Adhesive: Comply with MIL-A-3316C, Class 2, Grade A.

C. PVC Jacket Adhesive: Compatible with PVC jacket.

2.4 MASTICS AND COATINGS

A. Materials shall be compatible with insulation materials, jackets, and substrates.

B. Vapor-Retarder Mastic: Water based; suitable for indoor use on below-ambient services.

1. Water-Vapor Permeance: Comply with ASTM C755, Section 7.2.2, Table 2, for insulation type and service conditions.
2. Service Temperature Range: Minus 20 to plus 180 deg F.
3. Color: White.

C. Breather Mastic: Water based; suitable for indoor and outdoor use on above-ambient services.

1. Water-Vapor Permeance: ASTM E96, greater than 1.0 perm at manufacturer's recommended dry film thickness.
2. Service Temperature Range: Minus 20 to plus 180 deg F.
3. Color: White.

2.5 SEALANTS

A. FSK and Metal Jacket Flashing Sealants:

1. Materials shall be compatible with insulation materials, jackets, and substrates.
2. Fire- and water-resistant, flexible, elastomeric sealant.
3. Service Temperature Range: Minus 40 to plus 250 deg F.
4. Color: Aluminum.

B. ASJ Flashing Sealants and PVC Jacket Flashing Sealants:

1. Materials shall be compatible with insulation materials, jackets, and substrates.
2. Fire- and water-resistant, flexible, elastomeric sealant.
3. Service Temperature Range: Minus 40 to plus 250 deg F.
4. Color: White.

2.6 FACTORY-APPLIED JACKETS

A. Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:

1. ASJ: White, kraft-paper, fiberglass-reinforced scrim with aluminum-foil backing; complying with ASTM C1136, Type I.
2. ASJ-SSL: ASJ with self-sealing, pressure-sensitive, acrylic-based adhesive covered by a removable protective strip; complying with ASTM C1136, Type I.
3. FSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C1136, Type II.

2.7 FIELD-APPLIED JACKETS

A. Field-applied jackets shall comply with ASTM C921, Type I, unless otherwise indicated.

B. PVC Jacket: High-impact-resistant, UV-resistant PVC complying with ASTM D1784, Class 16354-C; thickness as scheduled; roll stock ready for shop or field cutting and forming. Thickness is indicated in field-applied jacket schedules.

1. Adhesive: As recommended by jacket material manufacturer.
2. Color: White.
3. Factory-fabricated fitting covers to match jacket if available; otherwise, field fabricate.

    a. Shapes: 45- and 90-degree, short- and long-radius elbows, tees, valves, flanges, unions, reducers, end caps, soil-pipe hubs, traps, mechanical joints, and P-trap and supply covers for lavatories.

2.8 TAPES

A. FSK Tape: Foil-face, vapor-retarder tape matching factory-applied jacket with acrylic adhesive; complying with ASTM C1136.

1. Width: 3 inches.
2. Thickness: 6.5 mils.
3. Adhesion: 90 ounces force/inch in width.
4. Elongation: 2 percent.
5. Tensile Strength: 40 lbf/inch in width.

B. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive; suitable for indoor and outdoor applications.

1. Width: 2 inches.
2. Thickness: 6 mils.
3. Adhesion: 64 ounces force/inch in width.
4. Elongation: 500 percent.
5. Tensile Strength: 18 lbf/inch in width.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions for compliance with requirements for installation tolerances and other conditions affecting performance of insulation application.

1. Verify that systems to be insulated have been tested and are free of defects.
2. Verify that surfaces to be insulated are clean and dry.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Surface Preparation: Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.

3.3 GENERAL INSTALLATION REQUIREMENTS

A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of piping including fittings, valves, and specialties.

B. Install insulation materials, forms, vapor barriers or retarders, jackets, and thicknesses required for each item of pipe system as specified in insulation system schedules.

C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.

D. Install insulation with longitudinal seams at top and bottom of horizontal runs.

E. Install multiple layers of insulation with longitudinal and end seams staggered.

F. Do not weld brackets, clips, or other attachment devices to piping, fittings, and specialties.

G. Keep insulation materials dry during application and finishing.

H. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

I. Install insulation with least number of joints practical.

J. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.

1. Install insulation continuously through hangers and around anchor attachments.
2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic.
3. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.
4. Cover inserts with jacket material matching adjacent pipe insulation. Install shields over jacket, arranged to protect jacket from tear or puncture by hanger, support, and shield.

K. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

L. Install insulation with factory-applied jackets as follows:

1. Draw jacket tight and smooth.
2. Cover circumferential joints with 3-inch-wide strips, of same material as insulation jacket. Secure strips with adhesive and outward clinching staples along both edges of strip, spaced 4 inches o.c.
3. Overlap jacket longitudinal seams at least 1-1/2 inches. Install insulation with longitudinal seams at bottom of pipe. Clean and dry surface to receive self-sealing lap. Staple laps with outward clinching staples along edge at 4 inches o.c.

    a. For below-ambient services, apply vapor-barrier mastic over staples.

4. Cover joints and seams with tape, according to insulation material manufacturer's written instructions, to maintain vapor seal.
5. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints and at ends adjacent to pipe flanges and fittings.

M. Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness.

N. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

O. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches similar to butt joints.

P. For above-ambient services, do not install insulation to the following:

1. Vibration-control devices.
2. Testing agency labels and stamps.
3. Nameplates and data plates.
4. Cleanouts.

3.4 PENETRATIONS

A. Insulation Installation at Roof Penetrations: Install insulation continuously through roof penetrations.

1. Seal penetrations with flashing sealant.
2. For applications requiring only indoor insulation, terminate insulation above roof surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
3. Extend jacket of outdoor insulation outside roof flashing at least 2 inches below top of roof flashing.
4. Seal jacket to roof flashing with flashing sealant.

B. Insulation Installation at Aboveground Exterior Wall Penetrations: Install insulation continuously through wall penetrations.

1. Seal penetrations with flashing sealant.
2. For applications requiring only indoor insulation, terminate insulation inside wall surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
3. Extend jacket of outdoor insulation outside wall flashing and overlap wall flashing at least 2 inches.
4. Seal jacket to wall flashing with flashing sealant.

C. Insulation Installation at Interior Wall and Partition Penetrations (That Are Not Fire Rated): Install insulation continuously through walls and partitions.

D. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Install insulation continuously through penetrations of fire-rated walls and partitions.

1. Comply with requirements in Section 07 for firestopping and fire-resistive joint sealers.

E. Insulation Installation at Floor Penetrations:

1. Pipe: Install insulation continuously through floor penetrations.
2. Seal penetrations through fire-rated assemblies. Comply with requirements in Section 07.

3.5 GENERAL PIPE INSULATION INSTALLATION

A. Requirements in this article generally apply to all insulation materials except where more specific requirements are specified in various pipe insulation material installation articles.

B. Insulation Installation on Fittings, Valves, Strainers, Flanges, and Unions:

1. Install insulation over fittings, valves, strainers, flanges, unions, and other specialties with continuous thermal and vapor-retarder integrity unless otherwise indicated.
2. Insulate pipe elbows using preformed fitting insulation or mitered fittings made from same material and density as adjacent pipe insulation. Each piece shall be butted tightly against adjoining piece and bonded with adhesive. Fill joints, seams, voids, and irregular surfaces with insulating cement finished to a smooth, hard, and uniform contour that is uniform with adjoining pipe insulation.
3. Insulate tee fittings with preformed fitting insulation or sectional pipe insulation of same material and thickness as used for adjacent pipe. Cut sectional pipe insulation to fit. Butt each section closely to the next and hold in place with tie wire. Bond pieces with adhesive.
4. Insulate valves using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. For valves, insulate up to and including the bonnets, valve stuffing-box studs, bolts, and nuts. Fill joints, seams, and irregular surfaces with insulating cement.
5. Insulate strainers using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. Fill joints, seams, and irregular surfaces with insulating cement. Insulate strainers so strainer basket flange or plug can be easily removed and replaced without damaging the insulation and jacket. Provide a removable reusable insulation cover. For below-ambient services, provide a design that maintains vapor barrier.
6. Insulate flanges and unions using a section of oversized preformed pipe insulation. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker.
7. Cover segmented insulated surfaces with a layer of finishing cement and coat with a mastic. Install vapor-barrier mastic for below-ambient services and a breather mastic for above-ambient services. Reinforce the mastic with fabric-reinforcing mesh. Trowel the mastic to a smooth and well-shaped contour.
8. For services not specified to receive a field-applied jacket, install fitted PVC cover over elbows, tees, strainers, valves, flanges, and unions. Terminate ends with PVC end caps. Tape PVC covers to adjoining insulation facing using PVC tape.
9. Stencil or label the outside insulation jacket of each union with the word "union." Match size and color of pipe labels.

C. Insulate instrument connections for thermometers, pressure gages, pressure temperature taps, test connections, flow meters, sensors, switches, and transmitters on insulated pipes. Shape insulation at these connections by tapering it to and around the connection with insulating cement and finish with finishing cement, mastic, and flashing sealant.

D. Install removable insulation covers at locations indicated. Installation shall conform to the following:

1. Make removable flange and union insulation from sectional pipe insulation of same thickness as that on adjoining pipe. Install same insulation jacket as adjoining pipe insulation.

2. When flange and union covers are made from sectional pipe insulation, extend insulation from flanges or union long at least two times the insulation thickness over adjacent pipe insulation on each side of flange or union. Secure flange cover in place with stainless-steel or aluminum bands. Select band material compatible with insulation and jacket.
3. Construct removable valve insulation covers in same manner as for flanges, except divide the two-part section on the vertical center line of valve body.
4. When covers are made from block insulation, make two halves, each consisting of mitered blocks wired to stainless-steel fabric. Secure this wire frame, with its attached insulation, to flanges with tie wire. Extend insulation at least 2 inches over adjacent pipe insulation on each side of valve. Fill space between flange or union cover and pipe insulation with insulating cement. Finish cover assembly with insulating cement applied in two coats. After first coat is dry, apply and trowel second coat to a smooth finish.
5. Unless a PVC jacket is indicated in field-applied jacket schedules, finish exposed surfaces with a metal jacket.

3.6 INSTALLATION OF MINERAL-FIBER INSULATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of preformed pipe insulation to pipe with wire or bands and tighten bands without deforming insulation materials.
2. Where vapor barriers are indicated, seal longitudinal seams, end joints, and protrusions with vapor-barrier mastic and joint sealant.
3. For insulation with factory-applied jackets on above-ambient surfaces, secure laps with outward clinched staples at 6 inches o.c.
4. For insulation with factory-applied jackets on below-ambient surfaces, do not staple longitudinal tabs. Instead, secure tabs with additional adhesive as recommended by insulation material manufacturer and seal with vapor-barrier mastic and flashing sealant.

B. Insulation Installation on Pipe Fittings and Elbows:

1. Install preformed sections of same material as straight segments of pipe insulation when available.
2. When preformed insulation elbows and fittings are not available, install mitered sections of pipe insulation, to a thickness equal to adjoining pipe insulation. Secure insulation materials with bands.

3.7 FIELD-APPLIED JACKET INSTALLATION

A. Where PVC jackets are indicated, install with 1-inch overlap at longitudinal seams and end joints. Seal with manufacturer's recommended adhesive.

1. Apply two continuous beads of adhesive to seams and joints, one bead under lap and the finish bead along seam and joint edge.

3.8 FINISHES

A. Insulation with ASJ or Other Paintable Jacket Material: Paint jacket with paint system identified below and as specified in Section 09.

1. Flat Acrylic Finish: Two finish coats over a primer that is compatible with jacket material and finish coat paint. Add fungicidal agent to render fabric mildew proof.

a. Finish Coat Material: Interior, flat, latex-emulsion size.

B. Color: Final color as selected by Architect. Vary first and second coats to allow visual inspection of the completed Work.

C. Do not field paint aluminum jackets.

3.9 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

B. Perform tests and inspections.

C. All insulation applications will be considered defective Work if sample inspection reveals noncompliance with requirements.

3.10 PIPING INSULATION SCHEDULE, GENERAL

A. Acceptable preformed pipe and tubular insulation materials and thicknesses are identified for each piping system and pipe size range.

B. Items Not Insulated: Unless otherwise indicated, do not install insulation on the following:

1. Drainage piping located in crawl spaces.
2. Underground piping.

3.11 INDOOR PIPING INSULATION SCHEDULE

A. Stormwater and Overflow:

1. All Pipe Sizes: Insulation shall be the following:

a. Mineral-Fiber, Preformed Pipe Insulation, Type I: 1 inch thick.

B. Roof Drain and Overflow Drain Bodies:

1. All Pipe Sizes: Insulation shall be the following:

a. Mineral-Fiber, Preformed Pipe Insulation, Type I: 1 inch thick.

3.12 INDOOR, FIELD-APPLIED JACKET SCHEDULE

A. Install jacket over insulation material. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.

B. If more than one material is listed, selection from materials listed is Contractor's option.

C. Piping, Concealed:

1. None.

D. Piping, Exposed:

1. None.

E. Piping, Exposed in wet or humid locations or next to free-standing sinks:

1. PVC: 20 mils thick.

END OF SECTION 22 07 19

SECTION 22 14 13 - STORM DRAINAGE PIPING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Pipe, tube, and fittings.
2. Specialty pipe and fittings.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.4 INFORMATIONAL SUBMITTALS

A. Coordination Drawings: Detail storm drainage piping. Show support locations, type of support, weight on each support, required clearances, and other details, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of the items involved:

1. Structural members to which drainage piping will be attached or suspended from.

B. Field quality-control reports.

1.5 QUALITY ASSURANCE

A. Piping materials shall bear label, stamp, or other markings of specified testing agency.

1.6 FIELD CONDITIONS

A. Interruption of Existing Storm Drainage Service: Do not interrupt service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary service according to requirements indicated:

1. Notify Construction Manager no fewer than two days in advance of proposed interruption of storm drainage service.
2. Do not proceed with interruption of storm drainage service without Construction Manager's written permission.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Components and installation shall be capable of withstanding the following minimum working pressure unless otherwise indicated:

1. Storm Drainage Piping: 10-foot head of water.

2.2 HUBLESS, CAST-IRON SOIL PIPE AND FITTINGS

A. Pipe and Fittings:

1. Standard: ASTM A 888 or CISPI 301.

B. CISPI, Hubless-Piping Couplings:

1. Standards: ASTM C 1277 and CISPI 310.
2. Description: Stainless-steel corrugated shield with stainless-steel bands and tightening devices; and ASTM C 564, rubber sleeve with integral, center pipe stop.

C. Heavy-Duty, Hubless-Piping Couplings:

1. Standard: ASTM C 1540.
2. Description: Stainless-steel shield with stainless-steel bands and tightening devices; and ASTM C 564, rubber sleeve with integral, center pipe stop.

2.3 PVC PIPE AND FITTINGS

A. NSF Marking: Comply with NSF 14, "Plastics Piping Systems Components and Related Materials," for plastic piping components. Include marking with "NSF-drain" for plastic storm drain and "NSF-sewer" for plastic storm sewer piping.

B. Solid-Wall PVC Pipe: ASTM D 2665; drain, waste, and vent.

C. PVC Socket Fittings: ASTM D 2665, made to ASTM D 3311, drain, waste, and vent patterns and to fit Schedule 40 pipe.

D. Adhesive Primer: ASTM F 656.

E. Solvent Cement: ASTM D 2564.

2.4 SPECIALTY PIPE FITTINGS

A. Transition Couplings:

1. General Requirements: Fitting or device for joining piping with small differences in ODs or of different materials. Include end connections same size as and compatible with pipes to be joined.

2. Fitting-Type Transition Couplings: Manufactured piping coupling or specified-piping-system fitting.
3. Shielded, Nonpressure Transition Couplings:

    a. Standard: ASTM C 1460.
    b. Description: Elastomeric or rubber sleeve with full-length, corrosion-resistant outer shield and corrosion-resistant-metal tension band and tightening mechanism on each end.
    c. End Connections: Same size as and compatible with pipes to be joined.

4. Pressure Transition Couplings:

    a. Standard: AWWA C219.
    b. Description: Metal, sleeve-type couplings same size as pipes to be joined, and with pressure rating at least equal to and ends compatible with pipes to be joined.
    c. Center-Sleeve Material: Manufacturer's standard.
    d. Gasket Material: Natural or synthetic rubber.
    e. Metal Component Finish: Corrosion-resistant coating or material.

PART 3 - EXECUTION

3.1 PIPING INSTALLATION

A. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems.

1. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations.
2. Install piping as indicated unless deviations from layout are approved on coordination drawings.

B. Install piping in concealed locations unless otherwise indicated and except in equipment rooms and service areas.

C. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.

D. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.

E. Install piping at indicated slopes.

F. Install piping free of sags and bends.

G. Install fittings for changes in direction and branch connections.

H. Install piping to allow application of insulation.

I. Make changes in direction for piping using appropriate branches, bends, and long-sweep bends.

1. Do not change direction of flow more than 90 degrees.
2. Use proper size of standard increasers and reducers if pipes of different sizes are connected.

    a. Reducing size of drainage piping in direction of flow is prohibited.

J. Install piping at the slope indicated on Contract Documents.

K. Install cast-iron soil piping according to CISPI's "Cast Iron Soil Pipe and Fittings Handbook.

L. Install aboveground PVC piping according to ASTM D 2665.

M. Plumbing Specialties:

1. Install drains in storm drainage gravity-flow piping.

    a. Comply with requirements for drains specified in Section 221423 "Storm Drainage Piping Specialties."

N. Do not enclose, cover, or put piping into operation until it is inspected and approved by authorities having jurisdiction.

O. Install sleeves for piping penetrations of walls, ceilings, floors, and sleeve seals for concrete walls and slabs.

1. Comply with requirements for sleeves specified in Section 220517 "Sleeves and Sleeve Seals for Plumbing Piping."

P. Install escutcheons for piping penetrations of walls, ceilings, and floors.

1. Comply with requirements for escutcheons specified in Section 220518 "Escutcheons for Plumbing Piping."

3.2 JOINT CONSTRUCTION

A. Hubless, Cast-Iron Soil Piping Coupled Joints:

1. Join according to CISPI 310 and CISPI's "Cast Iron Soil Pipe and Fittings Handbook" for hubless-piping coupling joints.

B. Plastic, Nonpressure-Piping, Solvent-Cemented Joints: Clean and dry joining surfaces. Join pipe and fittings according to the following:

1. Comply with ASTM F 402 for safe-handling practice of cleaners, primers, and solvent cements.
2. PVC Piping: Join according to ASTM D 2855 and ASTM D 2665 appendices.

3.3 SPECIALTY PIPE FITTING INSTALLATION

A. Transition Couplings:

1. Install transition couplings at joints of piping with small differences in ODs.
2. In Drainage Piping: Unshielded, nonpressure transition couplings.

3.4 INSTALLATION OF HANGERS AND SUPPORTS

A. Comply with requirements for hangers, supports, and anchor devices specified in Section 220529 "Hangers and Supports for Plumbing Piping and Equipment."

1. Install carbon-steel pipe hangers for horizontal piping in noncorrosive environments.
2. Install carbon-steel pipe support clamps for vertical piping in noncorrosive environments.
3. Vertical Piping: MSS Type 8 or Type 42, clamps.
4. Install individual, straight, horizontal piping runs:

    a. 100 Feet and Less: MSS Type 1, adjustable, steel clevis hangers.
    b. Longer Than 100 Feet: MSS Type 43, adjustable roller hangers.
    c. Longer Than 100 Feet if Indicated: MSS Type 49, spring cushion rolls.

5. Multiple, Straight, Horizontal Piping Runs 100 Feet or Longer: MSS Type 44, pipe rolls. Support pipe rolls on trapeze.
6. Base of Vertical Piping: MSS Type 52, spring hangers.

B. Install hangers for cast-iron piping, with maximum horizontal spacing and minimum rod diameters, to comply with MSS-58, locally enforced codes, and authorities having jurisdiction requirements, whichever are most stringent.

C. Install hangers for PVC piping, with maximum horizontal spacing and minimum rod diameters, to comply with manufacturer's written instructions, locally enforced codes, and authorities having jurisdiction requirements, whichever are most stringent.

D. Support horizontal piping and tubing within 12 inches of each fitting and coupling.

E. Support vertical cast-iron piping to comply with MSS-58, locally enforced codes, and authorities having jurisdiction requirements, whichever are most stringent, but as a minimum at base and at each floor.

F. Support vertical PVC piping with manufacturer's written instructions, locally enforced codes, and authorities having jurisdiction requirements, whichever are most stringent.

3.5 CONNECTIONS

A. Drawings indicate general arrangement of piping, fittings, and specialties.

B. Connect interior storm drainage piping to exterior storm drainage piping. Use transition fitting to join dissimilar piping materials.

C. Connect storm drainage piping to roof drains and storm drainage specialties.

1. Install test tees (wall cleanouts) in conductors near floor, and floor cleanouts with cover flush with floor.
2. Comply with requirements for cleanouts and drains specified in Section 221423 "Storm Drainage Piping Specialties."

D. Where installing piping adjacent to equipment, allow space for service and maintenance.

3.6 IDENTIFICATION

A. Identify exposed storm drainage piping.

B. Comply with requirements for identification specified in Section 220553 "Identification for Plumbing Piping and Equipment."

3.7 FIELD QUALITY CONTROL

A. During installation, notify authorities having jurisdiction at least 24 hours before inspection must be made. Perform tests specified below in presence of authorities having jurisdiction.

1. Roughing-in Inspection: Arrange for inspection of piping before concealing or closing-in after roughing-in.
2. Final Inspection: Arrange for final inspection by authorities having jurisdiction to observe tests specified below and to ensure compliance with requirements.

B. Test storm drainage piping according to procedures of authorities having jurisdiction or, in absence of published procedures, as follows:

1. Test for leaks and defects in new piping and parts of existing piping that have been altered, extended, or repaired.

    a. If testing is performed in segments, submit separate report for each test, complete with diagram of portion of piping tested.

2. Leave uncovered and unconcealed new, altered, extended, or replaced storm drainage piping until it has been tested and approved.

    a. Expose work that was covered or concealed before it was tested.

3. Test Procedure:

    a. Test storm drainage piping, except outside leaders, on completion of roughing-in.
    b. Close openings in piping system and fill with water to point of overflow, but not less than 10-foot head of water. From 15 minutes before inspection starts until completion of inspection, water level must not drop. Inspect joints for leaks.

4. Repair leaks and defects with new materials and retest piping, or portion thereof, until satisfactory results are obtained.
5. Prepare reports for tests and required corrective action.

C. Piping will be considered defective if it does not pass tests and inspections.

D. Prepare test and inspection reports.

3.8 CLEANING AND PROTECTION

A. Clean interior of piping. Remove dirt and debris as work progresses.

B. Protect drains during remainder of construction period to avoid clogging with dirt and debris and to prevent damage from traffic and construction work.

C. Place plugs in ends of uncompleted piping at end of day and when work stops.

3.9 PIPING SCHEDULE

A. Aboveground storm drainage piping located within a plenum space shall be the following:

1. Hubless, cast-iron soil pipe and fittings; CISPI, hubless-piping couplings; and coupled joints.
2. Dissimilar Pipe-Material Couplings: Shielded, nonpressure transition couplings.

B. Aboveground, storm drainage piping located not in a plenum space shall be any of the following:

1. Hubless, cast-iron soil pipe and fittings; CISPI, hubless-piping couplings; and coupled joints.
2. Solid-wall PVC pipe, PVC socket fittings, and solvent-cemented joints.
3. Dissimilar Pipe-Material Couplings: Shielded, nonpressure transition couplings.

END OF SECTION 221413

SECTION 22 14 23 - STORM DRAINAGE PIPING SPECIALTIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Metal roof drains.
2. Cleanouts.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.4 QUALITY ASSURANCE

A. Drainage piping specialties shall bear label, stamp, or other markings of specified testing agency.

PART 2 - PRODUCTS

2.1 METAL ROOF DRAINS

A. Cast-Iron, General-Purpose Roof Drains:

1. Standard: ASME A112.6.4.
2. Body Material: Cast iron.
3. Refer to Schedule on Contract Documents.

2.2 CLEANOUTS

A. Exposed Cleanouts:

1. Standard: ASME A112.36.2M.
2. Size: Same as connected branch.
3. Body Material: No-hub, test tee as required to match connected piping.
4. Closure: Countersunk, brass plug.
5. Closure Plug Size: Same as, or not more than, one size smaller than cleanout size.

B. Wall Cleanouts:

1. Standard: ASME A112.36.2M. Include wall access.
2. Size: Same as connected drainage piping.
3. Body: No-hub, pipe test tee as required to match connected piping.
4. Closure Plug:

    a. Brass.
    b. Countersunk head.
    c. Drilled and threaded for cover attachment screw.
    d. Size: Same as, or not more than, one size smaller than cleanout size.

5. Wall Access: Round, flat, chrome-plated brass or stainless-steel cover plate with screw.

C. Test Tees:

1. Standard: ASME A112.36.2M and ASTM A74, ASTM A888, or CISPI 301.
2. Size: Same as connected drainage piping.
3. Body Material: No-hub, pipe test tee as required to match connected piping.
4. Closure Plug: Countersunk, brass.
5. Closure Plug Size: Same as, or not more than, one size smaller than cleanout size.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install roof drains at low points of roof areas according to roof membrane manufacturer's written installation instructions.

1. Install flashing collar or flange of roof drain to prevent leakage between drain and adjoining roofing. Maintain integrity of waterproof membranes where penetrated.
2. Install expansion joints, if indicated, in roof drain outlets.
3. Position roof drains for easy access and maintenance.

B. Install cleanouts in aboveground piping and building drain piping according to the following instructions unless otherwise indicated:

1. Use cleanouts the same size as drainage piping up to NPS 4. Use sizes indicated for piping larger than NPS 4.
2. Locate cleanouts at each change in direction of piping greater than 45 degrees.
3. Locate cleanouts at minimum intervals of 50 feet for piping NPS 4 and smaller and 100 feet for larger piping or as indicated per Code.
4. Locate cleanouts at base of each vertical storm piping conductor.

C. For cleanouts located in concealed piping, install cleanout wall access covers, of types indicated, with cover flush with finished wall.

D. Install test tees in vertical conductors and near floor.

E. Install wall cleanouts in vertical conductors. Install access door in wall if indicated.

F. Install through-penetration firestop assemblies for penetrations of fire- and smoke-rated assemblies.

1. Comply with requirements in Section 07.

3.2 CONNECTIONS

A. Comply with requirements for piping specified in Section 221413 "Facility Storm Drainage Piping." Drawings indicate general arrangement of piping, fittings, and specialties.

3.3 FLASHING INSTALLATION

A. Fabricate flashing from single piece of metal unless large pans, sumps, or other drainage shapes are required.

B. Install sheet flashing on pipes, sleeves, and specialties passing through or embedded in floors and roofs with waterproof membrane.

C. Set flashing on floors and roofs in solid coating of bituminous cement.

D. Secure flashing into sleeve and specialty clamping ring or device.

3.4 PROTECTION

A. Protect drains during remainder of construction period to avoid clogging with dirt or debris and to prevent damage from traffic or construction work.

B. Place plugs in ends of uncompleted piping at end of each day or when work stops.

END OF SECTION 22 14 23

SECTION 230000 – GENERAL MECHANICAL REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SECTION INCLUDES

A. General requirements for Mechanical work including:

1. Definitions.
2. Regulatory Requirements.
3. References.
4. Quality Assurance.
5. Scope of Work and Coordination.
6. Submittals.
7. Cutting and Patching.
8. Waste Management.
9. Project Closeout.

B. General Product Requirements.

C. Housekeeping pads.

1.3 DEFINITIONS

A. The term “furnished” shall mean, to obtain and deliver to the job site.

B. The term “install” shall mean to unpack, store, assemble, fix in position, and connect for use.

C. The term “provide” shall mean to furnish and install.

D. The term “contractor” shall mean the mechanical contractor or any mechanical sub-contractor.

E. Listed: Equipment is "listed" if of a kind mentioned in a list which:

1. Is published by a nationally recognized laboratory which makes periodic inspection of production of such equipment.
2. States that such equipment meets nationally recognized standards or has been tested and found safe for use in a specified manner.

F. Labeled: Equipment is "labeled" if:

1. It embodies a valid label, symbol, or other identifying mark of a nationally recognized testing laboratory such as Underwriters Laboratories, Inc.
2. Production is periodically inspected by a nationally recognized testing laboratory.
3. The labeling indicates compliance with nationally recognized standards or tests to determine safe use in a specified manner.

G. Certified: Equipment is "certified" if:

1. Equipment has been tested and found by a nationally recognized testing laboratory to meet nationally recognized standards or to be safe for use in a specified manner.
2. Production is periodically inspected by a nationally recognized testing laboratory.
3. It bears a label, tag, or other record of certification.

H. Nationally recognized Testing Laboratory: A testing laboratory which complies with OSHA regulations.

1.4 REFERENCES

A. Unless the contract documents include more stringent requirements, applicable construction industry standards have the same force and effect as if copied directly into the contract documents to the extent referenced. Such standards are made part of the contract documents by reference.

B. Comply with standards and codes in effect as of the date of the contract documents unless otherwise indicated.

C. The contractor shall be familiar with industry standards applicable to the work. Where copies of standards are needed to perform a required construction activity, obtain copies directly from publication source.

D. The contractor shall comply with all laws, ordinances, statutes, and lawful orders issued by the authorities having jurisdiction and rules, conventions, and agreements within the construction industry that control the performance of the work.

1.5 QUALITY ASSURANCE:

A. Unless other standards and/or more stringent provisions are shown or specified, provide work meeting the following standards.

1. Air Conditioning and Refrigeration Institute (ARI).
2. Air Diffusion Council (ADC).
3. Air Movement and Control Association (AMCA).
4. American National Standards Institute (ANSI).
5. American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE).
6. American Society of Mechanical Engineers (ASME).
7. American Society for Testing and Materials (ASTM).
8. American Welding Society (AWS).

9. Associated Air Balance Council (AABC).
10. Copper Development Association (CDA).
11. Factory Mutual System (FM).
12. Institute of Boiler and Radiation, Hydronics Institute (IBR).
13. Manufacturers Standardization Society of Valve and Fitting Industry (MSS).
14. National Electrical Manufacturers Association (NEMA).
15. National Environmental Balancing Bureau (NEBB).
16. Sheet Metal and Air Conditioning Contractors National Association (SMACNA).
17. Underwriters Laboratories Inc. (UL): All material and equipment shall be listed, labeled or certified by Underwriters Laboratories, Inc., where such standards have been established. Equipment and material which are not covered by UL Standards are approved, provided equipment and materials are listed, labeled, certified or otherwise determined to meet safety requirements of a nationally recognized testing laboratory. Equipment of a class which no nationally recognized testing laboratory accepts, certifies, lists, labels, or determines to be safe, will be considered if inspected or tested in accordance with national industrial standards, such as NEMA, or ANSI. Evidence of compliance shall include certified test reports and definitive submittals.

1.6 SCOPE OR WORK AND COORDINATION

A. The work shall include, but not be limited to the following:

1. Provide new ductwork, dampers, diffusers, and accessories as indicated.
2. Provide new piping, valves, and accessories as indicated.
3. Provide new equipment where indicated.
4. Provide new controls as necessary to perform the sequences of operation.
5. Provide the services of an independent testing and balancing contractor to test, adjust, balance, and record system flows.

B. Existing building projects:

1. The contractor shall have limited use of the premises for construction operations as indicated on drawings by scope areas shown.
2. Maintain existing building in a weathertight condition throughout construction period. All work not specifically called for, but reasonably implied, including cutting, patching and repair of damage caused by construction operations shall be provided by the contractor.

C. The contractor shall ensure that all rules and regulations, including those which may be issued by the owner, are being observe. The contractor shall install signage, barriers, and other means to provide warnings and restrict access to construction areas.

D. Where physical interference cannot be readily resolved, prepare composite drawings at a scale of 1/4 inch = 1’-0” or larger. Show this work in relation to other work. Obtain written approval of proposed changes. Distribute drawings to other parties affected.

E. Subject to approval, provide modifications in layout as required to prevent conflict with existing conditions, other work, or for proper execution.

F. Each contractor shall coordinate operations with the work of other trades to ensure the proper installation, connection and operation of systems.

G. The contractor shall be responsible for taking field measurements as required to ensure the work will fit based on field conditions.

H. The contractor shall work with other trades to prepare project specific coordination drawings drawn accurately to scale. Do not base coordination drawings on reproductions of the contract documents or standard printed data. Indicate functional and spatial relationships of components of architectural, structural, civil, mechanical, and electrical systems. Indicate dimensions and make specific note of dimensions that appear to be in conflict with submitted equipment and minimum clearance requirements. Provide alternate sketches for resolution of such conflicts.

I. Contractor shall notify building owner in writing at least 3 days prior to shutdown of any building system. Contractor shall include coordination with owner for any work during an extended shutdown that may be necessary to perform outside of normal business hours.

1.7 SUBMITTALS

A. Contractor shall prepare and submit electronic submittals for work and items indicated. Submittal or submittal download notifications shall be sent via email. Provide submittals for the following: performance data and material specifications for all equipment listed in schedules, dimensioned sheet metal drawings, material specifications for items listed in products section, sequence of operations and material specifications for automatic temperature controls, and testing, adjusting, and balancing report.

B. The contractor shall allow for sufficient processing of submittals for review, prior to start dates of fabrication, purchasing, testing, and delivery necessary to meet project schedule. Contractor shall provide, and regularly update, a submittal and shop drawing schedule which is integrated into the construction schedule. Allow 7 days for the review and transmission of each submittal.

C. All submittals shall include the following information:

1. Rock Brook’s Project name and location.
2. Date.
3. Name and address of Engineer.
4. Name and address of Contractor and/or Sub-contractor.
5. Name and address of Supplier.
6. Name of Manufacturer.
7. Submittal number.
8. Description of the work or item submitted, including relevant specification section number(s).

D. Identify all deviations from the Contract Documents on the submittals.

E. Indicate partial submittals as such.

F. Transmit submittals with a transmittal form listing all included submittals.

G. Resubmittals shall follow the same procedures as for submittals. Identify resubmittals as such and include the following:

1. Note date and content of previous submittal.

2. Note date and content of revision in label or title block and clearly indicate the extent of the revision(s).

    a. Resubmittals that do not clearly identify what has been changed, may be returned without review until changes are identified.

3. Resubmit submittals until they are marked “Approved” or “Approved as Noted”.

H. Submittals and resubmittals marked “Approved as Noted” do not require resubmission for record.

I. Engineer shall review each submittal, make marks to indicate corrections or modifications, and stamp each submittal with an action stamp indicating action to be taken.

J. Approval of submittal shall neither relieve the contractor of the responsibility for compliance with the requirements of the plans and specifications, nor accuracy of measurements.

1.8 CUTTING AND PATCHING

A. The contractor shall remove, replace, patch, and repair materials and surfaces cut or damaged during installation by methods and with materials so as not to void existing warranties.

B. Cut in-place construction using methods least likely to damage elements to be retained, or adjoining construction. Cap, valve, or plug and seal remaining portions of pipe or conduit cut.

C. Patch construction by filling, repairing, refinishing, and similar operations following performance of the new work. Patch with durable seams that are as invisible as possible. Provide materials and comply with installation requirements specified by other sections.

D. The contractor shall repair or replace fireproofing and fire stopping disturbed during construction.

1.9 WASTE MANAGEMENT

A. The contractor shall perform the following for items salvaged for reuse during the work:

1. Clean salvaged items.
2. Protect and safely store salvaged items until reinstallation
3. Install salvaged items to comply with installation requirements for new materials and equipment.

B. The contractor shall perform the following for items salvaged for owner’s use:

1. Clean salvaged items.
2. Protect and safely store salvage items until delivery.
3. Transport salvaged items to the owner’s storage area.

C. The contractor shall follow the local ordinances for recycled materials.

D. The contractor shall dispose of all demolition and construction waste from the project site in a landfill or incinerator acceptable to authorities having jurisdiction.

1.10 PROJECT CLOSEOUT

A. The contractor shall complete or correct all work identified as not conforming to the contract documents.

B. The contractor shall prepare electronic record drawings in AutoCAD format to show actual as-built installation conditions.

C. The contractor shall prepare a complete set of operation and maintenance manuals indicating operation and maintenance requirements for each system, subsystem, and piece of equipment.

D. The contractor shall provide source information, manufacturer's maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list, maintenance service contracts, and warranty and bond information for each system, subsystem, and piece of equipment.

E. The contractor shall provide a complete set of emergency information indicating procedures for use by emergency personnel and by owner's operating personnel for types of emergencies indicated.

F. Unless otherwise specified in other Sections, the contractor shall develop an instruction program to train the owner's personnel in the operation, emergency procedures, adjustment, troubleshooting, maintenance, and repair of each system, subsystem, and piece of equipment.

PART 2 - PRODUCTS

2.1 GENERAL PRODUCT REQUIREMENTS

A. Where two or more items of the same type of equipment are required, the product of one manufacturer shall be used.

B. The products shown in the contract documents are based on a specific manufacturer, make, and model. These products establish significant qualities related to type, functions, dimension in-service performance, physical properties, appearance, and other characteristics for purposes of evaluating products of other named manufacturers.

C. The Engineer shall consider requests for substitution when the following conditions are satisfied:

1. Requested substitution offers owner a substantial advantage in cost, time, or energy conservation; after deducting additional responsibilities the substitution may present to the owner, including compensation to the Engineer for redesign and evaluation services and increased construction costs or related work.
2. Requested substitution does not require extensive revisions to the contract documents.
3. Requested substitution is consistent with the construction documents and shall produce the indicated results.

4. Requested substitution is fully documented and properly submitted.
5. Requested substitution shall not adversely affect the contractor's construction schedule and has been coordinated with the work or related trades.

D. The contractor shall deliver, store, and handle products using means and methods that shall prevent damage, deterioration, and loss. Comply with manufacturer's written instructions. The contractor shall store materials in a manner that shall not endanger project structure.

E. The contractor shall provide the manufacturer's standard written product warranty to the Owner. Where special warranties are indicated, the contractor shall provide the written warranty for the given time period to the Owner. Warranties specified in other sections shall be in addition to, and run concurrent with, other warranties required by the contract documents. Manufacturer's disclaimers and limitations on product warranties do not relieve the contractor of obligations under the requirements of the contract documents.

PART 3 - EXECUTION

3.1 HOUSEKEEPING PADS

A. Provide concrete pads for floor-mounted mechanical equipment in dimensions indicated. If thickness is not indicated, provide 4-inch-thick pad extending 4 inches beyond supported equipment outline, including skids, when appropriate. Equipment with support skids smaller than equipment outline shall not be exempt from these requirements.

B. Coordinate interface between pad and vibration isolation bases specified in Section 230529 "Hangers and Supports for HVAC Piping and Equipment".

C. Anchor equipment to pads, using equipment manufacturer's setting templates for anchor locations.

D. Roughen base slab to remove existing finish and to ensure mechanical bond.

E. Place concrete and install wire mesh one inch below finished surface.

F. Apply hand trowel finish.

G. The following equipment does not require housekeeping pads:

1. Equipment intended to be mobile; provided with wheels or casters.
2. Temporary equipment such as chemical storage tanks.

END OF SECTION 230000

SECTION 230500 – HVAC ROOF ACCESSORIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Roof curbs.
2. Equipment supports.
3. Pipe and duct supports.
4. Pipe portals.
5. Preformed flashing sleeves.

1.3 COORDINATION

A. Coordinate layout and installation of roof accessories with roofing membrane and base flashing and interfacing and adjoining construction to provide a leakproof, weathertight, secure, and noncorrosive installation.

B. Coordinate dimensions with rough-in information or Shop Drawings of equipment to be supported.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of roof accessory.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.

B. Shop Drawings: For roof accessories.

1. Include plans, elevations, keyed details, and attachments to other work. Indicate dimensions, loadings, and special conditions. Distinguish between plant- and field-assembled work.

C. Delegated-Design Submittal: For roof curbs equipment supports indicated to comply with performance requirements and design criteria, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1. Detail mounting, securing, and flashing of roof-mounted items to roof structure. Indicate coordinating requirements with roof membrane system.
2. Wind-Restraint Details: Detail fabrication and attachment of wind restraints. Show anchorage details and indicate quantity, diameter, and depth of penetration of anchors.

1.5 INFORMATIONAL SUBMITTALS

A. Coordination Drawings: Roof plans, drawn to scale, and coordinating penetrations and roof-mounted items. Show the following:

1. Size and location of roof accessories specified in this Section.
2. Method of attaching roof accessories to roof or building structure.
3. Other roof-mounted items including mechanical and electrical equipment, ductwork, piping, and conduit.
4. Required clearances.

B. Sample Warranties: For manufacturer's special warranties.

1.6 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For roof accessories to include in operation and maintenance manuals.

1.7 WARRANTY

A. Special Warranty on Painted Finishes: Manufacturer's standard form in which manufacturer agrees to repair finishes or replace roof accessories that show evidence of deterioration of factory-applied finishes within specified warranty period.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. General Performance: Roof accessories shall withstand exposure to weather and resist thermally induced movement without failure, rattling, leaking, or fastener disengagement due to defective manufacture, fabrication, installation, or other defects in construction.

B. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 "Quality Requirements," to design roof curbs and equipment supports to comply with wind performance requirements, including comprehensive engineering analysis by a qualified professional engineer, using performance requirements and design criteria indicated.

C. Wind-Restraint Performance: Meeting local windloads, approx. 114 mph.

2.2 ROOF CURBS

A. Roof Curbs: Internally reinforced roof-curb units capable of supporting superimposed live and dead loads, including equipment loads and other construction indicated on Drawings, bearing continuously on roof structure, and capable of meeting performance requirements; with welded or mechanically fastened and sealed corner joints, stepped integral metal cant raised the thickness of roof insulation, and integrally formed deck-mounting flange at perimeter bottom.

B. Size: Coordinate dimensions with roughing-in information or Shop Drawings of equipment to be supported.

C. Supported Load Capacity: Coordinate load capacity with information on Shop Drawings of equipment to be supported.

D. Material: Zinc-coated (galvanized) steel sheet, 0.052 inch thick.

1. Finish: Mill phosphatized.
2. Color: As indicated by manufacturer's designations.

E. Construction:

1. Curb Profile: Manufacturer's standard compatible with roofing system.
2. On ribbed or fluted metal roofs, form deck-mounting flange at perimeter bottom to conform to roof profile.
3. Fabricate curbs to minimum height of 12 inches above roofing surface unless otherwise indicated.
4. Top Surface: Level top of curb, with roof slope accommodated by sloping deck-mounting flange.
5. Sloping Roofs: Where roof slope exceeds 1:48, fabricate curb with perimeter curb height tapered to accommodate roof slope so that top surface of perimeter curb is level. Equip unit with water diverter or cricket on side that obstructs water flow.
6. Insulation: Factory insulated with 1-1/2-inch- thick glass-fiber board insulation.
7. Liner: Same material as curb, of manufacturer's standard thickness and finish.
8. Wind Restraint Straps and Base Flange Attachment: Provide wind restraint straps, welded strap connectors, and base flange attachment to roof structure at perimeter of curb, of size and spacing required to meet wind uplift requirements.
9. Platform Cap: Where portion of roof curb is not covered by equipment, provide weathertight platform cap formed from 3/4-inch-thick plywood covered with metal sheet of same type, thickness, and finish as required for curb.
10. Metal Counterflashing: Manufacturer's standard, removable, fabricated of same metal and finish as curb.

2.3 EQUIPMENT SUPPORTS

A. Equipment Supports: Rail-type metal equipment supports capable of supporting superimposed live and dead loads between structural supports, including equipment loads and other construction indicated on Drawings, spanning between structural supports; capable of meeting performance requirements; with welded corner joints, stepped integral metal cant raised the thickness of roof insulation, and integrally formed structure-mounting flange at bottom.

B. Size: Coordinate dimensions with roughing-in information or Shop Drawings of equipment to be supported.

C. Supported Load Capacity: Coordinate load capacity with information on Shop Drawings of equipment to be supported.

D. Material: Zinc-coated (galvanized) steel sheet, 0.052 inch thick.

1. Finish: Mill phosphatized.
2. Color: As indicated by manufacturer's designations.

E. Construction:

1. Curb Profile: Manufacturer's standard compatible with roofing system.
2. Nailer: Factory-installed continuous wood nailers 3-1/2 inches wide on top flange of equipment supports, continuous around support perimeter.
3. Wind Restraint Straps and Base Flange Attachment: Provide wind restraint straps, welded strap connectors, and base flange attachment to roof structure at perimeter of curb of size and spacing required to meet wind uplift requirements.
4. Platform Cap: Where portion of equipment support is not covered by equipment, provide weathertight platform cap formed from 3/4-inch-thick plywood covered with metal sheet of same type, thickness, and finish as required for curb.
5. Metal Counterflashing: Manufacturer's standard, removable, fabricated of same metal and finish as equipment support.
6. Fabricate equipment supports to minimum height of 12 inches above roofing surface unless otherwise indicated.
7. Sloping Roofs: Where roof slope exceeds 1:48, fabricate each support with height to accommodate roof slope so that tops of supports are level with each other. Equip supports with water diverters or crickets on sides that obstruct water flow.

2.4 PIPE AND DUCT SUPPORTS

A. Fixed-Height Cradle-Type Pipe Supports: Polycarbonate pipe stand accommodating up to 1-1/2-inch- diameter pipe or conduit; with provision for pipe retainer and with manufacturer's support pad or deck plate as recommended for penetration-free installation over roof membrane type; as required for quantity of pipe runs and sizes.

B. Adjustable-Height Structure-Mounted Pipe Supports: Extruded-aluminum tube, filled with urethane insulation; 2 inches in diameter; accommodating up to 7-inch- diameter pipe or conduit, with provision for pipe retainer; with aluminum baseplate, EPDM base seal, manufacturer's recommended hardware for mounting to structure or structural roof deck as indicated, stainless steel roller and retainer, and extruded-aluminum carrier assemblies; as required for quantity of pipe runs and sizes.

C. Curb-Mounted Pipe Supports: Galvanized steel support with welded or mechanically fastened and sealed corner joints, stepped integral metal cant raised the thickness of roof insulation, and integrally formed deck-mounting flange at perimeter bottom; with adjustable-height roller-bearing pipe support accommodating up to 20-inch- diameter pipe or conduit and with provision for pipe retainer; as required for quantity of pipe runs and sizes.

D. Duct Supports: Extruded-aluminum, urethane-insulated supports, 2 inches in diameter; with manufacturer's recommended hardware for mounting to structure or structural roof deck.

1. Finish: Manufacturer's standard.

2.5 METAL MATERIALS

A. Zinc-Coated (Galvanized) Steel Sheet: ASTM A653/A653M, G90 coating designation.

1. Mill-Phosphatized Finish: Manufacturer's standard for field painting.

2. Factory Prime Coating: Where field painting is indicated, apply pretreatment and white or light-colored, factory-applied, baked-on epoxy primer coat, with a minimum dry film thickness of 0.2 mil.
3. Exposed Coil-Coated Finish: Prepainted by the coil-coating process to comply with ASTM A755/A755M. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

    a. Two-Coat Fluoropolymer Finish: AAMA 621. System consisting of primer and fluoropolymer color topcoat containing not less than 70 percent polyvinylidene fluoride (PVDF) resin by weight.

4. Baked-Enamel or Powder-Coat Finish: After cleaning and pretreating, apply manufacturer's standard two-coat, baked-on finish consisting of prime coat and thermosetting topcoat to a minimum dry film thickness of 2 mils.
5. Concealed Finish: Pretreat with manufacturer's standard white or light-colored acrylic or polyester-backer finish consisting of prime coat and wash coat, with a minimum total dry film thickness of 0.5 mil.

B. Aluminum-Zinc Alloy-Coated Steel Sheet: ASTM A792/A792M, AZ50 coated.

1. Factory Prime Coating: Where field painting is indicated, apply pretreatment and white or light-colored, factory-applied, baked-on epoxy primer coat, with a minimum dry film thickness of 0.2 mil.
2. Exposed Coil-Coated Finish: Prepainted by the coil-coating process to comply with ASTM A755/A755M. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

    a. Two-Coat Fluoropolymer Finish: AAMA 621. System consisting of primer and fluoropolymer color topcoat containing not less than 70 percent polyvinylidene fluoride (PVDF) resin by weight.

3. Baked-Enamel or Powder-Coat Finish: After cleaning and pretreating, apply manufacturer's standard two-coat, baked-on finish consisting of prime coat and thermosetting topcoat to a minimum dry film thickness of 2 mils.
4. Concealed Finish: Pretreat with manufacturer's standard white or light-colored acrylic or polyester-backer finish consisting of prime coat and wash coat, with a minimum total dry film thickness of 0.5 mil.

C. Aluminum Sheet: ASTM B209, manufacturer's standard alloy for finish required, with temper to suit forming operations and performance required.

1. Mill Finish: As manufactured.
2. Factory Prime Coating: Where field painting is indicated, apply pretreatment and white or light-colored, factory-applied, baked-on epoxy primer coat, with a minimum dry film thickness of 0.2 mil.
3. Baked-Enamel or Powder-Coat Finish: AAMA 2603 except with a minimum dry film thickness of 1.5 mils. Comply with coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.
4. Concealed Finish: Pretreat with manufacturer's standard white or light-colored acrylic or polyester-backer finish consisting of prime coat and wash coat, with a minimum total dry film thickness of 0.5 mil.

D. Aluminum Extrusions and Tubes: ASTM B221, manufacturer's standard alloy and temper for type of use, finished to match assembly where used; otherwise mill finished.

E. Stainless Steel Sheet and Shapes: ASTM A240/A240M or ASTM A666, Type 304.

F. Steel Shapes: ASTM A36/A36M, hot-dip galvanized according to ASTM A123/A123M unless otherwise indicated.

G. Steel Tube: ASTM A500/A500M, round tube.

H. Galvanized-Steel Tube: ASTM A500/A500M, round tube, hot-dip galvanized according to ASTM A123/A123M.

I. Steel Pipe: ASTM A53/A53M, galvanized.

2.6 MISCELLANEOUS MATERIALS

A. Provide materials and types of fasteners, protective coatings, sealants, and other miscellaneous items required by manufacturers for a complete installation.

B. Acrylic Glazing: ASTM D4802, thermoformable, monolithic sheet, manufacturer's standard, Type UVA (formulated with UV absorber), Finish 1 (smooth or polished).

C. Polycarbonate Glazing: Thermoformable, monolithic polycarbonate sheets manufactured by extrusion process, burglar-resistance rated according to UL 972 with an average impact strength of 12 to 16 ft-lbf/in. of width when tested according to ASTM D256, Method A (Izod).

D. Cellulosic-Fiber Board Insulation: ASTM C208, Type II, Grade 1, thickness as indicated.

E. Glass-Fiber Board Insulation: ASTM C726, nominal density of 3 lb/cu. ft., thermal resistivity of 4.3 deg F x h x sq. ft./Btu x in. at 75 deg F, thickness as indicated.

F. Polyisocyanurate Board Insulation: ASTM C1289, thickness and thermal resistivity as indicated.

G. Wood Nailers: Softwood lumber, pressure treated with waterborne preservatives for aboveground use, acceptable to authorities having jurisdiction, and complying with AWPA C2; not less than 1-1/2 inches thick.

H. Fasteners: Roof accessory manufacturer's recommended fasteners suitable for application and metals being fastened. Match finish of exposed fasteners with finish of material being fastened. Provide nonremovable fastener heads to exterior exposed fasteners. Furnish the following unless otherwise indicated:

1. Fasteners for Zinc-Coated or Aluminum-Zinc Alloy-Coated Steel: Series 300 stainless steel or hot-dip zinc-coated steel according to ASTM A153/A153M or ASTM F2329.
2. Fasteners for Aluminum Sheet: Aluminum or Series 300 stainless steel.
3. Fasteners for Stainless Steel Sheet: Series 300 stainless steel.

I. Gaskets: Manufacturer's standard tubular or fingered design of neoprene, EPDM, PVC, or silicone or a flat design of foam rubber, sponge neoprene, or cork.

J. Elastomeric Sealant: ASTM C920, elastomeric polyurethane polymer sealant as recommended by roof accessory manufacturer for installation indicated; low modulus; of type, grade, class, and use classifications required to seal joints and remain watertight.

K. Butyl Sealant: ASTM C1311, single-component, solvent-release butyl rubber sealant; polyisobutylene plasticized; heavy bodied for expansion joints with limited movement.

L. Asphalt Roofing Cement: ASTM D4586/D4586M, asbestos free, of consistency required for application.

2.7 GENERAL FINISH REQUIREMENTS

A. Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

B. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, to verify actual locations, dimensions, and other conditions affecting performance of the Work.

B. Verify that substrate is sound, dry, smooth, clean, sloped for drainage, and securely anchored.

C. Verify dimensions of roof openings for roof accessories.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install roof accessories according to manufacturer's written instructions.

1. Install roof accessories level; plumb; true to line and elevation; and without warping, jogs in alignment, buckling, or tool marks.
2. Anchor roof accessories securely in place so they are capable of resisting indicated loads.
3. Use fasteners, separators, sealants, and other miscellaneous items as required to complete installation of roof accessories and fit them to substrates.
4. Install roof accessories to resist exposure to weather without failing, rattling, leaking, or loosening of fasteners and seals.

B. Metal Protection: Protect metals against galvanic action by separating dissimilar metals from contact with each other or with corrosive substrates by painting contact surfaces with bituminous coating or by other permanent separation as recommended by manufacturer.

1. Coat concealed side of uncoated aluminum roof accessories with bituminous coating where in contact with wood, ferrous metal, or cementitious construction.

2. Underlayment: Where installing roof accessories directly on cementitious or wood substrates, install a course of underlayment and cover with manufacturer's recommended slip sheet.
3. Bed flanges in thick coat of asphalt roofing cement where required by manufacturers of roof accessories for waterproof performance.

C. Roof Curb Installation: Install each roof curb so top surface is level.

D. Equipment Support Installation: Install equipment supports so top surfaces are level with each other.

E. Pipe Support Installation: Comply with MSS SP-58 and MSS SP-89. Install supports and attachments as required to properly support piping. Arrange for grouping of parallel runs of horizontal piping, and support together.

1. Pipes of Various Sizes: Space supports for smallest pipe size or install intermediate supports for smaller diameter pipes as specified for individual pipe hangers.

F. Seal joints with elastomeric sealant as required by roof accessory manufacturer.

3.3 REPAIR AND CLEANING

A. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing according to ASTM A780/A780M.

B. Touch up factory-primed surfaces with compatible primer ready for field painting.

C. Clean exposed surfaces according to manufacturer's written instructions.

D. Clean off excess sealants.

E. Replace roof accessories that have been damaged or that cannot be successfully repaired by finish touchup or similar minor repair procedures.

END OF SECTION 230500

SECTION 230713 - DUCT INSULATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes insulating the following duct services:

1. Outdoor, exposed supply and return.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product indicated. Include thermal conductivity, water-vapor permeance thickness, and jackets (both factory- and field-applied if any).

B. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.

1. Detail application of protective shields, saddles, and inserts at hangers for each type of insulation and hanger.
2. Detail insulation application at elbows, fittings, dampers, specialties and flanges for each type of insulation.
3. Detail application of field-applied jackets.
4. Detail application at linkages of control devices.

1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified Installer.

B. Material Test Reports: From a qualified testing agency acceptable to authorities having jurisdiction indicating, interpreting, and certifying test results for compliance of insulation materials, sealers, attachments, cements, and jackets, with requirements indicated. Include dates of tests and test methods employed.

C. Field quality-control reports.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Skilled mechanics who have successfully completed an apprenticeship program or another craft training program certified by the Department of Labor, Bureau of Apprenticeship and Training.

B. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products according to ASTM E84, by a testing agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing agency.

1. Insulation Installed Outdoors: Flame-spread index of 75 or less, and smoke-developed index of 150 or less.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Packaging: Insulation material containers shall be marked by manufacturer with appropriate ASTM standard designation, type and grade, and maximum use temperature.

1.7 COORDINATION

A. Coordinate clearance requirements with duct Installer for duct insulation application. Before preparing ductwork Shop Drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance.

B. Coordinate installation and testing of heat tracing.

1.8 SCHEDULING

A. Schedule insulation application after pressure testing systems and, where required, after installing and testing heat tracing. Insulation application may begin on segments that have satisfactory test results.

B. Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

PART 2 - PRODUCTS

2.1 INSULATION MATERIALS

A. Comply with requirements in "Duct Insulation Schedule, General," "Indoor Duct and Plenum Insulation Schedule," and "Aboveground, Outdoor Duct and Plenum Insulation Schedule" articles for where insulating materials shall be applied.

B. Products shall not contain asbestos, lead, mercury, or mercury compounds.

C. Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C871.

D. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C795.

E. Mineral-Fiber Board Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C612, Type IA or Type IB. For duct and plenum applications, provide insulation with factory-applied FSK jacket. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.

2.2 ADHESIVES

A. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated unless otherwise indicated.

B. FSK Jacket Adhesive: Comply with MIL-A-3316C, Class 2, Grade A for bonding insulation jacket lap seams and joints.

2.3 MASTICS AND COATINGS

A. Materials shall be compatible with insulation materials, jackets, and substrates.

B. Vapor-Retarder Mastic: Solvent based; suitable for outdoor use on below ambient services.

1. Water-Vapor Permeance: Comply with ASTM C755, Section 7.2.2, Table 2, for insulation type and service conditions.
2. Service Temperature Range: Minus 50 to plus 220 deg F.
3. Color: White.

C. Breather Mastic: Water based; suitable for indoor and outdoor use on above ambient services.

1. Water-Vapor Permeance: ASTM E96, greater than 1.0 perm at manufacturer's recommended dry film thickness.
2. Service Temperature Range: Minus 20 to plus 180 deg F.
3. Color: White.

2.4 LAGGING ADHESIVES

A. Description: Comply with MIL-A-3316C, Class I, Grade A and shall be compatible with insulation materials, jackets, and substrates.

1. Fire-resistant, water-based lagging adhesive and coating for use indoors to adhere fire-resistant lagging cloths over duct insulation.
2. Service Temperature Range: 0 to plus 180 deg F.
3. Color: White.

2.5 SEALANTS

A. FSK Flashing Sealants:

1. Materials shall be compatible with insulation materials, jackets, and substrates.
2. Fire- and water-resistant, flexible, elastomeric sealant.
3. Service Temperature Range: Minus 40 to plus 250 deg F.
4. Color: Aluminum.

2.6 FACTORY-APPLIED JACKETS

A. Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:

1. FSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C1136, Type II.

2.7 FIELD-APPLIED FABRIC-REINFORCING MESH

A. Woven Glass-Fiber Fabric: Approximately 6 oz./sq. yd. with a thread count of 5 strands by 5 strands/sq. in. for covering ducts.

B. Woven Polyester Fabric: Approximately 1 oz./sq. yd. with a thread count of 10 strands by 10 strands/sq. in., in a Leno weave, for ducts.

2.8 FIELD-APPLIED CLOTHS

A. Woven Glass-Fiber Fabric: Comply with MIL-C-20079H, Type I, plain weave, and presized a minimum of 8 oz./sq. yd.

2.9 FIELD-APPLIED JACKETS

A. Field-applied jackets shall comply with ASTM C921, Type I, unless otherwise indicated.

B. Self-Adhesive Outdoor Jacket: 60-mil-thick, laminated vapor barrier and waterproofing membrane for installation over insulation located aboveground outdoors; consisting of a rubberized bituminous resin on a cross-laminated polyethylene film covered with white aluminum-foil facing.

2.10 TAPES

A. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive; suitable for indoor and outdoor applications.

1. Width: 2 inches.
2. Thickness: 6 mils.
3. Adhesion: 64 ounces force/inch in width.
4. Elongation: 500 percent.
5. Tensile Strength: 18 lbf/inch in width.

2.11 SECUREMENTS

A. Insulation Pins and Hangers:

1. Capacitor-Discharge-Weld Pins: Copper- or zinc-coated steel pin, fully annealed for capacitor-discharge welding, 0.106-inch- diameter shank, length to suit depth of insulation indicated.

2. Cupped-Head, Capacitor-Discharge-Weld Pins: Copper- or zinc-coated steel pin, fully annealed for capacitor-discharge welding, 0.106-inch- diameter shank, length to suit depth of insulation indicated with integral 1-1/2-inch galvanized carbon-steel washer.

3. Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch-thick, aluminum sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches in diameter.

   a. Protect ends with capped self-locking washers incorporating a spring steel insert to ensure permanent retention of cap in exposed locations.

4. Nonmetal Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch-thick nylon sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches in diameter.

B. Staples: Outward-clinching insulation staples, nominal 3/4-inch-wide, stainless steel or Monel.

C. Wire: 0.080-inch nickel-copper alloy.

2.12 CORNER ANGLES

A. Aluminum Corner Angles: 0.040 inch thick, minimum 1 by 1 inch, aluminum according to ASTM B209, Alloy 3003, 3005, 3105, or 5005; Temper H-14.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions for compliance with requirements for installation tolerances and other conditions affecting performance of insulation application.

1. Verify that systems to be insulated have been tested and are free of defects.
2. Verify that surfaces to be insulated are clean and dry.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Surface Preparation: Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.

3.3 GENERAL INSTALLATION REQUIREMENTS

A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of ducts and fittings.

B. Install insulation materials, vapor barriers or retarders, jackets, and thicknesses required for each item of duct system as specified in insulation system schedules.

C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.

D. Install insulation with longitudinal seams at top and bottom of horizontal runs.

E. Install multiple layers of insulation with longitudinal and end seams staggered.

F. Keep insulation materials dry during application and finishing.

G. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

H. Install insulation with least number of joints practical.

I. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.

1. Install insulation continuously through hangers and around anchor attachments.
2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic.
3. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.

J. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

K. Install insulation with factory-applied jackets as follows:

1. Draw jacket tight and smooth.
2. Cover circumferential joints with 3-inch-wide strips, of same material as insulation jacket. Secure strips with adhesive and outward clinching staples along both edges of strip, spaced 4 inches o.c.
3. Overlap jacket longitudinal seams at least 1-1/2 inches. Clean and dry surface to receive self-sealing lap. Staple laps with outward clinching staples along edge at 2 inches o.c.

    a. For below ambient services, apply vapor-barrier mastic over staples.

4. Cover joints and seams with tape, according to insulation material manufacturer's written instructions, to maintain vapor seal.
5. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints and at ends adjacent to duct flanges and fittings.

L. Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness.

M. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

N. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches similar to butt joints.

3.4 PENETRATIONS

A. Insulation Installation at Roof Penetrations: Install insulation continuously through roof penetrations.

1. Seal penetrations with flashing sealant.
2. For applications requiring only indoor insulation, terminate insulation above roof surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
3. Extend jacket of outdoor insulation outside roof flashing at least 2 inches below top of roof flashing.
4. Seal jacket to roof flashing with flashing sealant.

3.5 INSTALLATION OF MINERAL-FIBER INSULATION

A. Board Insulation Installation on Ducts and Plenums: Secure with adhesive and insulation pins.

1. Apply adhesives according to manufacturer's recommended coverage rates per unit area, for 100 percent coverage of duct and plenum surfaces.
2. Apply adhesive to entire circumference of ducts and to all surfaces of fittings and transitions.
3. Install either capacitor-discharge-weld pins and speed washers or cupped-head, capacitor-discharge-weld pins on sides and bottom of horizontal ducts and sides of vertical ducts as follows:

    a. On duct sides with dimensions 18 inches and smaller, place pins along longitudinal centerline of duct. Space 3-inches maximum from insulation end joints, and 16 inches o.c.
    b. On duct sides with dimensions larger than 18 inches, space pins 16 inches o.c. each way, and 3 inches maximum from insulation joints. Install additional pins to hold insulation tightly against surface at cross bracing.
    c. Pins may be omitted from top surface of horizontal, rectangular ducts and plenums.
    d. Do not overcompress insulation during installation.
    e. Cut excess portion of pins extending beyond speed washers or bend parallel with insulation surface. Cover exposed pins and washers with tape matching insulation facing.

4. For ducts and plenums with surface temperatures below ambient, install a continuous unbroken vapor barrier. Create a facing lap for longitudinal seams and end joints with insulation by removing 2 inches from one edge and one end of insulation segment. Secure laps to adjacent insulation section with 1/2-inch outward-clinching staples, 1 inch o.c. Install vapor barrier consisting of factory- or field-applied jacket, adhesive, vapor-barrier mastic, and sealant at joints, seams, and protrusions.

    a. Repair punctures, tears, and penetrations with tape or mastic to maintain vapor-barrier seal.

b. Install vapor stops for ductwork and plenums operating below 50 deg F at 18-foot intervals. Vapor stops shall consist of vapor-barrier mastic applied in a Z-shaped pattern over insulation face, along butt end of insulation, and over the surface. Cover insulation face and surface to be insulated a width equal to two times the insulation thickness, but not less than 3 inches.

5. Install insulation on rectangular duct elbows and transitions with a full insulation section for each surface. Groove and score insulation to fit as closely as possible to outside and inside radius of elbows. Install insulation on round and flat-oval duct elbows with individually mitered gores cut to fit the elbow.
6. Insulate duct stiffeners, hangers, and flanges that protrude beyond insulation surface with 6-inch-wide strips of same material used to insulate duct. Secure on alternating sides of stiffener, hanger, and flange with pins spaced 6 inches o.c.

3.6 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

B. Perform tests and inspections.

C. Tests and Inspections:

1. Inspect ductwork, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to one location(s) for each duct system defined in the "Duct Insulation Schedule, General" Article.

D. All insulation applications will be considered defective Work if sample inspection reveals noncompliance with requirements.

3.7 DUCT INSULATION SCHEDULE, GENERAL

A. Plenums and Ducts Requiring Insulation:

1. Outdoor, exposed supply and return.

2. Outdoor, exposed lab hood exhaust

B. Items Not Insulated:

1. Fibrous-glass ducts.
2. Metal ducts with duct liner of sufficient thickness to comply with energy code and ASHRAE/IESNA 90.1.
3. Factory-insulated flexible ducts.
4. Factory-insulated plenums and casings.
5. Flexible connectors.
6. Vibration-control devices.
7. Factory-insulated access panels and doors.

3.8 ABOVEGROUND, DUCT AND PLENUM INSULATION SCHEDULE

A. Mineral-Fiber Board: R-12 (approx. 3 inches thick).

3.9 OUTDOOR, FIELD-APPLIED JACKET SCHEDULE

A. Install jacket over insulation material. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.

B. If more than one material is listed, selection from materials listed is Contractor's option.

C. Ducts and Plenums, Exposed, up to 48 Inches in Diameter or with Flat Surfaces up to 72 Inches:

1. Self-Adhesive Outdoor Jacket

END OF SECTION 230713

SECTION 230719 - HVAC PIPING INSULATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes insulation for HVAC piping systems.

B. Related Requirements:

1. Section 230713 "Duct Insulation" for duct insulation.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product. Include thermal conductivity, water-vapor permeance thickness, and jackets (both factory and field applied if any).

1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified Installer.

B. Material Test Reports: From a qualified testing agency acceptable to authorities having jurisdiction indicating, interpreting, and certifying test results for compliance of insulation materials, sealers, attachments, cements, and jackets, with requirements indicated. Include dates of tests and test methods employed.

C. Field quality-control reports.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Skilled mechanics who have successfully completed an apprenticeship program or another craft training program certified by the Department of Labor, Bureau of Apprenticeship and Training.

B. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products in accordance with ASTM E84, by a testing agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing agency.

1. Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.

2. Insulation Installed Outdoors: Flame-spread index of 75 or less, and smoke-developed index of 150 or less.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Packaging: Insulation material containers shall be marked by manufacturer with appropriate ASTM standard designation, type and grade, and maximum use temperature.

1.7 COORDINATION

A. Coordinate clearance requirements with piping Installer for piping insulation application. Before preparing piping Shop Drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance.

1.8 SCHEDULING

A. Schedule insulation application after pressure testing systems and, where required, after installing and testing heat tracing. Insulation application may begin on segments that have satisfactory test results.

B. Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

PART 2 - PRODUCTS

2.1 INSULATION MATERIALS

A. Comply with requirements in "Piping Insulation Schedule, General," "Indoor Piping Insulation Schedule," "Outdoor, Aboveground Piping Insulation Schedule," and "Outdoor, Underground Piping Insulation Schedule" articles for where insulating materials shall be applied.

B. Products shall not contain asbestos, lead, mercury, or mercury compounds.

C. Products that come into contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested in accordance with ASTM C871.

D. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable in accordance with ASTM C795.

E. Flexible Elastomeric: Closed-cell, sponge- or expanded-rubber materials. Comply with ASTM C534/C534M, Type I for tubular materials, Type II for sheet materials.

2.2 ADHESIVES

A. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated unless otherwise indicated.

B. Flexible Elastomeric Adhesive: Solvent-based adhesive.

1. Flame-spread index shall be 25 or less and smoke-developed index shall be 50 or less as tested in accordance with ASTM E84.
2. Wet Flash Point: Below 0 deg F.
3. Service Temperature Range: 40 to 200 deg F.
4. Color: Black.

2.3 LAGGING ADHESIVES

A. Adhesives shall comply with MIL-A-3316C, Class I, Grade A, and shall be compatible with insulation materials, jackets, and substrates.

1. Fire-resistant, water-based lagging adhesive and coating for use indoors to adhere fire-resistant lagging cloths over pipe insulation.
2. Service Temperature Range: 20 to plus 180 deg F.
3. Color: White.

2.4 SEALANTS

A. Materials shall be as recommended by the insulation manufacturer and shall be compatible with insulation materials, jackets, and substrates.

B. Joint Sealants:

1. Permanently flexible, elastomeric sealant.

a. Service Temperature Range: Minus 150 to plus 250 deg F.
b. Color: White or gray.

2.5 FIELD-APPLIED JACKETS

A. Field-applied jackets shall comply with ASTM C1136, Type I, unless otherwise indicated.

B. PVC Jacket: High-impact-resistant, UV-resistant PVC complying with ASTM D1784, Class 16354-C; thickness as scheduled; roll stock ready for shop or field cutting and forming. Thickness is indicated in field-applied jacket schedules.

1. Adhesive: As recommended by jacket material manufacturer.
2. Color: White.
3. Factory-fabricated fitting covers to match jacket if available; otherwise, field fabricate.

a. Shapes: 45- and 90-degree, short- and long-radius elbows, tees, valves, flanges, unions, reducers, end caps, soil-pipe hubs, traps, mechanical joints, and P-trap and supply covers for lavatories.

2.6 TAPES

A. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive; suitable for indoor and outdoor applications.

1. Width: 2 inches.
2. Thickness: 6 mils.
3. Adhesion: 64 ounces force/inch in width.
4. Elongation: 500 percent.
5. Tensile Strength: 18 lbf/inch in width.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions for compliance with requirements for installation tolerances and other conditions affecting performance of insulation application.

1. Verify that systems to be insulated have been tested and are free of defects.
2. Verify that surfaces to be insulated are clean and dry.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.

B. Clean and prepare surfaces to be insulated. Before insulating, apply a corrosion coating to insulated surfaces as follows:

1. Stainless Steel: Coat 300 series stainless steel with an epoxy primer 5 mils thick and an epoxy finish 5 mils thick if operating in a temperature range between 140 and 300 deg F. Consult coating manufacturer for appropriate coating materials and application methods for operating temperature range.
2. Carbon Steel: Coat carbon steel operating at a service temperature of between 32 and 300 deg F with an epoxy coating. Consult coating manufacturer for appropriate coating materials and application methods for operating temperature range.

C. Coordinate insulation installation with the tradesman installing heat tracing. Comply with requirements for heat tracing that apply to insulation.

D. Mix insulating cements with clean potable water; if insulating cements are to be in contact with stainless steel surfaces, use demineralized water.

3.3 GENERAL INSTALLATION REQUIREMENTS

A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of piping, including fittings, valves, and specialties.

B. Install insulation materials, forms, vapor barriers or retarders, jackets, and of thicknesses required for each item of pipe system, as specified in insulation system schedules.

C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.

D. Install insulation with longitudinal seams at top and bottom of horizontal runs.

E. Install multiple layers of insulation with longitudinal and end seams staggered.

F. Do not weld brackets, clips, or other attachment devices to piping, fittings, and specialties.

G. Keep insulation materials dry during storage, application, and finishing. Replace insulation materials that get wet.

H. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

I. Install insulation with least number of joints practical.

J. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

K. Cut insulation in a manner to avoid compressing insulation more than 25 percent of its nominal thickness.

L. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

M. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches in similar fashion to butt joints.

N. For above-ambient services, do not install insulation to the following:

1. Vibration-control devices.
2. Testing agency labels and stamps.
3. Nameplates and data plates.

3.4 PENETRATIONS

A. Insulation Installation at Roof Penetrations: Install insulation continuously through roof penetrations.

1. Seal penetrations with flashing sealant.

2. For applications requiring only indoor insulation, terminate insulation above roof surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
3. Extend jacket of outdoor insulation outside roof flashing at least 2 inches below top of roof flashing.
4. Seal jacket to roof flashing with flashing sealant.

B. Insulation Installation at Underground Exterior Wall Penetrations: Terminate insulation flush with sleeve seal. Seal terminations with flashing sealant.

C. Insulation Installation at Aboveground Exterior Wall Penetrations: Install insulation continuously through wall penetrations.

1. Seal penetrations with flashing sealant.
2. For applications requiring only indoor insulation, terminate insulation inside wall surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
3. Extend jacket of outdoor insulation outside wall flashing and overlap wall flashing at least 2 inches.
4. Seal jacket to wall flashing with flashing sealant.

3.5 GENERAL PIPE INSULATION INSTALLATION

A. Requirements in this article generally apply to all insulation materials, except where more specific requirements are specified in various pipe insulation material installation articles.

3.6 INSTALLATION OF FLEXIBLE ELASTOMERIC INSULATION

A. Seal longitudinal seams and end joints with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

B. Insulation Installation on Pipe Fittings and Elbows:

1. Install mitered sections of pipe insulation.
2. Secure insulation materials and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

C. Insulation Installation on Valves and Pipe Specialties:

1. Install preformed valve covers manufactured of same material as that of pipe insulation when available.
2. When preformed valve covers are not available, install cut sections of pipe and sheet insulation to valve body. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
3. Install insulation to flanges as specified for flange insulation application.
4. Secure insulation to valves and specialties, and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

3.7 INSTALLATION OF FIELD-APPLIED JACKETS

A. Where glass-cloth jackets are indicated, install directly over bare insulation or insulation with factory-applied jackets.

1. Draw jacket smooth and tight to surface with 2-inch overlap at seams and joints.
2. Embed glass cloth between two 0.062-inch-thick coats of lagging adhesive.
3. Completely encapsulate insulation with coating, leaving no exposed insulation.

B. Where PVC jackets are indicated and for horizontal applications, install with 1-inch overlap at longitudinal seams and end joints. Seal with manufacturer's recommended adhesive.

1. Apply two continuous beads of adhesive to seams and joints, one bead under lap and the finish bead along seam and joint edge.

3.8 FINISHES

A. Flexible Elastomeric Thermal Insulation: After adhesive has fully cured, apply two coats of insulation manufacturer's recommended protective coating.

3.9 FIELD QUALITY CONTROL

A. Owner will engage a qualified testing agency to perform tests and inspections.

B. Engage a qualified testing agency to perform tests and inspections.

C. Manufacturer's Field Service: Engage a factory-authorized service representative to test and inspect components, assemblies, and equipment installations, including connections.

D. Perform tests and inspections.

E. All insulation applications will be considered defective if they do not pass tests and inspections.

F. Prepare test and inspection reports.

3.10 PIPING INSULATION SCHEDULE, GENERAL

A. Insulation conductivity and thickness per pipe size shall comply with schedules in this Section or with requirements of authorities having jurisdiction, whichever is more stringent.

B. Acceptable preformed pipe and tubular insulation materials and thicknesses are identified for each piping system and pipe size range. If more than one material is listed for a piping system, selection from materials listed is Contractor's option.

C. Items Not Insulated: Unless otherwise indicated, do not install insulation on the following:

1. Underground piping.
2. Chrome-plated pipes and fittings unless there is a potential for personnel injury.

3.11 OUTDOOR, ABOVEGROUND PIPING INSULATION SCHEDULE

A. Refrigerant Piping:

1. All Pipe Sizes: Insulation shall be the following:

a. Flexible Elastomeric: 2 inches thick.

3.12 OUTDOOR, FIELD-APPLIED JACKET SCHEDULE

A. Install jacket over insulation material. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.

B. If more than one material is listed, selection from materials listed is Contractor's option.

C. Piping, Exposed:

1. PVC: 20 mils thick.

END OF SECTION 230719

SECTION 232300 - REFRIGERANT PIPING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Copper tube and fittings.
2. Refrigerants.

1.3 INFORMATIONAL SUBMITTALS

A. Field quality-control reports.

1.4 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For refrigerant valves and piping specialties to include in maintenance manuals.

1.5 QUALITY ASSURANCE

A. Comply with ASHRAE 15, "Safety Code for Refrigeration Systems."

B. Comply with ASME B31.5, "Refrigeration Piping and Heat Transfer Components."

1.6 PRODUCT STORAGE AND HANDLING

A. Store piping with end caps in place to ensure that piping interior and exterior are clean when installed.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Line Test Pressure for Refrigerant R-407C:

1. Suction Lines for Air-Conditioning Applications: 230 psig.
2. Suction Lines for Heat-Pump Applications: 380 psig.
3. Hot-Gas and Liquid Lines: 380 psig.

2.2 COPPER TUBE AND FITTINGS

A. Copper Tube: ASTM B 280, Type ACR.

B. Wrought-Copper Fittings, Brazed-Joint: ASME B16.50.

C. Wrought-Copper Unions: ASME B16.22.

D. Brazing Filler Metals: AWS A5.8/A5.8M.

2.3 REFRIGERANTS

A. ASHRAE 34, R-407C: Difluoromethane/Pentafluoroethane/1,1,1,2-Tetrafluoroethane.

PART 3 - EXECUTION

3.1 PIPING APPLICATIONS FOR REFRIGERANT R-407C

A. Suction Lines for Conventional Air-Conditioning Applications: Copper, Type ACR, annealed-temper tubing and wrought-copper fittings with brazed joints.

3.2 INSTALLATION OF PIPING, GENERAL

A. Install refrigerant piping according to ASHRAE 15.

B. Install piping adjacent to machines to allow service and maintenance.

C. Install piping free of sags and bends.

D. Install fittings for changes in direction and branch connections.

E. Select system components with pressure rating equal to or greater than system operating pressure.

F. Install piping as short and direct as possible, with a minimum number of joints, elbows, and fittings.

G. When brazing, remove solenoid-valve coils and sight glasses; also remove valve stems, seats, and packing, and accessible internal parts of refrigerant specialties. Do not apply heat near expansion-valve bulb.

3.3 PIPE JOINT CONSTRUCTION

A. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.

B. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.

C. Brazed Joints: Construct joints according to AWS's "Brazing Handbook," Chapter "Pipe and Tube."

1. Use Type BCuP (copper-phosphorus) alloy for joining copper socket fittings with copper pipe.
2. Use Type BAg (cadmium-free silver) alloy for joining copper with bronze or steel.

3.4 INSTALLATION OF HANGERS AND SUPPORTS

A. Install the following pipe attachments:

1. Adjustable steel clevis hangers for individual horizontal runs less than 20 feet long.
2. Roller hangers and spring hangers for individual horizontal runs 20 feet or longer.
3. Pipe Roller: MSS SP-58, Type 44 for multiple horizontal piping 20 feet or longer, supported on a trapeze.
4. Spring hangers to support vertical runs.
5. Copper-clad hangers and supports for hangers and supports in direct contact with copper pipe.

B. Install hangers for copper tubing, with maximum horizontal spacing and minimum rod diameters, to comply with MSS-58, locally enforced codes, and authorities having jurisdiction requirements, whichever are most stringent.

C. Support horizontal piping within 12 inches of each fitting.

D. Support vertical runs of copper tubing to comply with MSS-58, locally enforced codes, and authorities having jurisdiction requirements, whichever are most stringent.

3.5 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Comply with ASME B31.5, Chapter VI.
2. Test refrigerant piping, specialties, and receivers. Isolate compressor, condenser, evaporator, and safety devices from test pressure if they are not rated above the test pressure.
3. Test high- and low-pressure side piping of each system separately at not less than the pressures indicated in "Performance Requirements" Article.

    a. Fill system with nitrogen to the required test pressure.
    b. System shall maintain test pressure at the manifold gage throughout duration of test.
    c. Test joints and fittings with electronic leak detector or by brushing a small amount of soap and glycerin solution over joints.
    d. Remake leaking joints using new materials, and retest until satisfactory results are achieved.

B. Prepare test and inspection reports.

3.6 SYSTEM CHARGING

A. Charge system using the following procedures:

1. Install core in filter dryers after leak test but before evacuation.
2. Evacuate entire refrigerant system with a vacuum pump to 500 micrometers. If vacuum holds for 12 hours, system is ready for charging.
3. Break vacuum with refrigerant gas, allowing pressure to build up to 2 psig.
4. Charge system with a new filter-dryer core in charging line.

3.7 ADJUSTING

A. Adjust thermostatic expansion valve to obtain proper evaporator superheat.

B. Adjust high- and low-pressure switch settings to avoid short cycling in response to fluctuating suction pressure.

C. Adjust set-point temperature of air-conditioning or chilled-water controllers to the system design temperature.

D. Perform the following adjustments before operating the refrigeration system, according to manufacturer's written instructions:

1. Open shutoff valves in condenser water circuit.
2. Verify that compressor oil level is correct.
3. Open compressor suction and discharge valves.
4. Open refrigerant valves except bypass valves that are used for other purposes.
5. Check open compressor-motor alignment and verify lubrication for motors and bearings.

END OF SECTION 232300

SECTION 233113 - METAL DUCTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Single-wall rectangular ducts and fittings.
2. Sheet metal materials.
3. Sealants and gaskets.
4. Hangers and supports.

1.3 DEFINITIONS

A. OSHPD: Office of Statewide Health Planning and Development (State of California).

1.4 ACTION SUBMITTALS

A. Product Data: For each type of the following products:

1. Sealants and gaskets.

B. Shop Drawings:

1. Fabrication, assembly, and installation, including plans, elevations, sections, components, and attachments to other work.
2. Factory- and shop-fabricated ducts and fittings.
3. Duct layout indicating sizes, configuration, liner material, and static-pressure classes.
4. Fittings.
5. Reinforcement and spacing.
6. Seam and joint construction.
7. Penetrations through fire-rated and other partitions.
8. Equipment installation based on equipment being used on Project.
9. Locations for duct accessories, including dampers, turning vanes, and access doors and panels.
10. Hangers and supports, including methods for duct and building attachment and vibration isolation.

C. Delegated-Design Submittal:

1. Sheet metal thicknesses.
2. Joint and seam construction and sealing.
3. Reinforcement details and spacing.
4. Materials, fabrication, assembly, and spacing of hangers and supports.
5. Design Calculations: Calculations for selecting hangers and supports.

1.5 INFORMATIONAL SUBMITTALS

A. Coordination Drawings: A single set of plans or BIM model, drawn to scale, showing the items described in this Section, and coordinated with all building trades.

B. Welding certificates.

C. Field quality-control reports.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Delegated Duct Design: Duct construction, including sheet metal thicknesses, seam and joint construction, reinforcements, and hangers and supports, shall comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible" and with performance requirements and design criteria indicated in "Duct Schedule" Article.

B. Structural Performance: Duct hangers and supports shall withstand the effects of gravity loads and stresses within limits and under conditions described in SMACNA's "HVAC Duct Construction Standards - Metal and Flexible".

C. Airstream Surfaces: Surfaces in contact with airstream shall comply with requirements in ASHRAE 62.1.

D. ASHRAE Compliance: Applicable requirements in ASHRAE 62.1, Section 5 - "Systems and Equipment," and Section 7 - "Construction and System Startup."

E. ASHRAE/IES Compliance: Applicable requirements in ASHRAE/IES 90.1, Section 6.4.4 - "HVAC System Construction and Insulation."

F. Duct Dimensions: Unless otherwise indicated, all duct dimensions indicated on Drawings are inside clear dimensions and do not include insulation or duct wall thickness.

2.2 SINGLE-WALL RECTANGULAR DUCTS AND FITTINGS

A. General Fabrication Requirements: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible" based on indicated static-pressure class unless otherwise indicated.

1. Construct ducts of galvanized sheet steel unless otherwise indicated.

2. For ducts exposed to weather, cover with jacketing as noted in duct insulation specification.

B. Transverse Joints: Fabricate joints in accordance with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 2-1, "Rectangular Duct/Transverse Joints," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards - Metal and Flexible."

1. For ducts with longest side less than 36 inches, select joint types in accordance with Figure 2-1.
2. For ducts with longest side 36 inches or greater, use flange joint connector Type T-22, T-24, T-24A, T-25a, or T-25b. Factory-fabricated flanged duct connection system may be used if submitted and approved by engineer of record.

C. Longitudinal Seams: Select seam types and fabricate in accordance with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 2-2, "Rectangular Duct/Longitudinal Seams," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards - Metal and Flexible." All longitudinal seams shall be Pittsburgh lock seams unless otherwise specified for specific application.

D. Elbows, Transitions, Offsets, Branch Connections, and Other Duct Construction: Select types and fabricate in accordance with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Ch. 4, "Fittings and Other Construction," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards - Metal and Flexible."

2.3 SHEET METAL MATERIALS

A. General Material Requirements: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible" for acceptable materials, material thicknesses, and duct construction methods unless otherwise indicated. Sheet metal materials shall be free of pitting, seam marks, roller marks, stains, discolorations, and other imperfections.

B. Galvanized Sheet Steel: Comply with ASTM A653/A653M.

1. Galvanized Coating Designation: G90.
2. Finishes for Surfaces Exposed to View: Mill phosphatized.

C. Aluminum Sheets: Comply with ASTM B 209 (ASTM B 209M) Alloy 3003, H14 temper; with mill finish for concealed ducts, and standard, one-side bright finish for duct surfaces exposed to view.

D. Reinforcement Shapes and Plates: ASTM A36/A36M, steel plates, shapes, and bars; black and galvanized.

1. Where black- and galvanized-steel shapes and plates are used to reinforce aluminum ducts, isolate the different metals with butyl rubber, neoprene, or EPDM gasket materials.

E. Tie Rods: Galvanized steel, 1/4-inch-minimum diameter for lengths 36 inches or less; 3/8-inch-minimum diameter for lengths longer than 36 inches.

2.4 SEALANT AND GASKETS

A. General Sealant and Gasket Requirements: Surface-burning characteristics for sealants and gaskets shall be a maximum flame-spread index of 25 and a maximum smoke-developed index of 50 when tested in accordance with UL 723; certified by an NRTL.

B. Two-Part Tape Sealing System:

1. Tape: Woven cotton fiber impregnated with mineral gypsum and modified acrylic/silicone activator to react exothermically with tape to form hard, durable, airtight seal.
2. Tape Width: 3 inches.
3. Sealant: Modified styrene acrylic.
4. Water resistant.
5. Mold and mildew resistant.
6. Maximum Static-Pressure Class: 10-inch wg, positive and negative.
7. Service: Indoor and outdoor.
8. Service Temperature: Minus 40 to plus 200 deg F.
9. Substrate: Compatible with galvanized sheet steel (both PVC coated and bare), stainless steel, or aluminum.

C. Water-Based Joint and Seam Sealant:

1. Application Method: Brush on.
2. Solids Content: Minimum 65 percent.
3. Shore A Hardness: Minimum 20.
4. Water resistant.
5. Mold and mildew resistant.
6. VOC: Maximum 75 g/L (less water).
7. Maximum Static-Pressure Class: 10-inch wg, positive and negative.
8. Service: Indoor or outdoor.
9. Substrate: Compatible with galvanized sheet steel (both PVC coated and bare), stainless steel, or aluminum sheets.

D. Solvent-Based Joint and Seam Sealant:

1. Application Method: Brush on.
2. Base: Synthetic rubber resin.
3. Solvent: Toluene and heptane.
4. Solids Content: Minimum 60 percent.
5. Shore A Hardness: Minimum 60.
6. Water resistant.
7. Mold and mildew resistant.
8. Maximum Static-Pressure Class: 10-inch wg, positive or negative.
9. Service: Indoor or outdoor.
10. Substrate: Compatible with galvanized sheet steel (both PVC coated and bare), stainless steel, or aluminum sheets.

E. Flanged Joint Sealant: Comply with ASTM C920.

1. General: Single-component, acid-curing, silicone, elastomeric.
2. Type: S.
3. Grade: NS.

4. Class: 25.
5. Use: O.

F. Flange Gaskets: Butyl rubber, neoprene, or EPDM polymer with polyisobutylene plasticizer.

2.5 HANGERS AND SUPPORTS

A. Hanger Rods for Noncorrosive Environments: Galvanized-steel rods and nuts.

B. Hanger Rods for Corrosive Environments: Electrogalvanized, all-thread rods or galvanized rods with threads painted with zinc-chromate primer after installation.

C. Strap and Rod Sizes: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Table 5-1, "Rectangular Duct Hangers Minimum Size," and Table 5-2, "Minimum Hanger Sizes for Round Duct."

D. Steel Cables for Galvanized-Steel Ducts: Galvanized steel complying with ASTM A603.

E. Steel Cable End Connections: Galvanized-steel assemblies with brackets, swivel, and bolts designed for duct hanger service; with an automatic-locking and clamping device.

F. Duct Attachments: Sheet metal screws, blind rivets, or self-tapping metal screws; compatible with duct materials.

G. Trapeze and Riser Supports:

1. Supports for Galvanized-Steel Ducts: Galvanized-steel shapes and plates.
2. Supports for Stainless-Steel Ducts: Stainless-steel shapes and plates.
3. Supports for Aluminum Ducts: Aluminum or galvanized steel coated with zinc chromate.

PART 3 - EXECUTION

3.1 DUCT INSTALLATION

A. Drawing plans, schematics, and diagrams indicate general location and arrangement of duct system. Indicated duct locations, configurations, and arrangements were used to size ducts and calculate friction loss for air-handling equipment sizing and for other design considerations. Install duct systems as indicated unless deviations to layout are approved on Shop Drawings and coordination drawings.

B. Install ducts in accordance with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible" unless otherwise indicated.

C. Install ducts in maximum practical lengths with fewest possible joints.

D. Install factory- or shop-fabricated fittings for changes in direction, size, and shape and for branch connections.

E. Unless otherwise indicated, install ducts vertically and horizontally, and parallel and perpendicular to building lines.

F. Install ducts close to walls, overhead construction, columns, and other structural and permanent enclosure elements of building.

G. Install ducts with a clearance of 1 inch, plus allowance for insulation thickness.

H. Where ducts pass through exterior walls and are exposed to view, cover the opening between the partition and duct or duct insulation with sheet metal flanges of same metal thickness as the duct. Overlap openings on four sides by at least 1-1/2 inches.

I. Protect duct interiors from moisture, construction debris and dust, and other foreign materials both before and after installation. Comply with SMACNA's "IAQ Guidelines for Occupied Buildings Under Construction," Appendix G, "Duct Cleanliness for New Construction Guidelines."

J. Elbows: Use long-radius elbows wherever they fit.

1. Fabricate 90-degree rectangular mitered elbows to include turning vanes.
2. Fabricate 90-degree round elbows with a minimum of three segments for 12 inches and smaller and a minimum of five segments for 14 inches and larger.

K. Branch Connections: Use lateral or conical branch connections.

3.2 DUCT SEALING

A. Seal ducts for duct static-pressure, seal classes, and leakage classes specified in "Duct Schedule" Article in accordance with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible."

B. Seal ducts at a minimum to the following seal classes in accordance with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible":

1. Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible."
2. Seal Class A.

3.3 HANGER AND SUPPORT INSTALLATION

A. Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Chapter 5, "Hangers and Supports."

B. Building Attachments: Concrete inserts, powder-actuated fasteners, or structural-steel fasteners appropriate for construction materials to which hangers are being attached.

1. Where practical, install concrete inserts before placing concrete.
2. Install powder-actuated concrete fasteners after concrete is placed and completely cured.
3. Use powder-actuated concrete fasteners for standard-weight aggregate concretes or for slabs more than 4 inches thick.
4. Do not use powder-actuated concrete fasteners for lightweight-aggregate concretes or for slabs less than 4 inches thick.
5. Do not use powder-actuated concrete fasteners for seismic restraints.

C. Hanger Spacing: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Table 5-1, "Rectangular Duct Hangers Minimum Size," and Table 5-2, "Minimum

Hanger Sizes for Round Duct," for maximum hanger spacing; install hangers and supports within 24 inches of each elbow and within 48 inches of each branch intersection.

D. Hangers Exposed to View: Threaded rod and angle or channel supports.

E. Support vertical ducts with steel angles or channel secured to the sides of the duct with welds, bolts, sheet metal screws, or blind rivets; support at each floor and at a maximum intervals of 16 feet.

F. Install upper attachments to structures. Select and size upper attachments with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

3.4 FIELD QUALITY CONTROL

A. Perform tests and inspections.

B. Leakage Tests:

1. Comply with SMACNA's "HVAC Air Duct Leakage Test Manual." Submit a test report for each test.
2. Test the following systems:

    a. Ducts with a Pressure Class Higher Than 3-Inch wg: Test representative duct sections totaling no less than 25 percent of total installed duct area for each designated pressure class.

3. Disassemble, reassemble, and seal segments of systems to accommodate leakage testing and for compliance with test requirements.
4. Testing of each duct section is to be performed with access doors, coils, filters, dampers, and other duct-mounted devices in place as designed. No devices are to be removed or blanked off so as to reduce or prevent additional leakage.
5. Test for leaks before applying external insulation.
6. Conduct tests at static pressures equal to maximum design pressure of system or section being tested. If static-pressure classes are not indicated, test system at maximum system design pressure. Do not pressurize systems above maximum design operating pressure.
7. Give seven days' advance notice for testing.

C. Duct System Cleanliness Tests:

1. Visually inspect duct system to ensure that no visible contaminants are present.
2. Test sections of metal duct system, chosen randomly by Owner, for cleanliness in accordance with "Description of Method 3 - NADCA Vacuum Test" in NADCA ACR, "Assessment, Cleaning and Restoration of HVAC Systems."

    a. Acceptable Cleanliness Level: Net weight of debris collected on the filter media shall not exceed 0.75 mg/100 sq. cm.

D. Duct system will be considered defective if it does not pass tests and inspections.

E. Prepare test and inspection reports.

3.5 DUCT SCHEDULE

A. Fabricate ducts with galvanized sheet steel except as otherwise indicated and as follows:

1. Lab exhaust ductwork to be aluminum

2. Fabricate all ducts to achieve SMACNA pressure class, seal class, and leakage class as indicated below.

B. Ducts:

1. Pressure Class: Positive 3-inch wg.
2. Minimum SMACNA Seal Class: A.

C. Intermediate Reinforcement:

1. Galvanized-Steel Ducts: Galvanized steel.

D. Elbow Configuration:

1. Rectangular Duct: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-2, "Rectangular Elbows."

a. Velocity 1000 fpm or Lower:

1) Radius Type RE 1 with minimum 0.5 radius-to-diameter ratio.
2) Mitered Type RE 4 without vanes.

b. Velocity 1000 to 1500 fpm:

1) Radius Type RE 1 with minimum 1.0 radius-to-diameter ratio.
2) Radius Type RE 3 with minimum 0.5 radius-to-diameter ratio and two vanes.
3) Mitered Type RE 2 with vanes complying with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-3, "Vanes and Vane Runners," and Figure 4-4, "Vane Support in Elbows."

c. Velocity 1500 fpm or Higher:

1) Radius Type RE 1 with minimum 1.5 radius-to-diameter ratio.
2) Radius Type RE 3 with minimum 1.0 radius-to-diameter ratio and two vanes.
3) Mitered Type RE 2 with vanes complying with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-3, "Vanes and Vane Runners," and Figure 4-4, "Vane Support in Elbows."

2. Rectangular Duct: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-2, "Rectangular Elbows."

a. Radius Type RE 1 with minimum 1.5 radius-to-diameter ratio.
b. Radius Type RE 3 with minimum 1.0 radius-to-diameter ratio and two vanes.
c. Mitered Type RE 2 with vanes complying with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-3, "Vanes and Vane Runners," and Figure 4-4, "Vane Support in Elbows."

E. Branch Configuration:

1. Rectangular Duct: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-6, "Branch Connection."

a. Rectangular Main to Rectangular Branch: 45-degree entry.
b. Rectangular Main to Round Branch: Conical spin in.

END OF SECTION 233113

SECTION 260526 - GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes grounding and bonding systems and equipment.

B. Section includes grounding and bonding systems and equipment, plus the following special applications:

1. Underground distribution grounding.
2. Foundation steel electrodes.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

1.4 INFORMATIONAL SUBMITTALS

A. Coordination Drawings: Plans showing dimensioned locations of grounding features specified in "Field Quality Control" Article, including the following:

1. Test wells.
2. Ground rods.
3. Ground rings.
4. Grounding arrangements and connections for separately derived systems.

B. Qualification Data: For testing agency and testing agencies field supervisor.

C. Field quality-control reports.

1.5 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For grounding to include in emergency, operation, and maintenance manuals.

a. Plans showing as-built, dimensioned locations of grounding features specified in "Field Quality Control" Article, including the following:

1) Test wells.
2) Ground rods.
3) Ground rings.
4) Grounding arrangements and connections for separately derived systems.

b. Instructions for periodic testing and inspection of grounding features at test wells based on NFPA 70B.

1) Tests shall determine if ground-resistance or impedance values remain within specified maximums, and instructions shall recommend corrective action if values do not.
2) Include recommended testing intervals.

1.6 QUALITY ASSURANCE

A. Testing Agency Qualifications: Certified by NETA.

PART 2 - PRODUCTS

2.1 SYSTEM DESCRIPTION

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Comply with UL 467 for grounding and bonding materials and equipment.

2.2 CONDUCTORS

A. Insulated Conductors: Copper wire or cable insulated for 600 V unless otherwise required by applicable Code or authorities having jurisdiction.

B. Bare Copper Conductors:

1. Solid Conductors: ASTM B 3.
2. Stranded Conductors: ASTM B 8.
3. Bonding Cable: 28 kcmil, 14 strands of No. 17 AWG conductor, 1/4 inch in diameter.
4. Bonding Conductor: No. 4 AWG, stranded conductor.
5. Bonding Jumper: Copper tape, braided conductors terminated with copper ferrules; 1-5/8 inches wide and 1/16 inch thick.

C. Grounding Bus: Predrilled rectangular bars of annealed copper, 1/4 by 4 inches in cross section, with 9/32-inch holes spaced 1-1/8 inches apart. Stand-off insulators for mounting shall comply with UL 891 for use in switchboards, 600 V and shall be Lexan or PVC, impulse tested at 5000 V.

2.3 CONNECTORS

A. Listed and labeled by an NRTL acceptable to authorities having jurisdiction for applications in which used and for specific types, sizes, and combinations of conductors and other items connected.

B. Welded Connectors: Exothermic-welding kits of types recommended by kit manufacturer for materials being joined and installation conditions.

C. Bus-Bar Connectors: Mechanical type, cast silicon bronze, solderless exothermic-type wire terminals, and long-barrel, two-bolt connection to ground bus bar.

D. Bus-Bar Connectors: Compression type, copper or copper alloy, with two wire terminals.

E. Beam Clamps: Mechanical type, terminal, ground wire access from four directions, with dual, tin-plated or silicon bronze bolts.

F. Cable-to-Cable Connectors: Compression type, copper or copper alloy.

G. Conduit Hubs: Mechanical type, terminal with threaded hub.

H. Ground Rod Clamps: Mechanical type, copper or copper alloy, terminal with hex head bolt.

I. Service Post Connectors: Mechanical type, bronze alloy terminal, in short- and long-stud lengths, capable of single and double conductor connections.

J. Signal Reference Grid Clamp: Mechanical type, stamped-steel terminal with hex head screw.

K. Straps: Solid copper, copper lugs. Rated for 600 A.

L. Tower Ground Clamps: Mechanical type, copper or copper alloy, terminal one-piece clamp.

M. U-Bolt Clamps: Mechanical type, copper or copper alloy, terminal listed for direct burial.

N. Water Pipe Clamps:

1. Mechanical type, two pieces with zinc-plated bolts.

    a. Material: Die-cast zinc alloy.
    b. Listed for direct burial.

2. U-bolt type with malleable-iron clamp and copper ground connector.

2.4 GROUNDING ELECTRODES

A. Ground Rods: Copper-clad steel 3/4 inch by 10 feet.

B. Chemical-Enhanced Grounding Electrodes: Copper tube, straight or L-shaped, charged with nonhazardous electrolytic chemical salts.

1. Termination: Factory-attached No. 4/0 AWG bare conductor at least 48 inches long.
2. Backfill Material: Electrode manufacturer's recommended material.

C. Ground Plates: 1/4 inch thick, hot-dip galvanized.

PART 3 - EXECUTION

3.1 APPLICATIONS

A. Conductors: Install solid conductor for No. 8 AWG and smaller, and stranded conductors for No. 6 AWG and larger unless otherwise indicated.

B. Underground Grounding Conductors: Install bare copper conductor, No. 4/0 AWG minimum.

1. Bury at least 24 inches below grade.
2. Duct-Bank Grounding Conductor: Bury 12 inches above duct bank when indicated as part of duct-bank installation.

C. Grounding Bus: Install in electrical equipment rooms, in rooms housing service equipment, and elsewhere as indicated.

1. Install bus horizontally, on insulated spacers 2 inches minimum from wall, 6 inches above finished floor unless otherwise indicated.
2. Where indicated on both sides of doorways, route bus up to top of door frame, across top of doorway, and down; connect to horizontal bus.

D. Conductor Terminations and Connections:

1. Pipe and Equipment Grounding Conductor Terminations: Bolted connectors.
2. Underground Connections: Welded connectors except at test wells and as otherwise indicated.
3. Connections to Ground Rods at Test Wells: Bolted connectors.
4. Connections to Structural Steel: Welded connectors.

3.2 GROUNDING AT THE SERVICE

A. Equipment grounding conductors and grounding electrode conductors shall be connected to the ground bus. Install a main bonding jumper between the neutral and ground buses.

3.3 GROUNDING UNDERGROUND DISTRIBUTION SYSTEM COMPONENTS

A. Comply with IEEE C2 grounding requirements.

B. Grounding Manholes and Handholes: Install a driven ground rod through manhole or handhole floor, close to wall, and set rod depth so 4 inches will extend above finished floor. If necessary, install ground rod before manhole is placed and provide No. 1/0 AWG bare, tinned-copper conductor from ground rod into manhole through a waterproof sleeve in manhole wall. Protect ground rods passing through concrete floor with a double wrapping of pressure-sensitive insulating tape or heat-shrunk insulating sleeve from 2 inches above to 6 inches below concrete. Seal floor opening with waterproof, nonshrink grout.

C. Grounding Connections to Manhole Components: Bond exposed-metal parts such as inserts, cable racks, pulling irons, ladders, and cable shields within each manhole or handhole, to ground rod or grounding conductor. Make connections with No. 4 AWG minimum, stranded, hard-drawn copper bonding conductor. Train conductors level or plumb around corners and fasten to manhole walls. Connect to cable armor and cable shields according to written instructions by manufacturer of splicing and termination kits.

D. Pad-Mounted Transformers and Switches: Install two ground rods and ground ring around the pad. Ground pad-mounted equipment and noncurrent-carrying metal items associated with substations by connecting them to underground cable and grounding electrodes. Install tinned-copper conductor not less than No. 2 AWG for ground ring and for taps to equipment grounding terminals. Bury ground ring not less than 6 inches from the foundation.

3.4 EQUIPMENT GROUNDING

A. Install insulated equipment grounding conductors with all feeders and branch circuits.

B. Install insulated equipment grounding conductors with the following items, in addition to those required by NFPA 70:

1. Feeders and branch circuits.
2. Lighting circuits.
3. Receptacle circuits.
4. Single-phase motor and appliance branch circuits.
5. Three-phase motor and appliance branch circuits.
6. Flexible raceway runs.

C. Poles Supporting Outdoor Lighting Fixtures: Install grounding electrode and a separate insulated equipment grounding conductor in addition to grounding conductor installed with branch-circuit conductors.

3.5 INSTALLATION

A. Grounding Conductors: Route along shortest and straightest paths possible unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.

B. Ground Rods: Drive rods until tops are 2 inches below finished floor or final grade unless otherwise indicated.

1. Interconnect ground rods with grounding electrode conductor below grade and as otherwise indicated. Make connections without exposing steel or damaging coating if any.
2. For grounding electrode system, install at least three rods spaced at least one-rod length from each other and located at least the same distance from other grounding electrodes, and connect to the service grounding electrode conductor.

C. Test Wells: Ground rod driven through drilled hole in bottom of handhole. Handholes shall be at least 12 inches deep, with cover.

1. Install at least one test well for each service unless otherwise indicated. Install at the ground rod electrically closest to service entrance. Set top of test well flush with finished grade or floor.

D. Bonding Straps and Jumpers: Install in locations accessible for inspection and maintenance except where routed through short lengths of conduit.

1. Bonding to Structure: Bond straps directly to basic structure, taking care not to penetrate any adjacent parts.
2. Bonding to Equipment Mounted on Vibration Isolation Hangers and Supports: Install bonding so vibration is not transmitted to rigidly mounted equipment.
3. Use exothermic-welded connectors for outdoor locations; if a disconnect-type connection is required, use a bolted clamp.

E. Grounding and Bonding for Piping:

1. Metal Water Service Pipe: Install insulated copper grounding conductors, in conduit, from building's main service equipment, or grounding bus, to main metal water service entrances to building. Connect grounding conductors to main metal water service pipes; use a bolted clamp connector or bolt a lug-type connector to a pipe flange by using one of the lug bolts of the flange. Where a dielectric main water fitting is installed, connect grounding conductor on street side of fitting. Bond metal grounding conductor conduit or sleeve to conductor at each end.
2. Water Meter Piping: Use braided-type bonding jumpers to electrically bypass water meters. Connect to pipe with a bolted connector.
3. Bond each aboveground portion of gas piping system downstream from equipment shutoff valve.

F. Bonding Interior Metal Ducts: Bond metal air ducts to equipment grounding conductors of associated fans, blowers, electric heaters, and air cleaners. Install bonding jumper to bond across flexible duct connections to achieve continuity.

G. Ground Ring: Install a grounding conductor, electrically connected to each building structure ground rod and to each steel column, extending around the perimeter of building.

1. Install tinned-copper conductor not less than No. 2/0 AWG for ground ring and for taps to building steel.
2. Bury ground ring not less than 24 inches from building's foundation.

3.6 FIELD QUALITY CONTROL

A. Manufacturer's Field Service: Engage a factory-authorized service representative to test and inspect components, assemblies, and equipment installations, including connections.

B. Perform tests and inspections with the assistance of a factory-authorized service representative.

C. Tests and Inspections:

1. After installing grounding system but before permanent electrical circuits have been energized, test for compliance with requirements.
2. Inspect physical and mechanical condition. Verify tightness of accessible, bolted, electrical connections with a calibrated torque wrench according to manufacturer's written instructions.
3. Test completed grounding system at each location where a maximum ground-resistance level is specified, at service disconnect enclosure grounding terminal, at ground test wells, and at individual ground rods. Make tests at ground rods before any conductors are connected.

    a. Measure ground resistance no fewer than two full days after last trace of precipitation and without soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural ground resistance.
    b. Perform tests by fall-of-potential method according to IEEE 81.

4. Prepare dimensioned Drawings locating each test well, ground rod and ground-rod assembly, and other grounding electrodes. Identify each by letter in alphabetical order, and key to the record of tests and observations. Include the number of rods driven and their depth at each location, and include observations of weather and other phenomena that may affect test results. Describe measures taken to improve test results.

D. Grounding system will be considered defective if it does not pass tests and inspections.

E. Prepare test and inspection reports.

F. Report measured ground resistances that exceed the following values:

1. Power and Lighting Equipment or System with Capacity of 500 kVA and Less: 10 ohms.
2. Power and Lighting Equipment or System with Capacity of 500 to 1000 kVA: 5 ohms.
3. Power and Lighting Equipment or System with Capacity More Than 1000 kVA: 3 ohms.
4. Power Distribution Units or Panelboards Serving Electronic Equipment: 3 ohms.
5. Substations and Pad-Mounted Equipment: 5 ohms.
6. Manhole Grounds: 10 ohms.

G. Excessive Ground Resistance: If resistance to ground exceeds specified values, notify Architect promptly and include recommendations to reduce ground resistance.

END OF SECTION 260526

SECTION 260529 - HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Hangers and supports for electrical equipment and systems.
2. Construction requirements for concrete bases.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for the following:

a. Hangers.
b. Steel slotted support systems.
c. Nonmetallic support systems.
d. Trapeze hangers.
e. Clamps.
f. Turnbuckles.
g. Sockets.
h. Eye nuts.
i. Saddles.
j. Brackets.

2. Include rated capacities and furnished specialties and accessories.

B. Shop Drawings: For fabrication and installation details for electrical hangers and support systems.

1. Trapeze hangers. Include product data for components.
2. Steel slotted-channel systems.
3. Nonmetallic slotted-channel systems.
4. Equipment supports.
5. Vibration Isolation Base Details: Detail fabrication, including anchorages and attachments to structure and to supported equipment. Include adjustable motor bases, rails, and frames for equipment mounting.

C. Delegated-Design Submittal: For hangers and supports for electrical systems.

1. Include design calculations and details of trapeze hangers.
2. Include design calculations for seismic restraints.

1.4 INFORMATIONAL SUBMITTALS

A. Coordination Drawings: Reflected ceiling plan(s) and other details, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of the items involved:

1. Suspended ceiling components.
2. Structural members to which hangers and supports will be attached.

B. Welding certificates.

1.5 QUALITY ASSURANCE

A. Welding Qualifications: Qualify procedures and personnel according to the following:

1. AWS D1.1/D1.1M.
2. AWS D1.2/D1.2M.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Hangers and supports shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1. The term "withstand" means "the supported equipment and systems will remain in place without separation of any parts when subjected to the seismic forces specified and the system will be fully operational after the seismic event."
2. Component Importance Factor: 1.5.

B. Surface-Burning Characteristics: Comply with ASTM E 84; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

1. Flame Rating: Class 1.
2. Self-extinguishing according to ASTM D 635.

2.2 SUPPORT, ANCHORAGE, AND ATTACHMENT COMPONENTS

A. Steel Slotted Support Systems: Comply with MFMA-4 factory-fabricated components for field assembly.

1. Material: Galvanized steel.
2. Channel Width: 1-5/8 inches.
3. Metallic Coatings: Hot-dip galvanized after fabrication and applied according to MFMA-4.
4. Nonmetallic Coatings: Manufacturer's standard PVC, polyurethane, or polyester coating applied according to MFMA-4.
5. Painted Coatings: Manufacturer's standard painted coating applied according to MFMA-4.
6. Protect finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.
7. Channel Dimensions: Selected for applicable load criteria.

B. Conduit and Cable Support Devices: Steel hangers, clamps, and associated fittings, designed for types and sizes of raceway or cable to be supported.

C. Support for Conductors in Vertical Conduit: Factory-fabricated assembly consisting of threaded body and insulating wedging plug or plugs for nonarmored electrical conductors or cables in riser conduits. Plugs shall have number, size, and shape of conductor gripping pieces as required to suit individual conductors or cables supported. Body shall be made of malleable iron.

D. Structural Steel for Fabricated Supports and Restraints: ASTM A 36/A 36M steel plates, shapes, and bars; black and galvanized.

E. Mounting, Anchoring, and Attachment Components: Items for fastening electrical items or their supports to building surfaces include the following:

1. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete, steel, or wood, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.
2. Mechanical-Expansion Anchors: Insert-wedge-type, zinc-coated steel, for use in hardened portland cement concrete, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.
3. Concrete Inserts: Steel or malleable-iron, slotted support system units are similar to MSS Type 18 units and comply with MFMA-4 or MSS SP-58.
4. Clamps for Attachment to Steel Structural Elements: MSS SP-58 units are suitable for attached structural element.
5. Through Bolts: Structural type, hex head, and high strength. Comply with ASTM A 325.
6. Toggle Bolts: All-steel springhead type.
7. Hanger Rods: Threaded steel.

2.3 FABRICATED METAL EQUIPMENT SUPPORT ASSEMBLIES

A. Description: Welded or bolted structural-steel shapes, shop or field fabricated to fit dimensions of supported equipment.

PART 3 - EXECUTION

3.1 APPLICATION

A. Comply with NECA 1 and NECA 101 for application of hangers and supports for electrical equipment and systems unless requirements in this Section are stricter.

B. Comply with requirements for raceways and boxes specified in Section 260533 "Raceways and Boxes for Electrical Systems."

C. Maximum Support Spacing and Minimum Hanger Rod Size for Raceway: Space supports for EMTs, IMCs, and RMCs as required by NFPA 70. Minimum rod size shall be 1/4 inch in diameter.

D. Multiple Raceways or Cables: Install trapeze-type supports fabricated with steel slotted support system, sized so capacity can be increased by at least 25 percent in future without exceeding specified design load limits.

1. Secure raceways and cables to these supports with two-bolt conduit clamps.

E. Spring-steel clamps designed for supporting single conduits without bolts may be used for 1-1/2-inch and smaller raceways serving branch circuits and communication systems above suspended ceilings and for fastening raceways to trapeze supports.

3.2 SUPPORT INSTALLATION

A. Comply with NECA 1 and NECA 101 for installation requirements except as specified in this article.

B. Raceway Support Methods: In addition to methods described in NECA 1, EMTs, IMCs, and RMCs may be supported by openings through structure members, according to NFPA 70.

C. Strength of Support Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb.

D. Mounting and Anchorage of Surface-Mounted Equipment and Components: Anchor and fasten electrical items and their supports to building structural elements by the following methods unless otherwise indicated by code:

1. To Wood: Fasten with lag screws or through bolts.
2. To New Concrete: Bolt to concrete inserts.
3. To Masonry: Approved toggle-type bolts on hollow masonry units and expansion anchor fasteners on solid masonry units.
4. To Existing Concrete: Expansion anchor fasteners.
5. Instead of expansion anchors, powder-actuated driven threaded studs provided with lock washers and nuts may be used in existing standard-weight concrete 4 inches thick or greater. Do not use for anchorage to lightweight-aggregate concrete or for slabs less than 4 inches thick.

6. To Steel: Welded threaded studs complying with AWS D1.1/D1.1M, with lock washers and nuts.
7. To Light Steel: Sheet metal screws.
8. Items Mounted on Hollow Walls and Nonstructural Building Surfaces: Mount cabinets, panelboards, disconnect switches, control enclosures, pull and junction boxes, transformers, and other devices on slotted-channel racks attached to substrate by means that comply with seismic-restraint strength and anchorage requirements.

E. Drill holes for expansion anchors in concrete at locations and to depths that avoid the need for reinforcing bars.

3.3 INSTALLATION OF FABRICATED METAL SUPPORTS

A. Comply with installation requirements in Section 055000 "Metal Fabrications" for site-fabricated metal supports.

B. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor electrical materials and equipment.

C. Field Welding: Comply with AWS D1.1/D1.1M.

3.4 CONCRETE BASES

A. Construct concrete bases of dimensions indicated but not less than 4 inches larger in both directions than supported unit, and so anchors will be a minimum of 10 bolt diameters from edge of the base.

B. Use 3000-psi, 28-day compressive-strength concrete. Concrete materials, reinforcement, and placement requirements are specified in Section 033000 "Cast-in-Place Concrete."

C. Anchor equipment to concrete base as follows:

1. Place and secure anchorage devices. Use supported equipment manufacturer's setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
2. Install anchor bolts to elevations required for proper attachment to supported equipment.
3. Install anchor bolts according to anchor-bolt manufacturer's written instructions.

3.5 PAINTING

A. Touchup: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.

1. Apply paint by brush or spray to provide minimum dry film thickness of 2.0 mils.

B. Touchup: Comply with requirements in Section 099113 "Exterior Painting", Section 099123 "Interior Painting" and Section 099600 "High-Performance Coatings" for cleaning and touchup

painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal.

C. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A 780.

END OF SECTION 260529

SECTION 260533 - RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Metal conduits, tubing, and fittings.
2. Nonmetal conduits, tubing, and fittings.
3. Metal wireways and auxiliary gutters.
4. Nonmetal wireways and auxiliary gutters.
5. Surface raceways.
6. Boxes, enclosures, and cabinets.
7. Handholes and boxes for exterior underground cabling.

1.3 DEFINITIONS

A. ARC: Aluminum rigid conduit.

B. GRC: Galvanized rigid steel conduit.

C. IMC: Intermediate metal conduit.

1.4 ACTION SUBMITTALS

A. Product Data: For surface raceways, wireways and fittings, floor boxes, hinged-cover enclosures, and cabinets.

B. Shop Drawings: For custom enclosures and cabinets. Include plans, elevations, sections, and attachment details.

1.5 INFORMATIONAL SUBMITTALS

A. Coordination Drawings: Conduit routing plans, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of items involved:

1. Structural members in paths of conduit groups with common supports.
2. HVAC and plumbing items and architectural features in paths of conduit groups with common supports.

B. Qualification Data: For professional engineer.

C. Seismic Qualification Certificates: For enclosures, cabinets, and conduit racks and their mounting provisions, including those for internal components, from manufacturer.

1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.
4. Detailed description of conduit support devices and interconnections on which the certification is based and their installation requirements.

D. Source quality-control reports.

PART 2 - PRODUCTS

2.1 METAL CONDUITS, TUBING, AND FITTINGS

A. Listing and Labeling: Metal conduits, tubing, and fittings shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. GRC: Comply with ANSI C80.1 and UL 6.

C. ARC: Comply with ANSI C80.5 and UL 6A.

D. IMC: Comply with ANSI C80.6 and UL 1242.

E. PVC-Coated Steel Conduit: PVC-coated rigid steel conduit.

1. Comply with NEMA RN 1.
2. Coating Thickness: 0.040 inch, minimum.

F. EMT: Comply with ANSI C80.3 and UL 797.

G. FMC: Comply with UL 1; zinc-coated steel.

H. LFMC: Flexible steel conduit with PVC jacket and complying with UL 360.

I. Fittings for Metal Conduit: Comply with NEMA FB 1 and UL 514B.

1. Fittings for EMT:

a. Material: Steel.
b. Type: Compression.

2. Expansion Fittings: PVC or steel to match conduit type, complying with UL 651, rated for environmental conditions where installed, and including flexible external bonding jumper.

3. Coating for Fittings for PVC-Coated Conduit: Minimum thickness of 0.040 inch, with overlapping sleeves protecting threaded joints.

J. Joint Compound for IMC, GRC, or ARC: Approved, as defined in NFPA 70, by authorities having jurisdiction for use in conduit assemblies, and compounded for use to lubricate and protect threaded conduit joints from corrosion and to enhance their conductivity.

2.2 METAL WIREWAYS AND AUXILIARY GUTTERS

A. Description: Sheet metal, complying with UL 870 and NEMA 250, Type 1 unless otherwise indicated, and sized according to NFPA 70.

1. Metal wireways installed outdoors shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Fittings and Accessories: Include covers, couplings, offsets, elbows, expansion joints, adapters, hold-down straps, end caps, and other fittings to match and mate with wireways as required for complete system.

C. Wireway Covers: Screw-cover type unless otherwise indicated.

D. Finish: Manufacturer's standard enamel finish.

2.3 SURFACE RACEWAYS

A. Listing and Labeling: Surface raceways and tele-power poles shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Surface Metal Raceways: Galvanized steel with snap-on covers complying with UL 5. Manufacturer's standard enamel finish in color selected by Architect.

C. Surface Nonmetallic Raceways: Two- or three-piece construction, complying with UL 5A, and manufactured of rigid PVC with texture and color selected by Architect. Product shall comply with UL 94 V-0 requirements for self-extinguishing characteristics.

2.4 BOXES, ENCLOSURES, AND CABINETS

A. General Requirements for Boxes, Enclosures, and Cabinets: Boxes, enclosures, and cabinets installed in wet locations shall be listed for use in wet locations.

B. Sheet Metal Outlet and Device Boxes: Comply with NEMA OS 1 and UL 514A.

C. Cast-Metal Outlet and Device Boxes: Comply with NEMA FB 1, ferrous alloy, Type FD, with gasketed cover.

D. Nonmetallic Outlet and Device Boxes: Comply with NEMA OS 2 and UL 514C.

E. Metal Floor Boxes:

1. Material: Cast metal or sheet metal.

2. Type: Fully adjustable.
3. Shape: Rectangular.
4. Listing and Labeling: Metal floor boxes shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

F. Luminaire Outlet Boxes: Nonadjustable, designed for attachment of luminaire weighing 50 lb. Outlet boxes designed for attachment of luminaires weighing more than 50 lb shall be listed and marked for the maximum allowable weight.

G. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.

H. Cast-Metal Access, Pull, and Junction Boxes: Comply with NEMA FB 1 and UL 1773, galvanized, cast iron with gasketed cover.

I. Box extensions used to accommodate new building finishes shall be of same material as recessed box.

J. Device Box Dimensions: 4 inches square by 2-1/8 inches deep.

K. Gangable boxes are allowed.

L. Hinged-Cover Enclosures: Comply with UL 50 and NEMA 250, Type 1 with continuous-hinge cover with flush latch unless otherwise indicated.

1. Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.
2. Interior Panels: Steel; all sides finished with manufacturer's standard enamel.

M. Cabinets:

1. NEMA 250, Type 1 galvanized-steel box with removable interior panel and removable front, finished inside and out with manufacturer's standard enamel.
2. Hinged door in front cover with flush latch and concealed hinge.
3. Key latch to match panelboards.
4. Metal barriers to separate wiring of different systems and voltage.
5. Accessory feet where required for freestanding equipment.
6. Nonmetallic cabinets shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2.5 HANDHOLES AND BOXES FOR EXTERIOR UNDERGROUND WIRING

A. General Requirements for Handholes and Boxes:

1. Boxes and handholes for use in underground systems shall be designed and identified as defined in NFPA 70, for intended location and application.
2. Boxes installed in wet areas shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Polymer-Concrete Handholes and Boxes with Polymer-Concrete Cover: Molded of sand and aggregate, bound together with polymer resin, and reinforced with steel, fiberglass, or a combination of the two.

1. Standard: Comply with SCTE 77.

2. Configuration: Designed for flush burial with closed bottom unless otherwise indicated.
3. Cover: Weatherproof, secured by tamper-resistant locking devices and having structural load rating consistent with enclosure and handhole location.
4. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.
5. Cover Legend: Molded lettering, "ELECTRIC".
6. Conduit Entrance Provisions: Conduit-terminating fittings shall mate with entering ducts for secure, fixed installation in enclosure wall.
7. Handholes 12 Inches Wide by 24 Inches Long and Larger: Have inserts for cable racks and pulling-in irons installed before concrete is poured.

2.6 SOURCE QUALITY CONTROL FOR UNDERGROUND ENCLOSURES

A. Handhole and Pull-Box Prototype Test: Test prototypes of handholes and boxes for compliance with SCTE 77. Strength tests shall be for specified tier ratings of products supplied.

1. Tests of materials shall be performed by an independent testing agency.
2. Strength tests of complete boxes and covers shall be by either an independent testing agency or manufacturer. A qualified registered professional engineer shall certify tests by manufacturer.
3. Testing machine pressure gages shall have current calibration certification complying with ISO 9000 and ISO 10012 and traceable to NIST standards.

PART 3 - EXECUTION

3.1 RACEWAY APPLICATION

A. Outdoors: Apply raceway products as specified below unless otherwise indicated:

1. Exposed Conduit: GRC.
2. Concealed Conduit, Aboveground: GRC.
3. Underground Conduit: RNC, Type EPC-40-PVC, concrete encased.
4. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): LFNC.
5. Boxes and Enclosures, Aboveground: NEMA 250, Type 3R.

B. Indoors: Apply raceway products as specified below unless otherwise indicated:

1. Exposed, Not Subject to Physical Damage: EMT.
2. Exposed, Not Subject to Severe Physical Damage: EMT.
3. Exposed and Subject to Severe Physical Damage: GRC. Raceway locations include the following:

    a. Loading dock.
    b. Corridors used for traffic of mechanized carts, forklifts, and pallet-handling units.
    c. Mechanical rooms.

4. Concealed in Ceilings and Interior Walls and Partitions: EMT.
5. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): FMC, except use LFMC in damp or wet locations.

6. Damp or Wet Locations: GRC.
7. Boxes and Enclosures: NEMA 250, Type 1, except use NEMA 250, Type 4 stainless steel in institutional and commercial kitchens and damp or wet locations.

C. Minimum Raceway Size: 3/4-inch trade size.

D. Raceway Fittings: Compatible with raceways and suitable for use and location.

1. Rigid and Intermediate Steel Conduit: Use threaded rigid steel conduit fittings unless otherwise indicated. Comply with NEMA FB 2.10.
2. PVC Externally Coated, Rigid Steel Conduits: Use only fittings listed for use with this type of conduit. Patch and seal all joints, nicks, and scrapes in PVC coating after installing conduits and fittings. Use sealant recommended by fitting manufacturer and apply in thickness and number of coats recommended by manufacturer.
3. EMT: Use compression, steel fittings. Comply with NEMA FB 2.10.
4. Flexible Conduit: Use only fittings listed for use with flexible conduit. Comply with NEMA FB 2.20.

E. Install nonferrous conduit or tubing for circuits operating above 60 Hz. Where aluminum raceways are installed for such circuits and pass through concrete, install in nonmetallic sleeve.

F. Do not install aluminum conduits, boxes, or fittings in contact with concrete or earth.

G. Install surface raceways only where indicated on Drawings.

H. Do not install nonmetallic conduit where ambient temperature exceeds 120 deg F.

3.2 INSTALLATION

A. Comply with NECA 1 and NECA 101 for installation requirements except where requirements on Drawings or in this article are stricter. Comply with NFPA 70 limitations for types of raceways allowed in specific occupancies and number of floors.

B. Keep raceways at least 6 inches away from parallel runs of flues and steam or hot-water pipes. Install horizontal raceway runs above water and steam piping.

C. Complete raceway installation before starting conductor installation.

D. Comply with requirements in Section 260529 "Hangers and Supports for Electrical Systems" for hangers and supports.

E. Arrange stub-ups so curved portions of bends are not visible above finished slab.

F. Install no more than the equivalent of three 90-degree bends in any conduit run except for control wiring conduits, for which fewer bends are allowed. Support within 12 inches of changes in direction.

G. Conceal conduit and EMT within finished walls, ceilings, and floors unless otherwise indicated. Install conduits parallel or perpendicular to building lines.

H. Support conduit within 12 inches of enclosures to which attached.

I. Raceways Embedded in Slabs:

1. Run conduit larger than 1-inch trade size, parallel or at right angles to main reinforcement. Where at right angles to reinforcement, place conduit close to slab support. Secure raceways to reinforcement at maximum 10-foot intervals.
2. Arrange raceways to cross building expansion joints at right angles with expansion fittings.
3. Arrange raceways to keep a minimum of 2 inches of concrete cover in all directions.
4. Do not embed thread less fittings in concrete unless specifically approved by Architect for each specific location.
5. Change from ENT to GRC before rising above floor.

J. Stub-ups to Above Recessed Ceilings:

1. Use EMT, IMC, or RMC for raceways.
2. Use a conduit bushing or insulated fitting to terminate stub-ups not terminated in hubs or in an enclosure.

K. Threaded Conduit Joints, Exposed to Wet, Damp, Corrosive, or Outdoor Conditions: Apply listed compound to threads of raceway and fittings before making up joints. Follow compound manufacturer's written instructions.

L. Coat field-cut threads on PVC-coated raceway with a corrosion-preventing conductive compound prior to assembly.

M. Raceway Terminations at Locations Subject to Moisture or Vibration: Use insulating bushings to protect conductors including conductors smaller than No. 4 AWG.

N. Terminate threaded conduits into threaded hubs or with locknuts on inside and outside of boxes or cabinets. Install bushings on conduits up to 1-1/4-inch trade size and insulated throat metal bushings on 1-1/2-inch trade size and larger conduits terminated with locknuts. Install insulated throat metal grounding bushings on service conduits.

O. Install raceways square to the enclosure and terminate at enclosures with locknuts. Install locknuts hand tight plus 1/4 turn more.

P. Do not rely on locknuts to penetrate nonconductive coatings on enclosures. Remove coatings in the locknut area prior to assembling conduit to enclosure to assure a continuous ground path.

Q. Cut conduit perpendicular to the length. For conduits 2-inch trade size and larger, use roll cutter or a guide to make cut straight and perpendicular to the length.

R. Install pull wires in empty raceways. Use polypropylene or monofilament plastic line with not less than 200-lb tensile strength. Leave at least 12 inches of slack at each end of pull wire. Cap underground raceways designated as spare above grade alongside raceways in use.

S. Surface Raceways:

1. Install surface raceway with a minimum 2-inch radius control at bend points.
2. Secure surface raceway with screws or other anchor-type devices at intervals not exceeding 48 inches and with no less than two supports per straight raceway section. Support surface raceway according to manufacturer's written instructions. Tape and glue are not acceptable support methods.

T. Install raceway sealing fittings at accessible locations according to NFPA 70 and fill them with listed sealing compound. For concealed raceways, install each fitting in a flush steel box with a blank cover plate having a finish similar to that of adjacent plates or surfaces. Install raceway sealing fittings according to NFPA 70.

U. Install devices to seal raceway interiors at accessible locations. Locate seals so no fittings or boxes are between the seal and the following changes of environments. Seal the interior of all raceways at the following points:

1. Where conduits pass from warm to cold locations, such as boundaries of refrigerated spaces.
2. Where an underground service raceway enters a building or structure.
3. Where otherwise required by NFPA 70.

V. Comply with manufacturer's written instructions for solvent welding RNC and fittings.

W. Expansion-Joint Fittings:

1. Install in each run of aboveground RNC that is located where environmental temperature change may exceed 30 deg F and that has straight-run length that exceeds 25 feet. Install in each run of aboveground RMC and EMT conduit that is located where environmental temperature change may exceed 100 deg F and that has straight-run length that exceeds 100 feet.
2. Install type and quantity of fittings that accommodate temperature change listed for each of the following locations:

    a. Outdoor Locations Not Exposed to Direct Sunlight: 125 deg F temperature change.
    b. Outdoor Locations Exposed to Direct Sunlight: 155 deg F temperature change.
    c. Indoor Spaces Connected with Outdoors without Physical Separation: 125 deg F temperature change.
    d. Attics: 135 deg F temperature change.

3. Install fitting(s) that provide expansion and contraction for at least 0.00041 inch per foot of length of straight run per deg F of temperature change for PVC conduits. Install fitting(s) that provide expansion and contraction for at least 0.000078 inch per foot of length of straight run per deg F of temperature change for metal conduits.
4. Install expansion fittings at all locations where conduits cross building or structure expansion joints.
5. Install each expansion-joint fitting with position, mounting, and piston setting selected according to manufacturer's written instructions for conditions at specific location at time of installation. Install conduit supports to allow for expansion movement.

X. Flexible Conduit Connections: Comply with NEMA RV 3. Use a maximum of 72 inches of flexible conduit for recessed and semi-recessed luminaires. Maximum of 36 inches of flexible conduits for equipment subject to vibration, noise transmission, or movement; and for transformers and motors.

1. Use LFMC in damp or wet locations subject to severe physical damage.
2. Use LFMC or LFNC in damp or wet locations not subject to severe physical damage.

Y. Mount boxes at heights indicated on Drawings. If mounting heights of boxes are not individually indicated, give priority to ADA requirements. Install boxes with height measured to top of box unless otherwise indicated.

Z. Recessed Boxes in Masonry Walls: Saw-cut opening for box in center of cell of masonry block, and install box flush with surface of wall. Prepare block surfaces to provide a flat surface for a rain tight connection between box and cover plate or supported equipment and box.

AA. Horizontally separate boxes mounted on opposite sides of walls so they are not in the same vertical channel.

BB. Locate boxes so that cover or plate will not span different building finishes.

CC. Support boxes of three gangs or more from more than one side by spanning two framing members or mounting on brackets specifically designed for the purpose.

DD. Fasten junction and pull boxes to or support from building structure. Do not support boxes by conduits.

EE. Set metal floor boxes level and flush with finished floor surface.

FF. Set nonmetallic floor boxes level. Trim after installation to fit flush with finished floor surface.

3.3 INSTALLATION OF UNDERGROUND CONDUIT

A. Direct-Buried Conduit:

1. Excavate trench bottom to provide firm and uniform support for conduit. Prepare trench bottom as specified in Section 312000 "Earth Moving" for pipe less than 6 inches in nominal diameter.
2. Install backfill as specified in Section 312000 "Earth Moving."
3. After installing conduit, backfill and compact. Start at tie-in point, and work toward end of conduit run, leaving conduit at end of run free to move with expansion and contraction as temperature changes during this process. Firmly hand tamp backfill around conduit to provide maximum supporting strength. After placing controlled backfill to within 12 inches of finished grade, make final conduit connection at end of run and complete backfilling with normal compaction as specified in Section 312000 "Earth Moving."
4. Install manufactured duct elbows for stub-ups at poles and equipment and at building entrances through floor unless otherwise indicated. Encase elbows for stub-up ducts throughout length of elbow.
5. Install manufactured rigid steel conduit elbows for stub-ups at poles and equipment and at building entrances through floor.

    a. Couple steel conduits to ducts with adapters designed for this purpose, and encase coupling with 3 inches of concrete for a minimum of 12 inches on each side of the coupling.
    b. For stub-ups at equipment mounted on outdoor concrete bases and where conduits penetrate building foundations, extend steel conduit horizontally a minimum of 60 inches from edge of foundation or equipment base. Install insulated grounding bushings on terminations at equipment.

6. Warning Planks: Bury warning planks approximately 12 inches above direct-buried conduits but a minimum of 6 inches below grade. Align planks along centerline of conduit.

3.4 INSTALLATION OF UNDERGROUND HANDHOLES AND BOXES

A. Install handholes and boxes level and plumb and with orientation and depth coordinated with connecting conduits to minimize bends and deflections required for proper entrances.

B. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 1/2-inch sieve to No. 4 sieve and compacted to same density as adjacent undisturbed earth.

C. Elevation: In paved areas, set so cover surface will be flush with finished grade. Set covers of other enclosures 1 inch above finished grade.

D. Install handholes with bottom below frost line.

E. Install removable hardware, including pulling eyes, cable stanchions, cable arms, and insulators, as required for installation and support of cables and conductors and as indicated. Select arm lengths to be long enough to provide spare space for future cables but short enough to preserve adequate working clearances in enclosure.

F. Field-cut openings for conduits according to enclosure manufacturer's written instructions. Cut wall of enclosure with a tool designed for material to be cut. Size holes for terminating fittings to be used, and seal around penetrations after fittings are installed.

3.5 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS

A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies. Comply with requirements in Section 260544 "Sleeves and Sleeve Seals for Electrical Raceways and Cabling."

3.6 FIRESTOPPING

A. Install firestopping at penetrations of fire-rated floor and wall assemblies. Comply with requirements in Section 078413 "Penetration Firestopping."

3.7 PROTECTION

A. Protect coatings, finishes, and cabinets from damage and deterioration.

1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
2. Repair damage to PVC coatings or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION 260533

SECTION 260544 - SLEEVES AND SLEEVE SEALS FOR ELECTRICAL RACEWAYS AND CABLING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Sleeves for raceway and cable penetration of non-fire-rated construction walls and floors.
2. Sleeve-seal systems.
3. Sleeve-seal fittings.
4. Grout
5. Silicone sealants.

B. Related Requirements:

1. Section 078413 "Penetration Firestopping" for penetration firestopping installed in fire-resistance-rated walls, horizontal assemblies, and smoke barriers, with and without penetrating items.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

PART 2 - PRODUCTS

2.1 SLEEVES

A. Wall Sleeves:

1. Steel Pipe Sleeves: ASTM A 53/A 53M, Type E, Grade B, Schedule 40, zinc coated, plain ends.
2. Cast-Iron Pipe Sleeves: Cast or fabricated "wall pipe," equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop unless otherwise indicated.

B. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies: Galvanized-steel sheet; 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint, with tabs for screw-fastening the sleeve to the board.

C. PVC-Pipe Sleeves: ASTM D 1785, Schedule 40.

D. Molded-PVC Sleeves: With nailing flange for attaching to wooden forms.

E. Molded-PE or -PP Sleeves: Removable, tapered-cup shaped, and smooth outer surface with nailing flange for attaching to wooden forms.

F. Sleeves for Rectangular Openings:

1. Material: Galvanized sheet steel.
2. Minimum Metal Thickness:

a. For sleeve cross-section rectangle perimeter less than 50 inches and with no side larger than 16 inches, thickness shall be 0.052 inch.
b. For sleeve cross-section rectangle perimeter 50 inches or more and one or more sides larger than 16 inches, thickness shall be 0.138 inch.

2.2 SLEEVE-SEAL SYSTEMS

A. Description: Modular sealing device, designed for field assembly, to fill annular space between sleeve and raceway or cable.

1. Sealing Elements: EPDM rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
2. Pressure Plates: Stainless steel.
3. Connecting Bolts and Nuts: Stainless steel of length required to secure pressure plates to sealing elements.

2.3 SLEEVE-SEAL FITTINGS

A. Description: Manufactured plastic, sleeve-type, waterstop assembly made for embedding in concrete slab or wall. Units shall have plastic or rubber waterstop collar with center opening to match piping OD.

2.4 GROUT

A. Description: Nonshrink; recommended for interior and exterior sealing openings in non-fire-rated walls or floors.

B. Standard: ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.

C. Design Mix: 5000-psi, 28-day compressive strength.

D. Packaging: Premixed and factory packaged.

2.5 SILICONE SEALANTS

A. Silicone Sealants: Single-component, silicone-based, neutral-curing elastomeric sealants of grade indicated below.

1. Grade: Pourable (self-leveling) formulation for openings in floors and other horizontal surfaces that are not fire rated.

B. Silicone Foams: Multicomponent, silicone-based liquid elastomers that, when mixed, expand and cure in place to produce a flexible, non-shrinking foam.

PART 3 - EXECUTION

3.1 SLEEVE INSTALLATION FOR NON-FIRE-RATED ELECTRICAL PENETRATIONS

A. Comply with NECA 1.

B. Comply with NEMA VE 2 for cable tray and cable penetrations.

C. Sleeves for Conduits Penetrating Above-Grade Non-Fire-Rated Concrete and Masonry-Unit Floors and Walls:

1. Interior Penetrations of Non-Fire-Rated Walls and Floors:

a. Seal annular space between sleeve and raceway or cable, using joint sealant appropriate for size, depth, and location of joint. Comply with requirements in Section 079200 "Joint Sealants."
b. Seal space outside of sleeves with mortar or grout. Pack sealing material solidly between sleeve and wall so no voids remain. Tool exposed surfaces smooth; protect material while curing.

2. Use pipe sleeves unless penetration arrangement requires rectangular sleeved opening.
3. Size pipe sleeves to provide 1/4-inch annular clear space between sleeve and raceway or cable unless sleeve seal is to be installed.
4. Install sleeves for wall penetrations unless core-drilled holes or formed openings are used. Install sleeves during erection of walls. Cut sleeves to length for mounting flush with both surfaces of walls. Deburr after cutting.
5. Install sleeves for floor penetrations. Extend sleeves installed in floors 2 inches above finished floor level. Install sleeves during erection of floors.

D. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies:

1. Use circular metal sleeves unless penetration arrangement requires rectangular sleeved opening.
2. Seal space outside of sleeves with approved joint compound for gypsum board assemblies.

E. Roof-Penetration Sleeves: Seal penetration of individual raceways and cables with flexible boot-type flashing units applied in coordination with roofing work.

F. Aboveground, Exterior-Wall Penetrations: Seal penetrations using steel pipe sleeves and mechanical sleeve seals. Select sleeve size to allow for 1-inch annular clear space between pipe and sleeve for installing mechanical sleeve seals.

G. Underground, Exterior-Wall and Floor Penetrations: Install cast-iron pipe sleeves. Size sleeves to allow for 1-inch annular clear space between raceway or cable and sleeve for installing sleeve-seal system.

3.2 SLEEVE-SEAL-SYSTEM INSTALLATION

A. Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at raceway entries into building.

B. Install type and number of sealing elements recommended by manufacturer for raceway or cable material and size. Position raceway or cable in center of sleeve. Assemble mechanical sleeve seals and install in annular space between raceway or cable and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.

3.3 SLEEVE-SEAL-FITTING INSTALLATION

A. Install sleeve-seal fittings in new walls and slabs as they are constructed.

B. Assemble fitting components of length to be flush with both surfaces of concrete slabs and walls. Position waterstop flange to be centered in concrete slab or wall.

C. Secure nailing flanges to concrete forms.

D. Using grout, seal the space around outside of sleeve-seal fittings.

END OF SECTION 260544